Principles of Animal life 1

Final term start

Electron Transport Chain (I)

Electron Transport Chain (I)

Contents

1. Electron Transport chain
2. Location of Electron Transport Chain
3. Purpose of Electron Transport Chain
4. Components of Electron Transport Chain

Electron Transport Chain (I)

1. Electron Transport Chain

- It is an organized set of electron carriers and enzymes with enhanced oxidation reduction reactions which extract the energy of electrons.

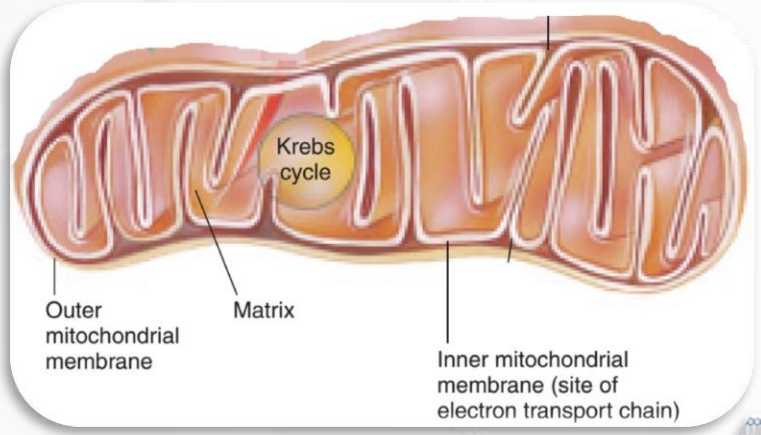
Electron Transport Chain (I)

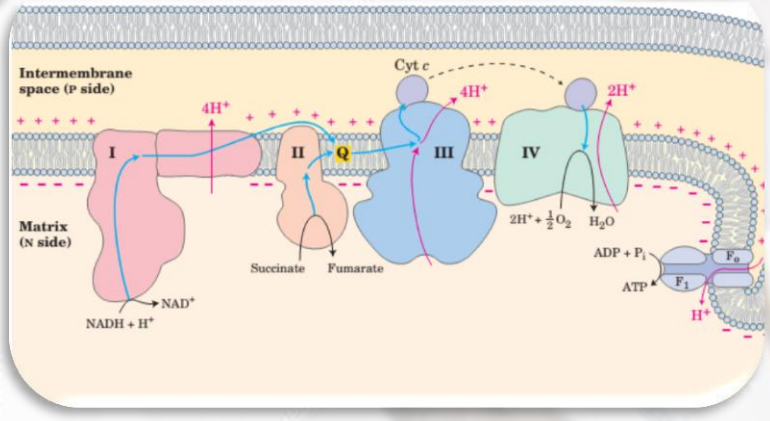
2. Location of Electron Transport Chain

- Inner mitochondrial membrane

Electron Transport Chain (I)

2. Location of Electron Transport Chain





Electron Transport Chain (I)

3. Purpose of Electron Transport Chain

- Accept electrons from NADH and $FADH_2$
- Pass electrons through redox center
- Take the energy of electrons
- Move H^+ across inner mitochondrial membrane

Electron Transport Chain (I)

4. Components of Electron Transport Chain

- NADH dehydrogenase
- Flavoproteins: FAD, FMN
- Ubiquinone: CoE Q
- Cytochromes
- Fe-S proteins

end

Principles of Animal life 1

Electron Transport Chain (II)

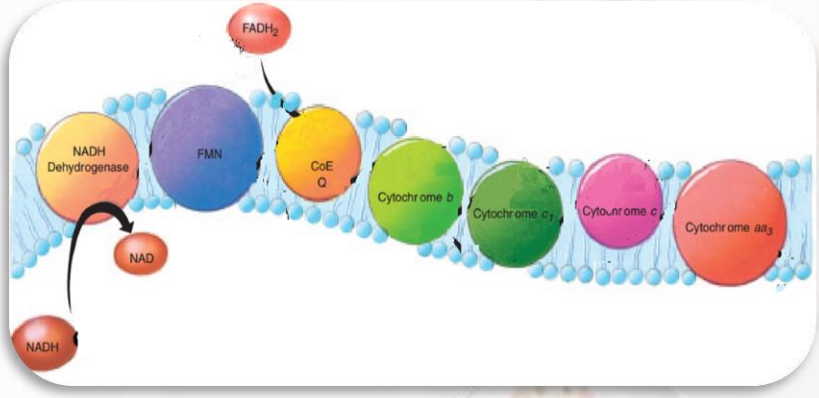
Electron Transport Chain (II)

Contents

1. Arrangement of Electron Carriers
2. Complexes of Electron Transport Chain
3. Movement of High Energy Electrons
4. Movement of Hydrogen Ions Through Electron Transport Chain

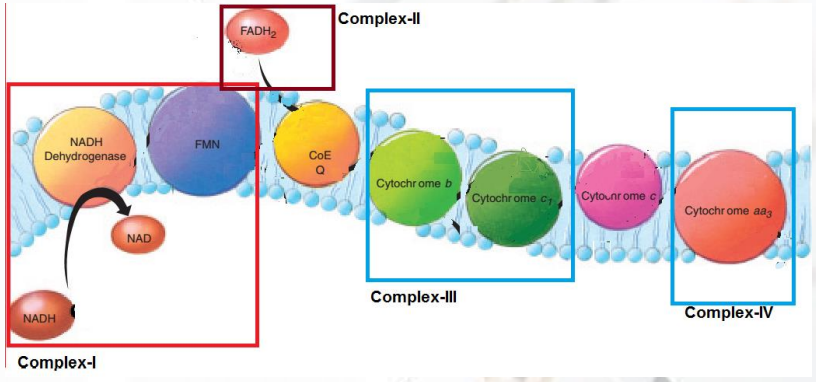
Electron Transport Chain (II)

1. Arrangement of Electron Carriers



Electron Transport Chain (II)

2. Complexes of Electron Transport Chain

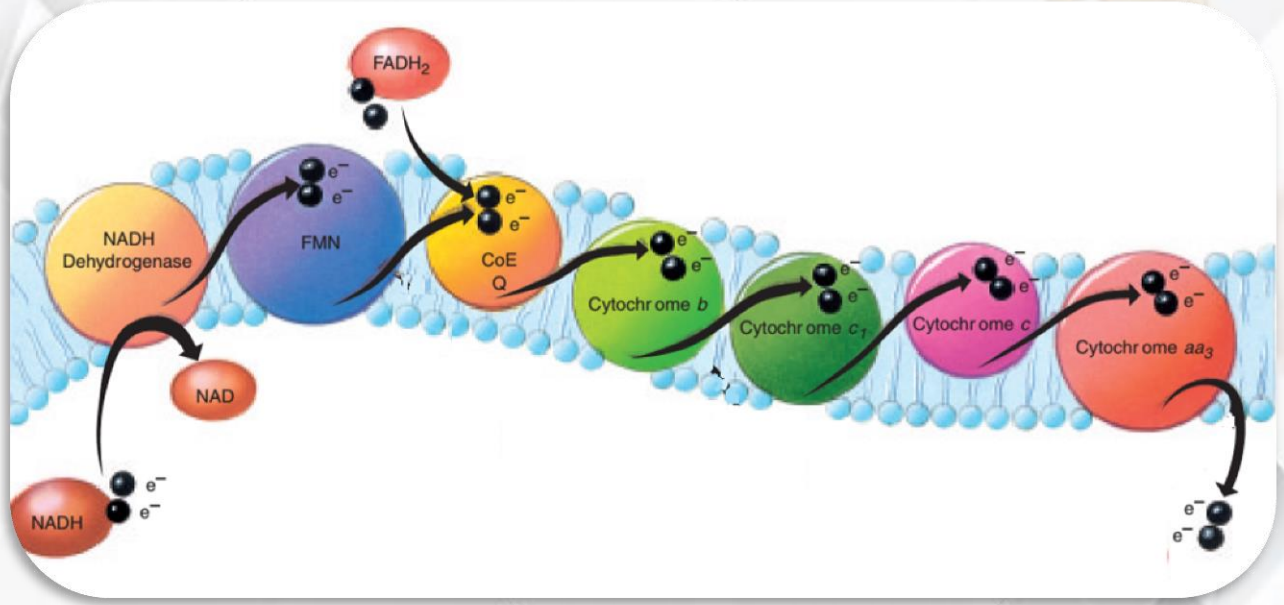


Electron Transport Chain (II)

3. Movement of High Energy Electrons

Electron Transport Chain (II)

3. Movement of High Energy Electrons



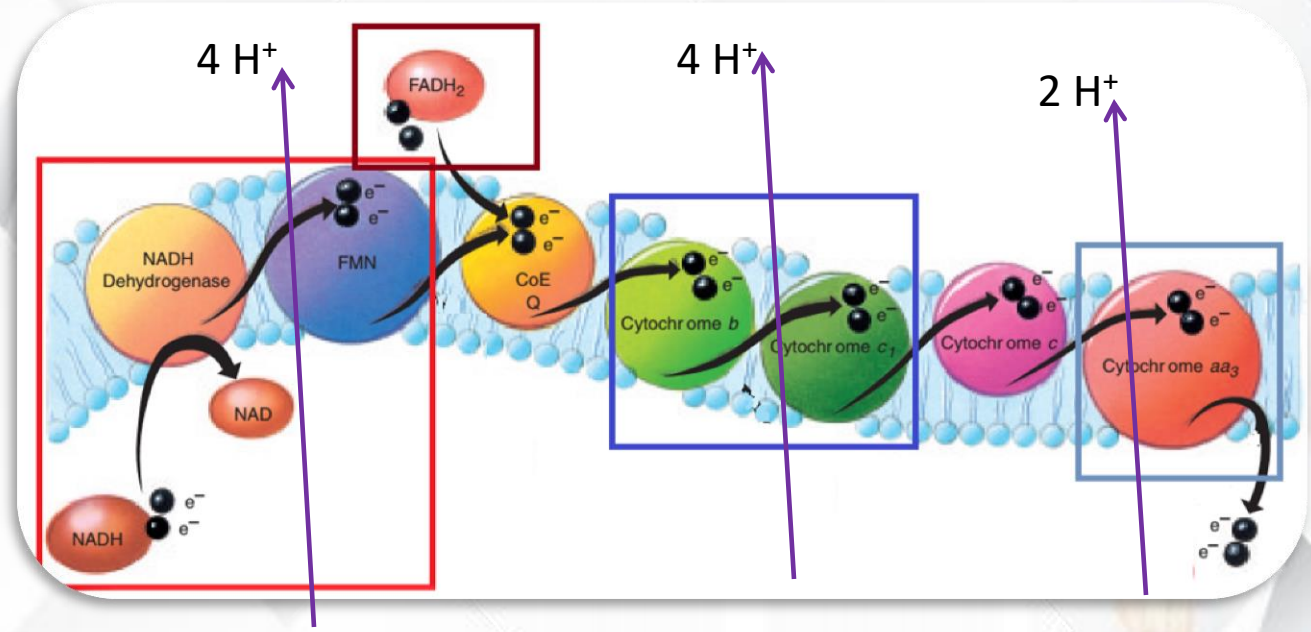
Ref. Miller and Harley. Zoology 4th Edition. p. 75

Electron Transport Chain (II)

4. Transport of H^+

Electron Transport Chain (II)

4. Transport of H^+



Principles of Animal life 1

ATP Synthesis

ATP Synthesis

Contents

1. H^+ Ion Gradient
2. ATP Synthetase
3. Mechanism of ATP Synthesis
4. Fate of e^- and H^+

ATP Synthesis

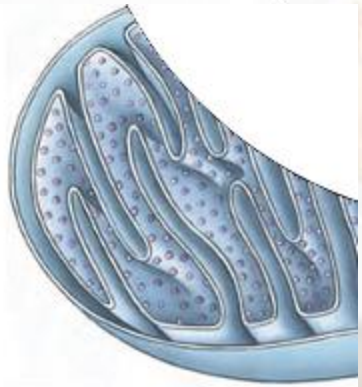
1. H^+ Gradient

- Electron transport chain pumps H^+ from mitochondria to inter membrane space
- H^+ concentration is higher in inter membrane space than mitochondrial matrix

ATP Synthesis

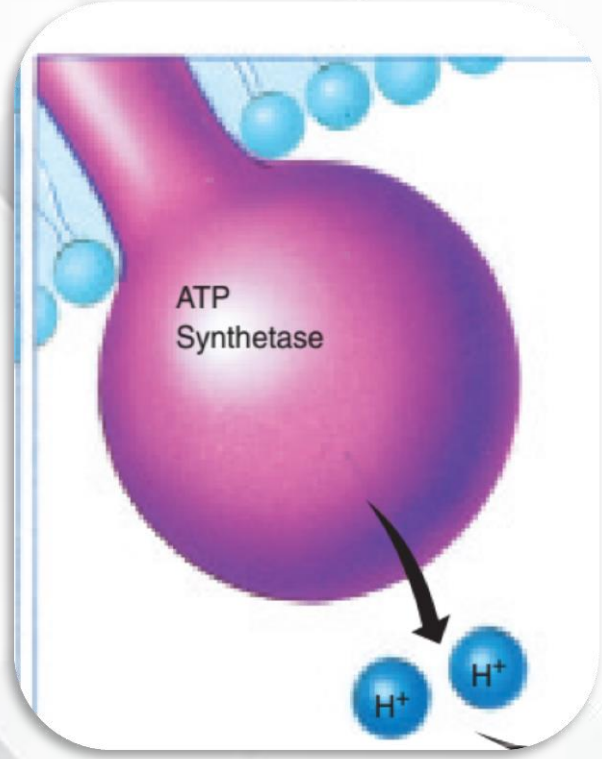
2. ATP Synthetase

- Inner mitochondrial membrane is selectively permeable
- H^+ can come back to matrix through a membrane protein "ATP Synthetase"

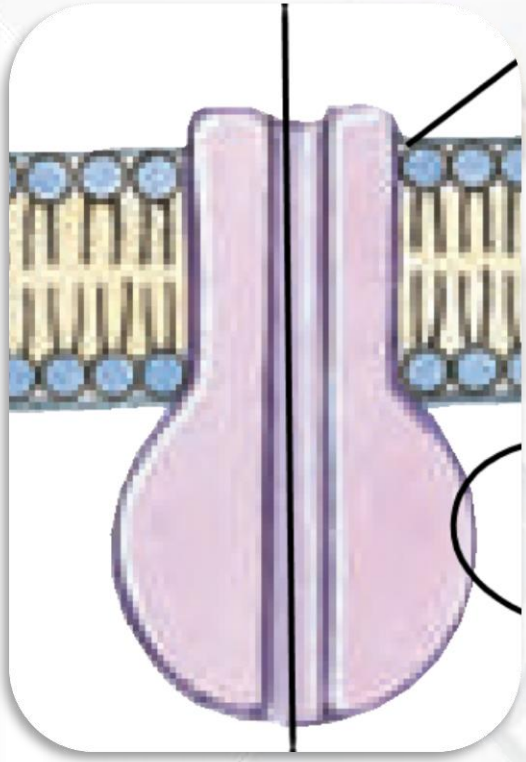


ATP Synthesis

2. ATP Synthetase



Ref. Miller and Harley. Zoology. 4th Edition. p. 75



Ref. Hickman. Integrated Principles of Zoology. 11th Edition. p. 68

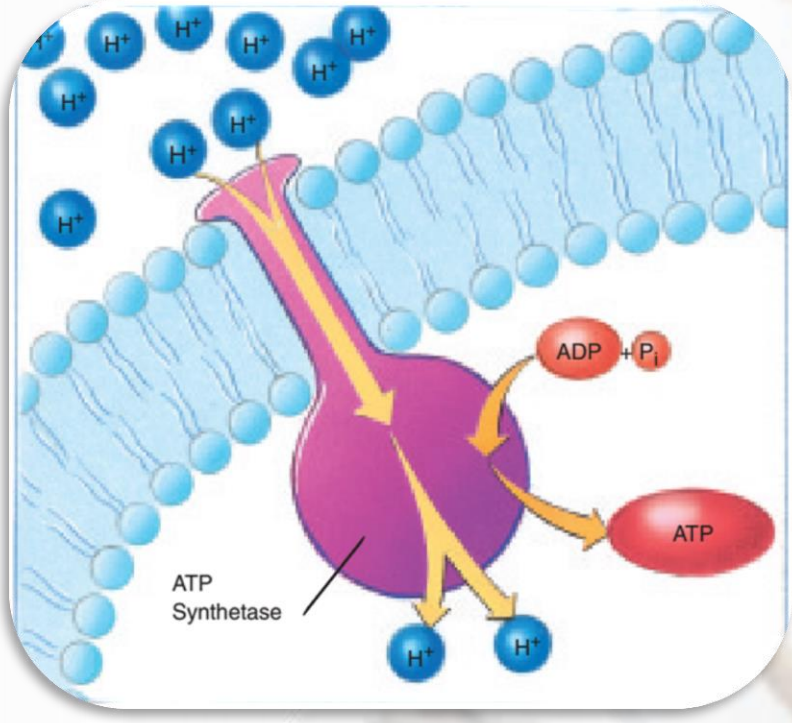
ATP Synthesis

3. Mechanism of ATP Synthesis

- H^+ come back to matrix through ATP synthetase
- Potential energy of H^+ is used for synthesis of ATP

ATP Synthesis

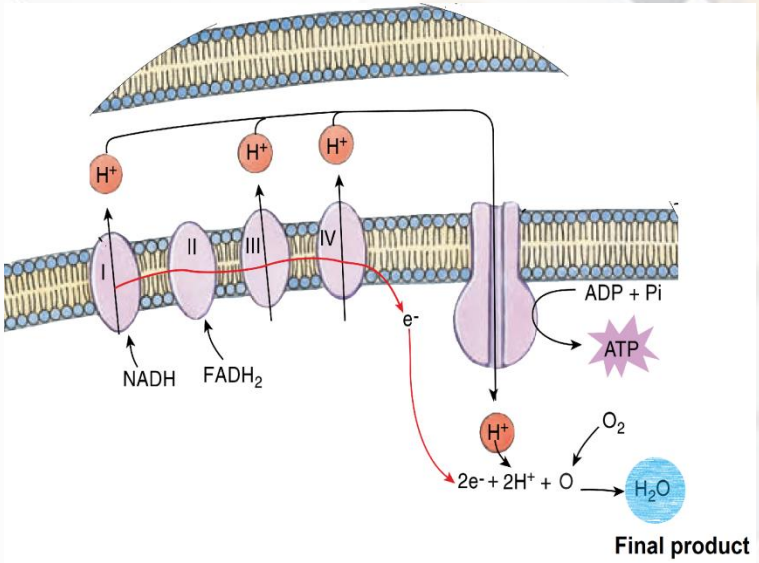
4. Mechanism of ATP Synthesis



Ref. Miller and Harley. Zoology. 4th Edition. p.75

ATP Synthesis

3. Fate of e^- and H^+



Ref. Hickman. Integrated Principles of Zoology. 11th Edition. p.68

end

Principles of Animal life 1

Energy Yield in Oxidation of Glucose

Energy Yield in Oxidation of Glucose

Contents

1. ATP Yield During Anaerobic Respiration
2. ATP Yield During Aerobic Respiration

Energy Yield in Oxidation of Glucose

1. ATP Yield During Anaerobic Respiration

- In the absence of oxygen
- Glucose is not completely oxidized
- Anaerobic respiration= Glycolysis+ fermentation
- Energy comes from glycolysis only

Energy Yield in Oxidation of Glucose

1. ATP Yield During Anaerobic Respiration

TABLE 32.1 APPROXIMATE ATP YIELD FROM THE COMPLETE ANAEROBIC OXIDATION OF ONE MOLECULE OF GLUCOSE

PROCESS	ATP PRODUCED	NADH, $FADH_2$ PRODUCED	ATP USED
Glycolysis	4 ATP	2 NADH	−2 ATP

Ref. Miller and Harley. Zoology. 4th Edition. p. 77

Energy Yield in Oxidation of Glucose

2. ATP Yield During Aerobic Respiration

- In the presence of oxygen
- Glucose is completely oxidized into CO_2 and H_2O
- Aerobic respiration= Glycolysis+ Krebs cycle+ electron transport chain

Energy Yield in Oxidation of Glucose

Process	ATP Produced	NADH, $FADH_2$ Produced	ATP Used
Glycolysis	4 ATP	2 NADH	−2 ATP
Entrance of NADH to Krebs cycle		2 NADH	−2 ATP
Krebs cycle	2 ATP	6 NADH, 2 $FADH_2$	
Electron transport chain:			
3 ATP for each of 2 NADH generated via glycolysis	6 ATP		
3 ATP for each of 8 NADH generated via Krebs cycle and entrance to the cycle	24 ATP		
2 ATP for each of 2 $FADH_2$ generated via Krebs cycle	4 ATP		
Total	40 ATP		−4 ATP
Net ATP	36 ATP		

2. ATP Yield During Aerobic Respiration

Ref. Miller and Harley. Zoology. 4th Edition. p. 77

Principles of Animal life 1

Metabolism of Fats (I)

Metabolism of Fats (I)

Contents

1. Importance of Fats
2. Digestion of Fats
3. Storage of Fats
4. Metabolism of Glycerol

Metabolism of Fats (I)

1. Importance of Fats

- Important part of our diet
- Structure: membranes
- Important source of energy to our body
- More than double amount of energy as compared to carbohydrates and proteins

Metabolism of Fats (I)

1. Importance of Fats

- 30% recommended
- Liver, heart and skeletal muscles meet half of energy requirement from fats
- Sole source of energy in hibernating animals
- Fats are actually triacylglycerides

Metabolism of Fats (I)

2. Digestion of fats

- Digestion occur in small intestine
- Bile…..Emulsification
- Lipase…Digested completed
- Absorb in small intestine and converted into triglycerides again

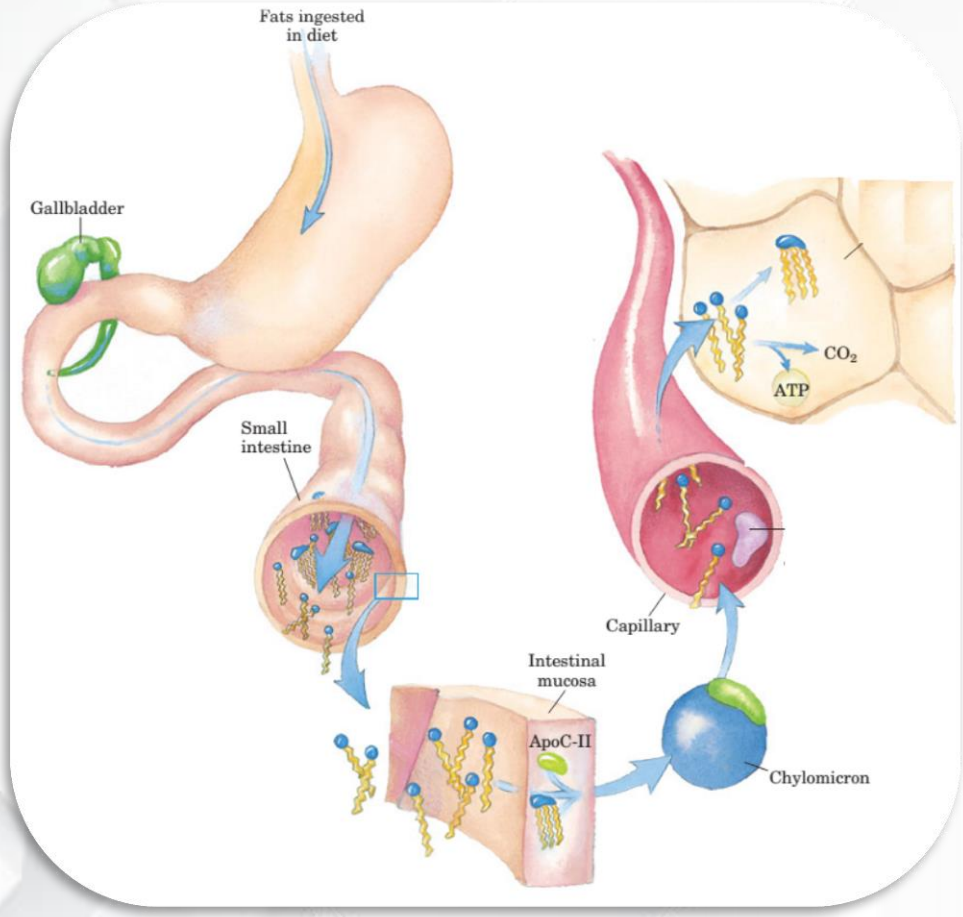
Metabolism of Fats (I)

3. Storage of Fats

- **Chylomicrons:** Fats are transported in body after packing with cholesterol and proteins to form chylomicron.
- Stored in adipose tissues and skeletal muscles

Metabolism of Fats (I)

3. Storage of Fats



Ref. Nelson and Cox. Lehninger Principles of Biochemistry. 5th Edition. p.648

Metabolism of Fats (I)

4. Metabolism of Glycerol

- When required fats are hydrolyzed by enzymes into glycerol and fatty acids
- Glycerol is converted to glyceraldehyde-3-phosphate
- Enters into glycolysis

end

Principles of Animal life 1

Metabolism of Fats (II)

Metabolism of Fats (II)

Contents

1. Metabolism of Fatty Acid
2. Why Fats Give More Energy?

Metabolism of Fats (II)

1. Metabolism of Fatty Acids

- In cell fatty acids oxidation mainly occurs in:
  - Mitochondria
- Some amount:
  - Endoplasmic reticulum
  - Peroxisome

Metabolism of Fats (II)

1. Metabolism of Fatty Acids

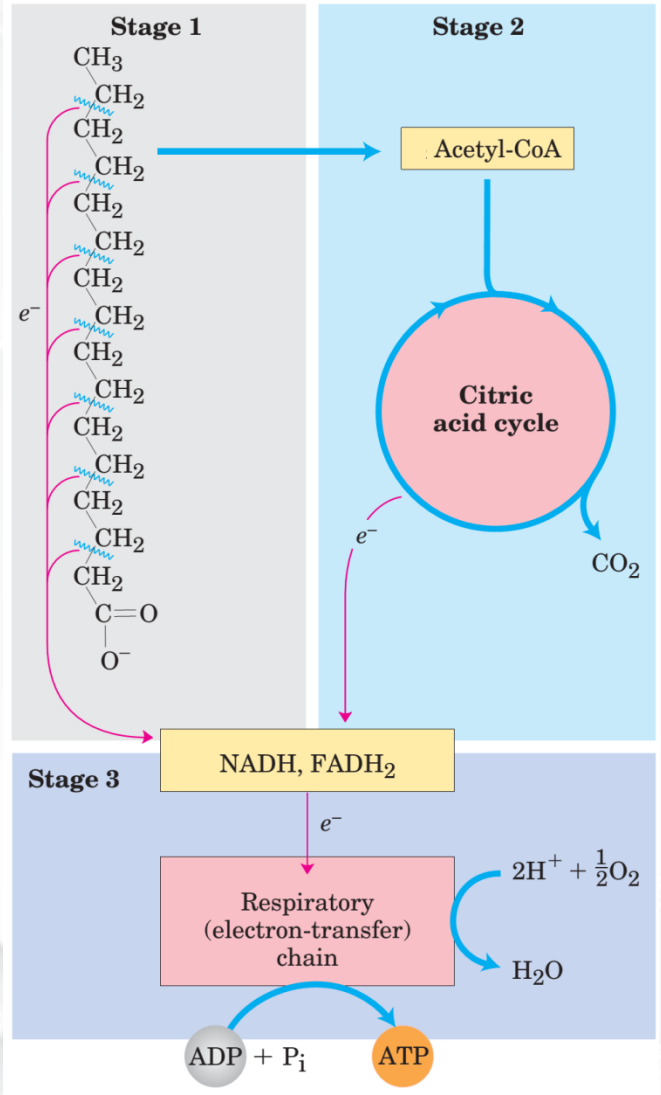
- Breakdown of fatty acids yield:
  - Acetyl CoA
  - NADH and $FADH_2$
- Acetyl CoA joins Krebs cycle

Metabolism of Fats (II)

1. Metabolism of Fatty Acids

- Acetyl CoA in Krebs cycle forms NADH and $FADH_2$
- NADH and $FADH_2$ enter into electron transport chain for ATP production

Metabolism of Fats (II)



1. Metabolism of Fatty Acids

Ref. Nelson and Cox. Lehninger Principles of Biochemistry. 5th Edition. p.653

Metabolism of Fats (II)

2. Why Fats Give More Energy?

- 1g of fats give 2.5 times more energy than 1g of carbohydrates and protein.
- This is because they contain much higher amount of C-H bonds

end

Principles of Animal life 1

Metabolism of Proteins (I)

Metabolism of Proteins (I)

Contents

1. Importance of Proteins
2. Digestion of Proteins
3. Amino Acids to Proteins
4. Breakdown of Amino Acids

Metabolism of Proteins (I)

1. Importance of proteins

- Muscle Fibers
- Bones
- Enzymes
- Hormones
- Antibodies
- Transporters
- Lens proteins of eyes
- Hair, nail, feather
- Spider web

Metabolism of Proteins (I)

2. Digestion of Proteins

- In Stomach and small intestine
- Proteins are finally broken down to amino acids
- In small intestine amino acids are absorbed

Metabolism of Proteins (I)

3. Amino Acids….. Protein

- Amino acids reach to cells through blood circulation.
- In cells amino acids are used for protein synthesis

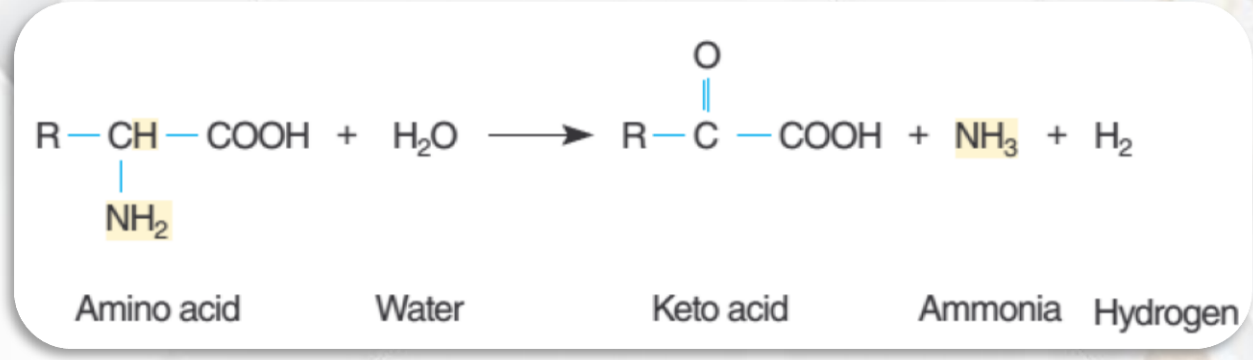
Metabolism of Proteins (I)

4. Break down of Amino Acids

- Amino acids pool
- Excessive amino acids are degraded mainly in liver and skeletal muscles
- Also during starvation
- Amino acid converted to Keto acid after deamination

Metabolism of Proteins (I)

4. Break down of Amino Acids



Principles of Animal life 1

Metabolism of Proteins (II)

Metabolism of Proteins (II)

Contents

1. Energy from Proteins
2. Excretion of Nitrogenous Wastes
3. Overview of Catabolism of Food Molecules

Metabolism of Proteins (II)

1. Energy from Proteins

- Keto acids are funneled to Krebs cycle to yield ATP
- 1g of proteins give approximately 4kcal energy
- Approximately equal to glucose

Metabolism of Proteins (II)

2. Excretion of Nitrogenous Wastes

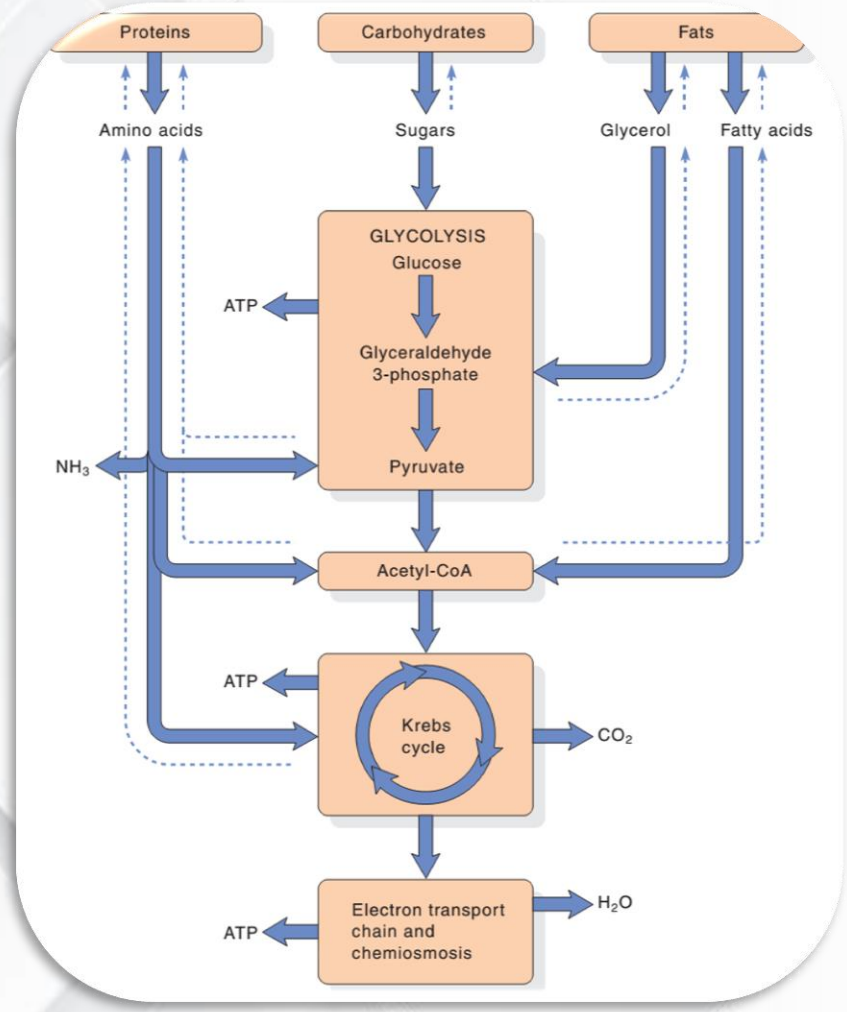
- Ammonia is formed during deamination of Amino acid
- In aquatic organisms ammonia is released from moist surfaces
- For release of ammonia large amount of water is required

Metabolism of Proteins (II)

2. Excretion of Nitrogenous Wastes

- In land animals ammonia is converted to urea
- Urea excreted through urinary system
- Moderate amount of water for urea
- Animals with acute shortage of water, release uric acid.

Metabolism of Proteins (II)



2. Overview of Catabolism of Food Molecules

Miller and Harley. Zoology. 4th Edition. p.76

Principles of Animal life 1

Control of Metabolism (I)

Control of Metabolism (I)

Contents

1. Cells are Very Efficient
2. Feed Back Inhibition
3. Control of Catabolism
4. Control of Anabolism

Control of Metabolism (I)

1. Cells are Very Efficient

- Cells do not waste energy
- If some molecule in excess, its synthesis is switched off

Control of Metabolism (I)

2. Feedback Inhibition

- The most common method for control of metabolism
- In this method the product inhibits enzyme that catalyzes the key step.
- e.g., ATP inhibits phosphofructokinase in glycolysis

Control of Metabolism (I)

3. Control of Catabolism

- If a cell is working hard its ATP is decreases
- Decrease in ATP start anaerobic respiration
- Anaerobic respiration slows when ATP is sufficient

Control of Metabolism (I)

3. Control of Anabolism

- In anabolism cells expends energy for biomolecules synthesis
- Anabolism is mainly controlled by regulating the enzyme activity at strategic points

end

Principles of Animal life 1

Control of Metabolism (II)

Control of Metabolism (II)

Contents

1. Example of control
2. Effect of ATP and ADP
3. Effect of Citric Acid
4. Control of Cellular Respiration: Flow Chart

Control of Metabolism (II)

1. Example of Control

- Phosphofructokinase
- By controlling the activity of this enzyme cell regulates its entire metabolic process
- Receptor sites for ATP, ADP, AMP, Citric acids

Control of Metabolism (II)

2. Effect of ATP, ADP and AMP

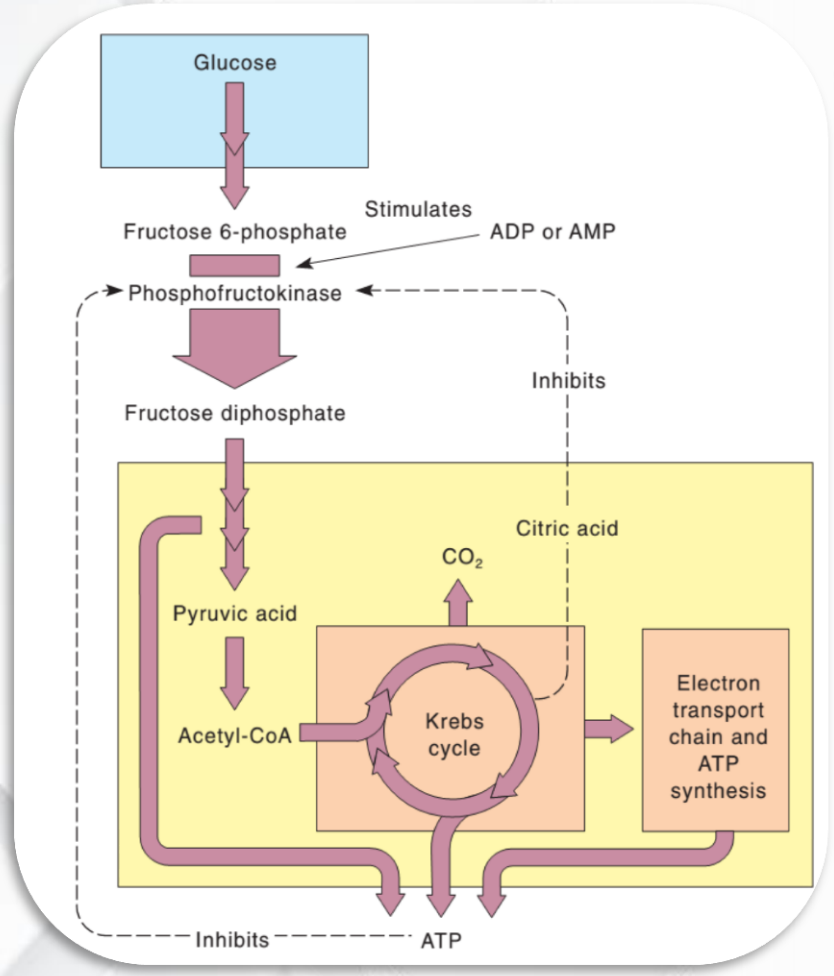
- When high ATP level
- Enzyme inhibited; resulting in inhibition of glycolysis
- If high ADP or AMP level
- Enzyme activated glycolysis starts

Control of Metabolism (II)

3. Effect of Citric Acid

- Citric acids produced in Krebs cycle
- citric acid in cytoplasm inhibits phosphofructokinase
- At low concentration of citric acid glycolysis accelerates

Control of Metabolism (II)



4. Control of Cellular Respiration: Flow Chart

Ref. Miller and Harley. Zoology. 4th Edition. p.77

Principles of Animal life 1

Metabolic Pool

Metabolic Pool

Contents

1. Metabolic Pool
2. Importance of Metabolic Pool
3. Open Systems

Metabolic Pool

1. Metabolic pool

- All the substances of metabolisms constitute metabolic pool

- It is actually reservoir of molecules upon which enzymes act

Metabolic Pool

2. Importance of Metabolic Pool

- All catabolic and anabolic reaction in cell use metabolic pool.
- Substrates
- Products
- Reaction intermediates

Metabolic Pool

3. Open Systems

- In which compound may enter or leave the biochemical pathway at various steps
- Almost all the biochemical pathways are open system
- Metabolic intermediates of Glycolysis and Krebs cycle may withdrawn or put into by other metabolic pathways

end

eproduction and Its Importance

Contents

1. Defining Reproduction
2. Why There is Need for Reproduction?
3. Reductive Potential
4. Types of Reproduction

1. Defining Reproduction

- The process of producing new individuals.
- It is a the basic feature of all forms of life

2. Why There is Need for Reproduction?

- With the passage of time animals get aged, new animals should take their place
- It is essential for existence of species
- Reproduction is essential for continuity of life on earth

3. Reproductive Potential

- Capability to reproduce.

- All the animals have great reproductive potential
- if resource are unlimited a single species of fishes may fill the all oceans.

4. Types of Reproduction

- Animals reproduce by two basic means

- Asexual reproduction
- Sexual reproduction

Principles of Animal life 1

Asexual Reproduction

Asexual Reproduction

Contents

1. Defining Asexual Reproduction
2. Identical Organisms
3. Method of early life on earth
4. Animal with Asexual Reproduction

Asexual Reproduction

1. Defining ASexual Reproduction

- It is the type of reproduction in which offspring are produced without fertilization of gametes.

Asexual Reproduction

2. Identical Organisms

- Basic method of cell division is mitosis
- No genetic recombination
- So the animals produced are identical to their parents and siblings

Asexual Reproduction

3. Method of Early Life on Earth

- Early organisms reproduced by asexual method
- This was the only method of reproduction in first 2 billion years of life
- During this life evolved slowly because of less genetic diversity

Asexual Reproduction

4. Animals With Asexual Reproduction

- It is common among:
- **Protozoa;** Amoeba, paramecium, Euglena
- **Lower invertebrates;** hydra, jellyfish, flatworms
- Asexual reproduction is correlated with ability to regenerate

Asexual Reproduction

5. Types of Asexual Reproduction in Animals

1) Fission
2) Budding
3) Fragmentation
4) Parthenogenesis

end

Principles of Animal life 1

Methods of Asexual Reproduction: Fission

Asexual Reproduction Methods: Fission

Contents

1. Fission
2. Types of Fission
3. Binary Fission
4. Multiple fission

Asexual Reproduction Methods: Fission

1. Fission

- L. *fissio, the act of splitting*
- It is type of asexual reproduction in which one organisms splits into two or more organisms
- The organisms divides by inward furrowing

Asexual Reproduction Methods: Fission

2. Types of Fission

- There are two basic types of fission
- Binary fission
- Multiple fission

Asexual Reproduction Methods: Fission

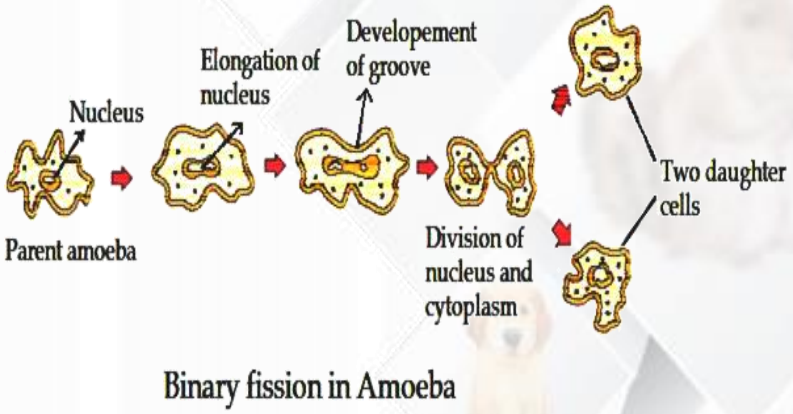
3. Binary Fission

- It is the process of fission in which one organism divided into two organisms.
- Both are approximately equal size
- Binary fission may be:
- Asymmetrical
- Transverse
- Longitudinal

Asexual Reproduction Methods: Fission

3. Binary Fission

- **Asymmetrical binary fission**



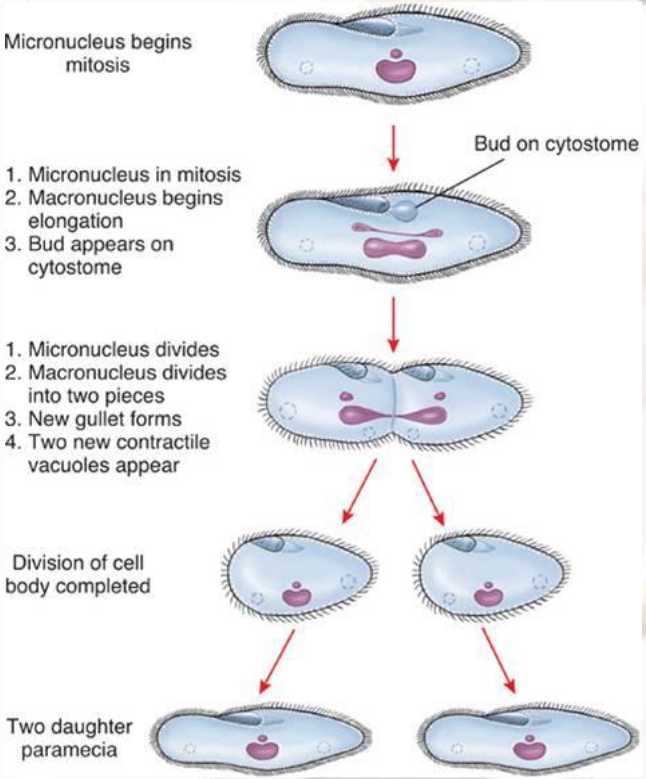
Binary fission in Amoeba

Ref. www.zigya.com

Asexual Reproduction Methods: Fission

3. Binary Fission

- Transverse fission



Ref. rowdysites.msudenver.edu

Asexual Reproduction Methods: Fission

3. Binary Fission

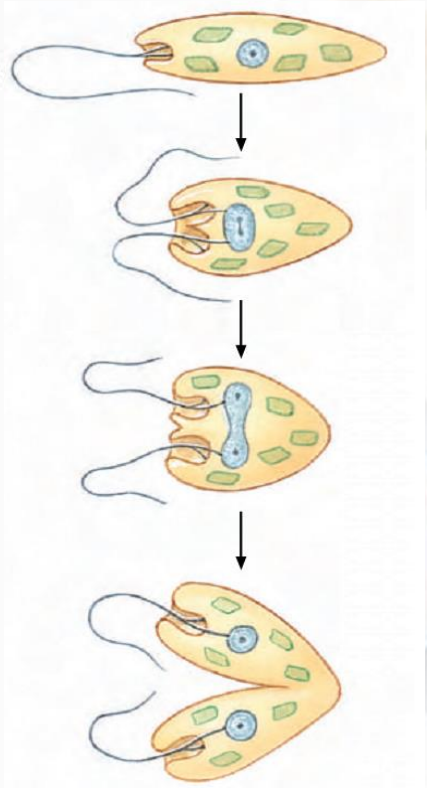
- Longitudinal fission

Miller and Harley. Zoology. 5th Edition. p. 476

Asexual Reproduction Methods: Fission

3. Binary Fission

- Longitudinal fission



Ref. Hickman. Integrated Principles of Zoology. 11th Edition. p.136

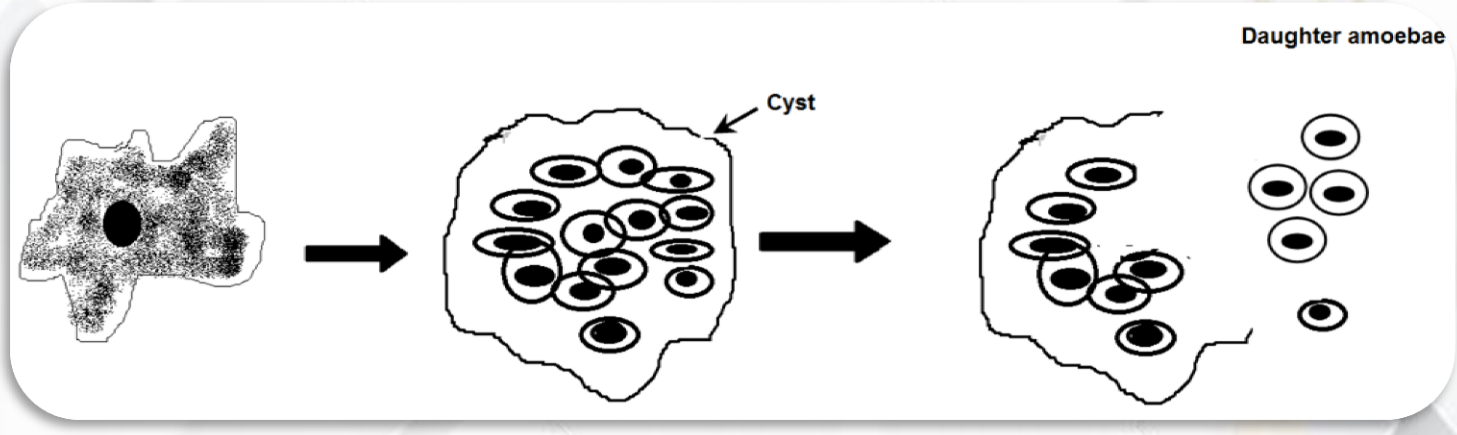
Asexual Reproduction Methods: Fission

4. Multiple Fission

- It is a number of successive ~~cell~~ divisions of nucleus followed by cytoplasm in which one cell divides into many cells
- E.g., amoeba, tapeworm

Asexual Reproduction Methods: Fission

4. Multiple Fission



Principles of Animal life 1

Methods of Asexual Reproduction: Budding

Asexual Reproduction Methods: Budding

Contents

1. Budding
2. Colony
3. Gemmule

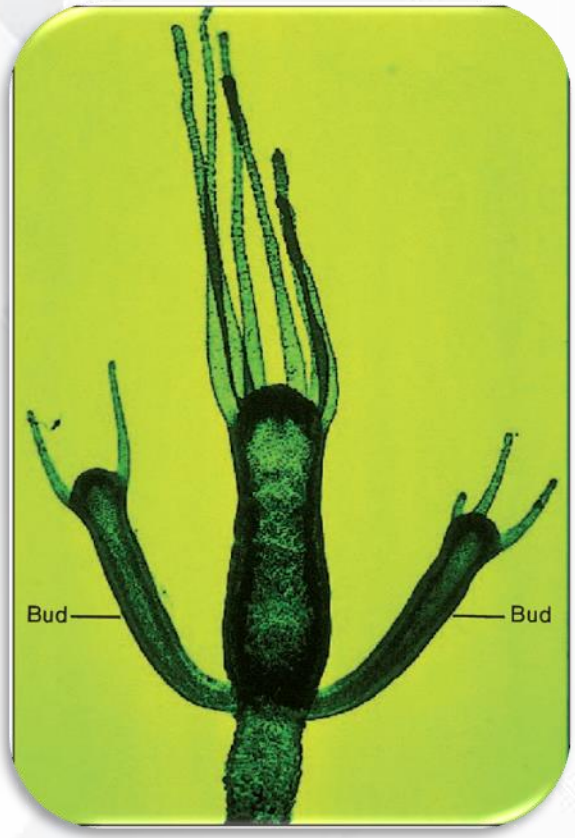
Asexual Reproduction Methods: Budding

1. Budding

- Asexual reproduction in which an outgrowth appears on the body of animal which grows into complete organism

- Hydra, Obelia, sponges

Asexual Reproduction Methods: Budding



Ref. Miller and Harley. Zoology. 5th Edition. p. 476

1. Budding

- A group of cells rapidly divides and form a small outgrowth called bud.
- This bud increases rapidly and form a complete organism

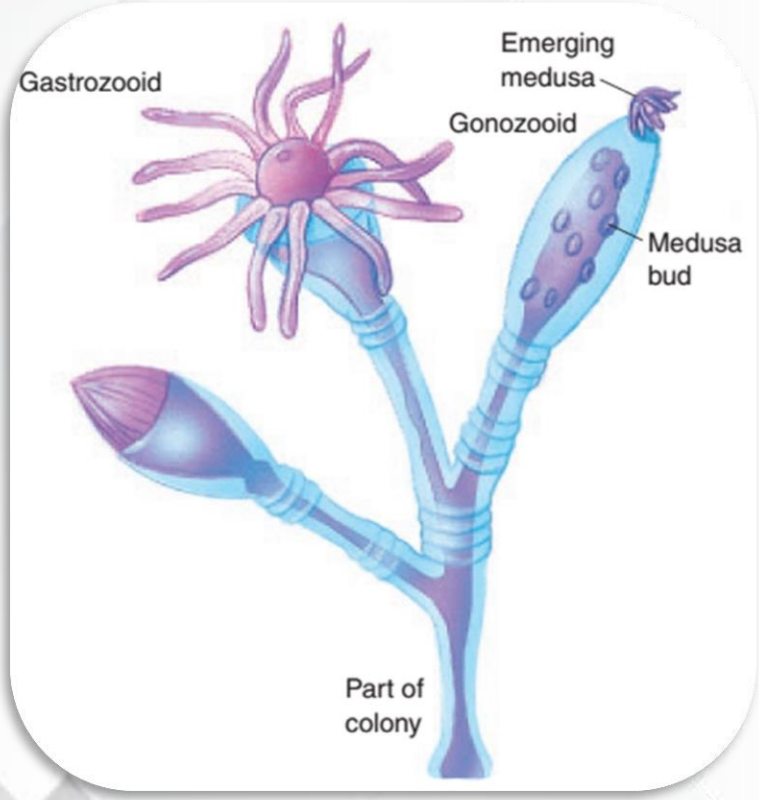
Asexual Reproduction Methods: Budding

2. Colony

- Sometimes the buds remain attached with the parental organism and form a closely associated group of individuals called a colony
- Members of a colony belong to same species

Asexual Reproduction Methods: Budding

2. Colony



Ref. Miller and Harley. Zoology. 5th Edition. p.678

Asexual Reproduction Methods: Budding

3. Gemmules

- In sponges internal buds are produces
- The internal buds are called as gemmules
- Gemmule is a collection of cells surrounded by body wall
- When parent die each gemmule gives rise to a new individual.

end

Principles of Animal life 1

Methods of Asexual Reproduction: Fragmentation

Asexual Reproduction Methods: Fragmentation

Contents

1. Fragmentation
2. Fragmentation in Different Groups of Animals

Asexual Reproduction Methods: Fragmentation

1. Fragmentation

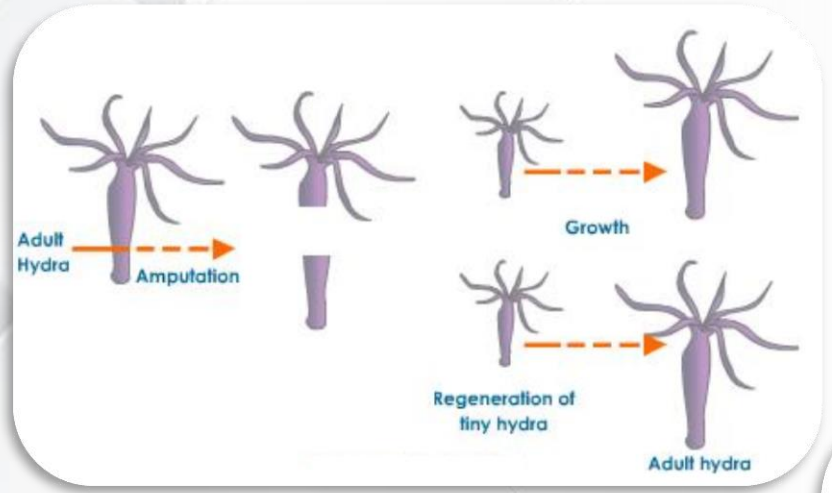
- It is the process of asexual reproduction in which an animal breaks into two or more pieces and each piece is capable of growing into a new organism

Asexual Reproduction Methods: Fragmentation

2. Fragmentation in Different Groups of Animals

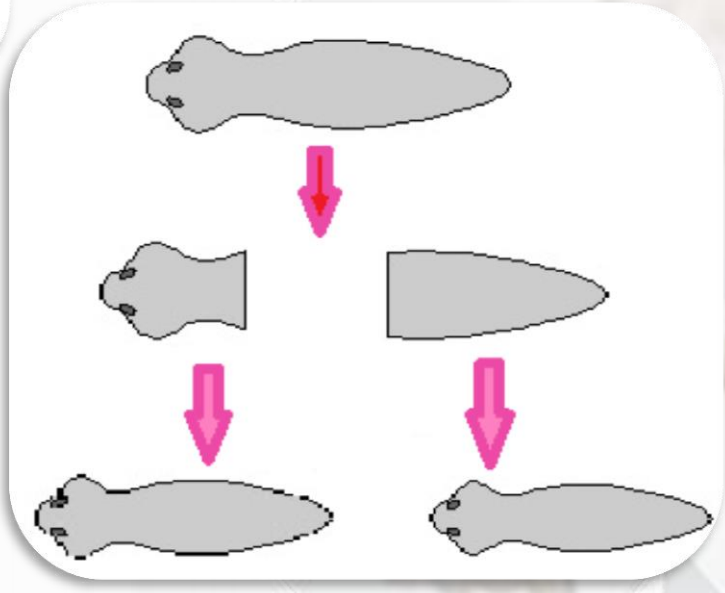
- Fragmentation is common in different groups of animals including cnidarians, Platyhelminthes and echinoderms

Asexual Reproduction Methods: Fragmentation



Ref. thescienceinfo.com

2. Fragmentation in Different Groups of Animals



Principles of Animal life 1

Methods of Asexual Reproduction: Parthenogenesis (I)

Asexual Methods: Parthenogenesis (I)

Contents

1. Parthenogenesis
2. Patterns of parthenogenesis
3. Ameiotic parthenogenesis
4. Meiotic parthenogenesis

Asexual Methods: Parthenogenesis (I)

1. Parthenogenesis

- The process of production of an organism from an unfertilized egg

Asexual Methods: Parthenogenesis (I)

2. Patterns of Parthenogenesis

- There are many patterns of parthenogenesis.
- Two most common:
- Meiotic parthenogenesis
- Ameiotic parthenogenesis

Asexual Methods: Parthenogenesis (I)

3. Ameiotic Parthenogenesis

- In this eggs are diploid
- They are produced without meiosis.
- Young ones are the clones of their parents
- This method is common in flatworm, rotifers, crustaceans and insects

Asexual Methods: Parthenogenesis (I)

4. Meiotic Parthenogenesis

- Eggs are produced by the meiosis process
- Haploid egg starts development
- Diploid number restored by chromosomal duplication
- In some insects remain haploid, no chromosomal duplication

Asexual Methods: Parthenogenesis (I)

4. Meiotic Parthenogenesis

- Egg may be activated by sperm to start parthenogenesis.
- Here sperm will not fertilize egg, just activates
- In other egg activated spontaneously

end

Principles of Animal life 1

Methods of Asexual Reproduction: Parthenogenesis (II)

Asexual Methods: Parthenogenesis (II)

Contents

1. Advantages of Parthenogenesis
2. Disadvantages of parthenogenesis
3. Parthenogenesis in insects
4. Prthenogenesis in reptiles

Asexual Methods: Parthenogenesis (II)

1. Advantages of Parthenogenesis

- When population consists of only females
- When environment is very stable
- Social organization of colonies of bees, wasps, ants (for males)

Asexual Methods: Parthenogenesis (II)

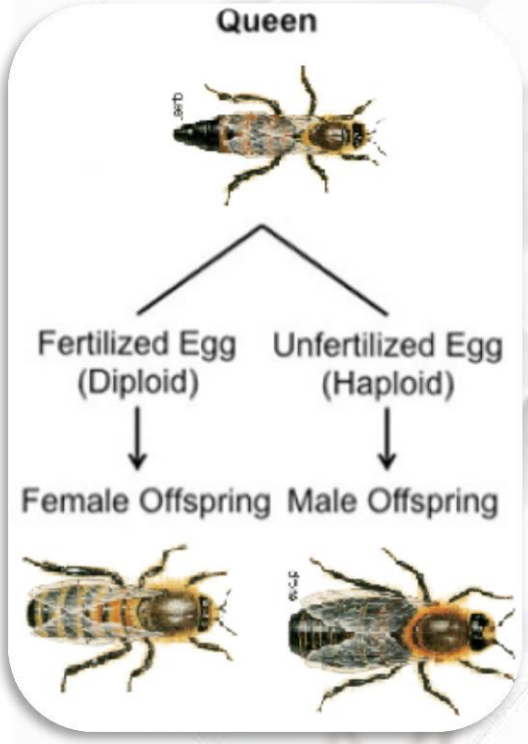
2. Disadvantages of Parthenogenesis

- Less genetic variability
- Little chances for organisms to shift gene combination in changing environment

Asexual Methods: Parthenogenesis (II)

3. Parthenogenesis in Insects

- Very common in wasps bees and ants



Asexual Methods: Parthenogenesis (II)

4. Parthenogenesis in Reptile

- Parthenogenesis have been observed in some lizards and snakes
- New Mexico Whiptail lizard are exclusively females

Asexual Methods: Parthenogenesis (II)

4. Parthenogenesis in Reptile

Ref. Reptilefact.com

Principles of Animal life 1

Advantages of Asexual Reproduction

Advantages of Asexual Reproduction

Contents

1. Species Survival
2. Minimal Energy Requirement
3. Positive Genetic Combination Passed on to Next Generation
4. Rapid increase in population

Advantages of Asexual Reproduction

1. Species Survival

- Asexual reproduction is specially the beneficial to those species have one sex only.

Advantages of Asexual Reproduction

2. Minimal Energy Requirements

- Only one sex is involved so less amount of energy required

Advantages of Asexual Reproduction

3. Positive Genetic Combinations Passed on to Next Generation

- Because there is no genetic recombination, the good traits acquired of an organism are transmitted to next generation

Advantages of Asexual Reproduction

4. Rapid Increase in Population

- Under stable conditions animals reproduce asexually causing a rapid increase in population

end

Principles of Animal life 1

Advantages of Asexual Reproduction

Disadvantages of Asexual Reproduction

Contents

1. Low Genetic Variability
2. Susceptibility Against Diseases
3. All Mutations Pass on to Next Generation

Disadvantages of Asexual Reproduction

1. Low Genetic Variability

- No genetic recombination
- Very little variations
- Population is genetically same

Disadvantages of Asexual Reproduction

2. Susceptibility Against Diseases

- As population is genetically same, so equally susceptible to disease or changing environmental conditions
- This may result in the death of whole population at once.

Disadvantages of Asexual Reproduction

3. All Mutations Pass on to Next Generation

- Most of the mutations are detrimental
- In sexual reproduction only germ cell mutation are passed on
- In Asexual reproduction all mutations are passed on to next generation

end

Principles of Animal life 1

Sexual Reproduction

Sexual Reproduction

Contents

1. Defining Sexual Reproduction
2. Two Sexes Involved
3. Types of Gametes
4. Production of gametes

Sexual Reproduction

1. Defining Sexual Reproduction

- It the types of reproduction in which new offspring are produced by the fusion of gametes.

Sexual Reproduction

2. Two Sexes are Involved

- Normally two sexes are involved:
- Male and female
- One set of chromosomes from male other from female
- New combinations formed

Sexual Reproduction

3. Types of Gametes

- Two types of gametes
- Male gametes: sperms
- Female gametes: eggs

- Both are haploid and produced by the process of meiosis

Sexual Reproduction

4. Production of Gametes

- Gametes are produced in gonads
- Male gonads: testes
- Female gonads: ovaries
- Sperms produced in testes: **Spermatogenesis**
- Eggs produced in ovaries: **Oogenesis**

end

Principles of Animal life 1

External and Internal Fertilization

External and Internal Fertilization

Contents

1. External Fertilization
2. Internal Fertilization
3. Spermatophores
4. Number of Gametes Produced

External and Internal Fertilization

1. External Fertilization

- In which fusion of gametes occurs outside the body of animals.
- Normally in water
- Broadcast spawning
- Fertilization is external, development is also external

External and Internal Fertilization

2. Internal Fertilization

- In which fusion of gametes occurs inside the body of animals.
- In mating process male introduces sperms into female body, sperms are motile they approach to female gametes

External and Internal Fertilization

3. Spermatophores

- Many sperms are packed into a ball like structure called spermatophore
- These spermatophores are introduced into female reproductive tract
- Sperms are protected in spermatophore
- Minimal loss of sperms

External and Internal Fertilization

4. Number of Gametes Produced

- Species with external fertilization produced large number of gametes
- Species with internal fertilization produce relatively few gametes

end

Principles of Animal life 1

Hermaphroditism

Hermaphroditism

Contents

1. Hermaphroditism
2. Monoecious
3. Fertilization in Hermaphrodites
4. Sequential Hermaphrodites
5. Protogynous
6. Protandrous

Hermaphroditism

1. Hermaphroditism

- It is the condition in which both male and female sex organs are present in the same animal.
- **Hermaphrodites**
- Beneficial to sessile animals

Hermaphroditism

2. Monoecious

- Gr. *Monos,* single and *oikos,* house
- One room for two sexes
- Means both sex organs are present in one animal
- Hermaphrodites are also called monoecious

Hermaphroditism

3. Fertilization in Hermaphrodites

- Some undergo self fertilization
- But mostly cross fertilization
- Two animals come close to each other, both donate and receive sperms

Hermaphroditism

4. Sequential Hermaphrodites

- When animal is one sex in first phase of life and opposite sex in other phase of life.
- Means animal is male in first phase of life and later becomes female, vice versa

Hermaphroditism

5. Protogyneous

- Those animal which are female in first half of life.
- In 2nd phase of life they are male

Hermaphroditism

6. Protandrous

- Those animals which are male in first half of life.
- In 2nd phase of life they are female

end

Principles of Animal life 1

Hermaphroditism: Examples

Hermaphroditism: Examples

Contents

1. Earthworm
2. Mangrove killifish
3. Hamlets
4. Sea Slugs
5. Clownfish
6. Wrasses

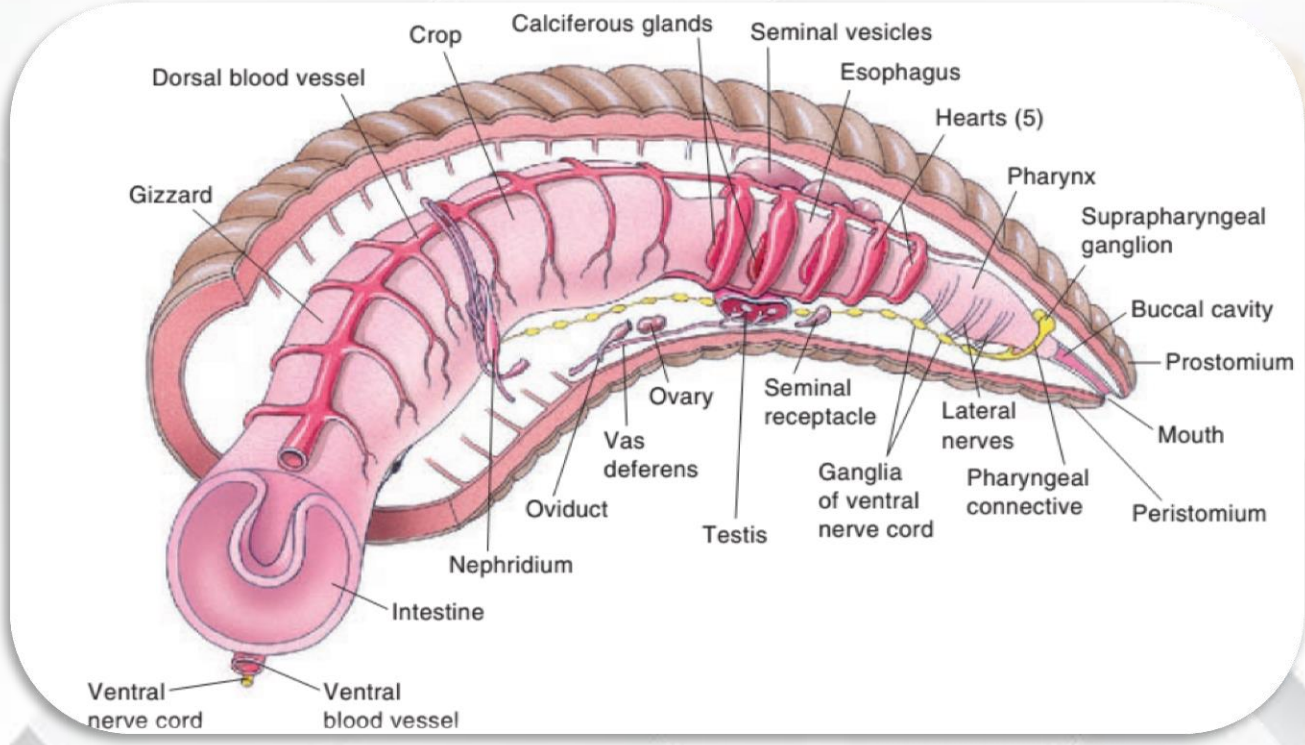
Hermaphroditism: Examples

1. Earthworm

- Earthworms and all oligochaetes are monoecious
- 2 pairs of testes one pair of ovaries
- During copulation two worms line up and facing opposite direction and exchange gametes

Hermaphroditism: Examples

1. Earthworm



Ref. Miller and Harley. Zoology. 5th Edition. p. 200

Hermaphroditism: Examples

2. Mangrove Killifish

- It lives along the costs of North, South and Central America
- They routinely reproduce by self fertilization

Berbel-Filho *et al.,* 2016

Hermaphroditism: Examples

3. Hamlet

- It is a fish mainly found in coral reefs of Crabbean sea and Gulf of Mexico
- They do not practice self fertilization

Ref. wikipedia.org

Hermaphroditism: Examples

4. Clone Fish

- Colorful reef fish found in symbiosis with sea anemone
- They are protandrous
- The large female and one reproductive male
- When female dies reproductive male turns to female

Hermaphroditism: Examples

4. Clone Fish

Ref. seahorseaquariums.com

Hermaphroditism: Examples

5. Wrasses

- Group of coral fishes in which protogyny is common

Ref. tfhmagazine.com

end

Principles of Animal life 1

Bisexual Reproduction

Bisexual Reproduction

Contents

1. Defining Bisexual Reproduction
2. Dioecious
3. Bisexual Reproduction is More Common
4. Sexual dimorphism

Bisexual Reproduction

1. Bisexual Reproduction

- It is the type of reproduction in which two separate individuals are involved
- Male and female

Bisexual Reproduction

2. Dioecious

- Gr. *Di,* two and *oikos*, room
- Two separate rooms for two sexes

Bisexual Reproduction

3. Bisexual Reproduction is More Common

- Most of the animals have bisexual reproduction method
- One benefit of this is that the offspring have new gene combination and new genotypes appear

Bisexual Reproduction

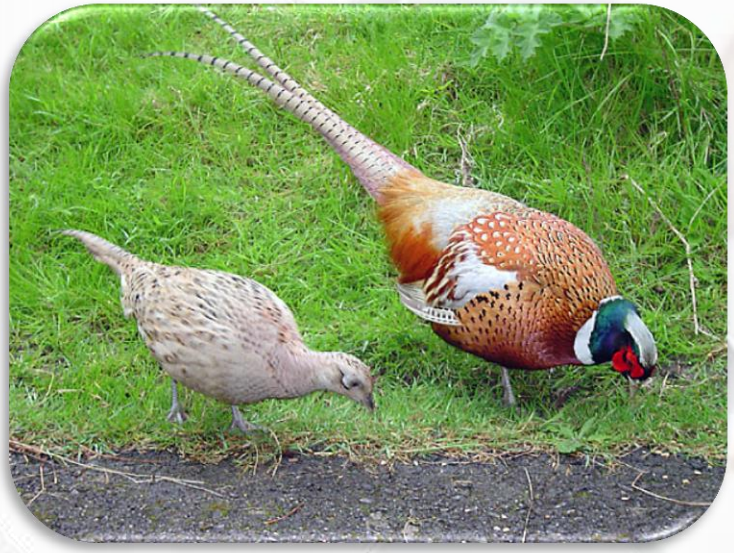
4. Sexual Dimorphism

- It is the condition in which both male and female are phenotypically different
- The difference is mainly in
  - Secondary sex characters
  - Size, colour, weight

Bisexual Reproduction

4. Sexual Dimorphism

Mandarin Ducks
Ref. Wikipedia.org



Common Pheasant
Ref. Wikipedia.org

Principles of Animal life 1

Advantages and Disadvantages of Sexual Reproduction

Sexual Reproduction :Advantages and Disadvantages

Contents

1. Advantages
    a) Genetic Variability
    b) Foundation for Evolution
    c) Elimination of Deleterious Mutation

Sexual Reproduction :Advantages and Disadvantages

Contents

2. Disadvantages
    a) Asexual Animals
    b) Adaptive Gene Combination may Dismantle
    c) Wastage of energy

Sexual Reproduction :Advantages and Disadvantages

1. Advantages

a) Genetic Variability

- Genetic recombination
- New combination of traits
- This may help in changing environment or in case of disease

Sexual Reproduction :Advantages and Disadvantages

1. Advantages

b) Foundation for Evolution

- Genetic variability is there
- Variation is the foundation of evolution

Sexual Reproduction :Advantages and Disadvantages

1. Advantages

c) Elimination of Deleterious Traits

- As there is genetic recombination
- Deleterious mutation are eliminated

Sexual Reproduction :Advantages and Disadvantages

2. Disadvantages

a) Asexual animals

- Not feasible for animal which rely on asexual means only
- Especially population with one sex only

Sexual Reproduction :Advantages and Disadvantages

2. Disadvantages

b) Adaptive Gene Combination Dismantle

- In sexual reproduction new genetic combination arise
- This will upset the already established adaptive gene combination

Sexual Reproduction :Advantages and Disadvantages

2. Disadvantages

c) Wastage of energy

- Many of the gametes are not fertilized
- This will cause a significant loss of energy

end

Principles of Animal life 1

Reproductive Strategies in Amphibians

Reproductive Strategies in Amphibians

Contents

1. Fertilization
2. Eggs
3. Breeding
4. Courtship behaviour
5. Development

Reproductive Strategies in Amphibians

1. Fertilization

- Order Caudata
  - Mostly internal fertilization
- Order Gymophiona
  - Internal fertilization
- Order Anura
  - External fertilization

Reproductive Strategies in Amphibians

2. Eggs

- No hard covering
- **Order Caudata**
- Single, clumps, strings
- **Order Gymophiona**
- Retained within body, some lay eggs
- **Order Anura**
- Laid in water, moist soil, tree trunks

Reproductive Strategies in Amphibians

3. Breeding

- **Temperate region**
- Linked to temperature
- **Tropical region**
- Linked to rainy season

Reproductive Strategies in Amphibians

4. Courtship Behviour

- Helps to locate breeding sites
- Identify potential mates
- Prepare individuals for reproduction
- ~~U~~ Ensure egg fertilization
- Olfactory, Visual, Vocalization and tactile cues

Reproductive Strategies in Amphibians

5. Development

- **Order Caudata**
- Larva aquatic, adult terrestrial
- **Order Gymophiona**
- In oviduct with fetal teeth to feed from lining, others aquatic larvae
- **Order Anura**
- Aquatic larvae, herbivores

end

Principles of Animal life 1

Reproductive Strategies in Reptiles and Birds

Reproductive Strategies in Reptiles and Birds

Contents

1. Fertilization
2. Eggs
3. Breeding
4. Courtship behaviour
5. Development

Reproductive Strategies in Reptiles and Birds

1. Fertilization

- Reptiles and birds are dioecious
- Fertilization is internal
- Fertilization in the upper portion of oviduct
- sperms may be stored in seminal vesicles up to 4years in some turtle and 6years in some snakes

Reproductive Strategies in Reptiles and Birds

2. Eggs

- Soft to Hard shells
- Desiccation resistant
- Enough amount of yolk
- Amniotic eggs with three extraembryonic membranes
  - Amnion
  - Chorion
  - Allantois

Reproductive Strategies in Reptiles and Birds

3. Breeding

- **Reptiles:**
- Copulation in winter during hibernacula
- Fertilization in spring
- **Birds:**
- In spring

Reproductive Strategies in Reptiles and Birds

4. Courtship Behviour

- **Reptiles:**
- Complex courtship behaviour
- Male actively seek out female
- Courtship function for sex recognition
- head-bobbing males lizards
- Tactile in snakes

Reproductive Strategies in Reptiles and Birds

4. Courtship Behviour

- **Birds:**
- Territorial behaviour
- Attract females by sounds (wood pecker), dancing (cranes, peacocks)
- >90% monogamous
- Polygamy in species where young ones require less parental care

Reproductive Strategies in Reptiles and Birds

4. Courtship Behviour

Ref. Hickman. Principles of integrated Zoology. 11th Edition. p. 601

Reproductive Strategies in Reptiles and Birds

5. Development

- **Reptiles:**
- External,
- Fertilized eggs remain unattended
- In some cases eggs are cared for humidity
- In crocodile post hatching care of dependent young ones.

Reproductive Strategies in Reptiles and Birds

5. Development

- **Birds:**
- External,
- Eggs are incubated, protected from predators
- Extensive parental care of young ones

end

Principles of Animal life 1

Reproductive Strategies in Mammals

Reproductive Strategies in Mammals

Contents

1. Eggs
2. Fertilization
3. Courtship Behaviour
4. Reproductive Cycle
5. Development

Reproductive Strategies in Mammals

1. Eggs

- Amniotic eggs
- Little amount of yolk except egg laying mammals

Reproductive Strategies in Mammals

2. Fertilization

- Mammals are dioecious
- Internal fertilization
- Some bats have delayed fertilization
- Fertilization in the upper portion of oviduct

Reproductive Strategies in Mammals

3. Courtship Behviour

- Complex courtship behaviour
- Male actively seek out female
- Territorial behaviour
- Vocalization is very common

Reproductive Strategies in Mammals

4. Reproductive Cycle

- Most mammals have definite time for egg maturation
- Females undergo estrus cycle: female is behaviorally and reproductively receptive to males
- Monoestrus and polyestrus

Reproductive Strategies in Mammals

5. Development

- Mostly viviparous
- **Montremes:** oviparous
- **Marsupial mammals:** uterine milk, primitive placenta, short gestation period
- **Eutherian mammals:** embryo implanted deeply, placenta formation, gestation period 20 days to 19 months

end

Principles of Animal life 1

Examples of Reproductive Function in Fishes and Amphibians

Reproductive Function in Fishes and Amphibians

Contents

1. Bony fishes

2. Amphibians

Reproductive Function in Fishes and Amphibians

1. Bony fishes

- 1000's of eggs are fertilized
  - Fungal and bacterial infection
  - Siltation
  - Predation
- Yolk only for short time development

Reproductive Function in Fishes and Amphibians

1. Bony fishes

- Hatched young ones actively feed
- Achieve maturity in short time
- In some fishes embryos are cared properly
- Sea horse and pipefish males carry embryo in their ventral pouches

Reproductive Function in Fishes and Amphibians

2. Amphibians

- Mostly external fertilization in water in few internal
- Eggs are cared aeration, and infected eggs are removed from clutches
- *Pipa* female carry eggs on back
- Little amount of yolk

Reproductive Function in Fishes and Amphibians

2. Amphibians

- Development in much longer time than ~~amphibians~~ fishes
- Larval stage to complete development
- Metamorphosis
- *Rheobatrachus* females carry egg and tadpole in stomach

end

Principles of Animal life 1

Examples of Reproductive Function in Reptiles and Birds

Reproductive Function in Reptiles and Birds

Contents

1. Reptiles
2. Birds

Reproductive Function in Reptiles and Birds

1. Reptiles

- True terrestrial vertebrates
- Shelled eggs with extraembryonic membranes
- Chorion amnion… protect and provide aqueous environment
- Allantois… gaseous exchange, excretory products

Reproductive Function in Reptiles and Birds

1. Reptiles

- Large amount of yolk
- Complete development within shelled egg
- No larval stage
- Some snakes are ovovivparous

Reproductive Function in Reptiles and Birds

2. Birds

- Except water fowl lack penis
- Deposit semen against cloaca
- Egg shells thicker than reptiles
- Fertilization before shell formation

Reproductive Function in Reptiles and Birds

2. Birds

- Hard shelled eggs are incubated by parents
- Extensive parental care
- Parental care and feeding more common than amphibians, reptiles and mammals

Reproductive Function in Reptiles and Birds

2. Birds

Ref. Hickman. Principles of integrated Zoology. 11th Edition. p. 601

Principles of Animal life 1

Examples of Reproductive Function in Mammals

Reproductive Function in Mammals

Contents

1. Monotremes
2. Marsupial Mammals
3. Placental Mammals

Reproductive Function in Mammals

1. Monotremes

- Duckbilled platypus, spiny anteater
- Lay eggs
- Duckbilled platypus…lay eggs in its nest/burrows
- Spiny anteater… incubate eggs in ventral pouch

Reproductive Function in Mammals

1. Monotremes

Ref. Miller and Harley. Zoology. 5th Edition. p.335

Reproductive Function in Mammals

2. Marsupial Mammals

- Koala, kangaroo
- Gestation period: 8 to 40 days
- Suckling….. 60 to 120 days

Reproductive Function in Mammals

2. Marsupial Mammals

Ref. Miller and Harley. Zoology. 5th Edition. p.335

Reproductive Function in Mammals

3. Placental Mammals

- Most of the mammals
- Deer, lion, monkeys etc.
- Internal development
- Placenta formation
- Young ones nourished through mammary glands
-

Reproductive Function in Mammals

Olive Baboon

3. Placental Mammals

Ref. Hickman. Principles of integrated Zoology. 11th Edition. p. 601

Principles of Animal life 1

Reproductive Patterns in Animals

Reproductive Patterns in Animals

Contents

1. Reproductive patterns
2. Oviparous
3. Ovoviviparous
4. Viviparous

Reproductive Patterns in Animals

1. Reproductive Patterns

- The pattern of egg laying and development of young ones.
- In all invertebrates and vertebrates one of the three patterns.
- Oviparity
- Ovoviviparity
- Viviparity

Reproductive Patterns in Animals

2. Oviparous

- Which lay eggs and development outside the body of animals
- Fertilization may be internal or external
- Majority of invertebrates and many vertebrates are oviparous
- Parental care may be or may not

Reproductive Patterns in Animals

3. Ovoviparous

- Which eggs are retained into the body but the young one gets all its food from yolk of egg
- Fertilization is internal
- Development also internal
- **Invertebrates:** Annelids, brachiopods, gastropods
- **Vertebrates:** Fishes and reptiles

Reproductive Patterns in Animals

4. Viviparous

- Which eggs develops into the body and the young one gets its nourishment from mother
- Fertilization is internal
- Development also internal
- **Invertebrates:** Scorpions
- **Vertebrates:** most mammals and elasmobranch fishes, few amphibians and reptiles

end

Principles of Animal life 1

Invertebrate Reproductive system

Invertebrate Reproductive Systems

Contents

1. Invertebrates with no reproductive system
2. Male reproductive system
3. Female reproductive system
4. Generalized Structure
5. Reproductive Structures of Polychaetes

Invertebrate Reproductive Systems

1. Invertebrates with no reproductive organs

- Some invertebrates like polychaetes have no permanent reproductive organs
- Body cavity lining cells produce gametes
- Which are released coelomic or nephridial duct, some times body wall may rupture

Invertebrate Reproductive Systems

2. Male reproductive system

- Parts of male reproductive system
- Testes
- Sperm duct/vas deferens
- Seminal vesicle
- Ejaculatory duct
- Accessory glands

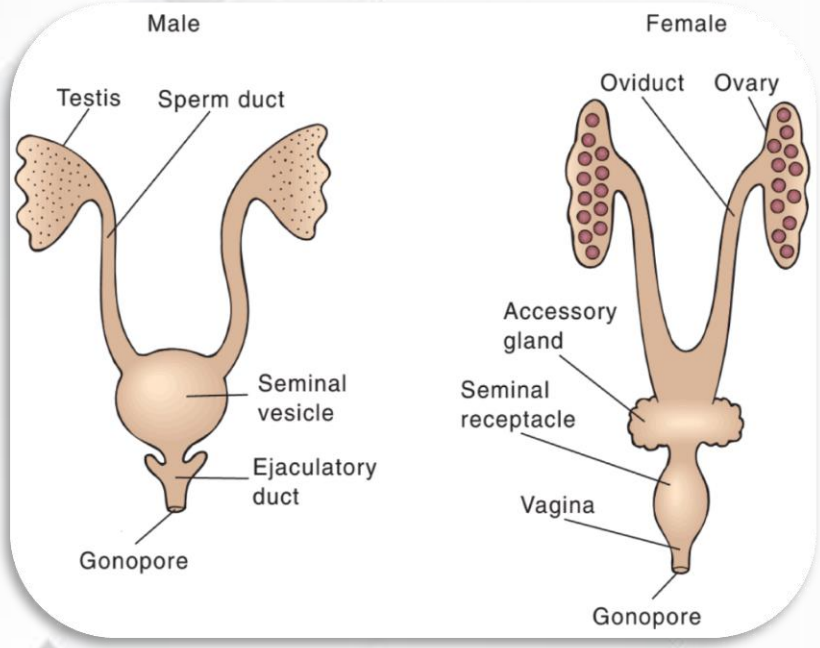
Invertebrate Reproductive Systems

3. Female reproductive system

- Parts of female reproductive system
- Ovaries
- Oviducts
- Seminal recepactacle
- Bursa
- Gonopore
- Oviposter

Invertebrate Reproductive Systems

4. Generalized Reproductive Structures

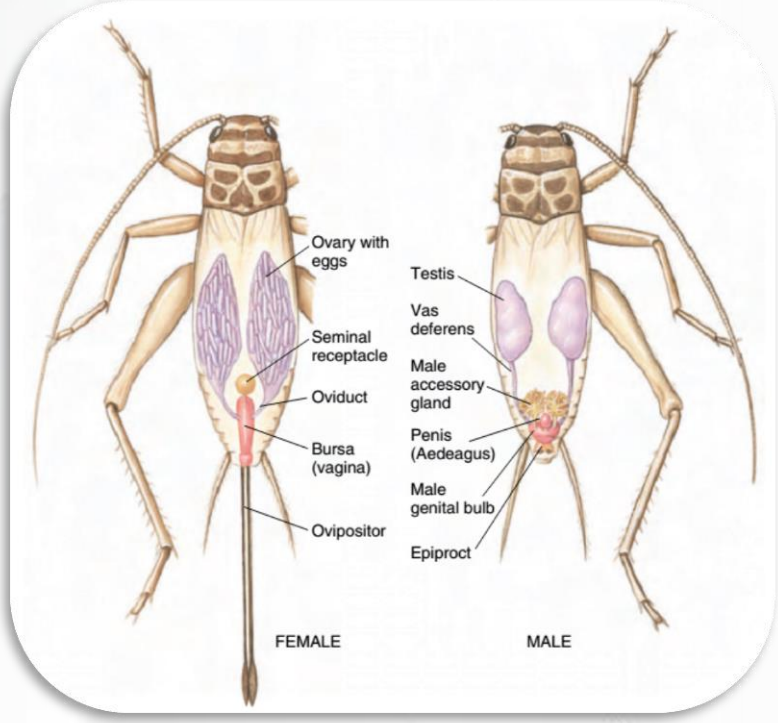


Generalized structure of reproductive systems in invertebrates

Ref. Miller and Harley. Zoology. 5th Edition. p.680

Invertebrate Reproductive Systems

5. Male and Female reproductive system of Cricket



Reproductive systems in crickets

Ref. Hickman. Integrated Principles of Zoology. 11th Edition. p.146

Principles of Animal life 1

Human Male Reproductive System

Human Male Reproductive System

Contents

1. Function of Male Reproductive System
2. Components of Human Male Reproductive System
3. Structure of Male Reproductive System

Human Male Reproductive System

1. Functions of Male Reproductive System

- Two basic function:
- Production of sperms
- Transfer of sperm to female reproductive tract

Human Male Reproductive System

2. Components of Male reproductive system

- Scrotum
- Testes
- Rete testis
- Epididymis
- Duct Deferens
- Ampulla
- Seminal Vesicle

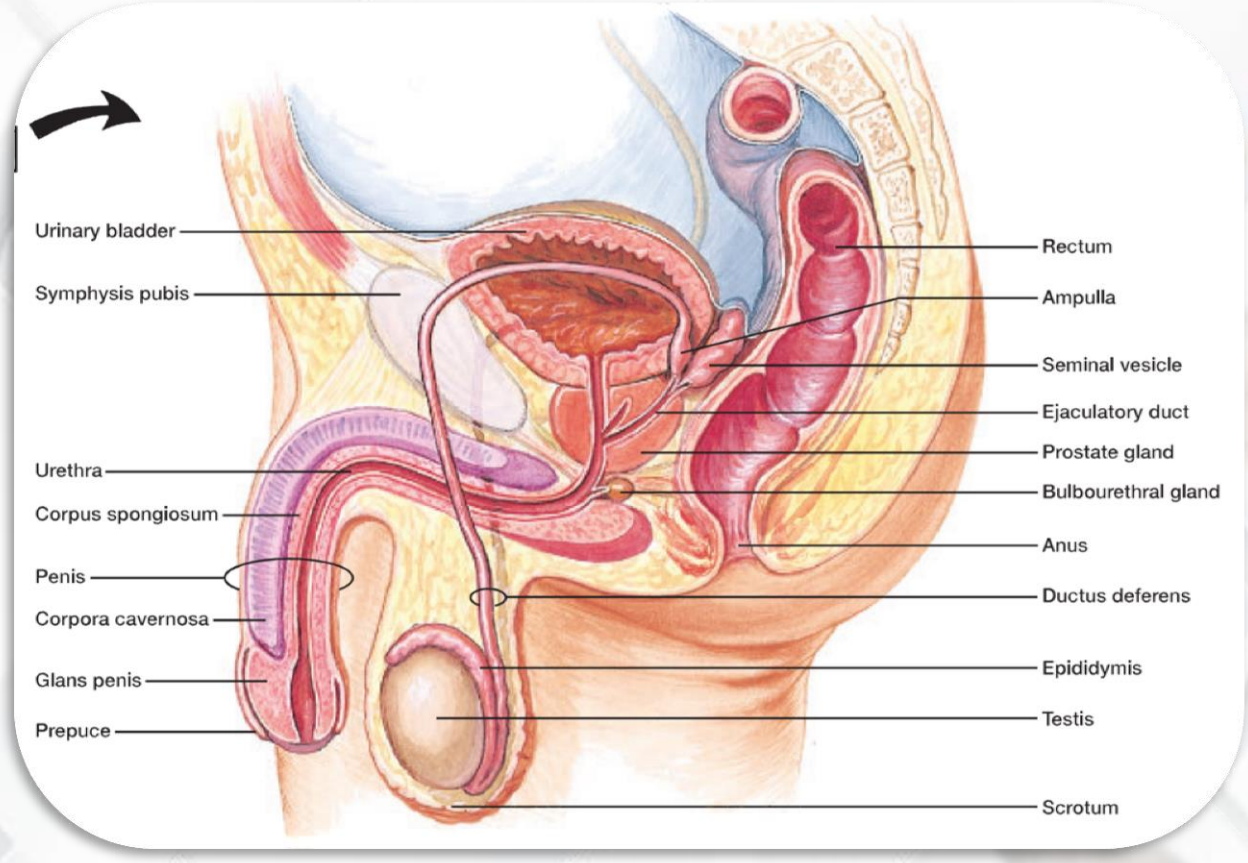
Human Male Reproductive System

2. Components of Male reproductive system

- Ejaculatory Duct
- Prostate Glands
- Bulbourethral Gland
- Penis

Human Male Reproductive System

3. Structure of Male Reproductive System



Ref. Miller and Harley. Zoology. 4th Edition. p.684

Principles of Animal life 1

Production of Sperms in human

Production of Sperms in Human

Contents

1. Testes
2. Seminiferous Tubules
3. Spermatogenesis
4. Structure of Sperm

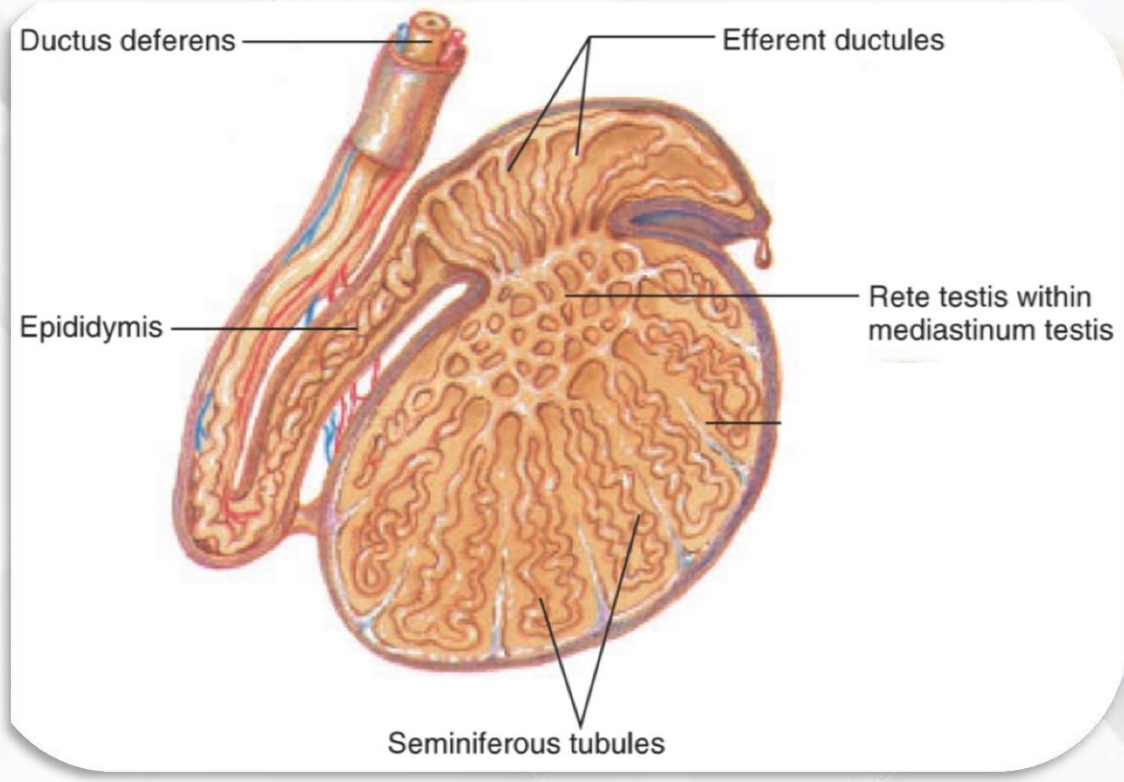
Production of Sperms in Human

1. Testes

- A pair of testes
- Scrotum
- 34^{o}C for sperm production and survival

Production of Sperms in Human

1. Testes



Ref. Miller and Harley. Zoology. 5th Edition. p. 481

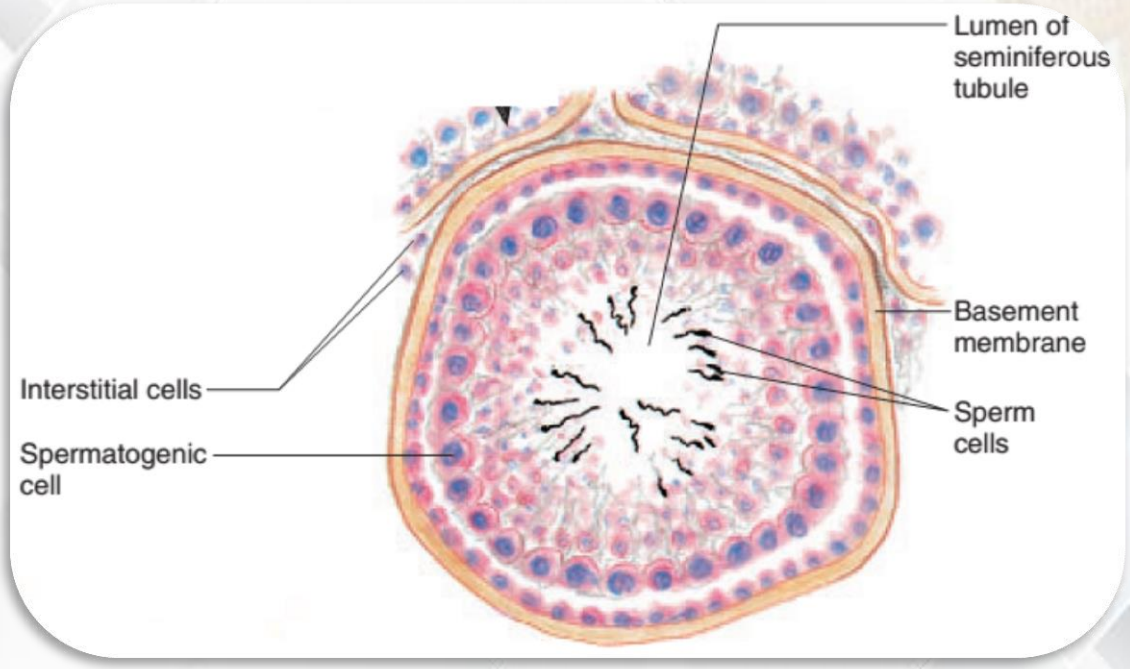
Production of Sperms in Human

2. Seminiferous Tubules

- About 800 in each testes
- Seminiferous tubules walls contain Two types of cells:
  - Spermatogenic cells
  - Sustentacular cells
- Clusters of interstitial cells between seminiferous tubules.

Production of Sperms in Human

2. Seminiferous Tubules



Ref. Miller and Harley. Zoology. 5th Edition. p. 481

Production of Sperms in Human

3. Spermatogenesis

- Thousands of sperms are produced in each second in seminiferous tubules
- The process of sperm production is called spermatogenesis
- Spermatogenic cells are present in the wall of seminiferous tubules

Production of Sperms in Human

3. Spermatogenesis

- Spermatogenic cells are also called spermatogonia
- Spermatogomium increase in size and becomes primary spermatocyte
- Meiosis-I results in two secondary spermatocytes

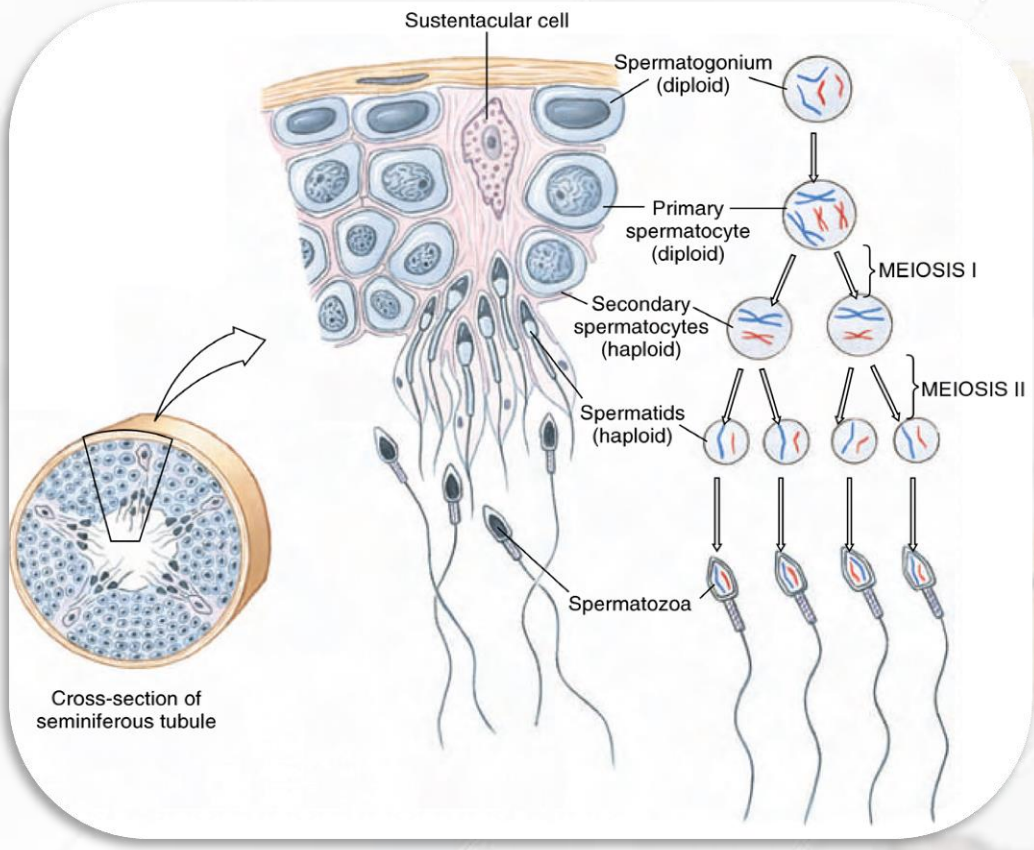
Production of Sperms in Human

3. Spermatogenesis

- After meiosis-II four haploid spermatids are formed
- Spermatid undergo morphological changes and are transformed into sperms which are motile

Production of Sperms in Human

3. Spermatogenesis



Ref. Hickman. Integrated Principles of Zoology. 11th Edition. p. 143

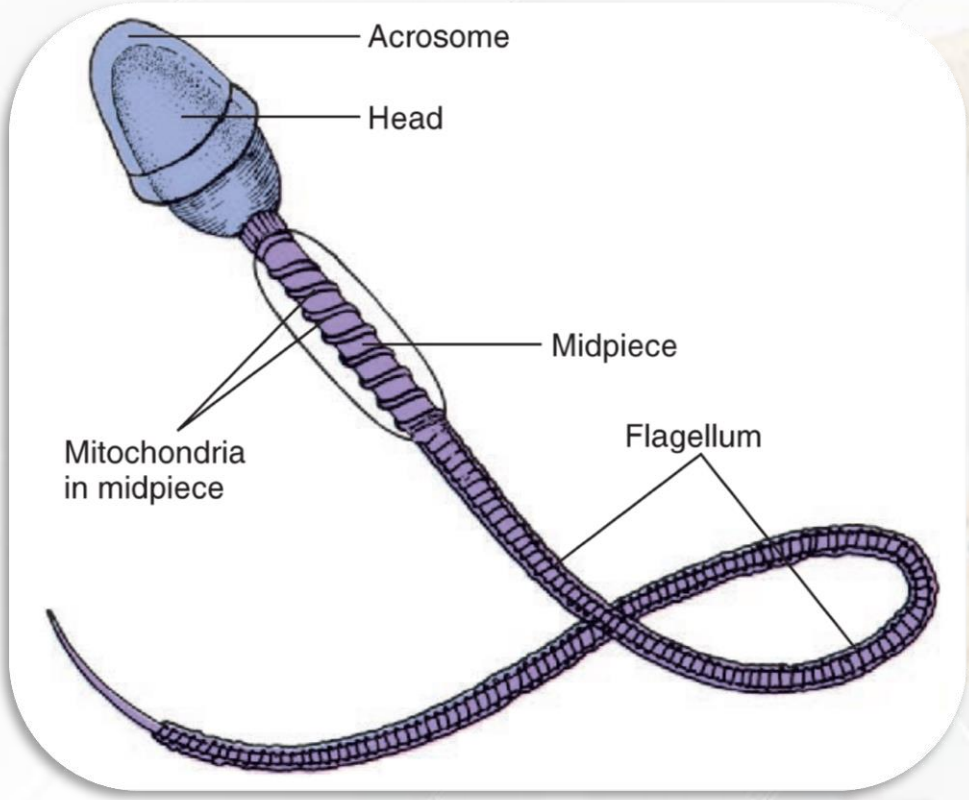
Production of Sperms in Human

4. Structure of Sperm

- A mature sperm consists of:
- **Head:** contain haploid nucleus and acrosome cap having acrosin.
- **Midpiece:** have mitochondria
- **Tail:** microtubules that bend for whiplike movements

Production of Sperms in Human

4. Structure of Sperm



Ref. Miller and Harley. Zoology. 5th Edition. p. 481

Principles of Animal life 1

Transport of Sperms in human

Transport of Sperms in Human

Contents

1. Seminiferous tubules
2. Rete Testis and Epididymis
3. Duct Deferens and Ampulla
4. Seminal Vesicles
5. Prostate Gland
6. Bulbourethral Glands
7. Semen

Transport of Sperms in Human

1. Seminiferous Tubules

- Seminiferous tubules contain permatogenic and sustentacular cells.
- Spermatogenic cells produce sperms while ustentacular cells secrete fluid for nourishment of sperms.

Transport of Sperms in Human

2. Rete Testis and Epididymis

- Sperms from seminiferous tubules comes into a network of called rete testis
- Then into epididymis where:
- Sperms are store
- Muscular contraction propel sperm forward

Transport of Sperms in Human

3. Duct Deferens and Ampulla

- Sperms through duct deferens are transported to ampulla where stored till ejaculation.

Transport of Sperms in Human

4. Seminal Vesicles

- Secrete water, fructose, prostaglandins and Vitamin C
- This is energy source for sperm motility and helps to neutralize pH of female reproductive tract

Transport of Sperms in Human

5. Prostate Gland

- Secretes water, enzymes, cholesterol, buffering salts and phospholipids

Transport of Sperms in Human

6. Bulbourethral Glands

- Secrete clear alkaline fluid that:
- Lubricates urethra to facilitate ejaculation

Transport of Sperms in Human

7. Semen

- Fluid that results from the combination of sperms and glandular secretions.

- Average human ejaculation contain 3 to 4 ml of semen having 300-400 million sperms.

end

Principles of Animal life 1

Hormonal Control of Male Reproductive Function

Hormonal Control of Male Reproductive Function

Contents

1. Androgens and gonadotropins
2. Testosterone and Inhibin
3. Hormonal Control
4. Feed back control

Hormonal Control of Male Reproductive Function

1. Androgens and Gonadotropins

- Male sex hormones are collectively called androgens e.g., Testosterones

- The hormones which travel from brain and pituitary to testis and ovaries are called gonadotropins e.g., FSH and LH

Hormonal Control of Male Reproductive Function

2. Testosterone and Inhibin

- **Testosterone:** secreted by interstitial cells
- Role in development of testis, prostate and secondary sex characteristics
- **Inhibin:** Produced by sustentacular cells
- It inhibits FSH

Hormonal Control of Male Reproductive Function

3. Hormonal Control

- Low level of testosterone is detected by hypothalamus
- Hypothalamus release GnRH
- GnRH effect the pituitary gland to release FSH and LH (ICSH)

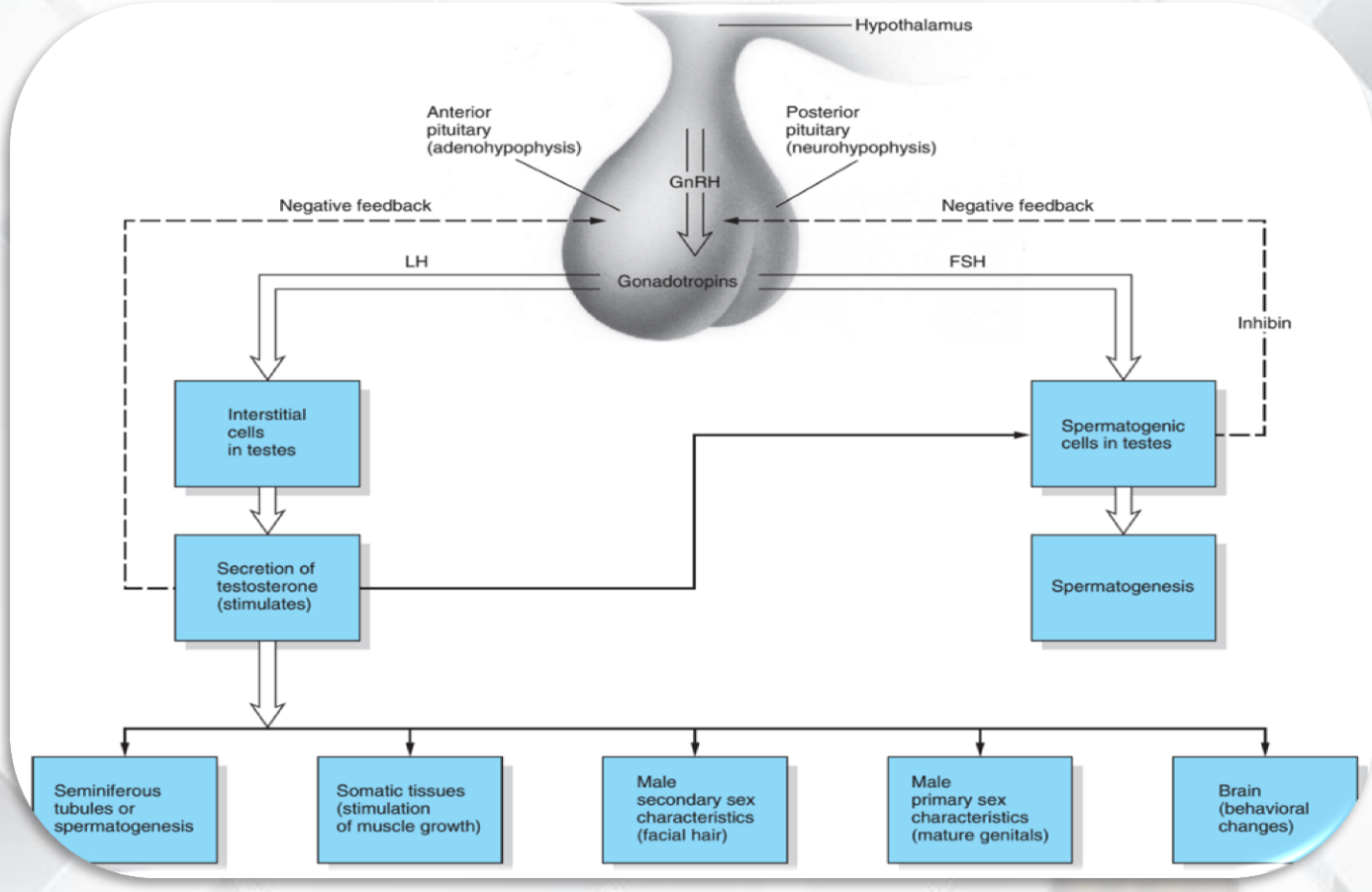
Hormonal Control of Male Reproductive Function

3. Hormonal Control

- FSH stimulate spermatogenesis
- LH stimulate the production of testosterone
- Testosterone inhibits the secretion of LH
- Inhibin inhibits the secretion of FSH

Hormonal Control of Male Reproductive Function

3. Hormonal Control



Ref. Miller and Harley. Zoology. 5th Edition. p. 482

Principles of Animal life 1

Human Female Reproductive System

Human Female Reproductive System

Contents

1. Functions of Female Reproductive System
2. Structures of Female Reproductive System

Human Female Reproductive System

1. Functions of Female Reproductive System

- Gamete production
- Nourish, carry and protect developing embryo
- Nursing young one after birth

Human Female Reproductive System

2. Structures of Female Reproductive System

- **Ovaries:** produce eggs, estrogen and progesterone
- **Uterine tubes:** carry eggs ovaries to uterus. Fertilization here.
- **Fimbriae:**
- **Uterus:** embryo implanted here

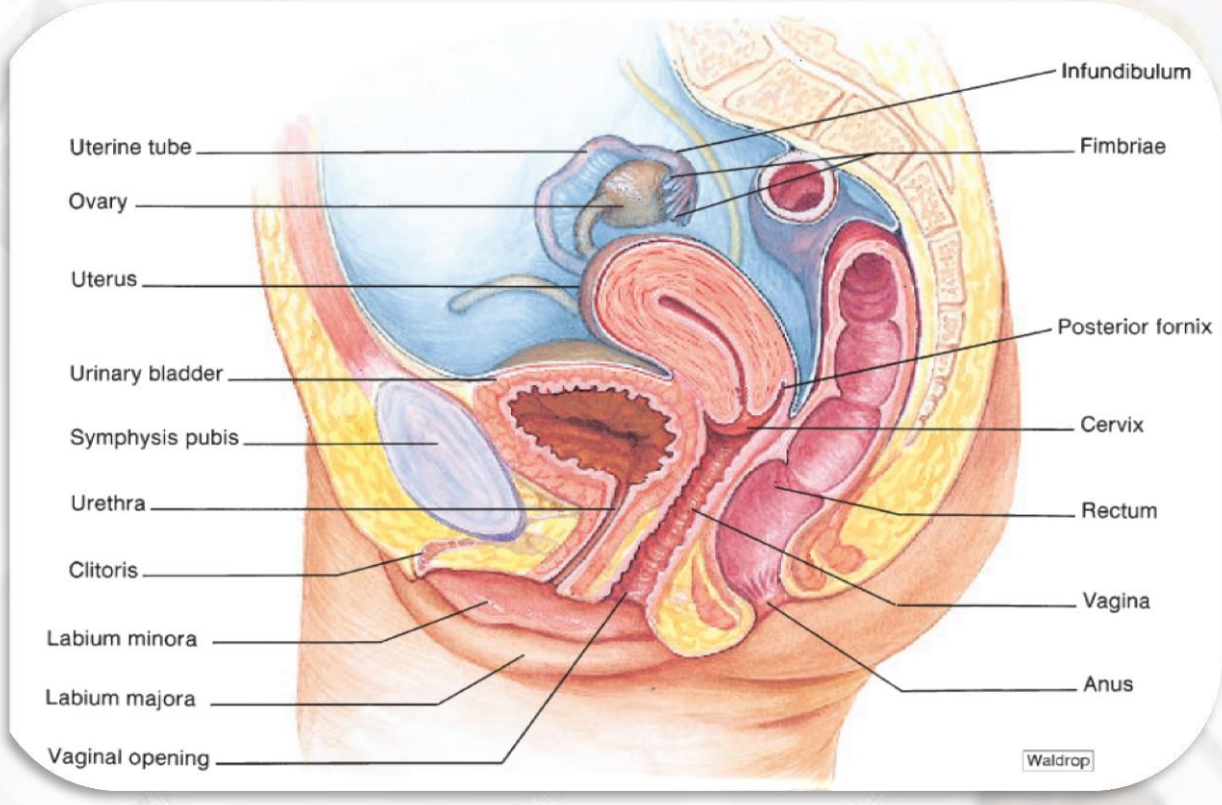
Human Female Reproductive System

2. Structures of Female Reproductive System

- **Vagina:** receives semen from male and baby passes through it during birth
- **External genitalia:** sexual arousal and protection
- **Mammary glands:** produce milk

Human Female Reproductive System

2. Structures of Female Reproductive System



Ref. Miller and Harley. Zoology. 5th Edition. p. 483

Principles of Animal life 1

Menstrual Cycle in Human

Menstrual Cycle in Human

Contents

1. The Menstrual Cycle
2. Follicular Phase
3. Ovulation
4. The Luteal Phase
5. Menstruation

Menstrual Cycle in Human

1. The Menstrual Cycle

- It is the monthly cycle of changes in ovaries and preparation of uterus for implantation of embryo

- The cycle begins at puberty and stops at the age of 45 to 55 years

Menstrual Cycle in Human

2. The Follicular Phase

- The controlling center is hypothalamus
- It releases GnRH which stimulate the release of FSH and LH from anterior pituitary
- FSH starts the development of oocyte in the ovarian follicles

Menstrual Cycle in Human

2. The Follicular Phase

- Developing follicle produces estrogen
- Estrogen builds up and prepares uterus lining to become thicker (form endometrium) for implantation of embryo

Menstrual Cycle in Human

3. Ovulation

- Elevated level of estrogen increases the release of LH from pituitary gland and inhibits FSH
- LH cause the mature follicle to enlarge rapidly and release secondary oocyte
- This process is ovulation

Menstrual Cycle in Human

4. The Luteal Phase

- Mature follicle after release of oocyte becomes corpus luteum
- Corpus luteum secretes estrogen and progesterone.
- They complete the development of endometrium and maintain it for 10-14 days

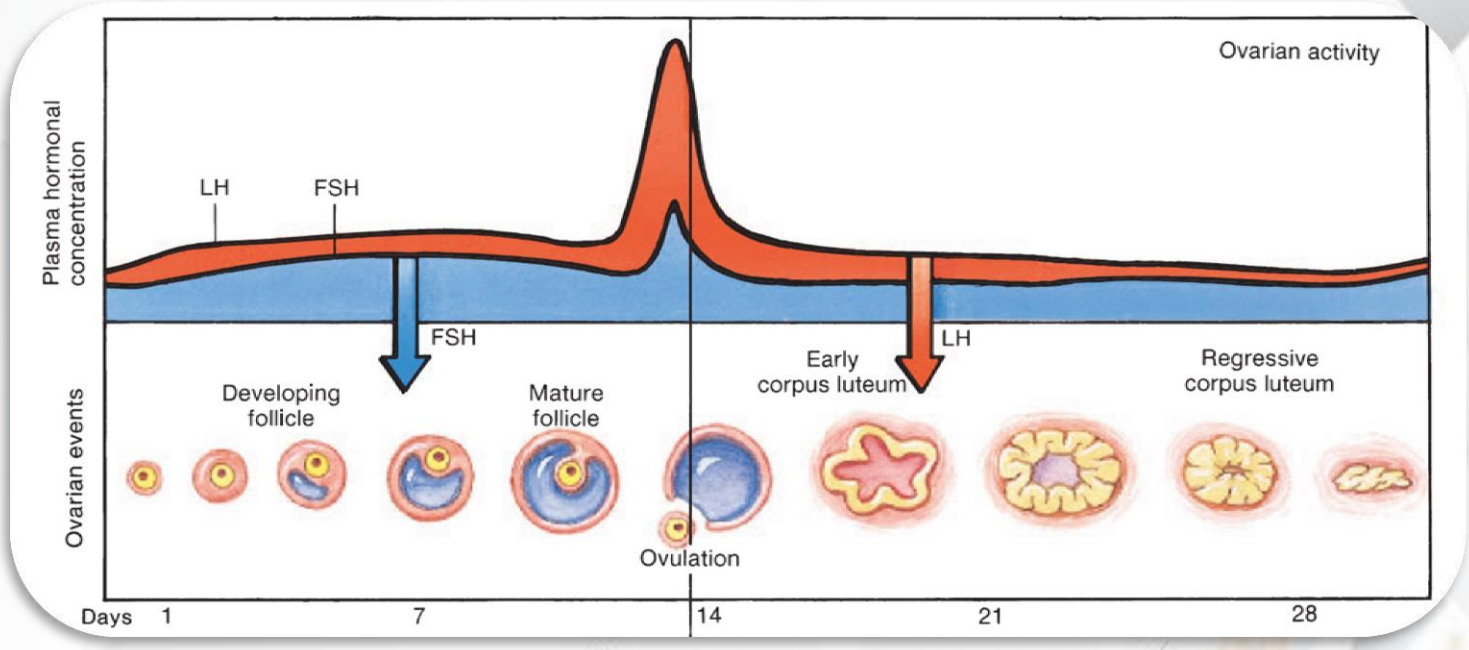
Menstrual Cycle in Human

5. Menstruation

- If the oocyte is not fertilized, corpus luteum disintegrates
- Estrogen and progesterone production stops
- As a result endometrium breaks and menstruation starts

Menstrual Cycle in Human

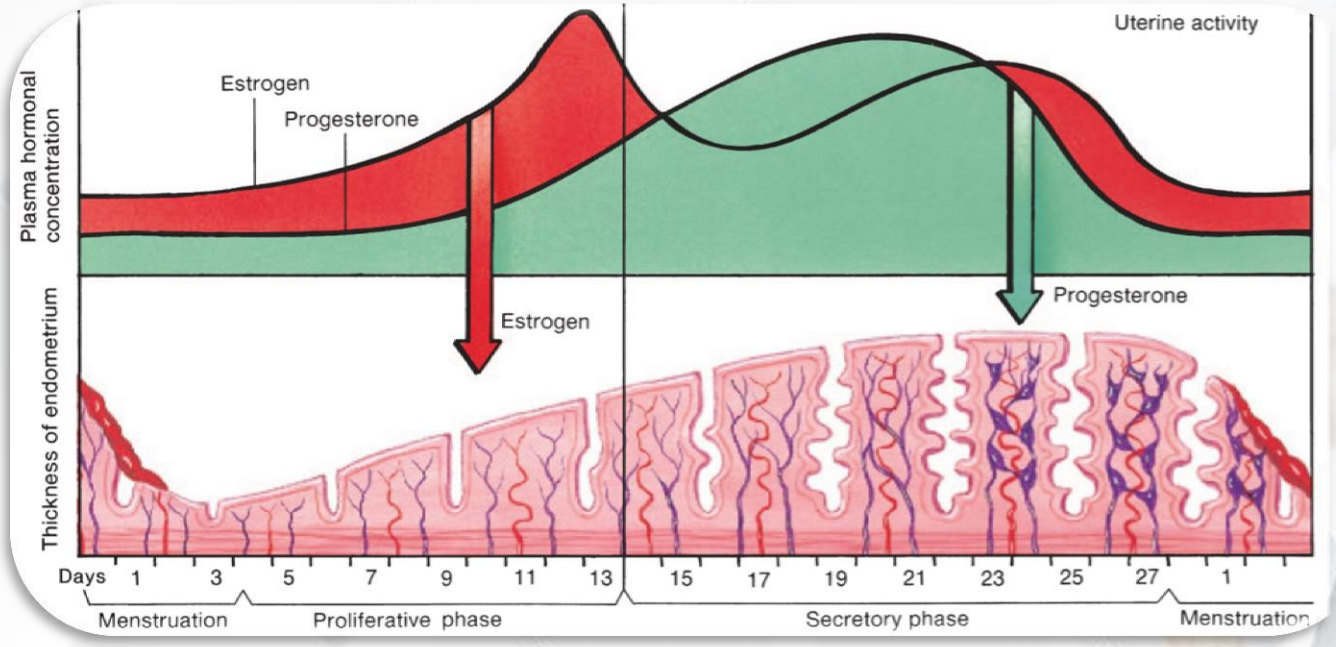
5. The Ovarian Cycle



Ref. Miller and Harley. Zoology. 5th Edition. p. 487

Menstrual Cycle in Human

5. The Menstrual Cycle



Ref. Miller and Harley. Zoology. 5th Edition. p. 487

Principles of Animal life 1

Follicle Development in Human

Follicle Development in Human

Contents

1. Structure of ovaries
2. Primordial Follicles
3. Primary Follicles
4. Ovulation
5. Corpus Luteum and Corpus Albicans
6. Atretic Follicles

Follicle Development in Human

1. Structure of Ovaries

- In cross section every ovary consists of:
- **Medulla:** have blood vessels and nerves
- **Cortex:** have developing follicles

Follicle Development in Human

2. Primordial Follicles

- During first five months of development
- Many small follicles are formed in the ovary of female fetus
- These are called primordial follicles

Follicle Development in Human

2. Primary Follicles

- At the time of puberty during each menstrual cycle some primordial follicles began to develope further and become primary follicles
- Then secondary follicles and graafian follicle

Follicle Development in Human

3. Ovulation

- Many follicle start development in each cycle but only one completes
- The follicle before release of egg is called graafian follicle
- LH hormone causes the rupture of follicle to release egg

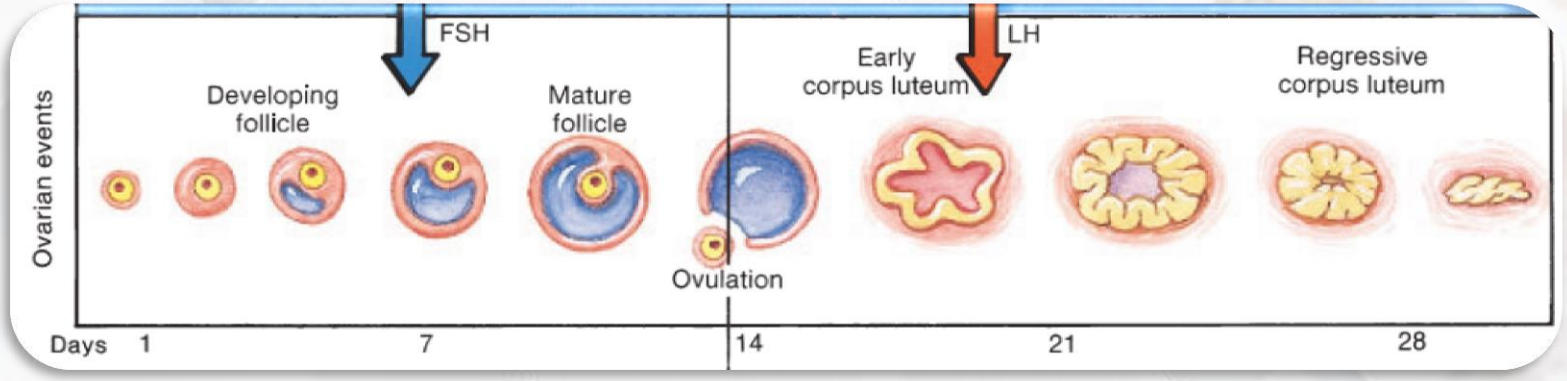
Follicle Development in Human

4. Corpus Luteum and Corpus Albicans

- Graafian follicle after release of oocyte becomes corpus luteum
- Corpus luteum remain active for 10-14 days
- If fertilization does not occur corpus luteum forms corpus albicans

Follicle Development in Human

4. Corpus Luteum and Corpus Albicans



Ref. Miller and Harley. Zoology. 5th Edition. p. 487

Follicle Development in Human

5. Atretic Follicles

- During each menstrual cycle several follicles start development but only one matures
- Others disintegrates by the process atresia
- During this cells are disintegrated by apoptosis.

end

Principles of Animal life 1

Production and Transport of Egg in Human

Production and Transport of Egg in Human

Contents

1. Egg Production
2. Ovulation
3. Fallopian tubes
4. Uterus

Production and Transport of Egg in Human

1. Egg Production

- Egg production starts in female while still at the developmental stages.
- A definite number of eggs development takes place in specific structures called follicles

Production and Transport of Egg in Human

1. Egg Production

- Egg development is halted at early stage of meiosis-I
- At puberty during menstrual cycle meiosis-I is completed and meiosis-II is started
- Meiosis-II will be completed after fertilization

Production and Transport of Egg in Human

2. Ovulation

- It is the release of oocyte from ovary.
- LH cause the release of oocyte from ovary

Production and Transport of Egg in Human

2. Egg Entry into Fallopian Tubes

- Ovary is encircled by finger like ciliated fimbriae
- The ciliary action and muscular contraction cause the entry of oocyte into fallopian tube also called uterine tube

Production and Transport of Egg in Human

3. Fertilization

- Fertilization takes place in upper third portion of uterine tube.
- Oocyte completes meiosis-II after fertilization
- Fertilization should take place within 24 hour of ovulation

Production and Transport of Egg in Human

4. Movement of Egg in Uterine Tube

- Egg moves due to ciliated cells and muscular contraction of uterine wall
- Takes about 5 days for uterine tube to uterus
- If fertilized, embryo implanted into uterus
- If unfertilized, released out

end

Principles of Animal life 1

Hormonal Control of Female Reproductive Function

Hormonal Control of Female Reproductive Function

Contents

1. Hypothalamus and Pituitary Gland
2. FSH and LH
3. Estrogen and Progesterone
4. Negative Feedback

Hormonal Control of Female Reproductive Function

1. Hypothalamus and Pituitary Gland

- Low level of estrogen stimulate hypothalamus to release Gonadotropin-releasing hormone (GnRH)
- GnRH acts on pituitary gland to release follicle stimulating hormone (FSH) and luteinizing hormone (LH)

Hormonal Control of Female Reproductive Function

2. Follicle Stimulating Hormone

- FSH and LH from pituitary act on ovary
- FSH start the development of follicle
- 15-20 follicles start development but only one matures
- Ovary also starts to produce estrogen

Hormonal Control of Female Reproductive Function

3. Estrogen

- Estrogen causes the thickness of endometrium of uterus
- Higher level of estrogen triggers LH and inhibit FSH form pituitary gland.

Hormonal Control of Female Reproductive Function

4. Luteinizing Hormone

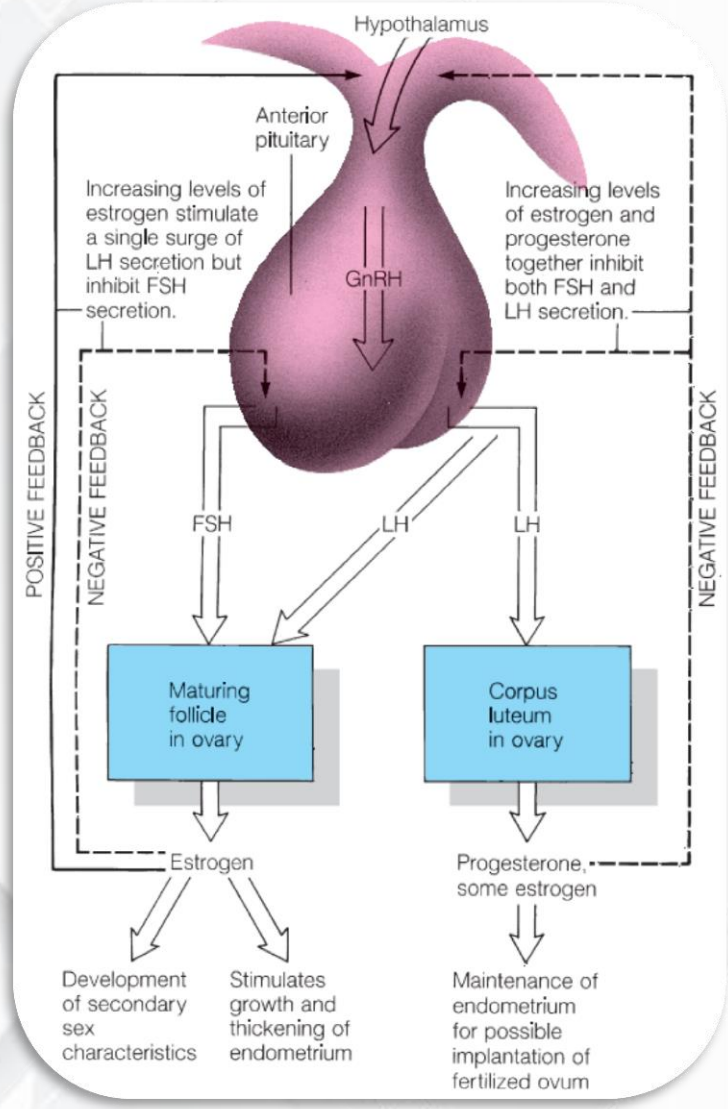
- LH from pituitary causes the ripening of ovarian follicle and release of oocyte from follicle
- LH also convert collapsed follicle into corpus luteum

Hormonal Control of Female Reproductive Function

5. Progesterone

- Corpus luteum secretes estrogen and progesterone.
- Both complete development and maintain endometrium
- Both also inhibit LH and FSH
- If no fertilization, FSH and estrogen level drop..new cycle starts

Hormonal Control of Female Reproductive Function



6. Positive and Negative Feedback

Ref. Miller and Harley. Zoology. 5th Edition. p. 485

Principles of Animal life 1

Historical Perspective of Embryology

Historical Perspective of Embryology

Contents

1. Preformation Theory
2. Epigenesis theory
3. Birth of modern biology
4. Developmental Biology

Historical Perspective of Embryology

1. Preformation Theory

- Most biologist of 17th and 18th century believe in it
- This theory says that gametes contain miniature version of all the element present in an adult

Historical Perspective of Embryology

2. Epigenesis Theory

- This theory gained popularity in mid 18th century
- This theory says that embryo contain material from which embryo is gradually build
- Much like a carpenter builds a building from nails, lumber and bricks

Historical Perspective of Embryology

3. Birth of Modern Embryology

- Modern embryology emerged with the proposed germ layer theory
- Embryo form 2-3 layers from which different body parts are formed

Historical Perspective of Embryology

4. Development Biology

- It the study by which animals and plants grow
- Developmental biology uses the tools of molecular biology to investigate the genetic control of development.

end

Principles of Animal life 1

Gametogenesis: Spermatogenesis

Gametogenesis: Spermatogenesis

Contents

1. Spermatogenesis
2. Location and Duration
3. Spermatogonium and Primary Spermatocyte
4. Secondary Spermatocyte and Spermatid
5. Spermatid to sperms
6. Overall Flow Chart

Gametogenesis: Spermatogenesis

1. Spermatogenesis

- The process of sperms production in male gonads through meiosis is called spermatogenesis

Gametogenesis: Spermatogenesis

2. Location and duration

- Spermatogenesis takes place in testes
- In human the entire process of spermatogenesis and its transportation takes about 3 months
- However, it is different in different animals

Gametogenesis: Spermatogenesis

3. Spermatogoium and Primary Spermatocytes

- Testes contain diploid stem cells
- Sperm mother cell is called spermatogomium
- It enlarges in its size and become primary spermatocyte

Gametogenesis: Spermatogenesis

4. Secondary Spermatocytes and Spermatids

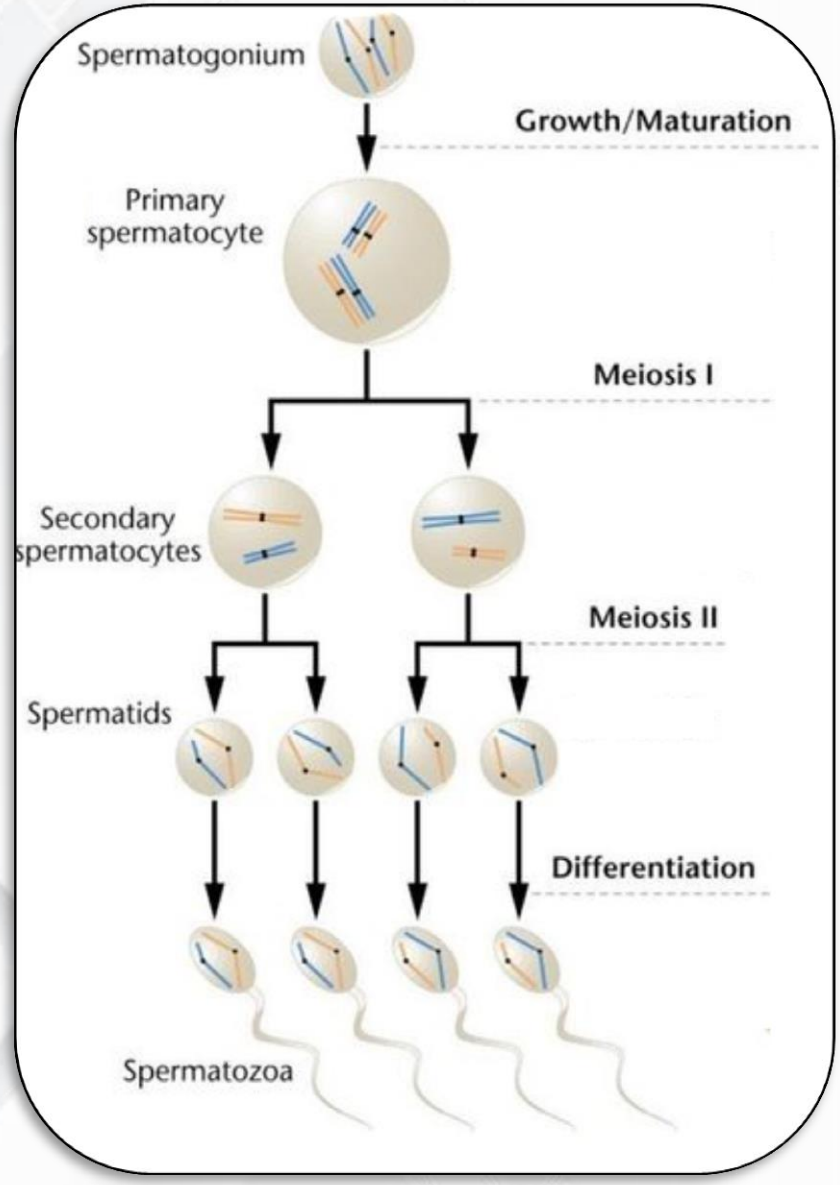
- Primary spermatocyte undergo meiotic cell division.
- After meiosis-I two secondary spermatocytes are formed
- Meiosis-II results in four haploid spermatids

Gametogenesis: Spermatogenesis

5. From spermatids to sperms

- Spermatids do not undergo any further cell division
- They undergo morphological changes to produce sperms

Gametogenesis: Spermatogenesis



6. Flow chart of Spermatogenesis

Ref. www.pinterest.com

Principles of Animal life 1

Gametogenesis: Oogenesis

Gametogenesis: Oogenesis

Contents

1. Oogenesis
2. Location of Oogenesis
3. Oogonium and Primary Oocyte
4. Secondary Oocyte and ovum
5. Overall Flow Chart

Gametogenesis: Oogenesis

1. Oogenesis

- The process of egg production in female gonads through meiosis is called oogenesis

Gametogenesis: Oogenesis

2. Location of Oogenesis

- Oogenesis takes place in ovaries
- Follicles are specific structures in which egg formation and maturation takes place

Gametogenesis: Oogenesis

3. Oogoium and Primary Oocytes

- Ovaries contain oogonia.
- Oogonium increase its size ….primary oocyte
- Primary oocyte is diploid cell with duplicated chromosome
- Each primary oocyte will form 1 ovum and polar bodies after reductive division

Gametogenesis: Oogenesis

4. Secondary Oocytes and Ovum

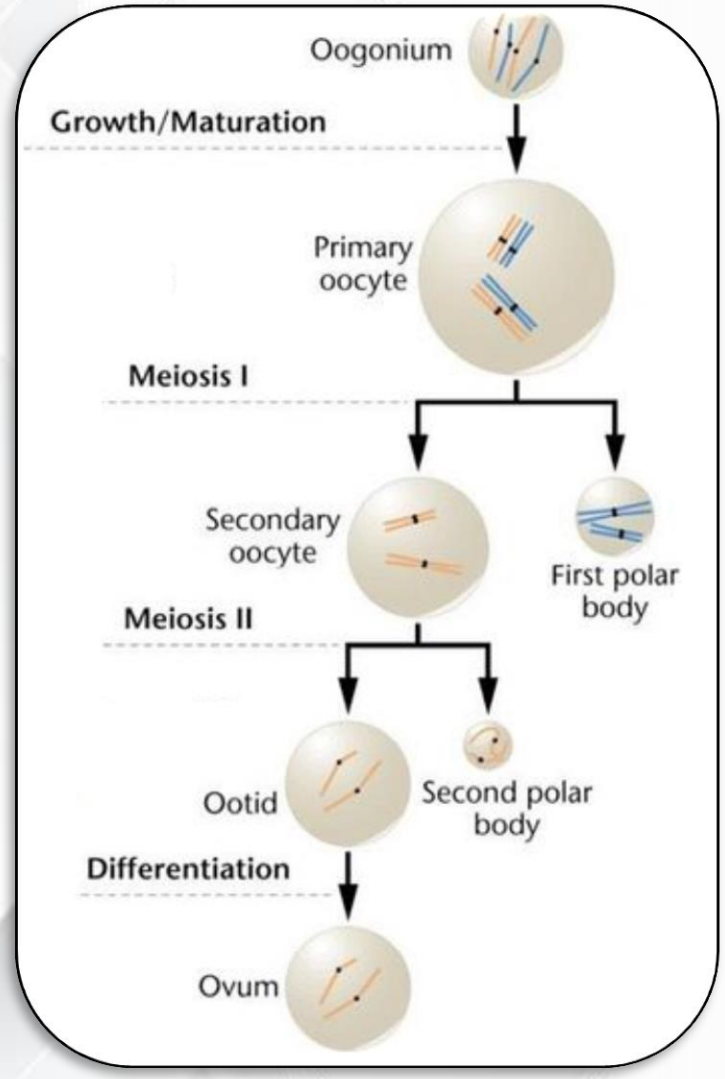
- Primary oocyte undergo meiotic cell division.
- After meiosis-I one secondary oocyte and one polar body is formed
- Meiosis-II results in a haploid ootid and another polar body

Gametogenesis: Oogenesis

4. Secondary Oocytes and Ovum

- Ootid undergo differentiation process to form ovum.

Gametogenesis: Oogenesis



6. Flow chart of Oogenesis

Ref. www.pinterest.com

Principles of Animal life 1

Types of Eggs

Types of Eggs

Contents

1. Classification of Eggs
2. Alecithal Eggs
3. Microlecithal Eggs
4. Mesolecithal Eggs
5. Macrolecithal Eggs
6. Homolecithal Eggs
7. Centrolecithal Eggs
8. Telolecithal Eggs

Types of Eggs

1. Classification of Eggs

- Eggs are classified on the basis of:
- Amount of yolk
  - Alecithal
  - Microlecithal
  - Mesolecithal
  - Macrolecithal

Types of Eggs

1. Classification of Eggs

- Eggs are classified on the basis of:

- Distribution of yolk
  - Homolecithal
  - Centrolecithal
  - Telolecithal

Types of Eggs

2. Alecithal Eggs

- They contain negligible amount of yolk
- e.g., mammals

Types of Eggs

3. Microlecithal Eggs

- They contain very little amount of yolk.
- e.g., sea urchin, amphioxus and marsupial mammals

Types of Eggs

4. Mesolecithal Eggs

- These are the eggs with moderate amount of yolk
- e.g., lung fish, amphibians

Types of Eggs

5. Macrolecithal Eggs

- Eggs with large amount of yolk
- e.g., reptiles and birds

Types of Eggs

6. Homolecithal eggs

- Yolk is equally distributed
- e.g., sea urchin, mammals

Types of Eggs

7. Centrolecithal Eggs

- These are the eggs with yolk in the center while cytoplasm in periphery
- e.g., insects

Types of Eggs

4. Telolecithal Eggs

- Yolk is present at one end of the egg while cytoplasm at other end.
- Animal pole and vegetal pole
- e.g., frogs, reptiles, birds

end

Principles of Animal life 1

Fertilization: The Process

Fertilization: The Process

Contents

1. Defining Fertilization
2. Methods of Fertilization
3. Vitelline Layer
4. Fusion of egg and sperm
5. Egg Activation
6. Diagram of egg fusion

Fertilization: The Process

1. Defining Fertilization

- Fertilization is the process in which egg and sperms fuse to form a diploid zygote
- Zygote is the first representative of next generation

Fertilization: The Process

2. Methods of Fertilization

- External fertilization:
- Internal Fertilization:
  - Copulation
  - Cloacal connection
  - Via spermatophore

Fertilization: The Process

3. Vitelline Membrane

- It is the structure surrounding plasma membrane of egg
- It is made up of fibrous proteins and receptors needed for sperm binding

Fertilization: The Process

4. Fusion of egg and sperm

- The sperm contain an acrosomal head
- Acrosome have specific enzymes called lysins which dissolve egg membrane during ferlization.
- Many sperms approach to egg but only one successful.

Fertilization: The Process

4. Fusion of egg and sperm

- In some species only nucleus fuses
- In others, mitochondria, centriole and flagella of sperm may get entry into egg.

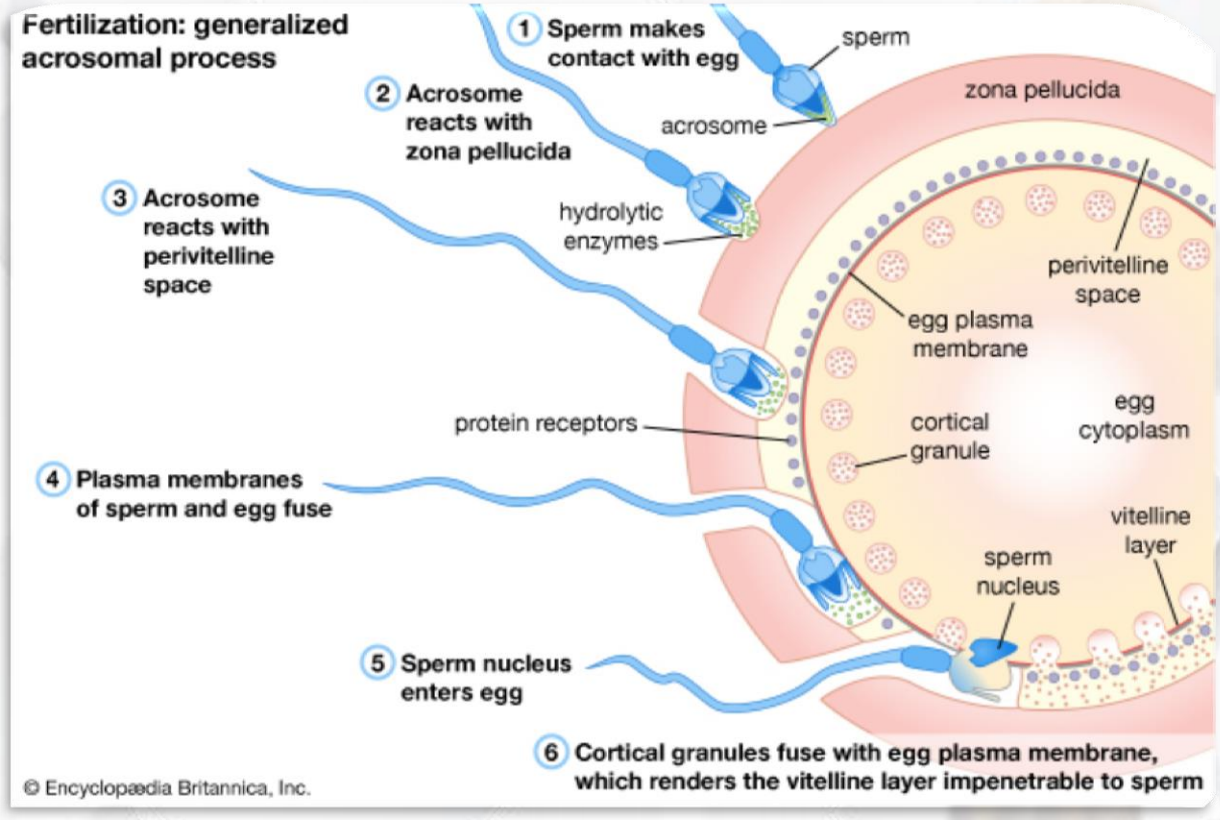
Fertilization: The Process

5. Egg Activation

- A series of biochemical changes in egg which ensure the completion of fertilization and initiate embryonic development.

Fertilization: The Process

6. Diagrammatic representation of Fertilization



Principles of Animal life 1

Fertilization: Membrane and Corticular Events

Fertilization: Membrane and Corticular Events

Contents

1. Single Fertilization
2. Microvilli surround sperm
3. Ionic Changes
4. Fertilization Membrane
5. Gray Crescent Formation

Fertilization: Membrane and Corticular Events

1. Single Fertilization

- Both haploid sperm and egg fuse to form a diploid zygote
- Normally many sperms approach to egg
- Only one sperm should fertilize the egg
- Second fertilization is stopped by changes in plasma membrane and outer region of cytoplasm

Fertilization: Membrane and Corticular Events

2. Microvilli Surround Sperm

- As sperm contacts, microvilli from egg plasma membrane wrap around the sperm.
- Contraction of microfilaments causes the sperm to draw into the egg

Fertilization: Membrane and Corticular Events

3. Ionic Changes

- Within milliseconds of sperm penetration, the charges on plasma membrane are changed
- This makes plasma membrane unresponsive to other sperms

Fertilization: Membrane and Corticular Events

4. Fertilization Membrane

- Fertilization membrane is formed by the discharge of granules of egg cortex
- These granules are discharged between plasma membrane and vitelline layer
- These granules loosen the contacts between plasma membrane and vitelline layer

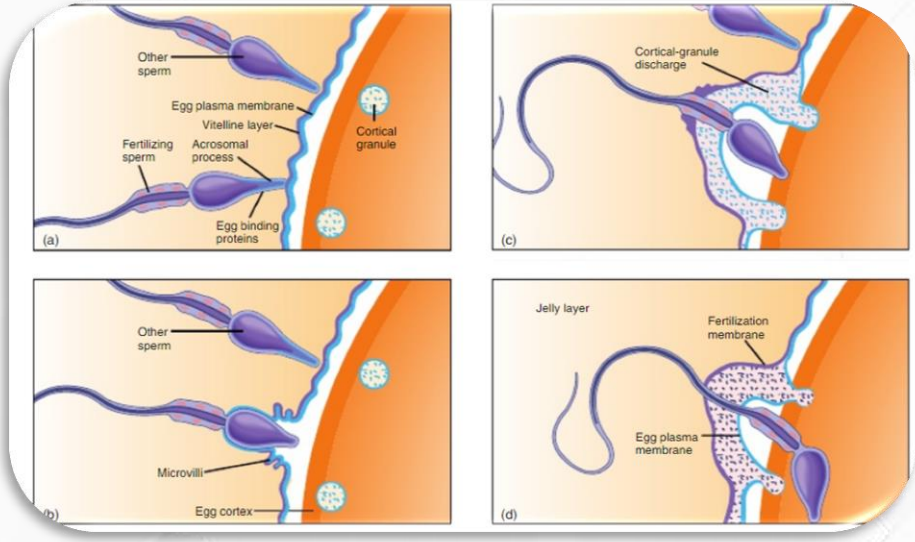
Fertilization: Membrane and Corticular Events

4. Fertilization Membrane

- Water molecules enter into the space
- Proteins of corticular granules thicken and strengthen the vitelline layer
- Fertilization membrane formed within 1 to 2 minutes

Fertilization: Membrane and Corticular Events

4. Fertilization Membrane



Ref. Miller and Harley. Zoology. 4th Edition. p. 136

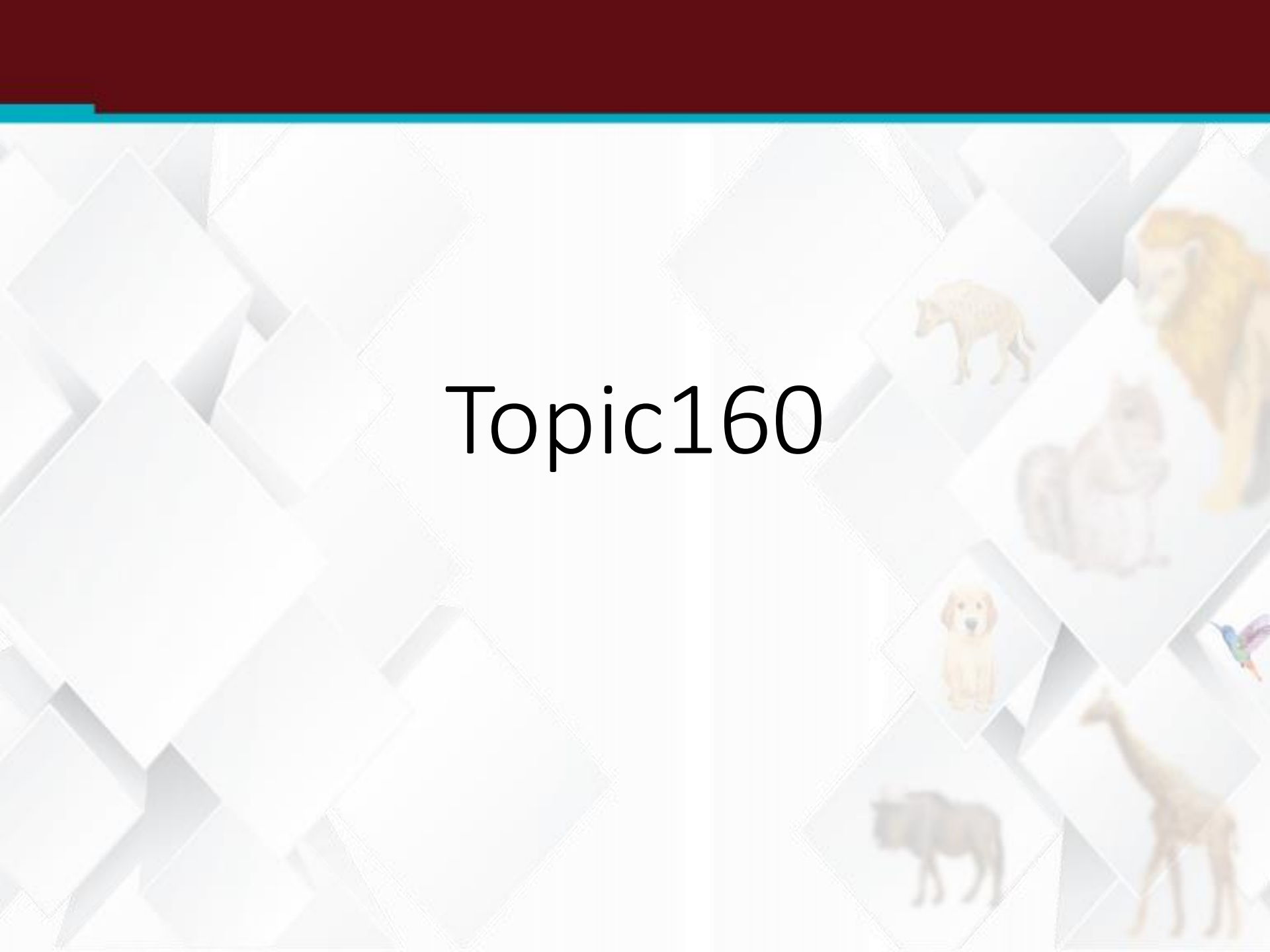
Fertilization: Membrane and Corticular Events

5. Gray Crescent formation

- After sperm penetration and corticular layer thickening, outer cytoplasm begins rotational and sliding movements
- In amphibians this results in the formation of gray crescent which is important in development

end

Topic160



- **LESSON 160**
- **Metabolic and nuclear events:** After fertilization ionic changes occur in cytoplasm causing an increase in **intracellular pH**. This triggers the developmental process in zygote. The zygote starts the cleavage (cell division) process. For this, **DNA replication** starts as two copies of DNA are required for two daughter cells. Rate of **protein synthesis** is also boosted. This is because more and more proteins and enzymes are required for cell division and daughter cells.

- mRNA already present in the cell is activated to start protein synthesis. The mRNA in zygote used to protein synthesis is maternal in origin. Thus bulk of initial proteins is synthesized by maternal mRNA. This influence is called **maternal dominance**.

Principles of Animal life 1

Cleavage and Its Patterns (I)

Cleavage and Its Patterns (I)

Contents

1. Defining Cleavage
2. Animal Pole and Vegetal Pole of Egg
3. Synchronous and asynchronous division
4. Distribution of Yolk Influence Cleavage Pattern

Cleavage and Its Patterns (I)

1. Defining Cleavage

- Cleavage refers to cell divisions that occur during embryonic development.

Cleavage and Its Patterns (I)

2. Animal Pole and Vegetal Pole of Egg

- **Animal pole:** The region of egg containing cytoplasm, its organelles and less amount of yolk
- **Vegetal Pole:** the region of egg containing more amount of yolk and less cytoplasm

Cleavage and Its Patterns (I)

3. Synchronous and Asynchronous Division

- Initially all the cells of embryo divide at the same time this is called synchrony
- As development proceeds, all cells don’t divide simultaneously. This is called asynchrony

Cleavage and Its Patterns (I)

4. Distribution of Yolk Influence Cleavage Pattern

- Eggs with evenly distributed yolk have cleavage pattern that result uniform size blastomeres
- Egg with unevenly distributed yolk result in unequal size blastomeres

end

Principles of Animal life 1

Cleavage and Its Patterns (II)

Cleavage and Its Patterns (II)

Contents

1. Holoblastic Cleavage
2. Meroblastic Cleavage
3. Spiral Cleavage
4. Radial Cleavage

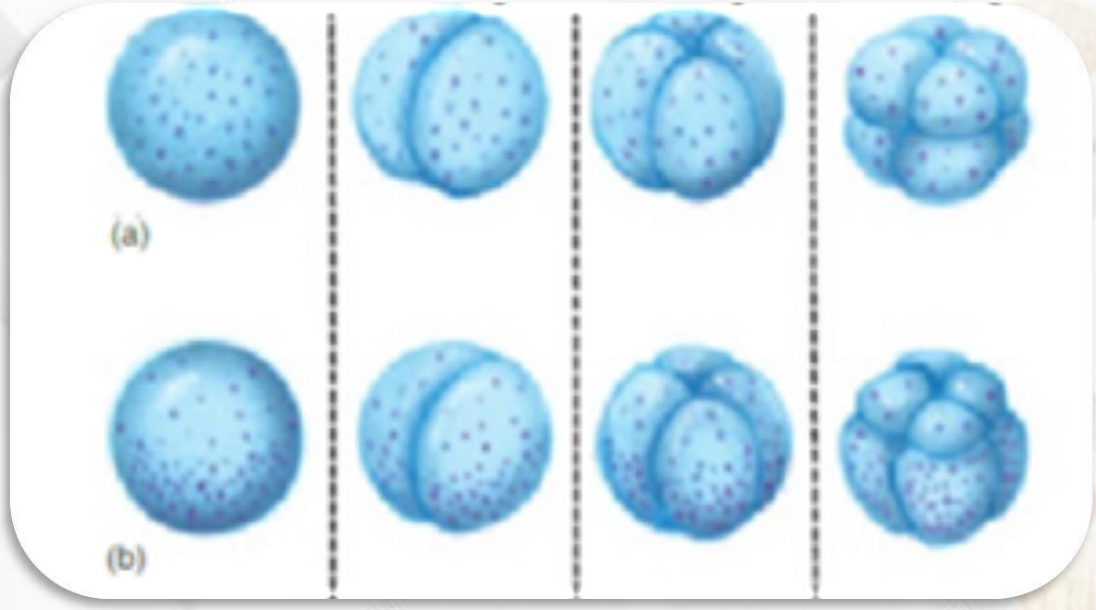
Cleavage and Its Patterns (II)

1. Holoblastic Cleavage

- Cleavage type in which egg is completely divided into two cells
- Egg contain little or moderate amount of yolk
- Eggs of Amphibians, mammals, echinoderms, flatworms, nematodes

Cleavage and Its Patterns (II)

1. Holoblastic Cleavage



Ref. Miller and Harley. Zoology. 4th Edition. p. 137

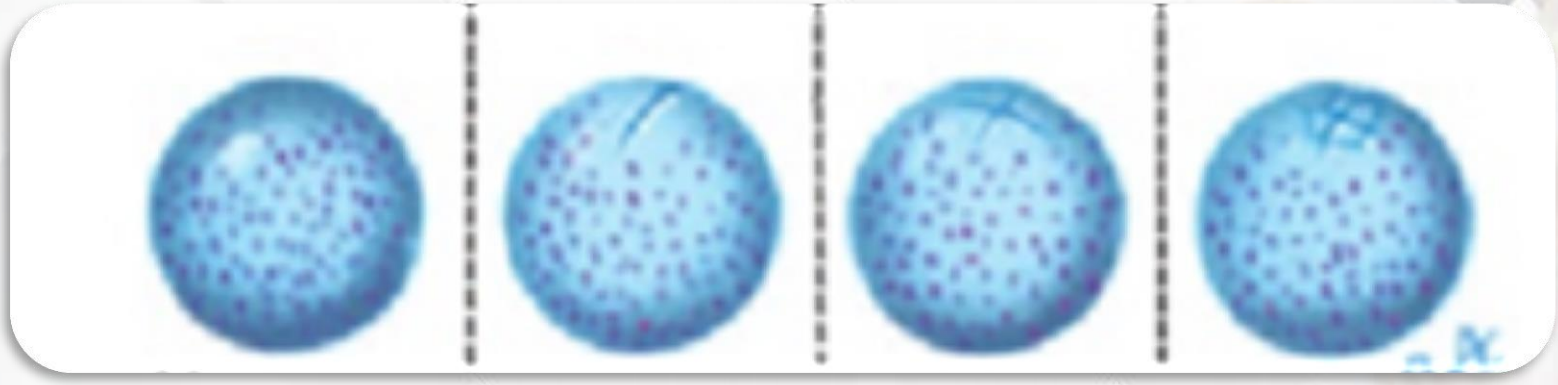
Cleavage and Its Patterns (II)

2. Meroblastic Cleavage

- Cleavage type in which egg is not completely divided into two cells
- This is due to large amount of yolk
- These embryo develop around or on top of yolk
- Eggs of birds and reptiles

Cleavage and Its Patterns (II)

2. Meroblastic Cleavage



Ref. Miller and Harley. Zoology. 4th Edition. p. 137

Cleavage and Its Patterns (II)

3. Spiral Cleavage

- It is pattern in which cells of animal pole are not exactly laying above the vegetal pole cells
- Cleavage takes place at oblique angle forming a spiral pattern
- Example: protostomes

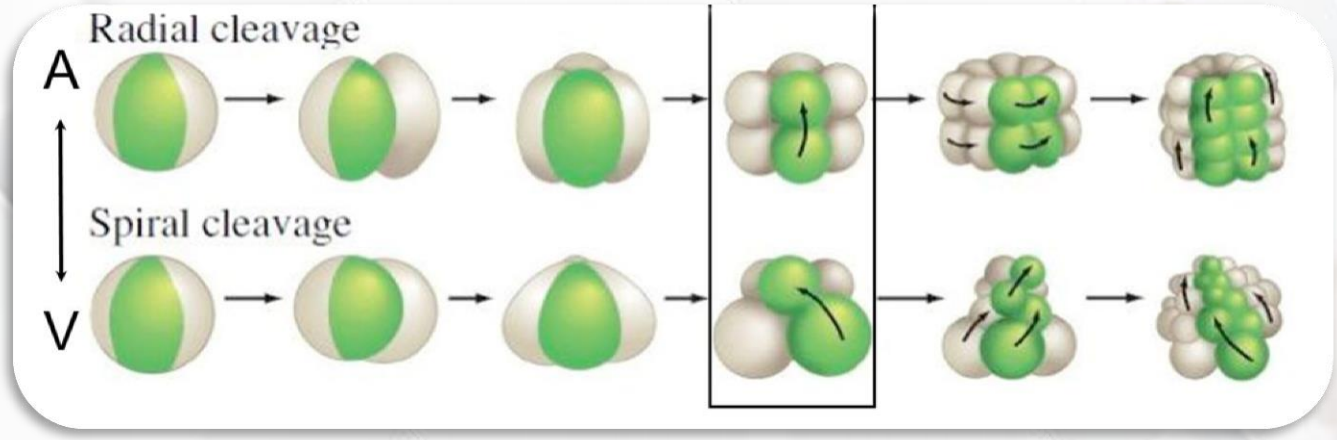
Cleavage and Its Patterns (II)

4. Radial Cleavage

- It is pattern in which cells of animal pole are exactly laying above the vegetal pole cells
- Upper tier of cells is exactly on the lower tier of cells
- Example: deurostomes

Cleavage and Its Patterns (II)

5. Radial Cleavage vs Spiral Cleavage



Ref. www.Slideplayer.com

Principles of Animal life 1

Echinoderm Embryology: General Concepts

Echinoderm Embryology: General Concepts

Contents

1. Extensively Studied
2. Egg type
3. Developmental Stages

Echinoderm Embryology: General Concepts

1. Extensively Studied

- Echinoderms include starfish, sea cucumber, brittle star and sea urchin
- Echinoderms are extensively studied group
- Easily maintained in laboratory

Echinoderm Embryology: General Concepts

2. Echinoderm Egg

- Eggs are laid in water
- Very little amount of yolk
- Microlecithal egg
- External fertilization

Echinoderm Embryology: General Concepts

3. Developmental Stages

- Fertilization in water
- Morula
- Gastrula
- Pluteus larva
- Development completed in approx. 30 hours

end

Principles of Animal life 1

Echinoderm Embryology: Cleavage

Echinoderm Embryology: Cleavage

Contents

1. Holoblastic Cleavage
2. Radial Cleavage
3. Plan of cell divisions

Echinoderm Embryology: Cleavage

1. Holoblastic Cleavage

- Echinoderms have microlecithal eggs
- Cleavage completely divide cell in two
- Initially same size cells produced

Echinoderm Embryology: Cleavage

2. Radial Cleavage

- Cleavage lines pass in such a way that cells lie exactly one above the other.
- Radial cleave is the characteristic of deuterostomes

Echinoderm Embryology: Cleavage

3. Plan of Division

- First two cleavage are from animal to vegetal pole (meridional) and perpendicular to each other
- Third cleavage is equatorial and right angle to first two. Separate animal pole and vegetal pole
- Fourth cleavage is medional

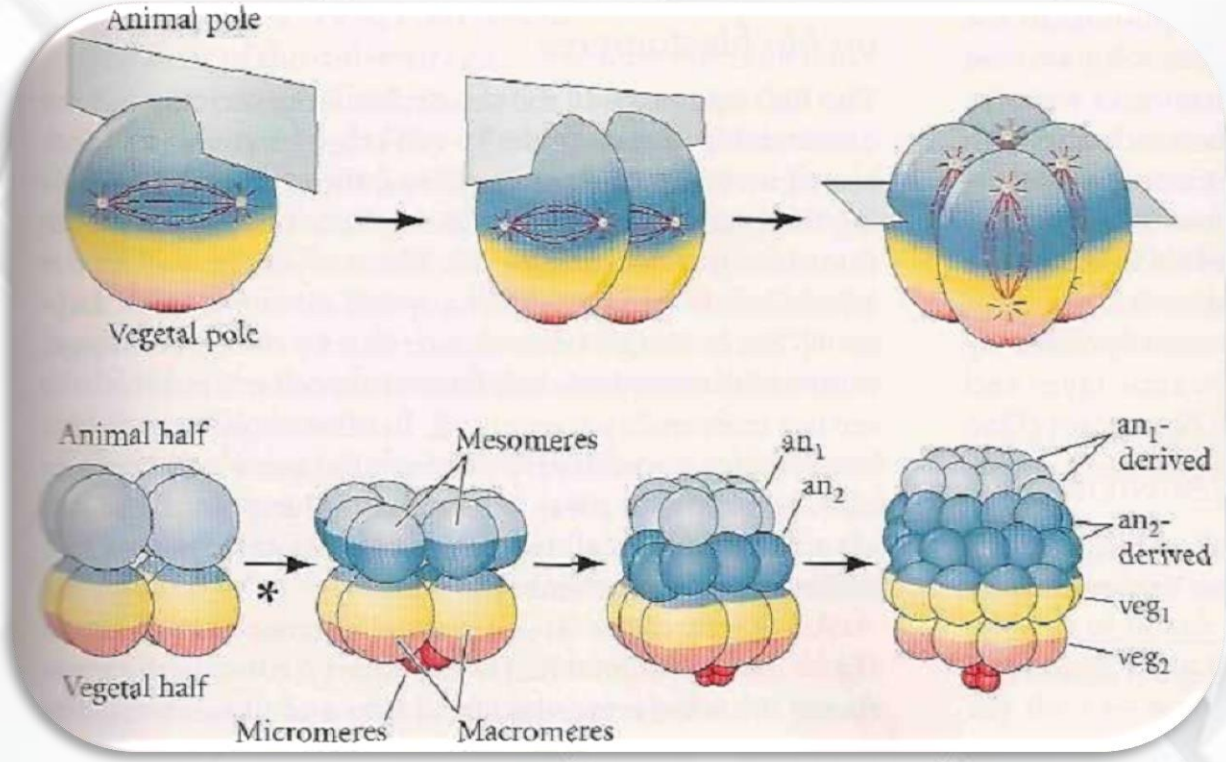
Echinoderm Embryology: Cleavage

3. Plan of Division

- At sixteen cell stage division is unequal
- Mesomeres
- Macromeres
- Micromeres

Echinoderm Embryology: Cleavage

3. Plan of Division



Ref. Gilbert. Developmental Biology. 10th Edition. p. 165

Principles of Animal life 1

Echinoderm Embryology: Morulation and Blastulation

Echinoderm : Morulation and Blastulation

Contents

1. Morula
2. Blastocoel Formation
3. Cilia Development
4. Primary Mesenchyme

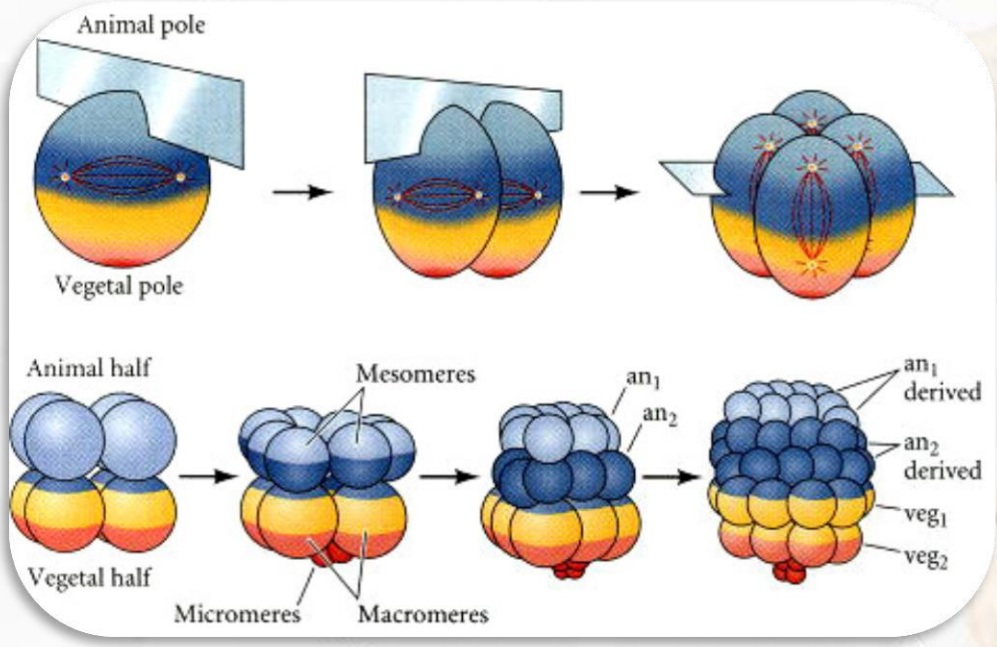
Echinoderm : Morulation and Blastulation

1. Morula

- Cleavage results in a ball of cells called morula
- Morula is a solid ball of cells without any cavity

Echinoderm : Morulation and Blastulation

1. Morula



Ref. www.ncbi.nlm.nih.gov

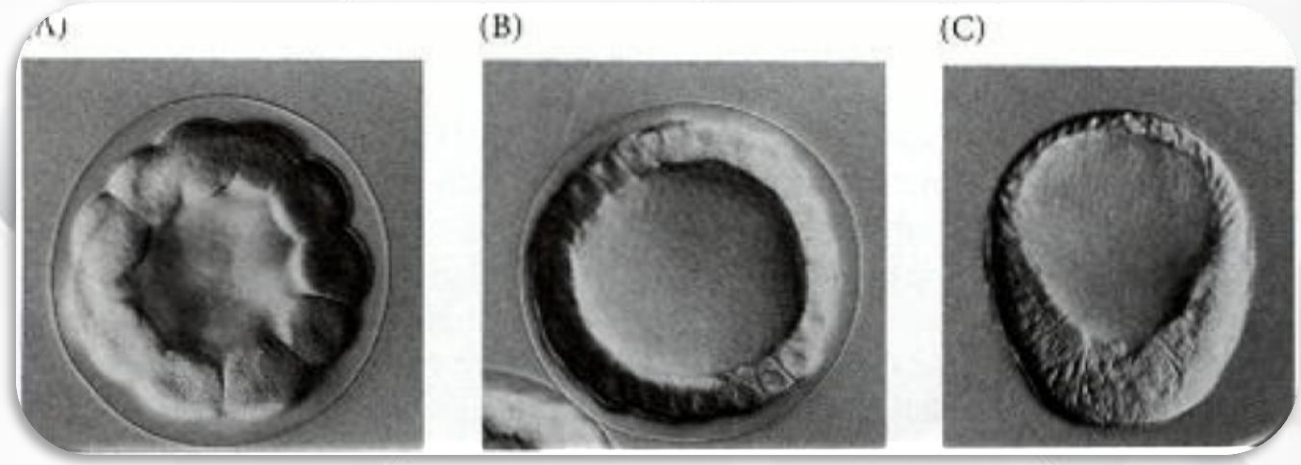
Echinoderm : Morulation and Blastulation

2. Blastocoel Formation

- After murula formation when embryo is about 120 cell stage, a fluid filled cavity **blastocoel** is formed in the embryo.
- Cell form a single layer around the blastocoel.
- Embryo at this stage is called blastula.

Echinoderm : Morulation and Blastulation

2. Blastocoel Formation



Ref. www.ncbi.nlm.nih.gov

Echinoderm : Morulation and Blastulation

3. Cilia Development

- Development up to blastula stage takes place within fertilization membrane.
- Cilia develop at blastula stage
- It comes out of fertilization membrane and start swiming

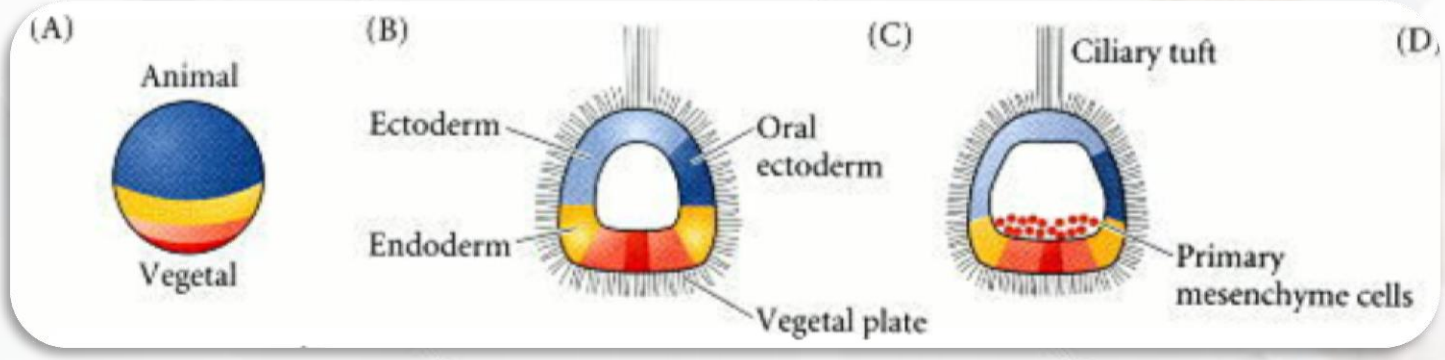
Echinoderm : Morulation and Blastulation

3. Primary Mesenchyme

- At late blastula stage, some cells breakoff and position themselves within blastocoel
- These cells are called primary mesenchyme
- They will form skeletal elements (spicules) of embryo.

Echinoderm : Morulation and Blastulation

4. Cilia and Primary Mesenchyme



Ref. www.ncbi.nlm.nih.gov

Principles of Animal life 1

Echinoderm Embryology: Gastrulation (I)

Echinoderm Embryology: Gastrulation (I)

Contents

1. Defining Gastrulation
2. Cells Invagination
3. Blastopore and Archenteron formation

Echinoderm Embryology: Gastrulation (I)

1. Defining Gastrulation

- Gastrulation is the process in which blastula is changed into gastrula.
- During this process primary germ layers are formed

Echinoderm Embryology: Gastrulation (I)

2. Cells Invagination

- Late blastula is about 1000 cells of different size
- Gastrulation begins at a point in vegetal half
- Where cells start invagination

Echinoderm Embryology: Gastrulation (I)

3. Blastopore and Archenteron

- The point where invagination starts is blastopore
- The invagination forms a new cavity in the embryo called archenteron
- Archenteron gradually replaces blatocoel

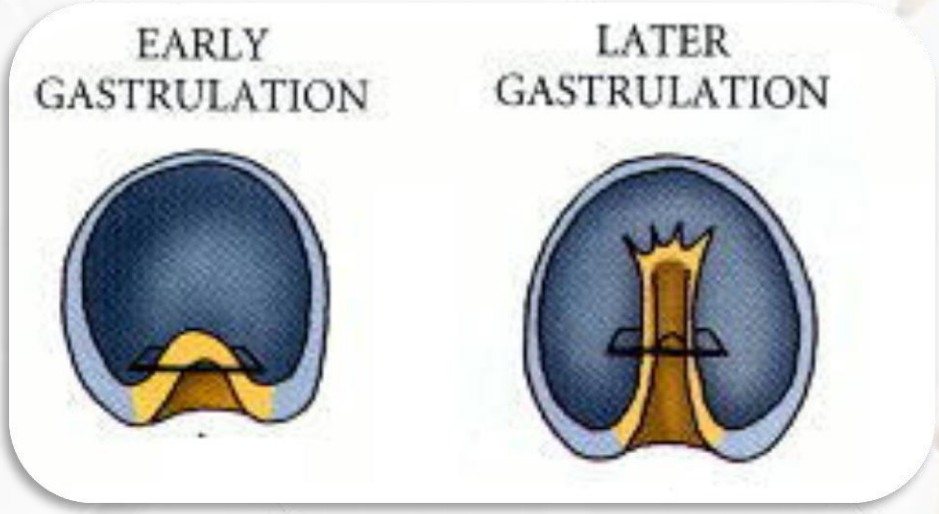
Echinoderm Embryology: Gastrulation (I)

3. Blastopore and Archenteron Formation

- Blastopore will form anus
- Archenteron is future gut of echinoderms

Echinoderm Embryology: Gastrulation (I)

3. Blastopore and Archenteron Formation



Ref. www.ncbi.nlm.nih.gov

Principles of Animal life 1

Echinoderm Embryology: Gastrulation (I)

Echinoderm Embryology: Gastrulation (I)

Contents

1. Secondary Mesenchyme Cells
2. Changes in Shape of Embryo
3. Pluteus Larva
4. Overall developmental stages

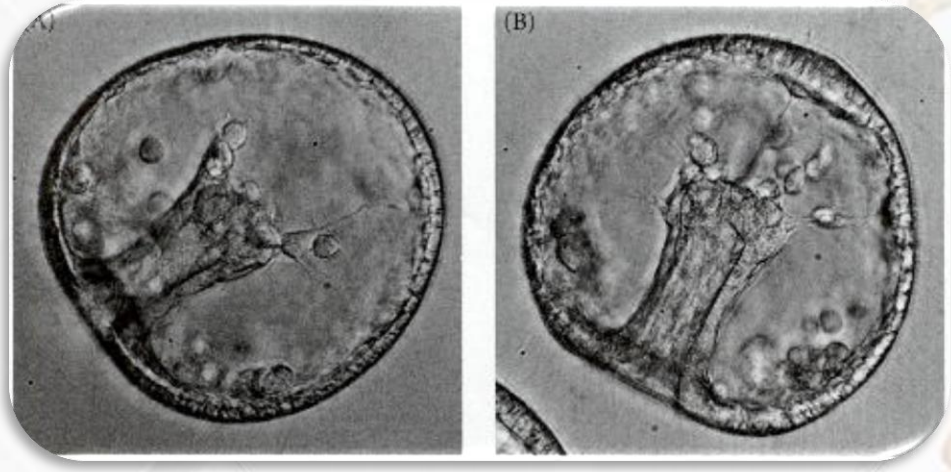
Echinoderm Embryology: Gastrulation (I)

1. Secondary Mesenchyme Cells

- Some cells at the tip of developing archenteron get apart and form secondary mesenchyme cells
- Secondary mesenchyme cells play a central role in pulling archenteron up to blastocoel wall

Echinoderm Embryology: Gastrulation (I)

1. Secondary Mesenchyme Cells



Ref. www.ncbi.nlm.nih.gov

Echinoderm Embryology: Gastrulation (I)

2. Changes in Shape of Embryo

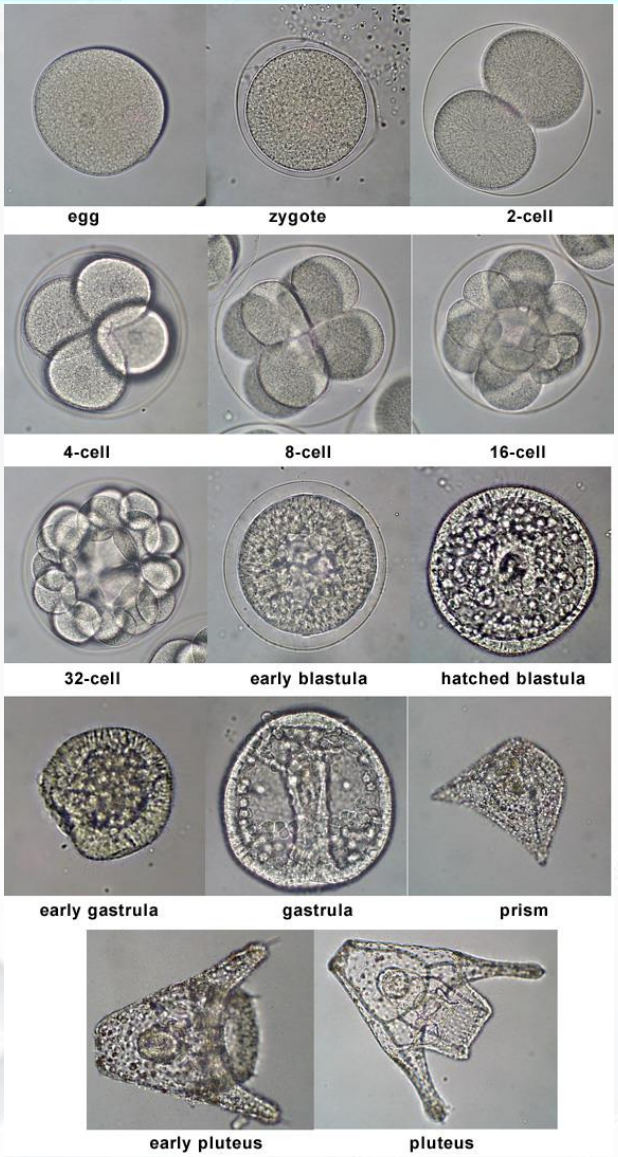
- Initially embryo is round in shape
- During gastrulation it begins to elongate
- It attains pyramidal shape (Prism)
- The larvae have a preferred direction of movement
- Front end is called anterior end

Echinoderm Embryology: Gastrulation (I)

3. Pluteus Larva

- After gastrulation the embryo attain prism shape then pluteus larva
- Pluteus larva is plaktonic
- It has bilateral symmetry
- Although adults have radial symmetry

Echinoderm Embryology: Gastrulation (I)



4. Overall Developmental Stages

Ref. www.swarthmore.edu

Principles of Animal life 1

Amphibian Embryology: General Concepts

Amphibian Embryology: General Concepts

Contents

1. Amphibian Eggs
2. Protection of Amphibian Egg
3. Developmental Stages

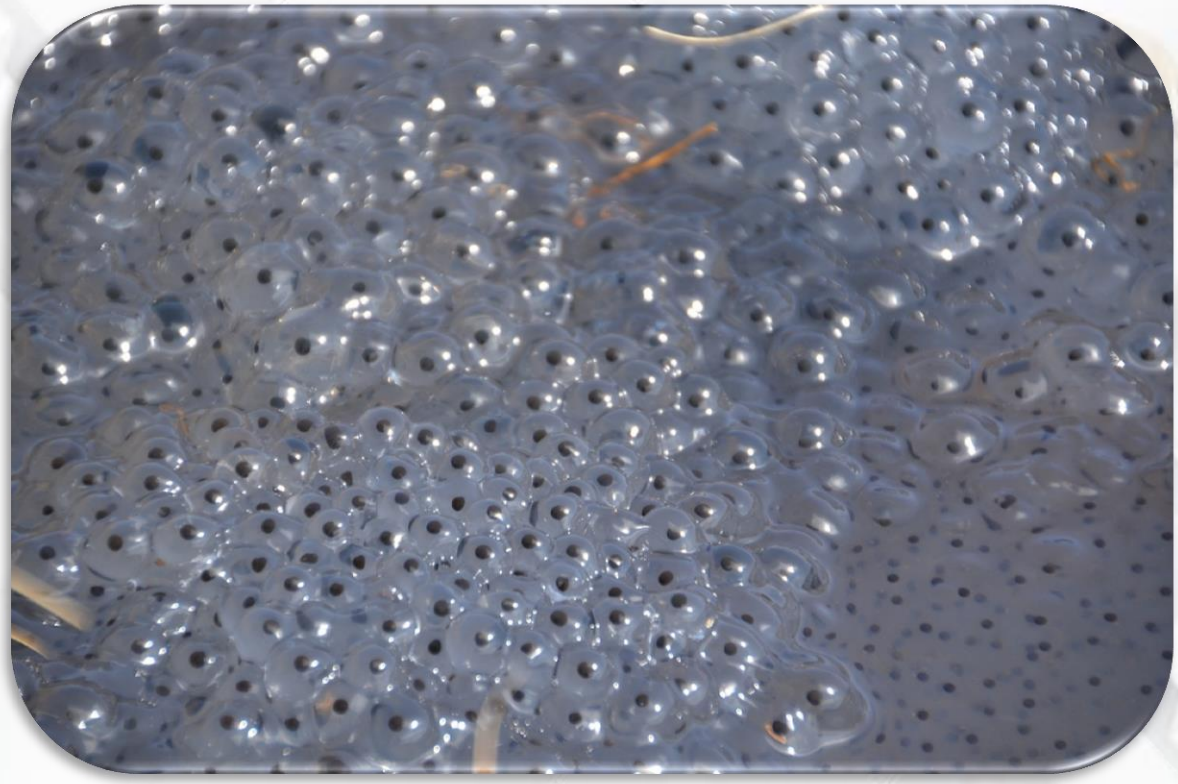
Amphibian Embryology: General Concepts

1. Amphibian Eggs

- Mostly laid in water
- External fertilization
- Mesolecithal egg
- Animal and vegetal poles
- Pigmented animal pole
- Heavy yolk in vegetal pole

Amphibian Embryology: General Concepts

1. Amphibian Eggs



Ref. needpix.com

Amphibian Embryology: General Concepts

2. Protection of Amphibian Egg

- Usually little care by parents
- Darkly pigmented animal pole on upper side camouflage from predators
- Light colour vegetal pole blends with sky when viewed form below

Amphibian Embryology: General Concepts

2. Protection of Amphibian Egg

- Dark color blends with bottom of water body
- Dark colour also absorbs heat
- This warming may promote development

Amphibian Embryology: General Concepts

3. Developmental Stages

- Fertilization in water
- Morula
- Blastula
- Gastrula
- Larval stage
- Adult

end

Principles of Animal life 1

Amphibian Embryology: Cleavage

Amphibian Embryology: Cleavage

Contents

1. Holoblastic Cleavage
2. Radial Cleavage
3. Plan of cell divisions

Amphibian Embryology: Cleavage

1. Holoblastic Cleavage

- Amphibians have mesolecithal eggs
- Cleavage completely divide cell in two
- Due to yolk at vegetal pole cleavage is slow
- Size of cells at animal pole is smaller while larger at vegetal pole

Amphibian Embryology: Cleavage

2. Radial Cleavage

- Cleavage lines pass in such a way that cells lie exactly one above the other.
- Radial cleavage is the characteristic of deuterostomes

Amphibian Embryology: Cleavage

3. Plan of Division

- First cleavage start from animal to vegetal pole
- Cleavage at vegetal pole is slow due to large amount of yolk
- It also divides gray crescent from middle
- Second cleavage is perpendicular to the first and divides embryo into four

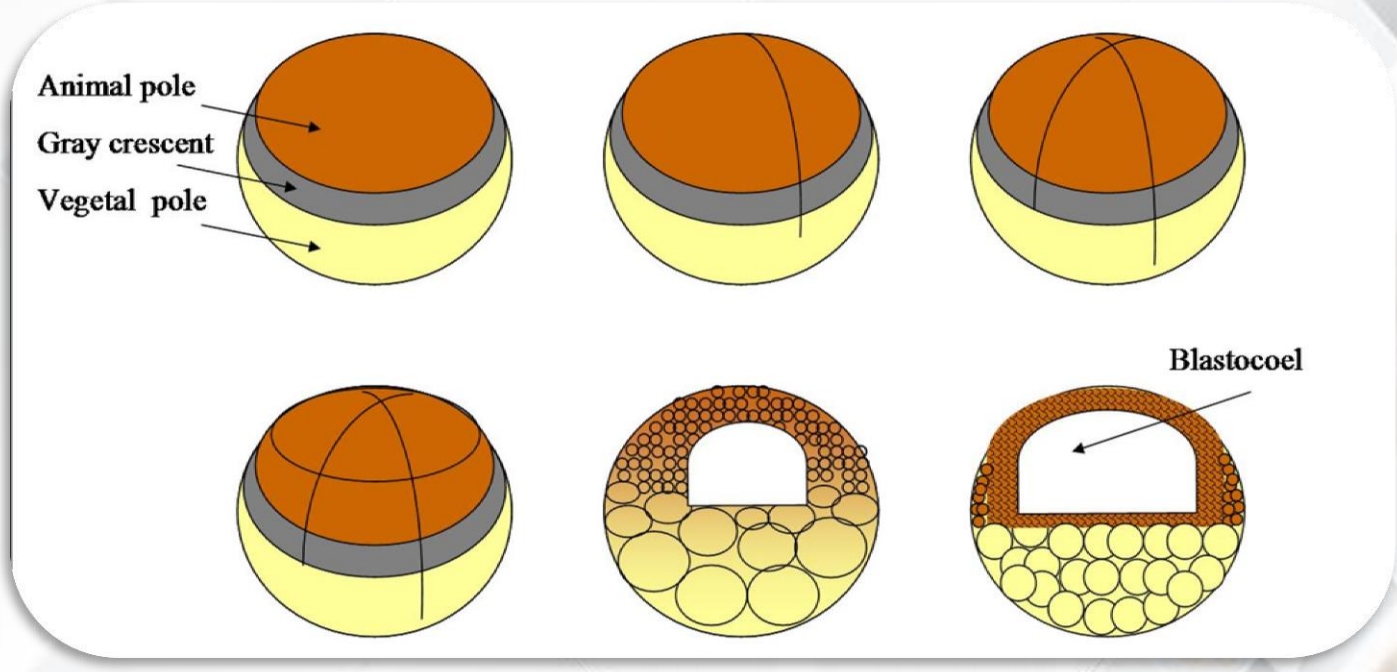
Amphibian Embryology: Cleavage

3. Plan of Division

- Third cleavage is equatorial and right angle to first two. Separate animal pole and vegetal pole
- Fourth cleavage is medional

Amphibian Embryology: Cleavage

3. Plan of Division



Ref. labs.bio.unc.edu

Principles of Animal life 1

Amphibian Embryology: Morulation and Blastulation

Amphibian : Morulation and Blastulation

Contents

1. Morula
2. Blastocoel Formation
3. Structure of Blastula

Amphibian : Morulation and Blastulation

1. Morula

- Cleavage results a ball of cells called morula
- Morula is a solid ball of cells without any cavity
- Cell towards animal pole are smaller while larger at vegetal pole due to presence of yolk

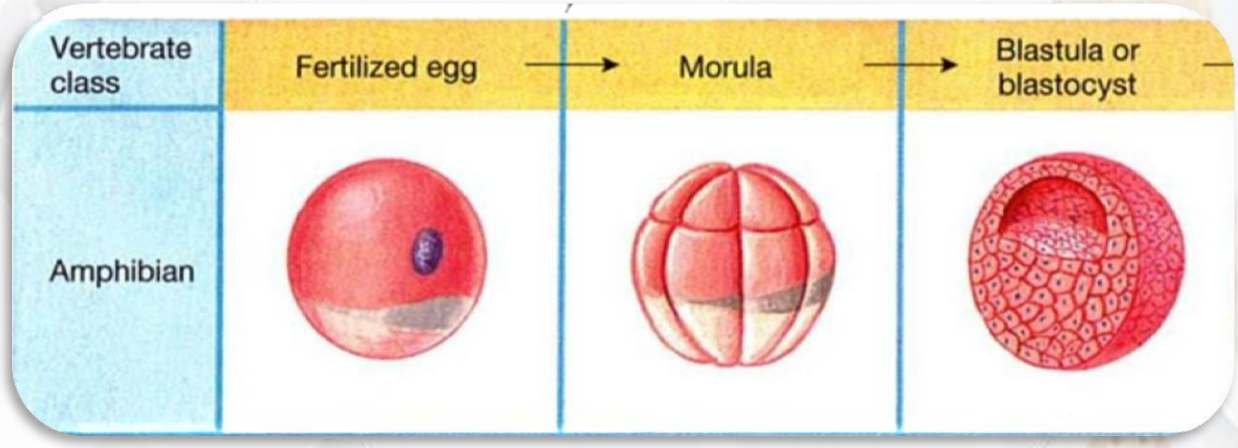
Amphibian : Morulation and Blastulation

2. Blastocoel Formation

- After morula formation, a fluid filled cavity **blastocoel** is formed in the embryo
- Cell form multiple layers around the blastocoel
- In amphibians blastocoel is formed towards animal pole due to yolk at vegetal pole

Amphibian : Morulation and Blastulation

3. Structure of Blastocoel



Ref. one-tab.com

Principles of Animal life 1

Amphibian Embryology: Gastrulation (I)

Amphibian Embryology: Gastrulation (I)

Contents

1. Defining Gastrulation
2. Cell are grouped on the surface of blastula
3. Blastopore formation
4. Involution
5. Yolk plug formation

Amphibian Embryology: Gastrulation (I)

1. Defining Gastrulation

- Gastrulation is the process during which cells arrange themselves into different layers called primary germ layers

Amphibian Embryology: Gastrulation (I)

2. Cells are grouped on the surface of blastula

- Cells forming different structures are grouped on the surface of blastula
- Embryologist use dyes and carbon particles to mark different cells
- Later these migrate to during gastrulation and form different structures

Amphibian Embryology: Gastrulation (I)

2. Cells are grouped on the surface of blastula

- So these cells are named according to their future structure such as presumptive notochord, presumptive mesoderm and so on.

Amphibian Embryology: Gastrulation (I)

3. Blastopore Formation

- Gastrulation begins with a slit like opening **blastopore** between gray crescent and ventral pole.
- Animal pole margin of blastopore is called its dorsal lip
- Cells at the bottom of groove move anteriorly and groove spreads transversely

Amphibian Embryology: Gastrulation (I)

4. Involution

- Cell at the surface begin to roll over the dorsal lip
- This process is involution
- Migrating cells spread across the slit like blastopore transversely and downward to the vegetal pole

Amphibian Embryology: Gastrulation (I)

4. Involution

- Due to this migration one end of slit meets the other end

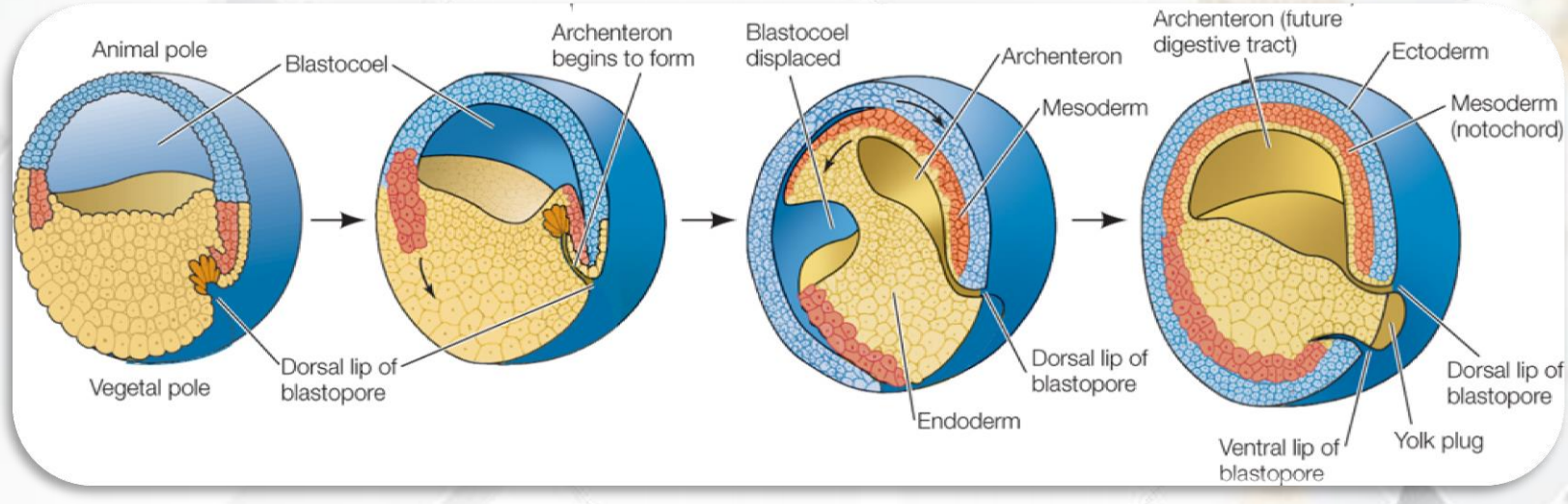
Amphibian Embryology: Gastrulation (I)

5. Yolk Plug Formation

- The ring like blastopore surrounds the yolk cells and they protrude out of the blastopore
- This is called yolk plug
- Later blastopore cells enclose yolk plug and blastopore is considered as closed

Amphibian Embryology: Gastrulation (I)

3. Gastrulation in amphibians



Ref. macmillanhighered.com

Principles of Animal life 1

Amphibian Embryology: Gastrulation (II)

Amphibian Embryology: Gastrulation (II)

Contents

1. Archenteron
2. Epiboly
3. Mesoderm Formation

Amphibian Embryology: Gastrulation (II)

1. Archenteron

- Rolling over cells from blastopore form archenteron
- Blastocoel becomes smaller and archenteron become larger
- Archenteron is future gut.
- Yolk cell make floor of gut

Amphibian Embryology: Gastrulation (II)

2. Epiboly

- Some of the animal pole cell are rolling into the blastopore, they will form endoderm and mesoderm
- The remaining animal pole cells are called ectoderm
- These ectodermal cells towards blastopore and spread over entire embryo.
- This is called **epiboly**

Amphibian Embryology: Gastrulation (II)

3. Mesoderm Formation

- Some of the last cells rolling over the blastopore are presumptive notochord and presumptive mesoderm
- Initially they form dorsal lining of archenteron near blastopore

Amphibian Embryology: Gastrulation (II)

3. Mesoderm Formation

- Later they detach from endoderm and move between endoderm and ectoderm in the region of blastopore
- This is called **chordamesoderm** which differentiates into notochord

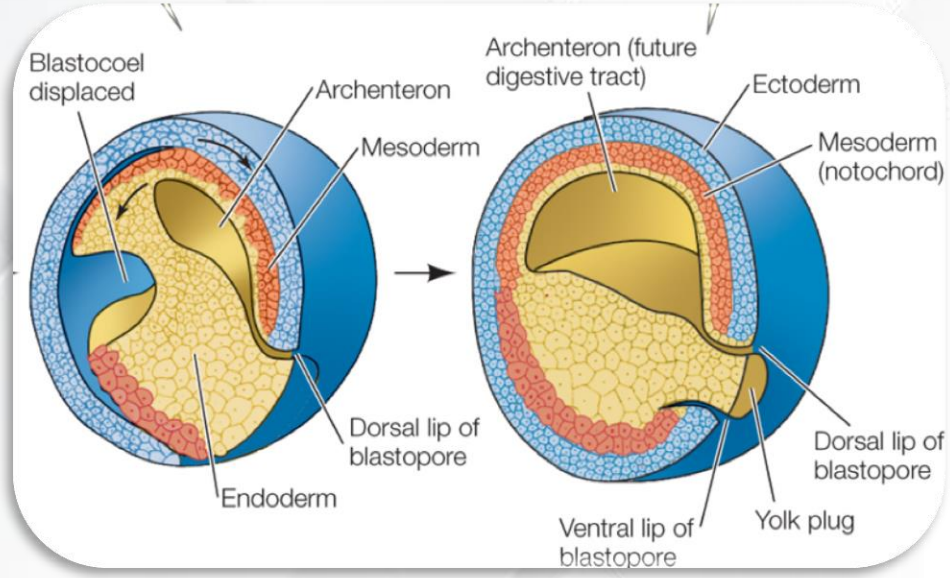
Amphibian Embryology: Gastrulation (II)

3. Mesoderm Formation

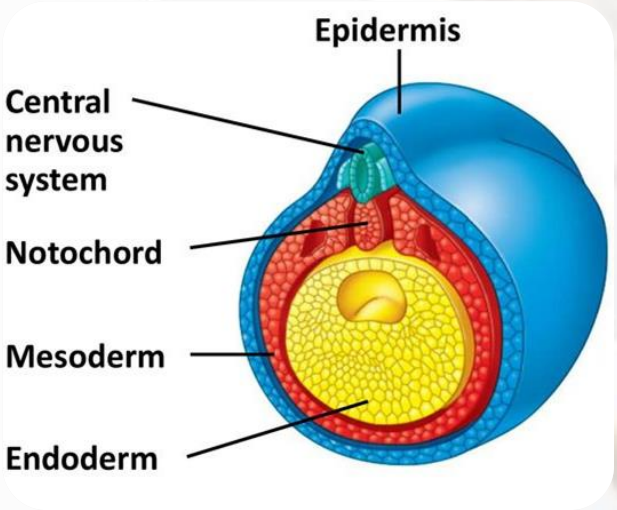
- Lateral to notochord mesodermal cell spread and thicken along sides of embryo.
- These thickenings are called **somites**
- Somites appear as row of pumps from side view
- Eventually somites spilt to form coelom

Amphibian Embryology: Gastrulation (II)

3. Mesoderm Formation



Ref. macmillanhighered.com



Ref. slideplayer.com

Principles of Animal life 1

Amphibian Embryology: Neurulation

Amphibian Embryology: Neurulation

Contents

1. Neurulation
2. Neural Plate
3. Neural Folds
4. Neural Tube
5. Brain
6. Neurulation process

Amphibian Embryology: Neurulation

1. Neurulation

- The process of forming neural tube is called neurulation

Amphibian Embryology: Neurulation

2. Neural Plate

- An oval shape area on the dorsal surface of embryo is formed
- Microfilament flatten and thicken this area
- This region is neural plate

Amphibian Embryology: Neurulation

3. Neural Folds

- The edges of the neural plate roll up and over the midline of neural plate
- These raised edges in longitudinal direction are called neural folds

Amphibian Embryology: Neurulation

4. Neural tube

- The raised neural folds
- Meet with each other longitudinally over the midline of neural plate
- Now neural plate is converted into a tubular structure called neural tube

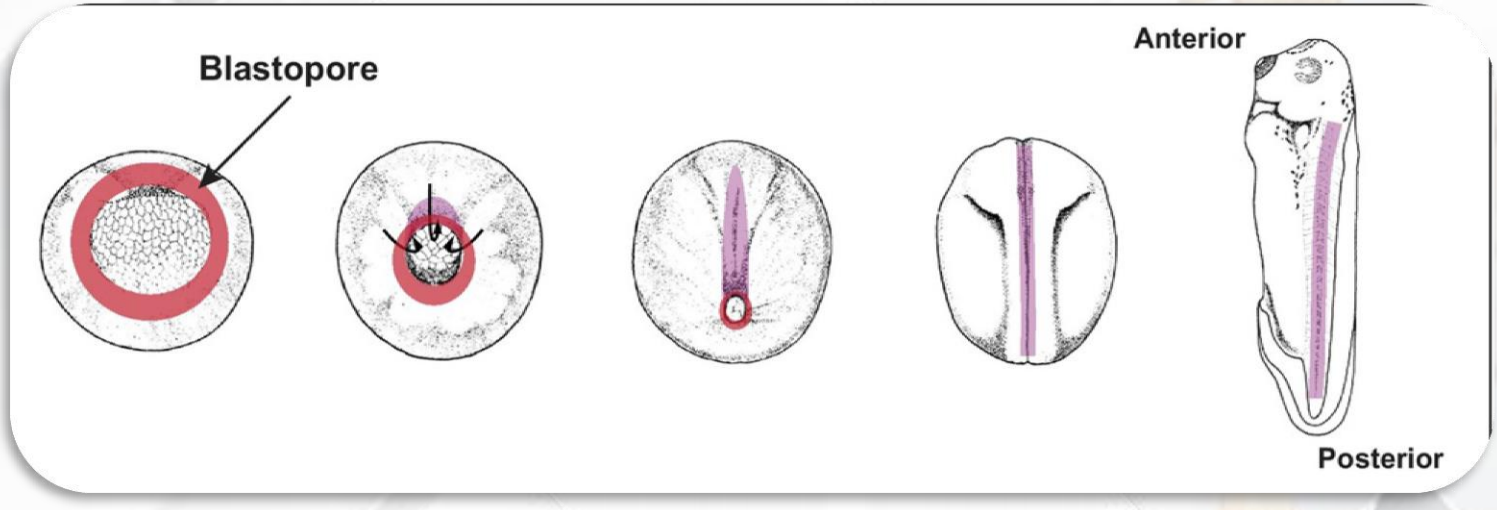
Amphibian Embryology: Neurulation

5. Brain

- The portion of neural tube that is close at the end will form brain

Amphibian Embryology: Neurulation

6. Neurulation Process



Ref. elifesciences.org

Principles of Animal life 1

Avian Embryology: General Concepts

Avian Embryology: General Concepts

Contents

1. Chick Embryo
2. Egg Type
3. Fertilization
4. Developmental Stages

Avian Embryology: General Concepts

1. Chick Embryo

- Chicken development is used as a model for the development of birds and reptiles

Avian Embryology: General Concepts

2. Egg Types

- Chicken egg contain large amount yolk
- They are macrolecithal eggs
- They are also telolecithal eggs
- Yolk is also protected by albumin $CaCO_3$ shell

Avian Embryology: General Concepts

3. Fertilization

- Fertilization is internal
- In upper portion of oviduct

Avian Embryology: General Concepts

4. Developmental Stages

- Development starts right after fertilization while egg is still in oviduct
- All developmental stage are completed within shelled egg
- In chicken development is completed in approximately 3 weeks

end

Principles of Animal life 1

Structure of Avian Egg

Structure of Avian Egg

Contents

1. Egg Formation
2. Egg White
3. Egg shell
4. Egg Structure

Structure of Avian Egg

1. Egg Formation

- In ovaries
- Large amount of yolk
- Mature egg released from ovaries
- Comes into oviduct
- Fertilization takes place in upper portion of oviduct

Structure of Avian Egg

2. Egg White

- Membranes and fluid collected around the egg while moving through oviduct
- Water and albumin is deposited around the egg
- This protect from desiccation and mechanical damage

Structure of Avian Egg

2. Egg White

- Albumin is used for protection and source of nutrition
- Two denser strands of albumin attached to the egg shell

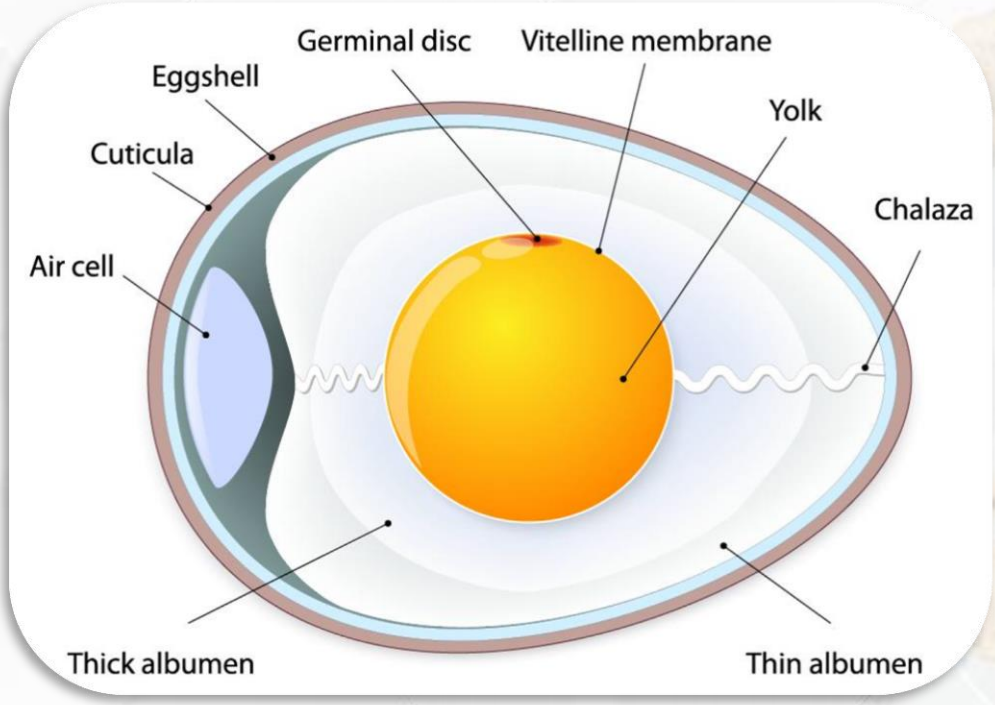
Structure of Avian Egg

3. Egg Shell

- Shell gland secrete $CaCO_3$ which deposit around the egg
- It is also impregnated with proteins
- Thousands of tiny pores (40-50µm)are for exchange of gases
- Inside shell two membranes
- Air pocket

Structure of Avian Egg

3. Egg Structure



Ref. chickens.allotment-garden.org

Principles of Animal life 1

Avian Embryology: Cleavage

Avian Embryology: Cleavage

Contents

1. Meroblastic Cleavage
2. Radial Cleavage
3. Plan of cell divisions

Avian Embryology: Cleavage

1. Meroblastic Cleavage

- Chicken eggs have large amount of yolk
- Cleavage do not divide cell in two
- There is no division in the large amount of yolk
- Discoidal cleavage

Avian Embryology: Cleavage

2. Radial Cleavage

- Cleavage lines pass in such a way that cells lie exactly one above the other.
- Radial cleave is the characteristic of deuterostomes

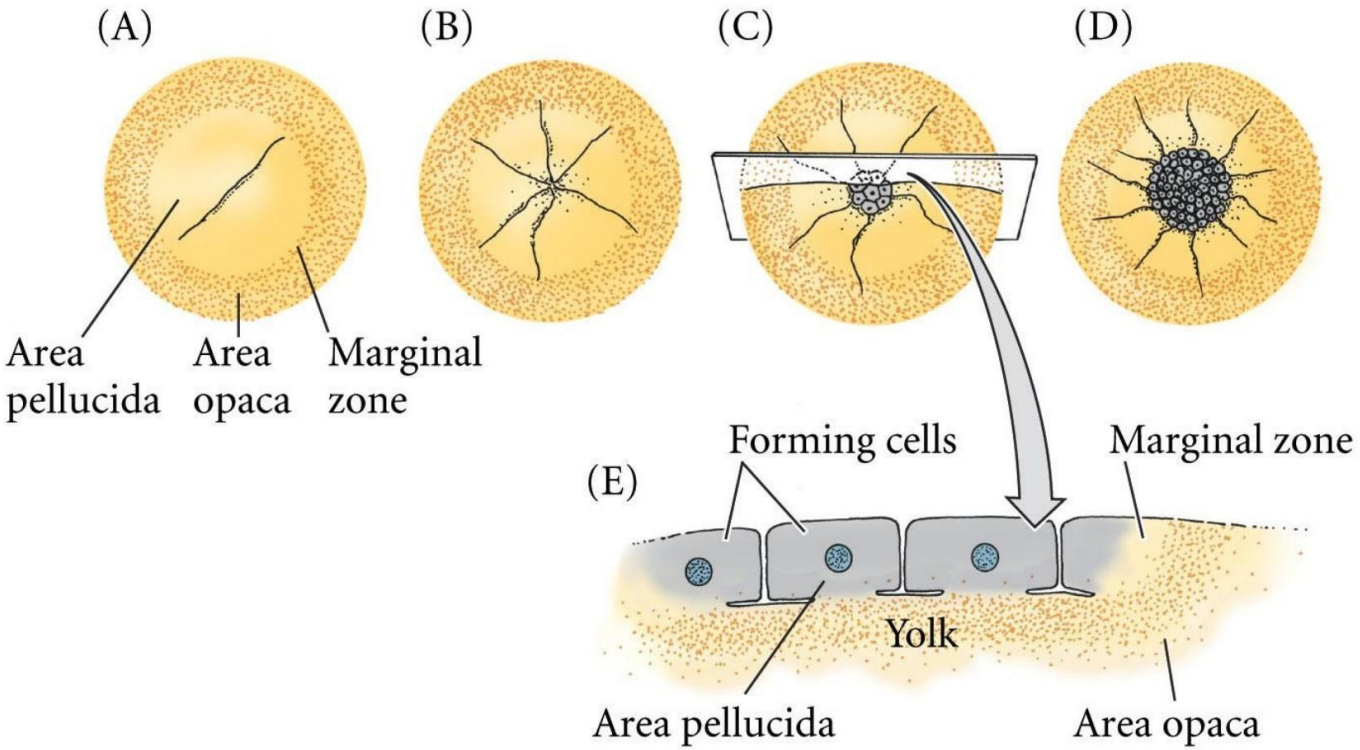
Avian Embryology: Cleavage

3. Plan of Division

- Same like other deuterostomes
- First two cleavage are vertical and perpendicular to each other
- Third cleavage is equatorial and right angle to first two. Fourth cleavage is vertical

Avian Embryology: Cleavage

Discoidal meroblastic cleavage in a chick egg



DEVELOPMENTAL BIOLOGY, Eighth Edition, Figure 11.14 © 2006 Sinauer Associates, Inc.

Principles of Animal life 1

Avian Embryology: Blastulation

Avian Embryology: Blastulation

Contents

1. Blastoderm
2. Blastocoel
3. Epiblast and Hypoblast
4. Chick Blastulation

Avian Embryology: Blastulation

1. Blastoderm

- The cells in the animal pole dived
- Approximately 60,000 cell form blastoderm

Avian Embryology: Blastulation

2. Blastocoel

- The yolk in below blastoderm is used and a fluid filled cavity appears
- This cavity is called blastocoel

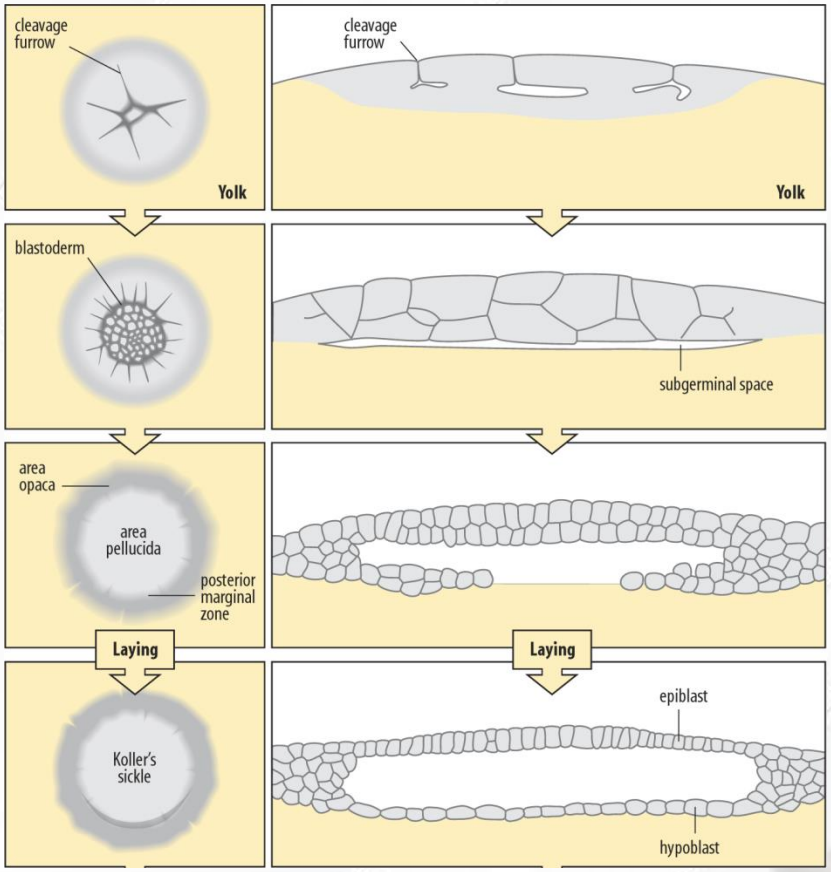
Avian Embryology: Blastulation

3. Epiblast and Hypoblast

- Blastoderm splits into two layers
- Outer layer is called epiblast
- Inner layer is hypoblast

Avian Embryology: Blastulation

4. Chick Blastulation



Ref. mun.ca/biology

Principles of Animal life 1

Avian Embryology: Gastrulation (I)

Avian Embryology: Gastrulation (I)

Contents

1. Primitive Streak
2. Henson’s Node
3. Endoderm Formation
4. Mesoderm Formation

Avian Embryology: Gastrulation (I)

1. Primitive Streak

- The dorsal cells of blastoderms form a medial linear invagination called primitive streak
- Primitive streak gradually extends anteriorly
- It defines the axis of embryo
- It also separate right and left side

Avian Embryology: Gastrulation (I)

2. Henson's Node

- The anterior margin of primitive streak there is a regional thickening called Henson's node.
- Henson's node contains funnel shaped depression through which cell enter between epiblast and hypoblast

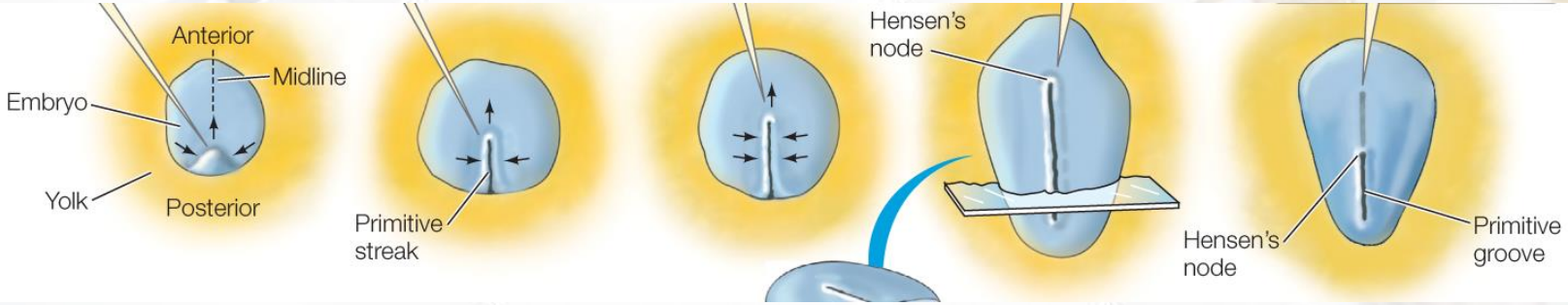
Avian Embryology: Gastrulation (I)

2. Henson's Node

- Henson's node is functionally equivalent to blastopore of amphibians

Avian Embryology: Gastrulation (I)

Primitive Streak and Henson's Node



Ref. macmillanhighered.com

Avian Embryology: Gastrulation (I)

3. Endoderm Formation

- First cells migrating through Henson's node move anteriorly and ventrally to form pharyngeal endoderm of foregut
- Hypoblast cell are moved anteriorly

Avian Embryology: Gastrulation (I)

4. Mesoderm Formation

- Next migrating cells move anteriorly, but not far away
- They remain between endoderm and epiblast to form prechordal plate mesoderm
- Next migrating cells form chordamesoderm to form notochord

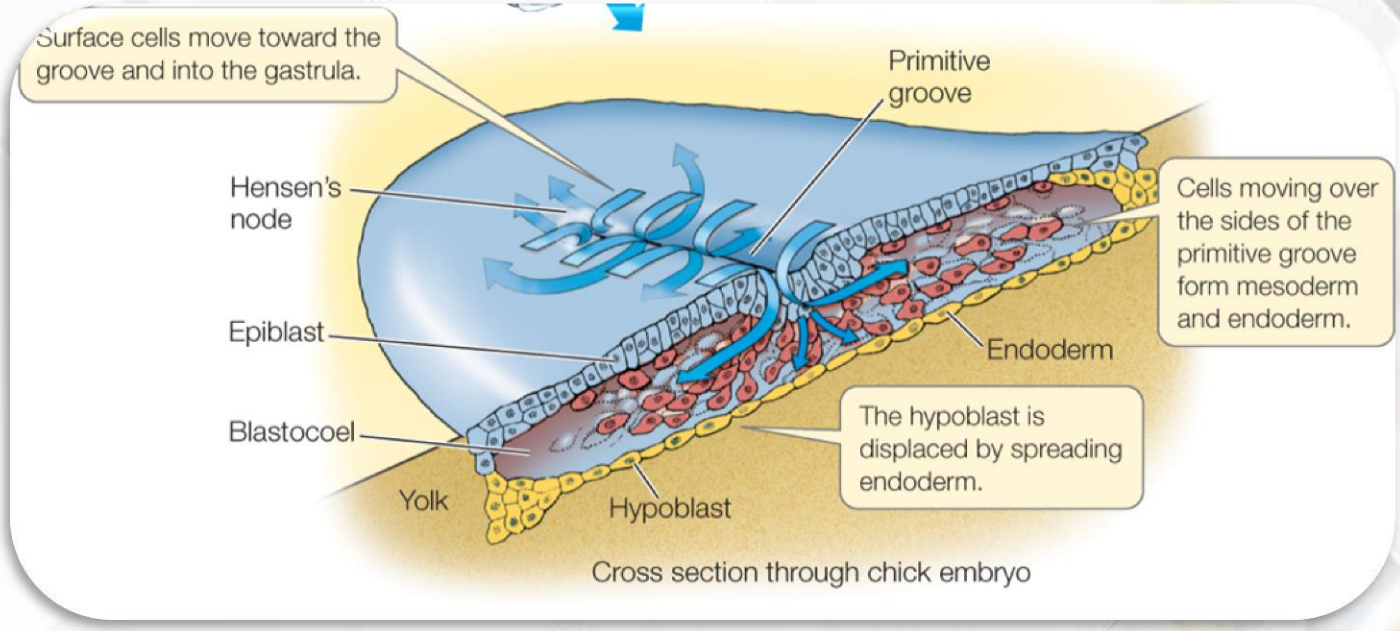
Avian Embryology: Gastrulation (I)

4. Mesoderm Formation

- Cells that ingress through middle part of streak give rise to middle part of somites and to heart and kidney
- Cell which ingress through posterior end give rise to lateral plate and extra embryonic mesoderm

Avian Embryology: Gastrulation (I)

4. Mesoderm Formation



Ref. macmillanhighered.com

Principles of Animal life 1

Avian Embryology: Gastrulation (II)

Avian Embryology: Gastrulation (II)

Contents

1. Ectoderm Formation
2. Regression of Primitive Streak and Epiboly

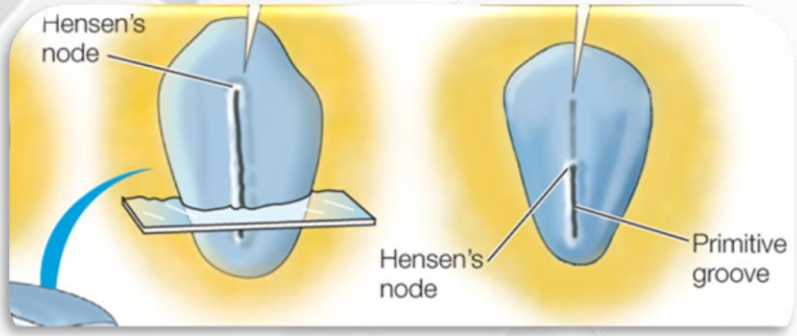
Avian Embryology: Gastrulation (II)

1. Ectoderm Formation

- After migration, the cell remaining outside form ectoderm
- Ectodermal cells close to streak form neural tube
- Ectodermal cell farther from streak will form epidermis

Avian Embryology: Gastrulation (II)

2. Regression of Primitive Streak and Epiboly



Ref. macmillanhighered.com

Avian Embryology: Gastrulation (II)

2. Regression of Primitive Streak and Epiboly

- As cells migrate, primitive streak regresses
- Henson's node attains a more posterior position
- By time endodermal and mesodermal cells have attain their position

Avian Embryology: Gastrulation (II)

2. Regression of Primitive Streak and Epiboly

- Now the epiblast cells are all ectodermal cells
- The ectodermal cell migrate and enclose the yolk
- This process is called epiboly

Principles of Animal life 1

Mammalian Embryology: General Concepts

Mammalian Embryology: General Concepts

Contents

1. Mammalian Egg
2. Fertilization and Development
3. Developmental Stages

Mammalian Embryology: General Concepts

1. Mammalian Eggs

- Egg laying mammals… large eggs
- Marsupial mammals… microlecithal eggs
- Placental mammals… alecithal eggs
- Placental mammals egg is smallest in animals kingdom

Mammalian Embryology: General Concepts

2. Fertilization and Development

- Internal fertilization
- Placental and marsupial mammals have internal development
- Both give birth to young ones

Mammalian Embryology: General Concepts

2. Fertilization and Development

- Marsupial mammals have premature birth
- Keep in marsupium
- Placental have long gestation period
- Fetus is fed through placenta

Mammalian Embryology: General Concepts

3. Developmental Stages

- Zygote
- Embryo
- Fetus
- newborn

end

Principles of Animal life 1

Mammalian Embryology: Cleavage and Blastulation

Mammalian Embryology: Cleavage and Blastulation

Contents

1. Cleavage
2. Compaction
3. Morulation
4. Totipotent and Pluripotent
5. Blastula
6. Diagrammatic Representation

Mammalian Embryology: Cleavage and Blastulation

1. Cleavage

- Mammalian cleavage is slowest in animal kingdom
- Blastomeres have unique orientation
- Rotational cleavage
- First cleavage is meridional while in second cleavage one is meridional and other horizontal

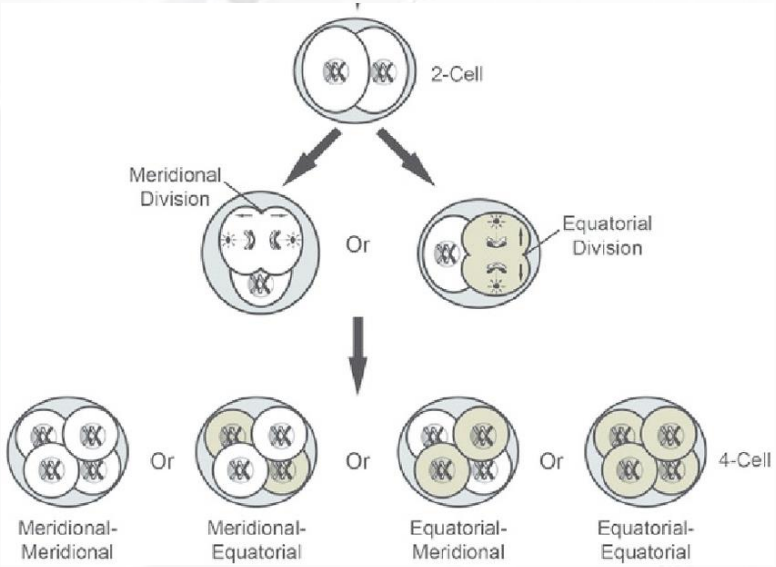
Mammalian Embryology: Cleavage and Blastulation

1. Cleavage

- Cell division is asynchronous in the very beginning
- Cell are not 2-4-8, but may be odd number
- For proteins synthesis cell switches from maternal to zygote control at 2 cell stage

Mammalian Embryology: Cleavage and Blastulation

1. Cleavage



Ref. Hasley et al., 2017. Vertebrate Embryonic Cleavage Pattern Determination

Mammalian Embryology: Cleavage and Blastulation

2. Compaction

- Through 8 cell stage, loose arrangement with plenty of space
- After third cleavage the cells tightly huddle together to form a compact ball.

Mammalian Embryology: Cleavage and Blastulation

3. Morula

- 8 cells divide to produce 16 cell morula
- A small group of internal cells is called **inner cell mass** (ICM) while the outer cells form **trophoblast**
- ICM form embryo while trophoblast forms chorion for placenta formation

Mammalian Embryology: Cleavage and Blastulation

4. Totipotent and Pluripotent

- 8 cells stage blastomere can form both ICM and trophoblast, they are called totipotent
- ICM cells can form any type of embryo cells, they are called pluripotent

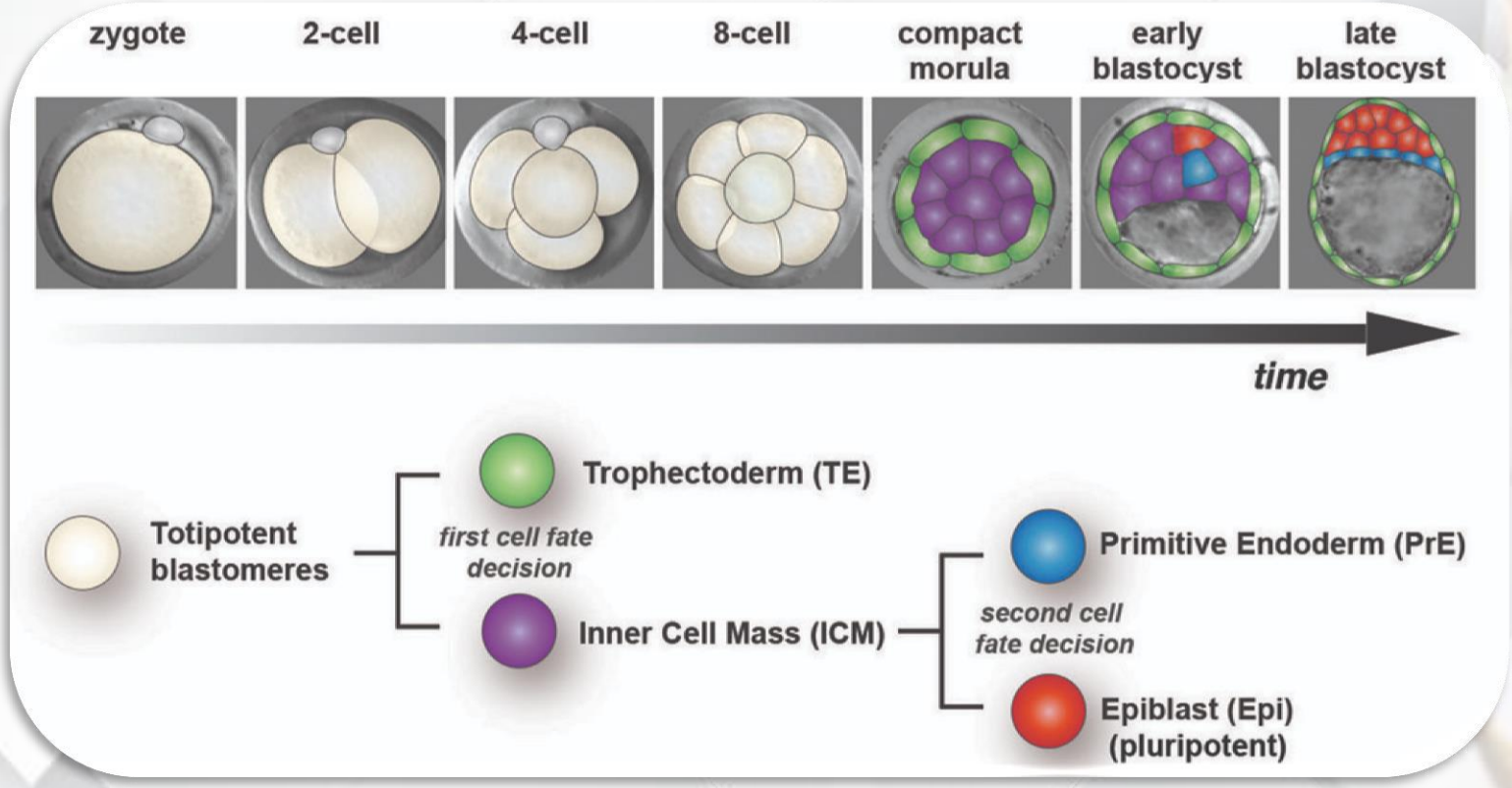
Mammalian Embryology: Cleavage and Blastulation

5. Blastula

- Trophoblast cells of morula secrete fluid that create blastocoel
- As blastocoel expands ICM and trophoblast take their positions
- At this stage blastula is called blastocyts

Mammalian Embryology: Cleavage and Blastulation

6. Mammalian Blastulation



Ref. mskcc.org

Principles of Animal life 1

Mammalian Embryology: Gastrulation

Mammalian Embryology: Gastrulation

Contents

1. Similarity with Birds and Reptiles
2. Epiblast and Hypoblast
3. Embryomic Layers Formation

Mammalian Embryology: Gastrulation

1. Similarity with Reptiles and Birds

- Both mammals and birds originate from reptile
- Despite difference in amount of yolk both have striking similarity in development

Mammalian Embryology: Gastrulation

2. Epiblast and Hypoblast

- First segregation of cells within inner cell mass to form hypoblast and hypoblast
- Both epiblast and hypoblast form a structure **bilaminar germ disc**

Mammalian Embryology: Gastrulation

2. Epiblast and Hypoblast

- Hypoblast cells form extraembryonic endoderm which form yolk sac
- Epiblast form embryonic endoderm and amnion cavity

Mammalian Embryology: Gastrulation

3. Embryonic layers Formation

- Epiblast cells move through primitive streak for form endoderm and mesoderm
- The cell anterior and lateral to primitive streak form ectoderm

end

Principles of Animal life 1

Hormonal Regulation in Pregnant Female

Hormonal Regulation in Pregnant Female

Contents

1. Human Chorionic Gonadotropin
2. Progesterone
3. Prolactin
4. Oxytocin
5. Prostaglandins

Hormonal Regulation in Pregnant Female

1. Human Chorionic Gonadotropin (hCG)

- The cells developing embryo and placenta release hCG
- hCG keeps the corpus luteum from disintegration
- High level of hCG is used as immunological for pregnancy by 2 weeks

Hormonal Regulation in Pregnant Female

2. Progesterone

- Progesterone is released by corpus luteum
- Progesterone maintains the endometrium lining
- This retains pregnancy

Hormonal Regulation in Pregnant Female

3. Prolactin

- It is secreted by pituitary gland.
- It is produced in pregnant female to start the process of milk production and eject milk after childbirth

Hormonal Regulation in Pregnant Female

4. Oxytocin

- It is also produced by pituitary gland
- It caused contraction of smooth muscles to start labour and expulsion of young one
- It also stimulate nipples to release milk

Hormonal Regulation in Pregnant Female

5. Prostaglandins

- Prostaglandins secretion increase during labour
- It also causes smooth muscle contraction

end

Principles of Animal life 1

Prenatal Development in Human: 1st Trimester

Prenatal Development in Human: 1st Trimester

Contents

1. Embryo, Fetus and Newborn
2. Trimesters
3. First Trimester
4. Implantation of embryo
5. Gastrulation
6. Organogenesis and Growth

Prenatal Development in Human: 1st Trimester

1. Embryo, Fetus and Newborn

- After two weeks of fertilization till the end of 8th week is embryo
- From 9th week till birth is fetus
- After the birth is newborn or baby

Prenatal Development in Human: 1st Trimester

2. Timesters

- Total gestation period is 280 days
- This duration is divided into 3 months periods called trimesters

Prenatal Development in Human: 1st Trimester

3. First Trimester

- Starts from fertilization
- Zygote divides to become morula
- By 4th day it is converted into 50- 120 cell blastocyst

Prenatal Development in Human: 1st Trimester

4. Implantation

- As embryo reaches into uterus, it implants itself onto uterine wall
- Trophoblast invade the endometrium

Prenatal Development in Human: 1st Trimester

5. Gastrulation

- During gastrulation embryonic layers are formed
- Inner cell mass give rise to embryo
- Gastrulation is similar to birds and reptiles

Prenatal Development in Human: 1st Trimester

6. Organogenesis and Growth

- Once gastrulation completed organogenesis starts
- Different organ systems are shaped
- By middle of first trimester all body organ begin to develop
- Embryo also grows in size

end

Principles of Animal life 1

Prenatal Development in Human: 2nd Trimester

Prenatal Development in Human: 2nd Trimester

Contents

1. Second trimester
2. Major Events of Second Trimester

Prenatal Development in Human: 2nd Trimester

1. Second Trimester

- It ranges from 14 to 27 week

Prenatal Development in Human: 2nd Trimester

2. Major Events of Second Trimester

- Rapid growth of fetus
- 14 week… RBC formation, sex apparent
- 16 week….eyes move
- 17 week….Toenail develop, heart beat
- 18 week…. hearing

Prenatal Development in Human: 2nd Trimester

2. Major Events of Second Trimester

- 20 week….. Feel movement, sleep wake
- 21 week… baby sucks finger
- 22 week…. Eyebrow and hair visible
- 23 week…. Finger prints form
- 26 week… Lung develop

end

Principles of Animal life 1

Prenatal Development in Human: 3rd Trimester

Prenatal Development in Human: 3rd Trimester

Contents

1. Third Trimester
2. Developmental Changes

Prenatal Development in Human: 3rd Trimester

1. Third Trimester

- It start from 7th month till birth
- Fetus has potential to survive if born premature

Prenatal Development in Human: 3rd Trimester

2. Developmental Changes

- 27 week….. Eyes partially open
- 30 week… Baby hair grow
- 31 week…. Rapid weight gain
- 32 week…. Breathing practice
- 34 week…. Fingernails grow

Prenatal Development in Human: 3rd Trimester

2. Developmental Changes

- 35 week…. Skin pink and smooth
- 37 week… Head turn down
- 38 week….. Toe nail grow
- 39 week….. Chest prominent, testis continue to descend
- 40 week….Birth

end

Principles of Animal life 1

The Placenta

The Placenta

Contents

1. Placenta
2. Placenta Formation
3. Role of Placenta
4. Umbilical Cord
5. Diagrammatic Representation

The Placenta

1. Placenta

- Placenta is s temporary organ connects developing fetus to mother via umbilical cord

The Placenta

2. Placenta Formation

- Human have placenta formed form chorion and allantois
- The tiny chorionic villi sent out from blastocyst form placenta
- Placenta formation starts from day 6 and is completed by 1st trimester

The Placenta

2. Placenta Formation

- Embryonic blood vessels of placenta do not merge with maternal blood vessels
- It is about 22cm in length

The Placenta

3. Role of Placenta

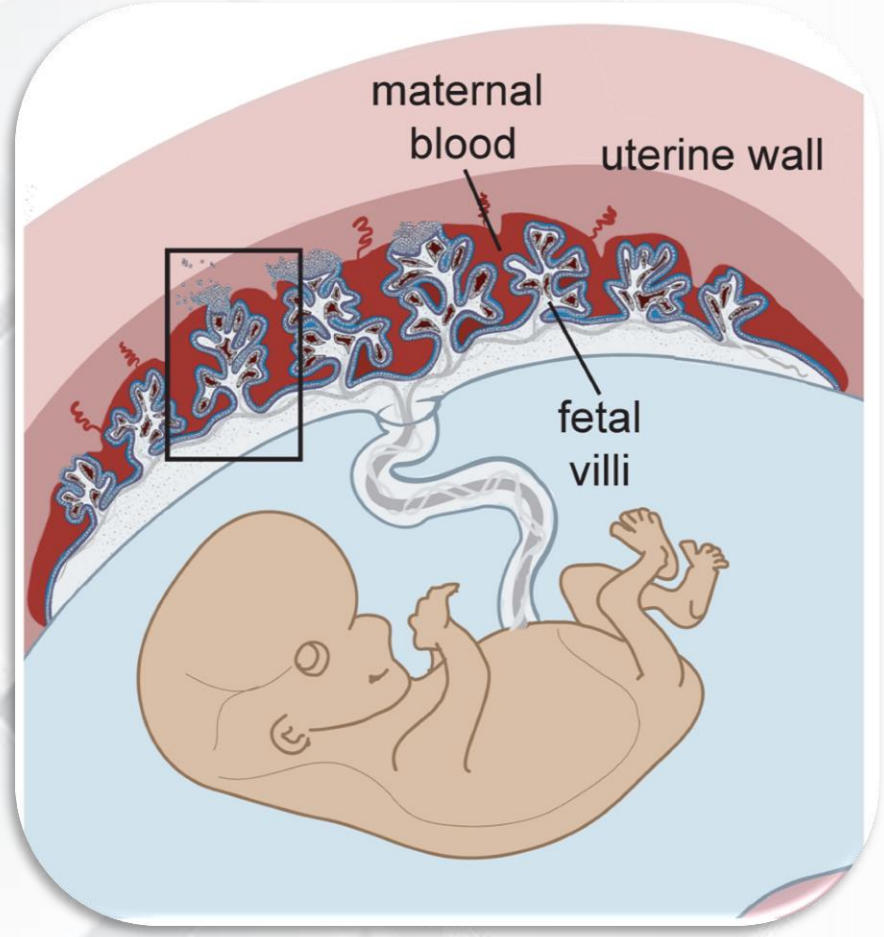
- Placenta is used for:
  - Nutrient uptake
  - Thermoregulation
  - Waste elimination
  - Gaseous exchange
  - Fight against internal infection
  - Placental hormone production

The Placenta

4. Umbilical cord

- Placenta is connected to fetus through umbilical cord
- It contain two fetal umbilical arteries and one fetal umbilical vein
- They spiral about each other in umbilical cord
- Average size is 55cm with 1-2cm diameter

The Placenta



5. Diagrammatic Representation

Ref. pmgbiology.com

Principles of Animal life 1

Birth in Human

Birth in Human

Contents

1. Gestation Period
2. Parturition
3. Relaxin Hormone
4. ACTH Stimulation
5. Oxytocin
6. Birth and Afterbirth
7. Stages of Labor and Parturition

Birth in Human

1. Total Gestation

- It is the total time period which embryo or fetus spends within body of mother
- In human it is 280 days or 40 weeks

Birth in Human

2. Parturition

- The process of birth is called parturition
- During parturition, mother's uterine muscles begin to contract and cervix begins to dilate

Birth in Human

3. Relaxin Hormone

- Mother ovaries and placenta produce relaxin hormone
- Relaxin causes mother pelvic bone to slightly separate, so the baby may pass from birth canal

Birth in Human

4. ACTH Stimulation

- During birth baby pituitary gland secretes Adrenocorticotropic hormone (ACTH)
- ACTH stimulates adrenal glands to secrete prostaglandins
- Prostaglandins work along with oxytocin

Birth in Human

5. Oxytocin

- Stretch in uterine wall stimulate pituitary to secrete oxytocin hormone in mother
- Oxytocin and prostaglandins cause powerful muscular contractions
- Muscular contraction increases over time

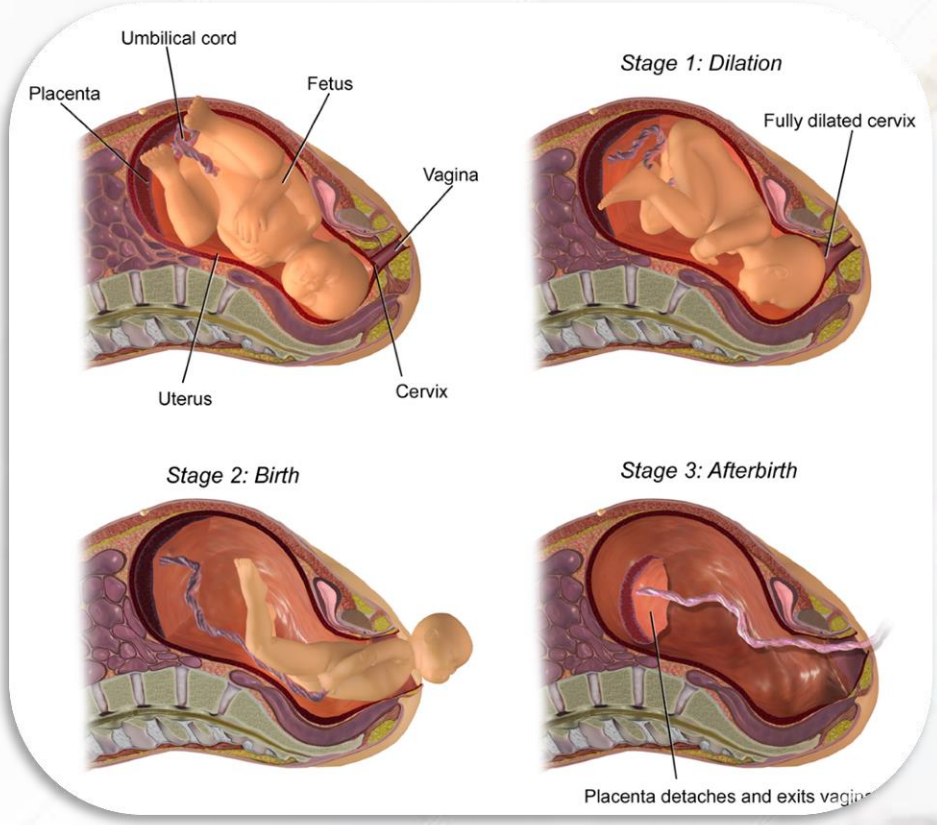
Birth in Human

6. Birth and Afterbirth

- Usually within an hour of muscular contraction baby is expelled out from uterus
- After baby comes out the process of contraction continue to expel placenta, the process is called **Afterbirth**

Birth in Human

7. Stages of Labor and Parturition



Ref. Miller and Harley. Zoology. 4th Edition. p. 694

Principles of Animal life 1

Multiple Births

Multiple Births

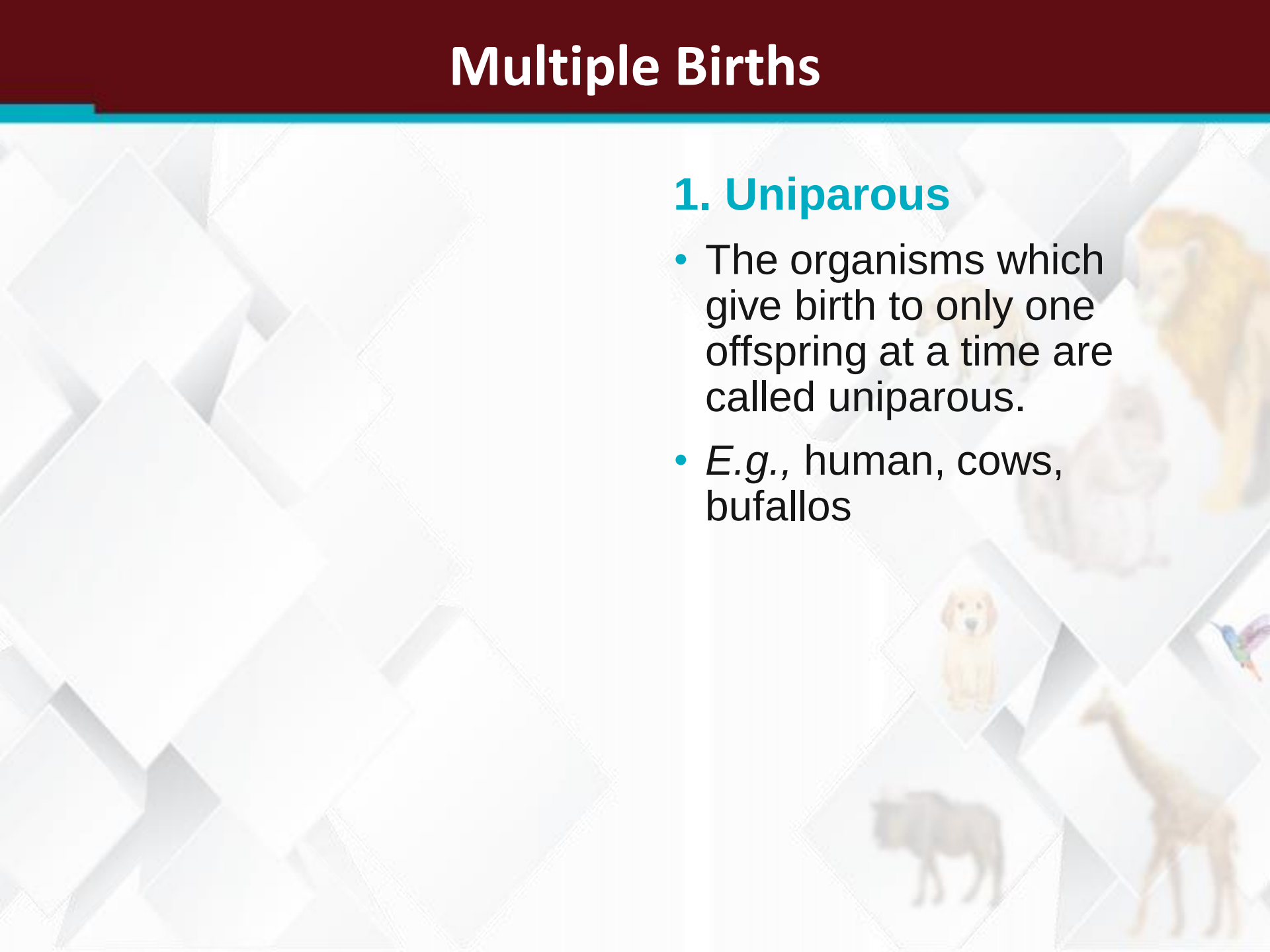
Contents

1. Uniparous
2. Multiparous
3. Monozygotic Twins
4. Dizygotic Twins

Multiple Births

1. Uniparous

- The organisms which give birth to only one offspring at a time are called uniparous.
- *E.g.,* human, cows, bufallos



Multiple Births

2. Multiparous

- These are animals giving birth to more than one young ones
- *E.g.,* mice, rat, dogs, cats
- Usually the young ones are both male and female. However armadillo give birth to four young ones at same time, all males or females

Multiple Births

3. Monozygotic Twins

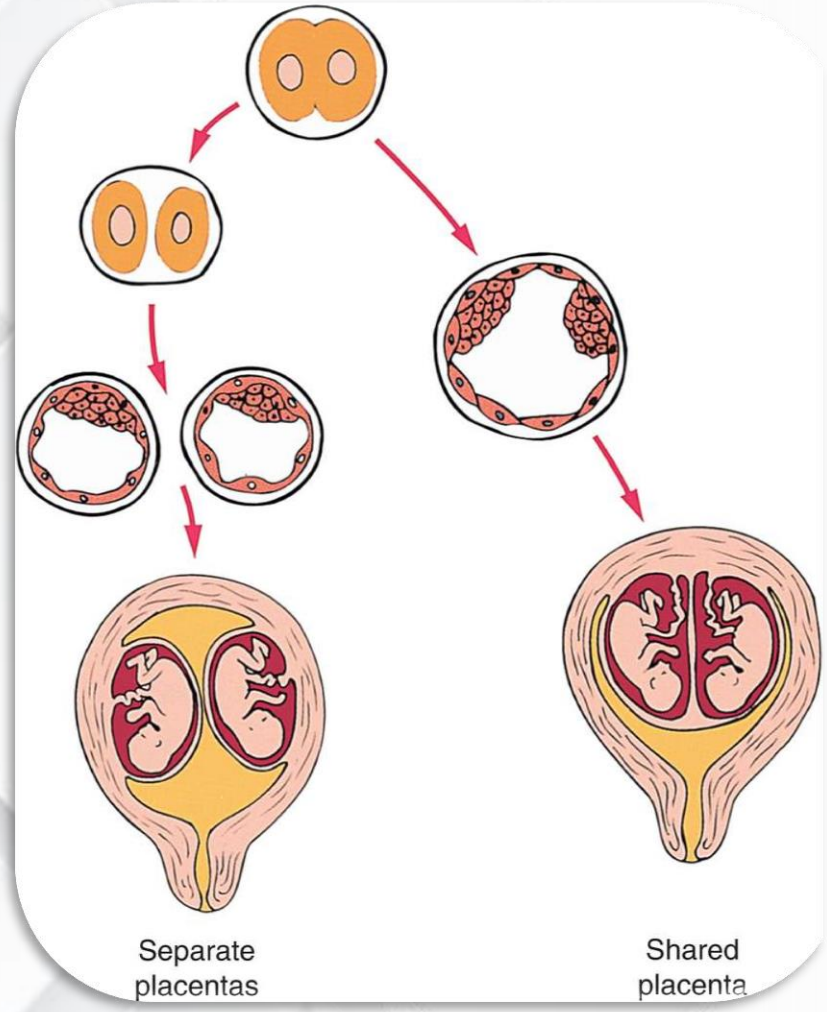
- Produced from one zygote (identical twins)
- They are strikingly alike and of same sex
- About 33% have separate placenta
- Majority have separate amniotic sac

Multiple Births

3. Monozygotic Twins

- Small proportion have one placenta and one amniotic sac
- Indicate separation occur after 9 day of pregnancy
- Conjoining risk, condition is Siamese twining

Multiple Births



3. Monozygotic Twins

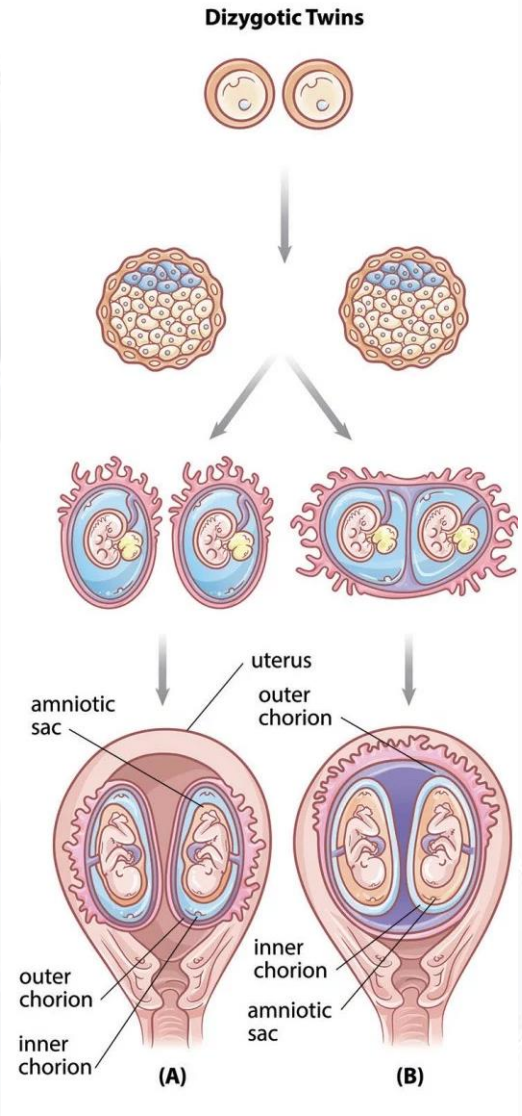
Ref. Carlson, 2009

Multiple Births

4. Dizygotic Twins

- Produced from two zygote (Fraternal twins)
- Do not have striking similarities
- May be of same or different sex
- They have separate placenta and separate amniotic sac

Multiple Births



4. Dizygotic Twins

Ref. b2bpodcast.home.blog

Principles of Animal life 1

Milk Production and Lactation

Milk Production and Lactation

Contents

1. Mammary Glands
2. Lactation
3. Hormonal control of Milk Production
4. Colostrum

Milk Production and Lactation

1. Mammary Glands

- Mammary glands are unique feature of mammals
- Mammary glands are evolved from sweat glands in the skin

Milk Production and Lactation

2. Lactation

- It is the process of milk production and release by the mammary glands

Milk Production and Lactation

3. Hormonal Control of Milk Production

- During pregnancy breasts enlarges due to hormone prolactin
- Estrogen and progesterone inhibit the secretion of milk before birth
- After birth both decrease and milk production starts

Milk Production and Lactation

3. Hormonal Control of Milk Production

- Newborn stimulate pituitary to release oxytocin and prolactin
- Prolactin produce milk while oxytocin triggers its release from mammary glands

Milk Production and Lactation

4. Colostrum

- It is a higher protein fluid in breast at the time of birth
- It contains maternal antibodies to boost immune system
- It works as laxative to remove meconium
- Mother’s breast don’t release milk for 1-3 days

end

Principles of Animal life 1

Extraembryonic Membranes

Extraembryonic Membranes

Contents

1. Amniotes and Anamniotes
2. Role of Extraembryonic Membranes
3. Development of Extraembryonic Membranes

Extraembryonic Membranes

1. Amniotes and Anamniotes

- Reptiles, birds and mammals embryo have amnion, chorion and allantois are called amniotes
- Eggs of fishes and amphibians do not have amnion, chorion and allantois are called anamniotes

Extraembryonic Membranes

2. Role of Extraembryonic Membranes

- Amnion: encloses embryo in a fluid filled sac. Protects against shock and desiccation
- Chorion: near to shell, mechanical support, highly vascular, gaseous exchange

Extraembryonic Membranes

2. Role of Extraembryonic Membranes

- Allantois: storage and excretion of toxic wastes, also for gaseous exchange
- Yolk sac: distributes nutrients to developing embryo

Extraembryonic Membranes

3. Development of Extraembryonic Membrane

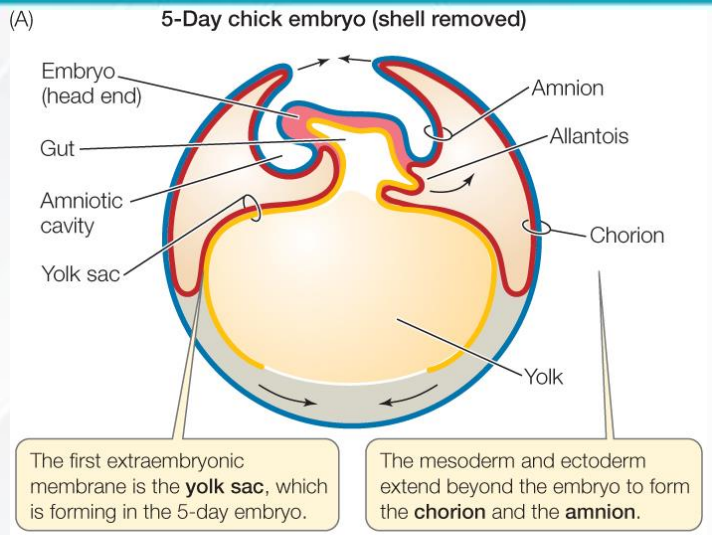
- Ectoderm and mesoderm on both side of embryo lift off the yolk and grow dorsally, they fuse to form chorion (outer side) and amnion (inner side)

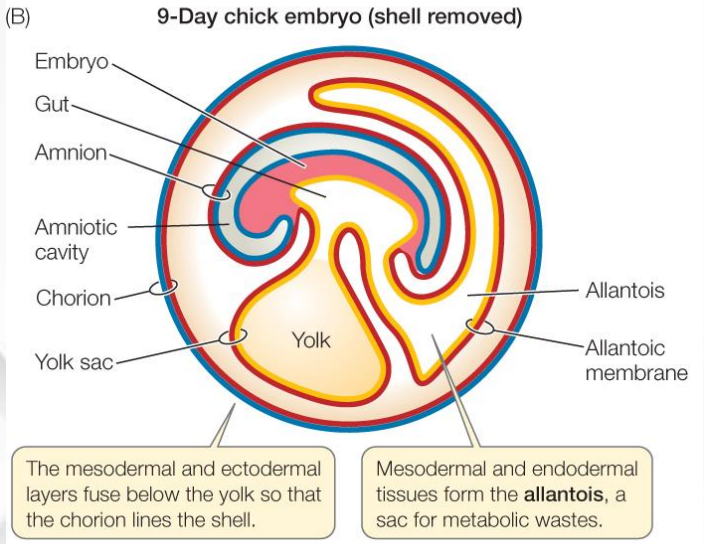
Extraembryonic Membranes

3. Development of Extraembryonic Membrane

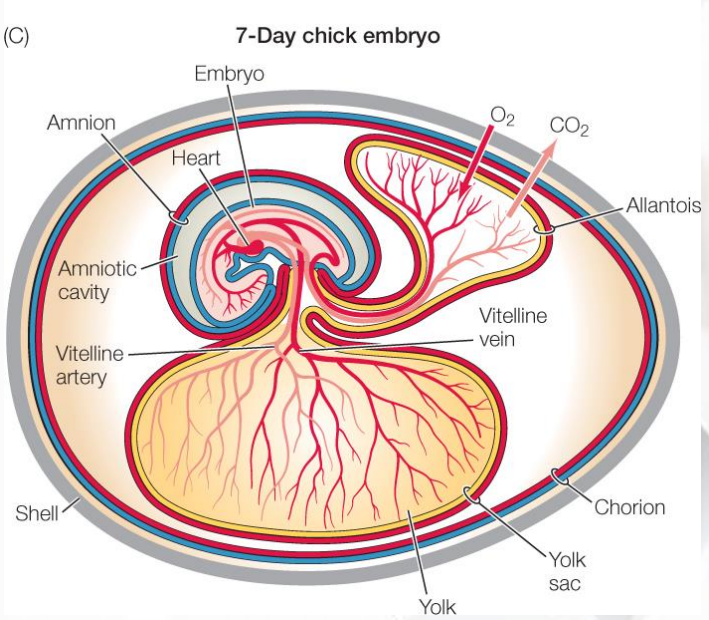
- A ventral outgrowth of gut tract give rise to allantois. It gradually increase in its size along with development
- Yolk sac develops from proliferation of endoderm and mesoderm around yolk

Extraembryonic Membranes





3. Development of Extraembryonic Membrane



Ref. macmillanhighered.com

Principles of Animal life 1

Primary Germ Layers: Ectoderm Derivatives

Primary Germ Layers: Ectoderm Derivatives

Contents

1. Primary germ layers
2. Formation of Ectoderm
3. Development of nervous system
4. Development of epidermis

Primary Germ Layers: Ectoderm Derivatives

1. Primary Germ Layers

- During gastrulation three germ layers are formed
  - Ectoderm
  - Mesoderm
  - Endoderm
- These germ layers give rise to different body parts

Primary Germ Layers: Ectoderm Derivatives

2. Formation of Ectoderm

- It is the outer most layer of cells formed during gastrulation
- Ectoderm is most exterior layer of embryo

Primary Germ Layers: Ectoderm Derivatives

3. Development of Nervous System

- The ectodermal cells aligned themselves into a columnar pattern to form **neural plate**
- The edges of neural plate are raised to form **neural folds**
- Neural plate is converted into a depression called **neural groove**

Primary Germ Layers: Ectoderm Derivatives

3. Development of Nervous System

- Both neural folds merge with each other in a longitudinal fashion to form neural tube.
- Neural tube will form central nervous system
- While cells of neural crest will form peripheral nervous system

Primary Germ Layers: Ectoderm Derivatives

4. Development of Epidermis

- Epidermis is the outer most layer of skin
- Below epidermis is endodermis and hypodermis
- After neurulation all ectodermal cells form epidermis.
- Initially single layered, quickly form two layers

Primary Germ Layers: Ectoderm Derivatives

4. Development of Epidermis

- The basal layer or germinal epithelium give rise to all epidermal cells
- Epidermis also give rise to:
  - hair,
  - sebaceous glands,
  - sweat glands,
  - Nails, claws, hoofs

end

Principles of Animal life 1

Primary Germ Layers: Endoderm Derivatives

Primary Germ Layers: Endoderm Derivatives

Contents

1. Formation of Endoderm
2. Digestive Tract
3. Respiratory Tract
4. Endocrine Glands
5. Auditory System
6. Urinary System

Primary Germ Layers: Endoderm Derivatives

1. Endoderm

- Endoderm is the inner most primary germ layer formed during gastrulation

Primary Germ Layers: Endoderm Derivatives

2. Digestive Tract

- The entire gastrointestinal tract except mouth, pharynx and anus develop from endoderm
- The lining of all glands which are part digestive system
- Liver, gall bladder and pancreas

Primary Germ Layers: Endoderm Derivatives

3. Respiratory Tract

- Lungs
  - Trachea
  - Bronchi
  - Alveoli

Primary Germ Layers: Endoderm Derivatives

4. Endocrine Glands

- Thyroid
- Parathyroid
- Thymus
- Pancreas

Primary Germ Layers: Endoderm Derivatives

5. Auditory System

- Auditory tube
- Tympanic cavity

Primary Germ Layers: Endoderm Derivatives

6. Urinary System

- Urinary bladder
- Vagina
- Urethra

end

Principles of Animal life 1

Primary Germ Layers: Mesoderm Derivatives

Primary Germ Layers: Mesoderm Derivatives

Contents

1. Formation of Mesoderm
2. Coelom Formation
3. Skeletal System
4. Urinogenital Structures
5. Circulatory System
6. Muscular System
7. Skin

Primary Germ Layers: Mesoderm Derivatives

1. Mesoderm

- Mesoderm is one of the primary germ layer formed during gastrulation
- It is present between ectoderm and endoderm

Primary Germ Layers: Mesoderm Derivatives

2. Coelom Formation

- It is the body cavity surrounded by mesodermal cells
- Coelom may be formed by out pouching or splitting of mesoderm

Primary Germ Layers: Mesoderm Derivatives

3. Skeletal System

- Cartilage
- Bones

Primary Germ Layers: Mesoderm Derivatives

4. Urinogenital Structures

- Kidneys
- Gonads
- Associated ducts
- Adrenal glands

Primary Germ Layers: Mesoderm Derivatives

5. Circulatory System

- Blood cells
  - RBCs
  - WBCs
  - Platelets
- Blood vessels
  - Arteries
  - Veins
  - Capillaries
- Heart

Primary Germ Layers: Mesoderm Derivatives

6. Muscular Tissues

- Skeletal muscles
- Smooth muscles
- Cardiac muscles

Primary Germ Layers: Mesoderm Derivatives

7. Skin

- Dermis
- Hypodermis

Primary Germ Layers: Mesoderm Derivatives

7. Other Tissues

- Tendons
- Ligaments
- Adipose tissus

end

Principles of Animal life 1

Mechanism of Development: Nuclear equivalence

Mechanism of Development: Nuclear equivalence

Contents

1. Roux-Weismann Hypothesis
2. Hans Driesch Discovery
3. Hans Spemann Discovery

Mechanism of Development: Nuclear equivalence

1. Roux-Weismann Hypothesis

- Nineteenth century
- How developing embryo develop multiple types of cells?
- Heredity material is distributed unequally
- Genome is broken down into smaller and smaller fragment

Mechanism of Development: Nuclear equivalence

2. Hans Driesch Discovery

- 1892
- If two cell stage sea urchin embryo is mechanically shaken to get cells apart, both develop into normal larvae
- Conclusion: both cells contain same genetic material

Mechanism of Development: Nuclear equivalence

3. Hans Spemann Discovery

- Placed a minute ligature of hair around salamander zygote
- Nucleus containing portion divides
- Later at 16 cell stage, one nucleus move to anucleated portion and it also start cell division
- When separated both developed normally

Mechanism of Development: Nuclear equivalence

3. Hans Spemann

- Both halves developed normally only if both receive gray crescent
- This means cytoplasmic contents paly their role in formation of different types of cells

end

Principles of Animal life 1

Mechanism of Development: Cytoplasmic Specification

Mechanism of Development: Cytoplasmic Specification

Contents

1. Morphogenetic Determinants
2. Tunicates Example
3. Vertebrate Embryo
4. Mosaic Development
5. Regulative Development

Mechanism of Development: Cytoplasmic Specification

1. Morphogenetic Determinants

- Cytoplasmic components that are unequally distributed within egg
- These determinants control the development of particular cell type
- Fate of blastomeres is specific by type of cytopalsm

Mechanism of Development: Cytoplasmic Specification

2. Tunicate Example

- Some tunicates contain different colours in cytoplasm
- **Yellow:** muscle cells
- **Gray equatorial:** notochord and neural tube
- **Clear:** larval epidermis
- **Gray vegetal:** gut

Mechanism of Development: Cytoplasmic Specification

3. Vertebrate Embryo

- Cytoplasmic specification is relatively less important
- Gray crescent form dorsal lip of blastopore, is essential for gastrulation

Mechanism of Development: Cytoplasmic Specification

4. Mosaic Development

- When a particular blastomere is isolated form embryo, it forms characteristic structure
- In the absence of that cell the structure is missing
- Mosaic development is characteristic of most protostomes

Mechanism of Development: Cytoplasmic Specification

5. Regulative Development

- In many animals, fate of cell depends upon interaction with neighbors rather than cytoplasm itself
- Each cell is able to produce entire embryo
- If a blastomere is removed, the other change their fate to compensate this.

end

Principles of Animal life 1

Mechanism of Development: Embryonic Induction

Mechanism of Development: Embryonic Induction

Contents

1. Embryonic Induction
2. Spemann and Mangold experiment
3. Primary induction
4. Secondary Induction

Mechanism of Development: Embryonic Induction

1. Embryonic Induction

- The capacity of some cells to evoke developmental process in others
- It is wide spread phenomenon in development

Mechanism of Development: Embryonic Induction

2. Spemann and Mangold Experiment

- Embryonic induction was first reported
- The dorsal lip of blastopore when transplanted to later or ventral side of other embryo, it developed notochord, somite
- Also ectoderm of host
- Whole system formed

Mechanism of Development: Embryonic Induction

3. Primary Induction

- Dorsal lip of blastopore can induce the development of whole embryo, it is called as primary organizer
- The phenomenon of induction by blastopore is called primary induction.

Mechanism of Development: Embryonic Induction

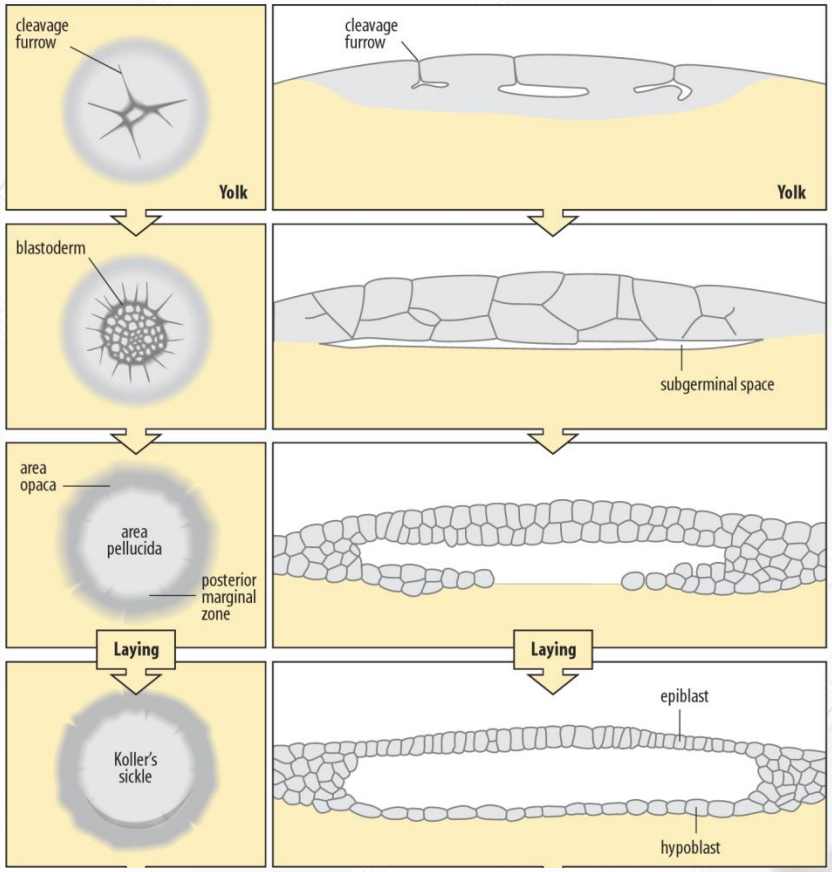
4. Secondary Induction

- Formation of other type of cell by inductive process is termed as secondary induction

end

Mechanism of Development: Embryonic Induction

4. Chick Blastulation



Ref. mun.ca/biology

Principles of Animal life 1

Development: Protostome Vs Deuterostome (I)

Development: Protostome Vs Deuterostome (I)

Contents

1. Protostomes
2. Deuterostomes
3. Cleavage Pattern
4. Determinate and Indeterminate Cleavage

Development: Protostome Vs Deuterostome (I)

1. Protostomes

- Protostomes belong to group protostomia of animal kingdom
- Together with deuterostomes,they are part of grade bilateria
- Member phyla: Platyhelminthes, Rotifera, Nematoda, Annelida, arthropoda, Mollusca,

Development: Protostome Vs Deuterostome (I)

2. Deuterostomes

- Deuterstomes belong to group deuterostomia of animal kingdom
- Together with protostomes,they are part of grade bilateria
- Member phyla: Echinodermata, Hemichordata, Chordara

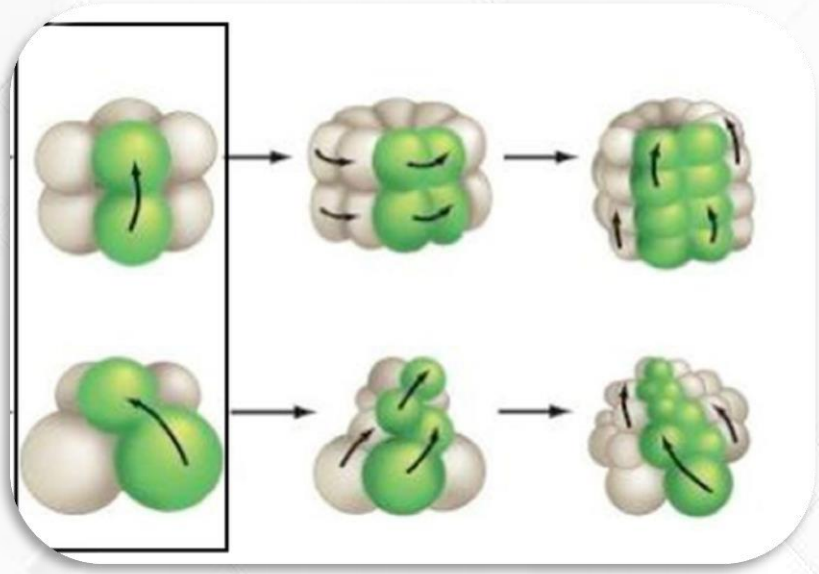
Development: Protostome Vs Deuterostome (I)

3. Cleavage Pattern

- Both differ in cleavage pattern.
- **Protostomes:** Spiral cleavage as spindles fibers are oriented obliquely to animal vegetal axis
- **Deuterostomes:** Radial cleavage as spindles fibers are oriented perpendicular to animal vegetal axis

Development: Protostome Vs Deuterostome (I)

3. Cleavage Pattern



Ref. Slideplayer.com

Development: Protostome Vs Deuterostome (I)

4. Determinate and Indeterminate Cleavage

- **Determinate:** Fate of blastomeres is established at very early stage *i.e,* 2 or 4 cell stage. If some cells detach embryo fails to develop.
- **Indeterminate:** Fate of blastomeres is established at later stages of development

end

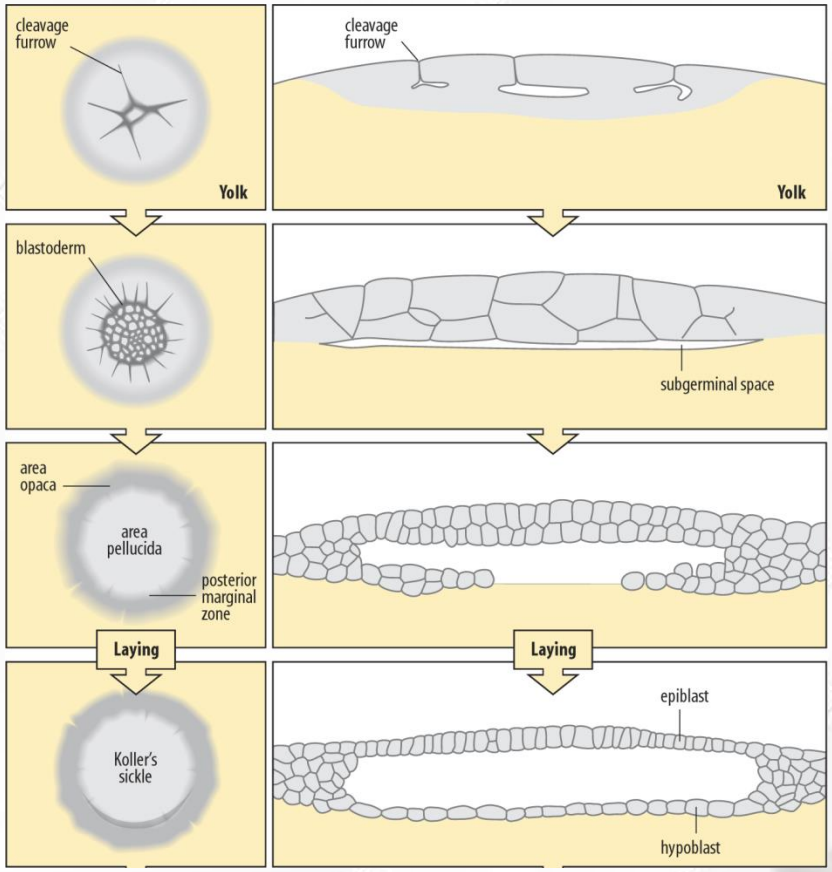
Development: Protostome Vs Deuterostome (I)

4. Secondary Induction

- Formation of other type of cell by inductive process is termed as secondary induction

Development: Protostome Vs Deuterostome (I)

4. Chick Blastulation



Ref. mun.ca/biology

Principles of Animal life 1

Development: Protostome vs Deuterostome (II)

Development: Protostome vs Deuterostome (II)

Contents

1. Development of Mouth and Anus
2. Coelom Formation
3. Larval Types

Development: Protostome vs Deuterostome (II)

1. Development of Mouth and Anus

- **Protostomes:** The blastopore formed during gastrulation remain open and forms mouth while anus develops from opposite side of cavity.
- **Deuterostome:** blastopore develops into anus while mouth develops from distal end.

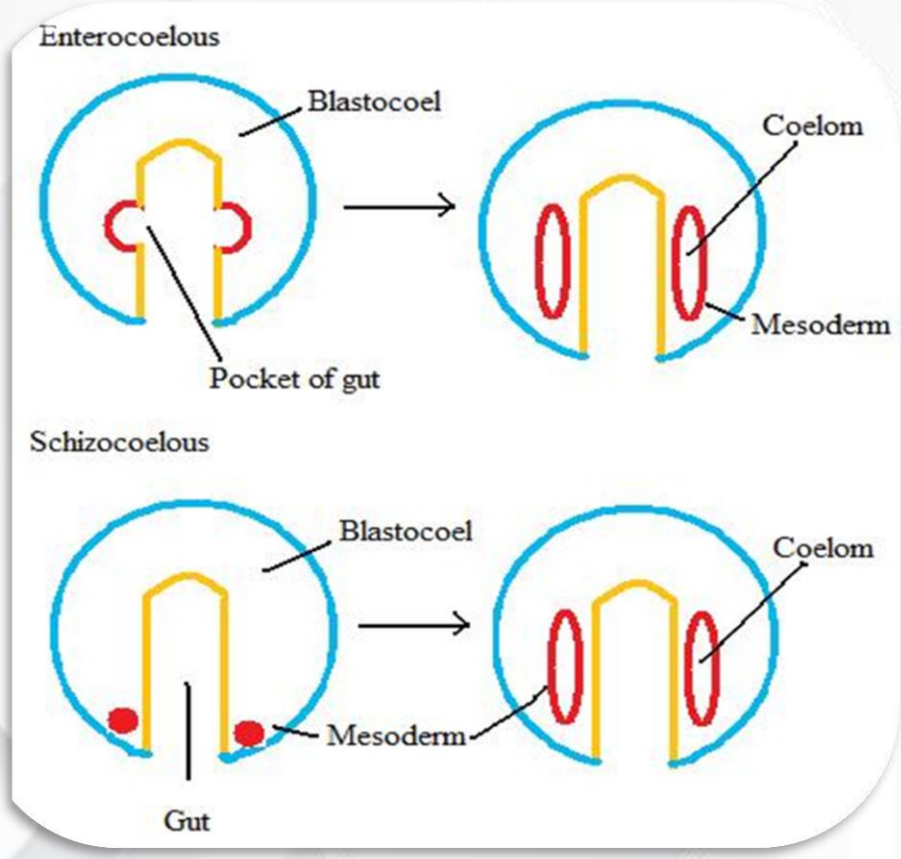
Development: Protostome vs Deuterostome (II)

2. Coelom Formation

- **Protostomes:** Coelom is formed by splitting of mesoderm. This pattern is called schizocoelous.
- **Deuterostome:** Coelom is formed by out pouching of primitive gut (archenteron). This pattern is called enterocoelous.

Development: Protostome vs Deuterostome (II)

2. Coelom Formation



Ref. www.chegg.com

Development: Protostome vs Deuterostome (II)

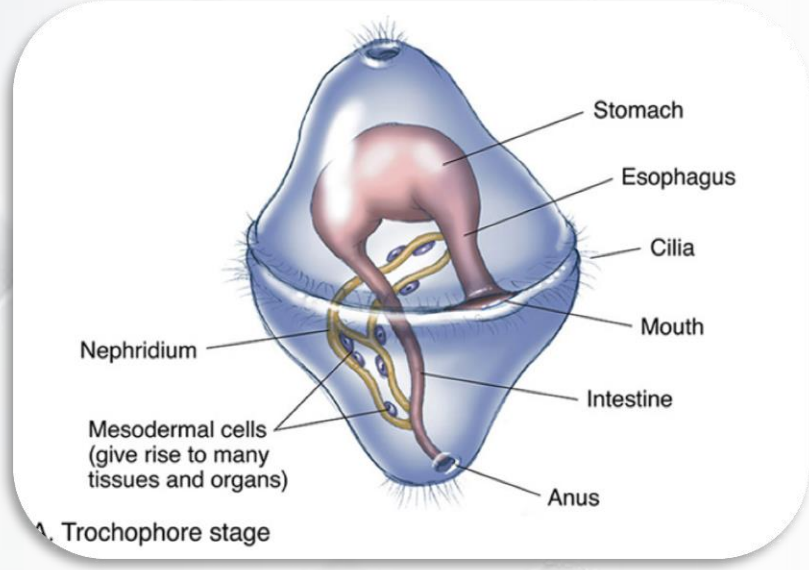
3. Larval Stages

- **Protostomes:** Development of many protostome include a top shape larva called trochophore larva

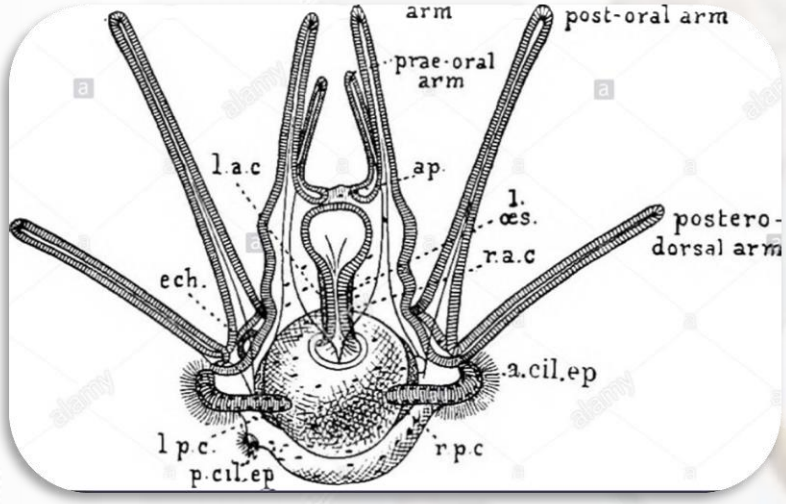
- **Deuterostome:** Some deuterostome have a Kidney-bean shaped larva called dipleurula.

Development: Protostome vs Deuterostome (II)

3. Larval Stages



Ref. carlsonstockart.com



Ref. alamy.com

Principles of Animal life 1

Metamorphosis

Metamorphosis

Contents

1. Metamorphosis
2. Instars
3. Advantage of metamorphosis
4. Disadvantages of Metamorphosis
5. Hormonal Control of Metamorphosis
6. Types

Metamorphosis

1. Metamorphosis

- Metamorphosis is an abrupt change in structure and function from immature stage to adult
- It takes place in insects, mollusks, crustaceans, cnidarians, echinoderms, tunicates, fishes and amphibians.

Metamorphosis

2. Instars

- During metamorphosis animals undergo a series of molts to grow, and each stage between molts is called an instar.
- Mosquito larvae have four instars
- Butterfly larvae have five instars

Metamorphosis

3. Advantages of Metamorphosis

- To complete development with food stage
- Reduced competition: Juvenile and adult stages have different mode of nutrition. Tadpole of frog is herbivore while adult is insectivore

Metamorphosis

4. Disadvantages of Metamorphosis

- Short adult life while time spent on metamorphosis is pretty long
- Larva and adult do not share same food. This may be disadvantageous in food poor environment

Metamorphosis

5. Hormonal Control of Metamorphosis

- **Vertebrates:** Thyroid hormones
- **Insects:** Prothoracicotropic hormone from insect brain
- This stimulates Ecdysone hormone released by prothoracic gland

Metamorphosis

5. Hormonal Control of Metamorphosis

- At larval stage juvenile hormone is also releases~~s~~d this results in next larval instar
- At final stage of larva, juvenile hormone is not released and ecdysone converts larva into pupa

Metamorphosis

5. Types

- Complete metamorphosis
- Incomplete metamorphosis

end

Principles of Animal life 1

Holometabolous Metamorphosis

Holometabolous Metamorphosis

Contents

1. Holometabolus Metamorphosis
2. Egg Stage
3. Larval Stage
4. Pupa Stage
5. Adult Stage

Holometabolous Metamorphosis

1. Holometabolous Metamorphosis

- This is also called complete metamorphosis
- In about 88% insects
- It is the type of metamorphosis which separates the physiological process of growth from differentiation and sexual stages

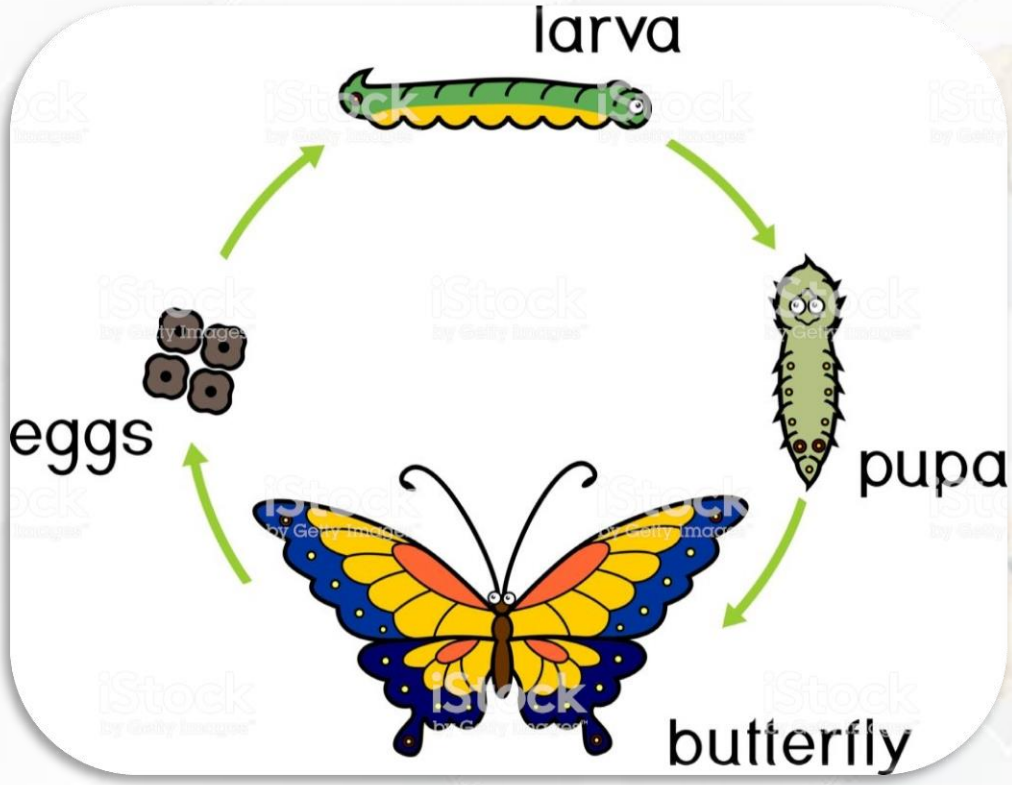
Holometabolous Metamorphosis

1. Holometabolous Metamorphosis

- Each process is completed without intervention of other stage
- Four stages
  - Egg
  - Larva
  - Pupa
  - Adult
- *E.g.*, Insects

Holometabolous Metamorphosis

1. Holometabolous Metamorphosis



Ref. istockphoto.com

Holometabolous Metamorphosis

2. Egg Stage

- First stage of insect life cycle
- Development starts after fertilization or sometime without fertilization
- Early embryonic development completed at this stage
- Egg stage is very short but in some cases undergo diapause

Holometabolous Metamorphosis

3. Larval Stage

- Second stage of holometabolous life cycle
- Larvae are hatched out from eggs
- Motile and worm like form
- Actively feed and grow in size
- Never possess wings
- Simple eyes

Holometabolous Metamorphosis

3. Larval Stage

- There are several larval stages or instars
- Larva molts for passing from one instar to the next one

Holometabolous Metamorphosis

4. Pupa Stage

- Third stage of holometabolous metamorphosis
- Larva undergo metamorphosis to become pupa
- Larva form a case or cocoon around itself
- It is quiescent, non feeding developmental stage

Holometabolous Metamorphosis

4. Pupa Stage

- Mostly pupa move very little, however some are mobile
- Some insects undergo diapause as pupa
- It is the stage of drastic changes in morphology and physiology

Holometabolous Metamorphosis

5. Adult Stage

- Final stage of metamorphosis
- Adults have wings and reproductive organs
- Most insects adult grow very little
- Some do not feed and entirely focus on mating and reproduction

end

Principles of Animal life 1

Hemimetabolous Metamorphosis

Hemimetabolous Metamorphosis

Contents

1. Hemimetabolous Metamorphosis
2. Egg Stage
3. Nymph Stage
4. Adult stage

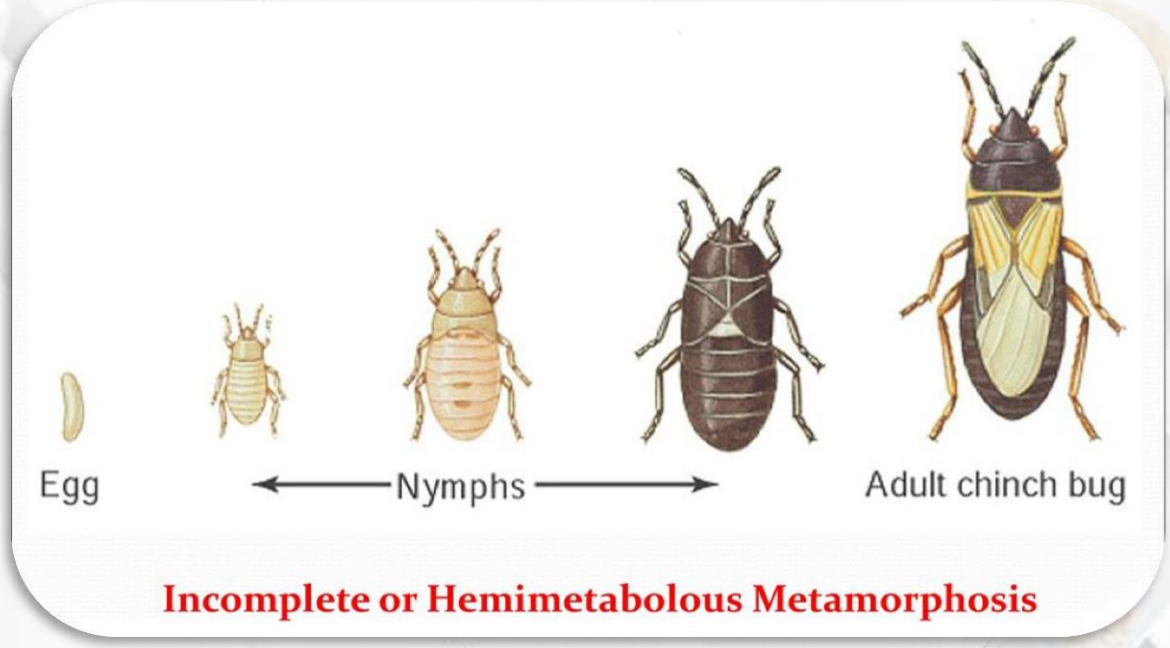
Hemimetabolous Metamorphosis

1. Hemimetabolous Metamorphosis

- It is also called gradual incomplete metamorphosis
- It is the mode of development of certain insects with three distinct stages
  - Egg
  - Nymph
  - Adult
- E.g., Aphid, grasshopper, cockroach

Hemimetabolous Metamorphosis

1. Hemimetabolous Metamorphosis



Ref. slideplayer.com

Hemimetabolous Metamorphosis

2. Egg Stage

- First stage of development
- Initial stages of development are completed at egg stage
- Egg stage varies in different groups but in some cases undergo diapause
- Eggs of aphids are hatched before they are layed

Hemimetabolous Metamorphosis

3. Nymph Stage

- Nymph is an immature form hatched out from the eggs
- Nymph lacks wings
- They have underdeveloped reproductive organs
- Nymph undergo gradual changes called instars
- Aquatic insect may have tracheal gills

Hemimetabolous Metamorphosis

4. Adult Stage

- The apparent difference from nymph is presence of wings
- They are also reproductively mature

end

Principles of Animal life 1

Hierarchy of Ecology

Hierarchy of Ecology

Contents

1. Ecology
2. Ecological Hierarchy
3. Organism
4. Population
5. Community
6. Ecosystem
7. Biosphere

Hierarchy of Ecology

1. Ecology

- It is the branch of biology that deals with the relationship of organisms to one another and with their physical environment

Hierarchy of Ecology

2. Ecological Hierarchy

- It is the way in which ecological components interact with each other
- There are five levels
  - Organism
  - Population
  - Community
  - Ecosystem
  - Biosphere

Hierarchy of Ecology

3. Organism

- Organism is the first level in ecological hierarchy
- All the living things come at this level
- E.g., human, lion, tiger, cow, tree, shrub
- The way how organism interact with its environment is studied at this level

Hierarchy of Ecology

4. Population

- Group of organism of same species living at same geographical location at same time
- At this level we study how the members of group live together and interact with each other

Hierarchy of Ecology

5. Community

- Groups of different species living together in a given area
- The number of species represent the diversity of community
- Community ecology include
  - Predation
  - Parasitism
  - Competition

Hierarchy of Ecology

6. Ecosystem

- It is the study of all population in a community and their physical environment
- Biotic components and abiotic components
- How living organism make the use of abiotic components
- How they get energy
- Effect of physical factors on organisms

Hierarchy of Ecology

7. Biosphere

- Biggest of all levels
- All the area of earth and atmosphere occupied by living organisms
- Main focus is on:
  - Geology
  - Climate
  - Water bodies
  - Latitude
  - Longitude

end

Principles of Animal life 1

Biotic and Abiotic Factors of an Ecosystem

Biotic and Abiotic Factors of an Ecosystem

Contents

1. Ecosystem
2. Components of Ecosystem
3. Biotic Factors
4. Abiotic Factors

Biotic and Abiotic Factors of an Ecosystem

1. Ecosystem

- It is the unit of ecology where living organisms interact with their physical environment
- Transfer of energy and material between organisms takes place within an ecosystem
- The incorporation of energy and materials in an ecosystem is called its productivity

Biotic and Abiotic Factors of an Ecosystem

1. Ecosystem

- Ecosystems are very dynamic
- They are controlled by different external and internal factors

Biotic and Abiotic Factors of an Ecosystem

2. Components of Ecosystem

- Ecosystem has two main components/factors
- Biotic factors/components
- Abiotic factors/components

Biotic and Abiotic Factors of an Ecosystem

3. Biotic Factors

- All living organisms in ecosystem make biotic factors.
  - Producers
  - Consumers
  - Decomposers

Biotic and Abiotic Factors of an Ecosystem

4. Abiotic Factors

- All nonliving things in ecosystem make abiotic factors.
  - Temperature
  - Moisture
  - Light
  - Soil
  - Air
  - water

end

Principles of Animal life 1

Abiotic Factors: Temperature

Abiotic Factors: Temperature

Contents

1. Temperature
2. Loss of Heat to Environment
3. Gain of Heat
4. Effect on Habitat Requirement
5. Body Size and Metabolism

Abiotic Factors: Temperature

1. Temperature

- It is the measure of heat present in a body
- Animals extend a part of their energy on regulating temperature
- Temperature influence metabolic rate

Abiotic Factors: Temperature

2. Heat Loss to Environment

- Conduction
- Convection
- Radiations
- Evaporation

Abiotic Factors: Temperature

3. Heat Gain

- Radiations
- Conduction
- Metabolism

Abiotic Factors: Temperature

4. Effect on Habitat Requirement

- Thermoregulatory needs influence habitat requirement:
- Availability of food
- Availability of water
- Availability of shelter

end

Abiotic Factors: Temperature

5. Body Size and Metabolism

- Larger animals have low metabolic rates
- Smaller animals have higher metabolic rate
- Mouse have 20 times higher metabolic rate than elephant
- This is probably due to large surface to volume ratio for heat exchange

end

Principles of Animal life 1

Abiotic Factors: Moisture

Abiotic Factors: Moisture

Contents

1. Moisture
2. Gain of Water in Animals
3. Loss of Water in Animals
4. Measures to Reduce Water Loss
5. Water Regulation in Aquatic Animals

Abiotic Factors: Moisture

1. Moisture

- It is the amount of water diffused in environment as vapours, within solid or condensed on surface
- In air the amount of water is also called humidity
- High moisture….low water loss
- Less moisture….more water loss

Abiotic Factors: Moisture

2. Gain of Water in Animals

- Animals acquire water from:
- Food
- Drinking
- Metabolism

Abiotic Factors: Moisture

3. Loss of Water in Animals

- Water is lost along with
- Urine
- Feces
- Gaseous exchanges
- Evaporative cooling

Abiotic Factors: Moisture

4. Measures to Reduce Water Loss

- Animals opt following strategies to reduce water loss
- Dry feces
- Concentrated urine or semisolid excretory wastes
- Nocturnal behavior
- Ability to store water

Abiotic Factors: Moisture

5. Water Regulation in Aquatic Animals

- Freshwater animals live in hypotonic environment: conserve ionic loss
- Marine animals live in hypertonic environment: avoid salt accumulation and dehydration

end

Principles of Animal life 1

Abiotic Factors: Soil

Abiotic Factors: Soil

Contents

1. Soil
2. Soil Characteristics
3. Soil as Habitat
4. Effect on Plant and Animal Life

Abiotic Factors: Soil

1. Soil

- Soil reflects the geological and biological history
- It effects the animals life directly and indirectly

Abiotic Factors: Soil

2. Soil Characteristics

- Characters include
  - Texture
  - Amount of organic matter
  - Fertility
  - Water holding ability
- They directly affect the diversity of life on or in the soil

Abiotic Factors: Soil

3. Soil as Habitat

- Soil is the habitat of large variety of animals including
  - Protozoa
  - Nematodes
  - Earthworms
  - Arthropods
  - Salamanders
  - Snakes
  - Moles

Abiotic Factors: Soil

4. Effect on Plant and Animal Life

- Soil have obvious effect on plant growing in that area
- Different kinds of plants influence the animal life whether they are living in or on the soil

end

Principles of Animal life 1

Population Growth

Population Growth

Contents

1. Populations change over time
2. Survivorship Curves
3. Types I Curve
4. Type II Curve
5. Type III Curve

Population Growth

1. Population Change Over Time

- Population are dynamic in nature
- With the passage of time they are subjected to change due to:
  - Birth
  - Deaths
  - Migration

Population Growth

2. Survivorship Curves

- Individuals within population are estimated by survivorship curve
- It is the logarithmic plot between survivors and their age
- Curves types are:
  - Type I
  - Type II
  - Type III

Population Growth

3. Type I Curve

- Also called convex population survivorship curve
- Individuals of type I, population survive to an old age, then rapidly die
- Environmental factors rarely cause mortality
- Individuals live their potential life span
- E.g. human population

Population Growth

4. Type II Curve

- Also called diagonal population survivorship curve
- Individuals of type II, population have constant probability of death throughout their life
- Environmental factors influence mortality
- E.g., populations of birds and rodents

Population Growth

4. Type II Curve

- Mortality in natural populations is often high
- For Example, each cohort of birds experience 50% mortality each year
- Potential life span of American robin is 11 years, however average life is 1.4 years

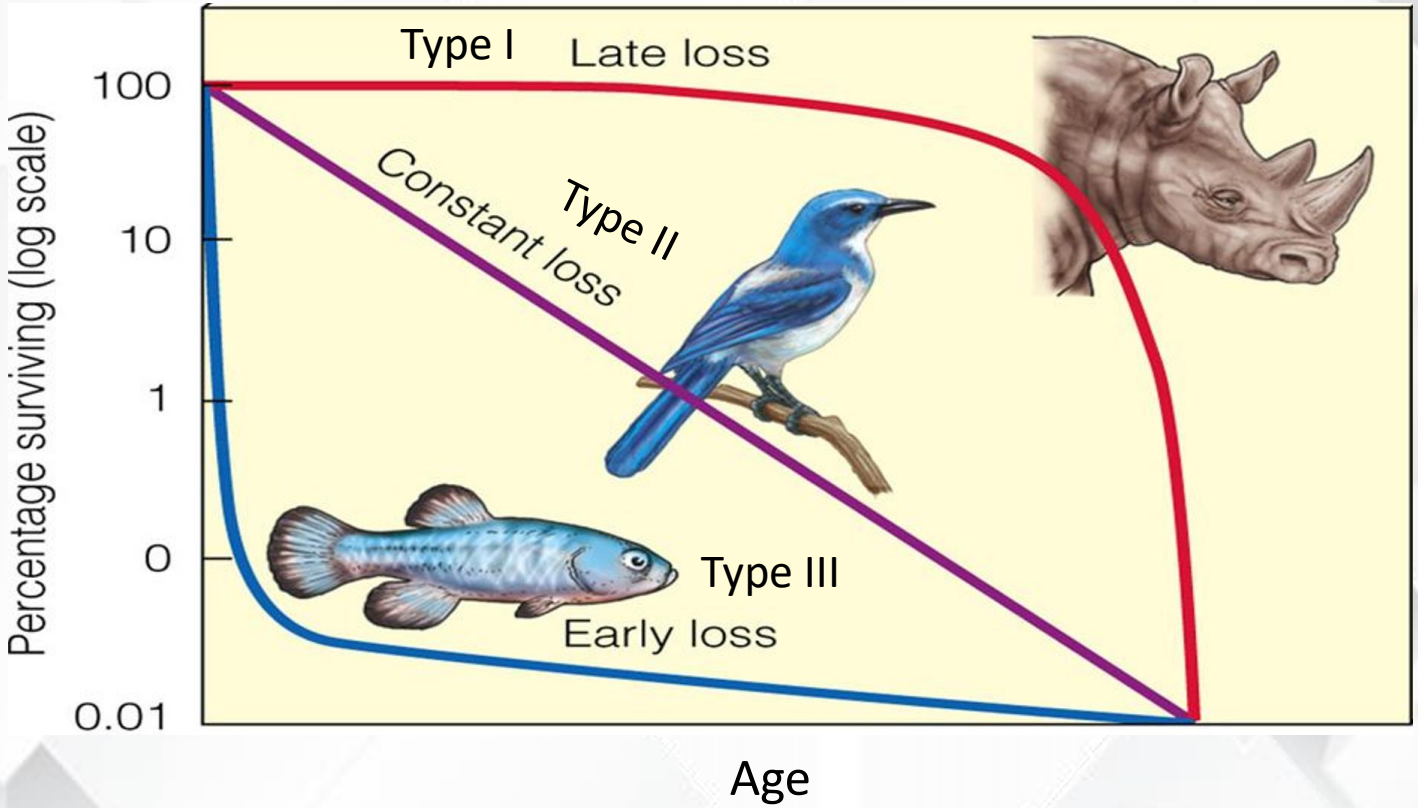
Population Growth

5. Type III Curve

- Also called concave population survivorship curve
- Individuals of type III, population have very high juvenile mortality
- Adults have much lower mortality rate
- E.g., fishes and many invertebrates

end

Population Growth



Ref. Brooks/cole- Thomson 2005. p. 177

Principles of Animal life 1

Exponential Growth

Exponential Growth

Contents

1. Exponential Growth
2. Exponential Growth in Fruit Flies
3. Factors Affecting Reproductive Potential
4. Biotic potential

Exponential Growth

1. Exponential Growth

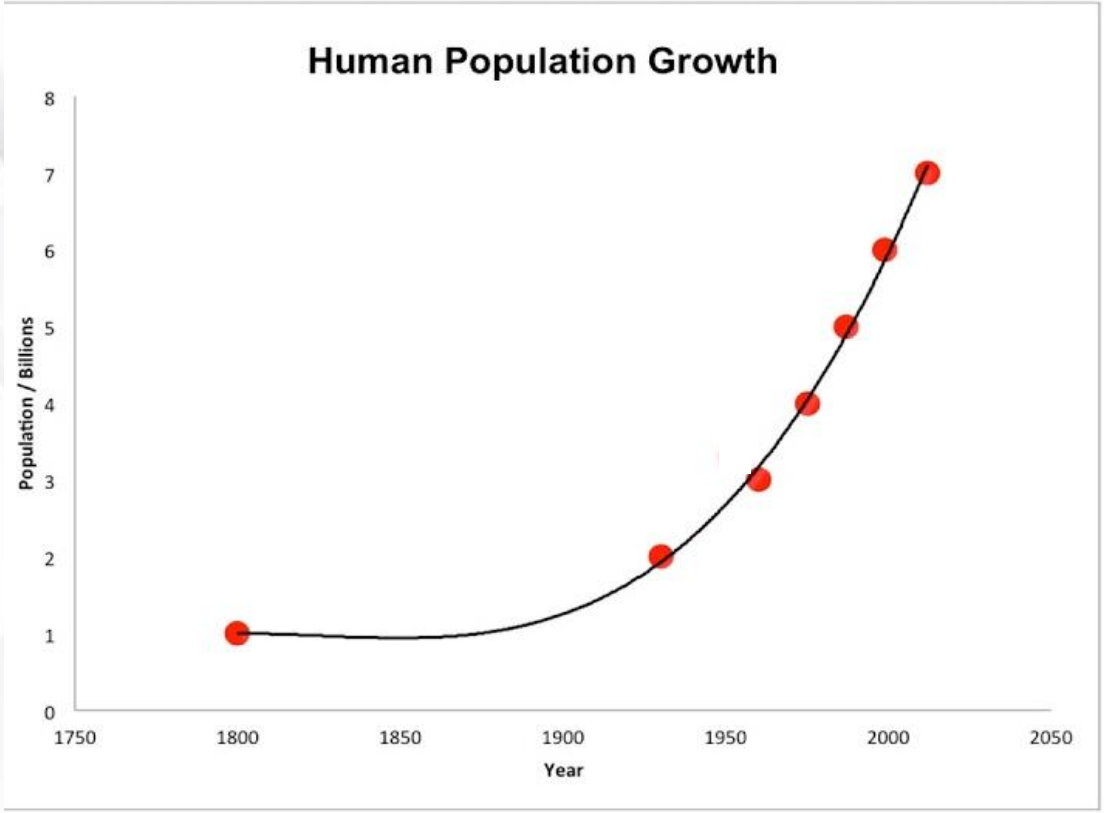
- Population growth in which number of individuals increase by a constant multiple in each generation
- Population increases by same ratio per unit time

Exponential Growth

2. Exponential Growth in Fruit Flies

- A few fruit flies in a large jar with abundant food
- 1 female lay 50 eggs
- 2 weeks for maturity
- Each female of 2nd generation produce 50 more flies and so on

Exponential Growth



Exponential Growth

3. Factors Affecting Reproductive Potential

- Number of offspring produced
- Survival to reproductive age
- Reproductive period duration
- Length of time taken to maturity

Exponential Growth

4. Biotic Potential

- The capacity of a population to increase maximally is called biotic potential
- It is also called intrinsic rate of growth

end

Principles of Animal life 1

Logistic Population Growth

Logistic Population Growth

Contents

1. Environmental Resistance
2. Carrying Capacity
3. Logistic Population Growth

Logistic Population Growth

1. Environmental Resistance

- The constrain of climate, food, space and other environmental factors is called environmental resistance
- Due to these factors exponential growth can not occur indefinitely

Logistic Population Growth

2. Carrying Capacity

- The size of population that a particular environment can support is called its carrying capacity
- It is symbolized by *K*

Logistic Population Growth

3. Logistic Population Growth

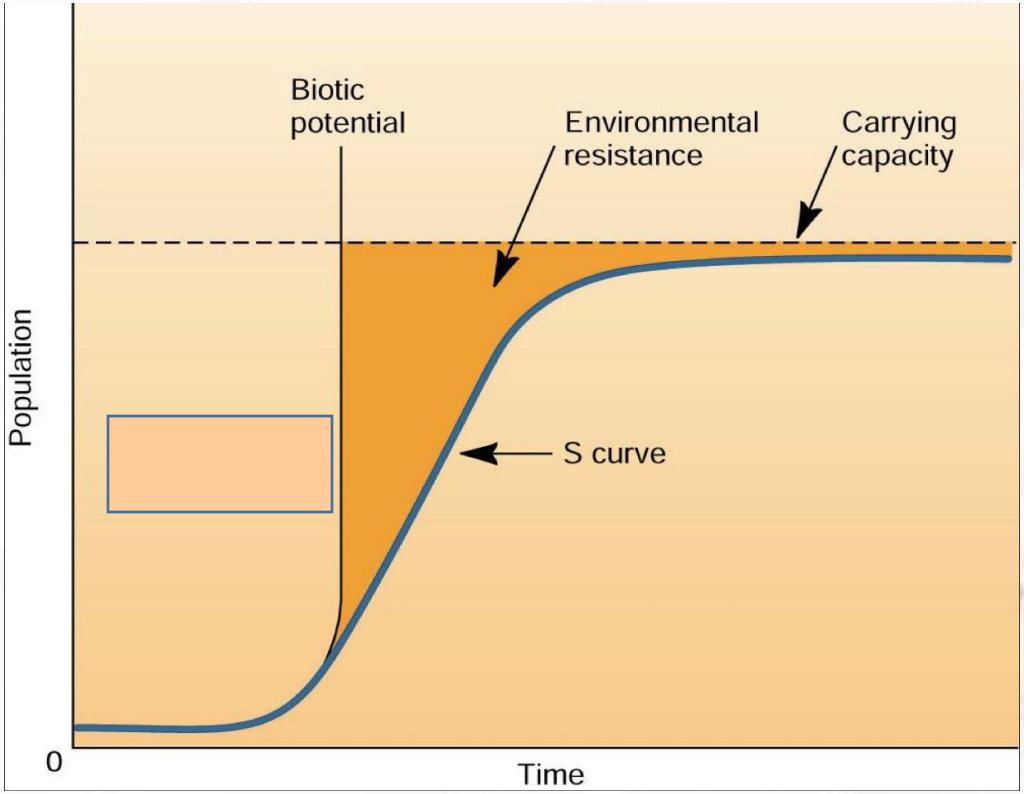
- When a population reaches to carrying capacity of a particular environment, its growth is decreased.
- This results in logistic growth curve
- Logistic growth curves reflect limited resources

Logistic Population Growth

3. Logistic Population Growth

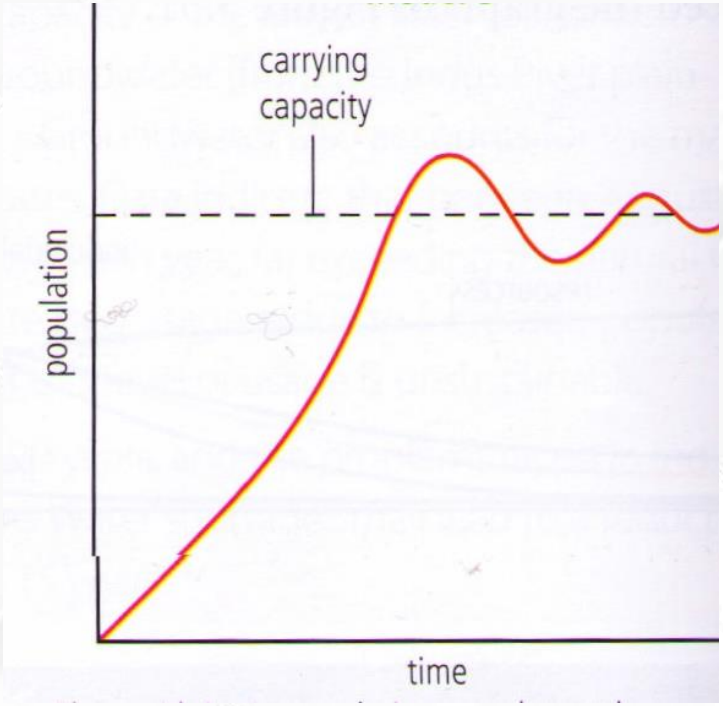
- S-shaped curve: At carrying capacity population growth levels off
- J-shaped curve: when population overshoot carrying capacity. Growth curve may fluctuate around K or assume J-shape

Logistic Population Growth



Ref. memorangapp.com

Logistic Population Growth



Ref. yesitsyomoma.wordpress.com

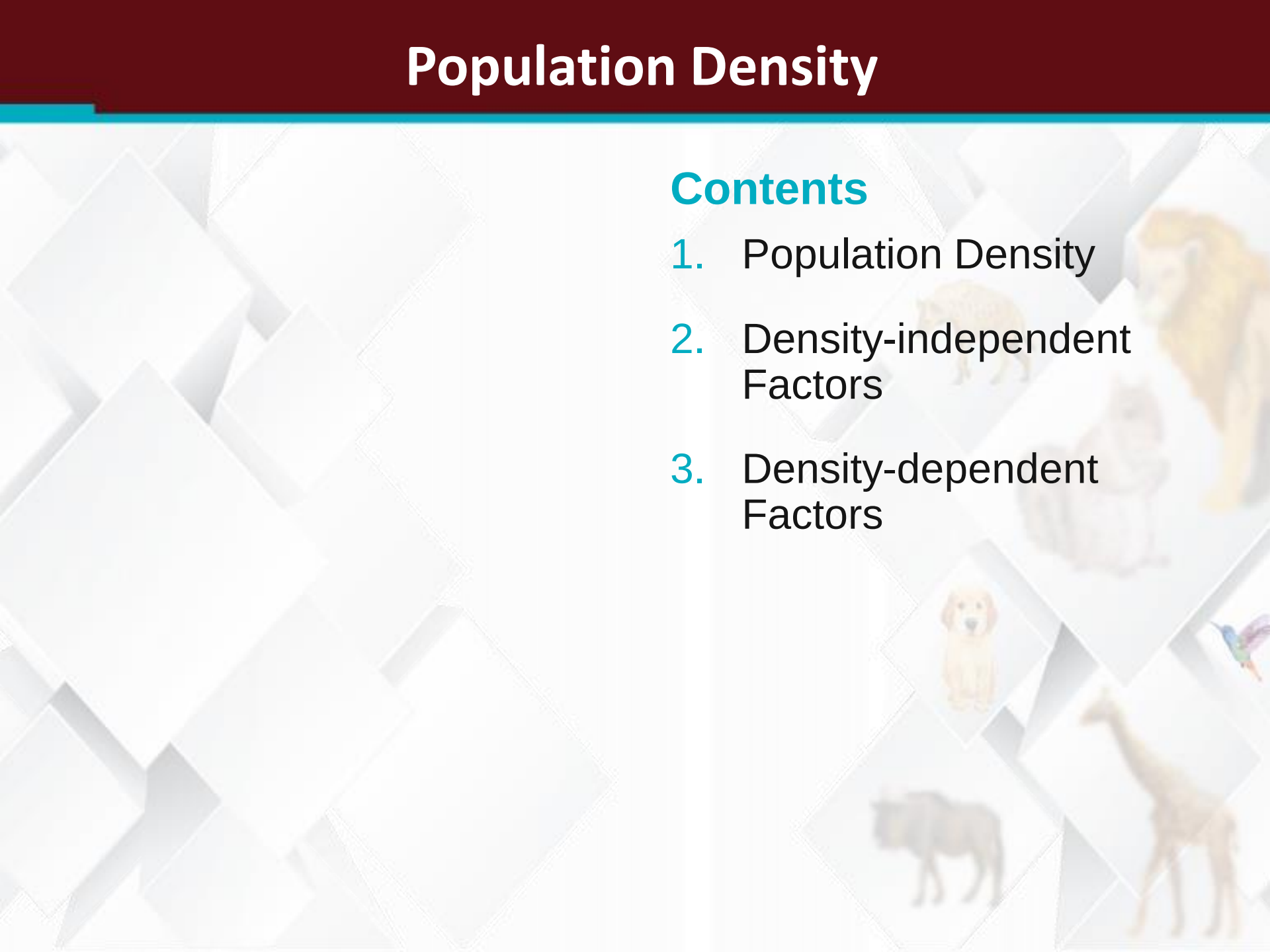
Principles of Animal life 1

Population Density

Population Density

Contents

1. Population Density
2. Density-independent Factors
3. Density-dependent Factors



Population Density

1. Population Density

- It is the number of individuals living per unit area.
- Population density impact populations in predictable ways

Population Density

2. Density-independent Factors

- These factors affect population regardless of their number
- Weather conditions
- Earth quake
- Pollutants
- Deforestation
- Hurricane
- Fire

Population Density

3. Density-dependent Factors

- These are the factors which are more severe when population density is high
- Competition for resources
- Disease
- Predation
- Parasitism

end

Principles of Animal life 1

Animal Inactivity

Animal Inactivity

Contents

1. Animal Inactivity
2. Torpor
3. Hibernation
4. Winter Sleep
5. Aestivation

Animal Inactivity

1. Animal Inactivity

- When food is scarce or environmental conditions are severe, the animals stop feeding and metabolic activities are decreased.
- Torpor
- Hibernation
- Winter Sleep
- Aestivation

Animal Inactivity

2. Torpor

- Period of decreased metabolism and lower body temperature in daily activity cycle
- Hummingbird and bat should eat constantly
- At the end of torpor body temperature and metabolism increase quickly

Animal Inactivity

3. Hibernation

- Time of low body temperature and decreased metabolism from weeks to months
- The set point of thermoregulatory center drop to 20^{o}C
- Thermoregulation not suspended in homeotherms
- Rodents, shrews and bats
- Ectotherms may not regulate

Animal Inactivity

4. Winter Sleep

- Period of winter inactivity
- Body temperature do not drop substantially
- Animals can wake and become active
- e.g., Bears

Animal Inactivity

5. Aestivation

- inactivity period of some animals to withstand in drought
- Animals live in burrows
- Do not eat or drink
- Emerge after moisture returns
- E.g., many invertebrates and frogs

end

Principles of Animal life 1

Interspecific Interaction: Competitive Exclusion

Interspecific Interaction: Competitive Exclusion

Contents

1. Competitive Exclusion
2. Universality of Competitive Exclusion
3. Example of Competitive Exclusion

Interspecific Interaction: Competitive Exclusion

1. Competitive Exclusion

- It states that two species competing for same requirement of food, habitat, nesting sites and other conditions of life can not coexist

Interspecific Interaction: Competitive Exclusion

2. Universality of Competitive Exclusion

- It influenced the scientists for more than 50 years
- Numerous studies in nature failed to document its universality
- This is because no two species have identical requirements

Interspecific Interaction: Competitive Exclusion

3. Example of Competitive Exclusion

- In California Golden-naveled scalesucker is a wasp parasite of red scale insect.
- Scientists introduced lingnan scalesucker from China
- Within 2 years lingnan replaced golden-naveled scalesucker

end

Principles of Animal life 1

Interspecific Interaction: Coexistence

Interspecific Interaction: Coexistence

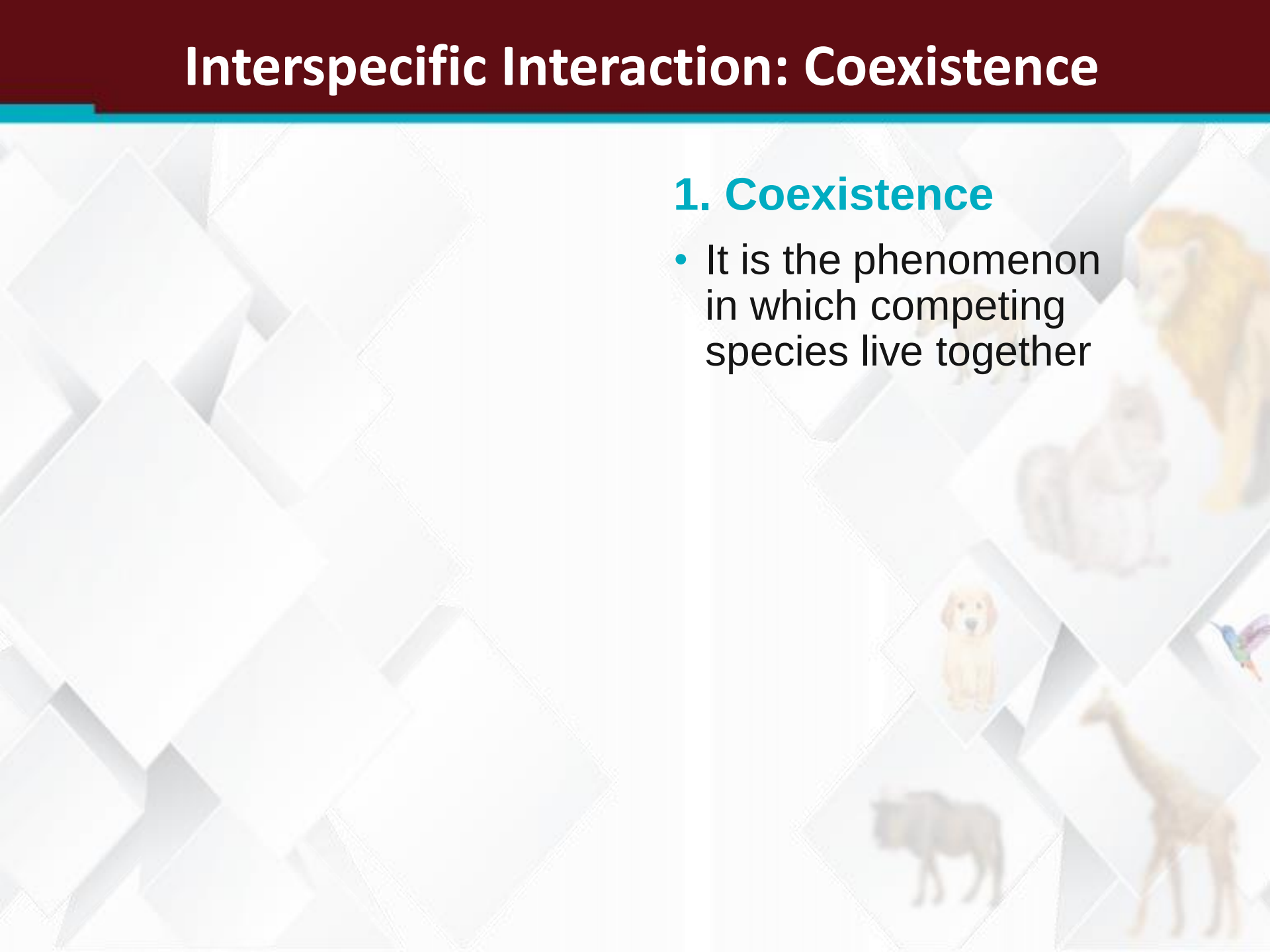
Contents

1. Coexistence
2. How Coexistence Occurs?
3. Coexistence May Turn into Competitions

Interspecific Interaction: Coexistence

1. Coexistence

- It is the phenomenon in which competing species live together



Interspecific Interaction: Coexistence

2. How Coexistence Occurs

- Coexistence can occur when species utilize resources in slightly different way
- MacArthur studied 5 species of warblers that use same caterpillar from different region of trees

Interspecific Interaction: Coexistence

2. How Coexistence Occurs

Ref. allaboutbirds.org

Interspecific Interaction: Coexistence

3. Coexistence May Turn into Competition

- By partitioning resource can be divided
- However, when resources are not enough then similarities become limiting factors and competitive exclusion may occur

end

Principles of Animal life 1

Interspecific Interaction: Coevolution

Interspecific Interaction: Coevolution

Contents

1. Coevolution
2. Role of Natural Selection
3. Flowering plants and Pollinators
4. Hummingbird and Passion Flowers

Interspecific Interaction: Coevolution

1. Coevolution

- It is a phenomenon in which ecologically related species sometimes coordinated such that each species exert a strong selective influence of the other

Interspecific Interaction: Coevolution

2. Role of Natural Selection

- Coevolution…..when competing for same resources or …... predator-prey relationships
- Natural selection favours the development of selective characters
- If balanced, no winner

Interspecific Interaction: Coevolution

3. Flowering Plants and Pollinators

- Plants display adaptation to attract specific pollinators
- Elaborate olfactory and visual adaptations such as bright colour, petal arrangement
- Flowers red and tubular while hummingbird has long beak and see red colour

Interspecific Interaction: Coevolution

4. Hummingbird and Passion Flower

- *Ensifera ensifera* is a hummingbird with extremely long beak
- Passion flowers store nectar deep within tubular flower
- Competition for nectar feeding is limited

end

Principles of Animal life 1

Interspecific Interaction: Symbiosis

Interspecific Interaction: Symbiosis

Contents

1. Symbiosis
2. Facultative and Obligate Symbionts
3. Ectosymbiosis
4. Endosymbiosis

Interspecific Interaction: Symbiosis

1. Symbiosis

- Symbiosis is a close and long term association between two different organisms
- Organism may get benefit or harm
- Mutualism
- Commensalism
- Parasitism

Interspecific Interaction: Symbiosis

2. Facultative and Obligate Symbionts

- Organisms are entirely dependent on symbiosis are obligate. *E.g.,* Fungus of lichen
- Which can survive as independent and in association. *E.g.,* algae of lichen

Interspecific Interaction: Symbiosis

3. Ectosymbiosis

- It is a symbiosis in which symbionts live on the body surface of the host or in the digestive tact
- E.g., Lice, barnacles and cleaner fish

Interspecific Interaction: Symbiosis

4. Endosymbiosis

- It is the relationship in which one symbiont live within the tissue of other.
- E.g., nitrogen fixing bacteria in root nodules, algae inside the corals

end

Principles of Animal life 1

Interspecific Interaction: Parasitism

Interspecific Interaction: Parasitism

Contents

1. Parasitism
2. Ectoparasites and Endoparasites
3. Intermediate Host and Definite Host
4. Parasitoids

Interspecific Interaction: Parasitism

1. Parasitism

- It is a relationship in which on organism (parasite) live in or on other organism (host).
- Parasite gets benefit while host is harmed.
- Host usually survives in this interaction

Interspecific Interaction: Parasitism

2. Ectoparasites and Endoparsites

- Parasites living on the body surface of host are ectoparasites. E.g., lice, leeches

- Parasites living within the body of host are endoparasites. E.g., tape worm, liver fluke, plasmodium

Interspecific Interaction: Parasitism

3. Intermediate Host and Definite Host

- Some parasites have complex life histories with multiple host
- Intermediate host: Immature stages and asexual reproduction of parasite
- Definite host: host where parasite spends its mature life and reproduce sexually

Interspecific Interaction: Parasitism

4. Parasitoids

- These are the animals that deposit eggs or spend developmental stages on the other animals.
- As development proceed they kill the host.
- They have properties of predator and parasite
- E.g., parasitoid wasps

end

Principles of Animal life 1

Interspecific Interaction: Predation

Interspecific Interaction: Predation

Contents

1. Predation
2. Predator Prey Population
3. Prey Distribution

Interspecific Interaction: Predation

1. Predation

- It is type of interaction in which on animal kills and eat other animals.
- The animal which eats is called predator and the animal being eaten is called prey

Interspecific Interaction: Predation

2. Predator Prey Population

- Interaction between predator and prey are complex
- A large population of prey supports more number of predators.
- A decreased population of prey results in a decreased population of predators

Interspecific Interaction: Predation

3. Prey Distribution

- Predators also effect the distribution of prey.
- For example, the lake trout of North America has never moved to freshwater of Siberian freshwater system.
- This is probably due to predation by Pacific sea lamprey

end

Principles of Animal life 1

Interspecific Interaction: Commensalism

Interspecific Interaction: Commensalism

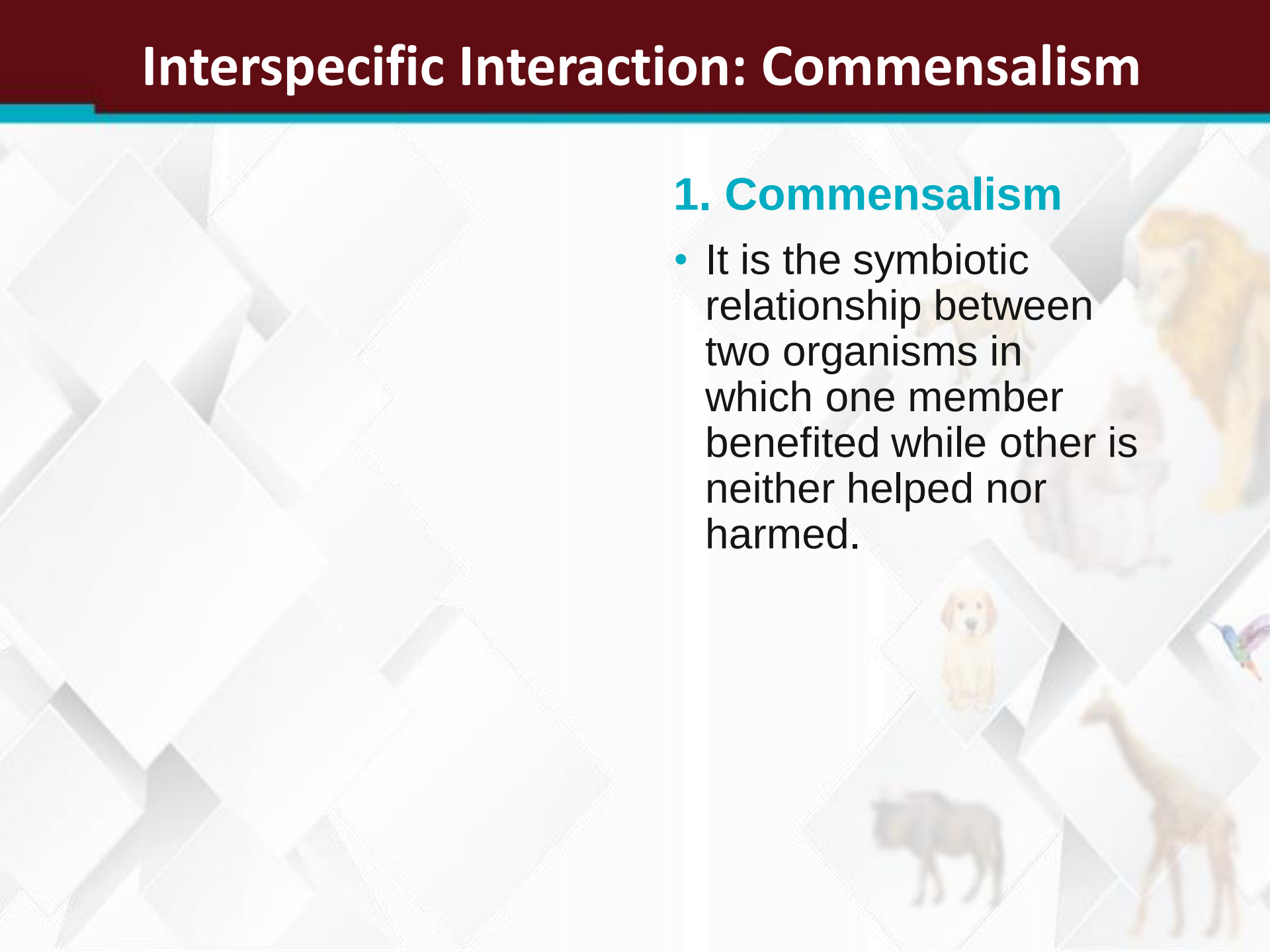
Contents

1. Commensalism
2. Remora Fish and Shark
3. Tree Frogs and Plants
4. Commensal May Turn Parasite

Interspecific Interaction: Commensalism

1. Commensalism

- It is the symbiotic relationship between two organisms in which one member benefited while other is neither helped nor harmed.



Interspecific Interaction: Commensalism

2. Remora Fish and Shark

- It is a small fish with disk on head.
- By these disks they get attached with large sharks and whales
- When shark or whale feed they eat the scrapes of food.

Interspecific Interaction: Commensalism

2. Remora Fish and Whale Shark

Ref. http://awesomeocean.com

Interspecific Interaction: Commensalism

3. Tree Frog and Plant

- Tree frogs live in green leaves of tree.
- They feed on insects and are protected from predators in green leaves of plants

Interspecific Interaction: Commensalism

3. Tree Frog and Plant

Ref. sciencing.com

Interspecific Interaction: Commensalism

4. Commensal May Become a Parasite

- A commensal do not harm its host.
- Sometimes environmental factor change and a commensal becomes parasite
- E.g., *Entamoeba gingivalis* lives in mouth of human may cause teeth and gum decay occasionally

end

Principles of Animal life 1

Interspecific Interaction: Mutualism

Interspecific Interaction: Mutualism

Contents

1. Mutualism
2. Clownfish and Sea Anemone
3. Cleaning Mutualism
4. Coevolution in Mutualism

Interspecific Interaction: Mutualism

1. Mutualism

- It is the type of symbiotic association in which both animals get benefit from each other.

Interspecific Interaction: Mutualism

2. Clownfish and Sea Anemone

- Both live together in a close association
- Sea anemone protect the clownfish
- Clownfish provide food to sea anemone

Interspecific Interaction: Mutualism

2. Clownfish and Sea Anemone

Ref. nationalgeographic.com

Interspecific Interaction: Mutualism

3. Cleaning Mutualism

- One animal cleans other animal
- Small fishes or invertebrates may swim into the mouth of larger fishes to remove ectoparasites
- Cleaning fishes find safety from predators

Interspecific Interaction: Mutualism

3. Cleaning Mutualism

Ref. thewildcoastblog.com

Principles of Animal life 1

Interspecific Interaction: Camouflage

Interspecific Interaction: Camouflage

Contents

1. Camouflage
2. Cryptic Coloration
3. Countershading

Interspecific Interaction: Camouflage

1. Camouflage

- It is the phenomenon in which the colour pattern of animal helps to hide the animal from others
- Two major types
  - Cryptic coloration
  - Countershading

Interspecific Interaction: Camouflage

2. Cryptic Coloration

- It is the type of camouflage in which animals takes pattern according to surroundings.
- Some prey animals are camouflage from predators
- E.g., arctic hare hare white fur coat in winter and brown in spring

Interspecific Interaction: Camouflage

2. Cryptic Coloration

Ref. pinterest.com

Ref. .coolantarctica.com

Interspecific Interaction: Camouflage

2. Cryptic Coloration

- Predators use camouflage to avoid detection by prey
- Prey use camouflage to avoid detection by predators
- Asian tiger's rufous-red coat with black strips help it to undetected during stalking

Interspecific Interaction: Camouflage

3. Countershading

- It is a camouflage common in eggs of frog and toad
- Darkly pigmented on top while light pigmented on the bottom
- From top dark top mixes with dark bottom
- From bottom light colour blends with light air water interface

end

Principles of Animal life 1

Interspecific Interaction: Aposematic Coloration and Mimicry

Aposematic Coloration and Mimicry

Contents

1. Aposematic Coloration
2. Dendrobatid Frogs of America
3. Mimicry
4. Batesian Mimicry
5. Mullerian Mimicry

Aposematic Coloration and Mimicry

1. Aposematic Coloration

- Phenomenon in which animals protect themselves by advertising themselves dangerous or distasteful.
- They actually warn the predators to not attack on them

Aposematic Coloration and Mimicry

2. Dandrobatid Frogs of America

- Dandrobatid frogs of central and south America have poisonous secretion and are brightly colored.
- Their bright colour is a warning to the predators

Aposematic Coloration and Mimicry

2. Dandrobatid Frogs of America

Ref. kids.nationalgeographic.com

Aposematic Coloration and Mimicry

3. Mimicry

- It is the phenomenon in which one species gains resemblance with other species and gets protection from predators
- This resemblance may be with one or more than one species

Aposematic Coloration and Mimicry

4. Batesian Mimicry

- When one species resembles with other species and protected by aposematic coloration.
- Mimic and model
- Flies belonging to family Syrphidae resemble with bees or wasps, thus protected.

Aposematic Coloration and Mimicry

4. Batesian Mimicry

Ref. .inaturalist.org

Aposematic Coloration and Mimicry

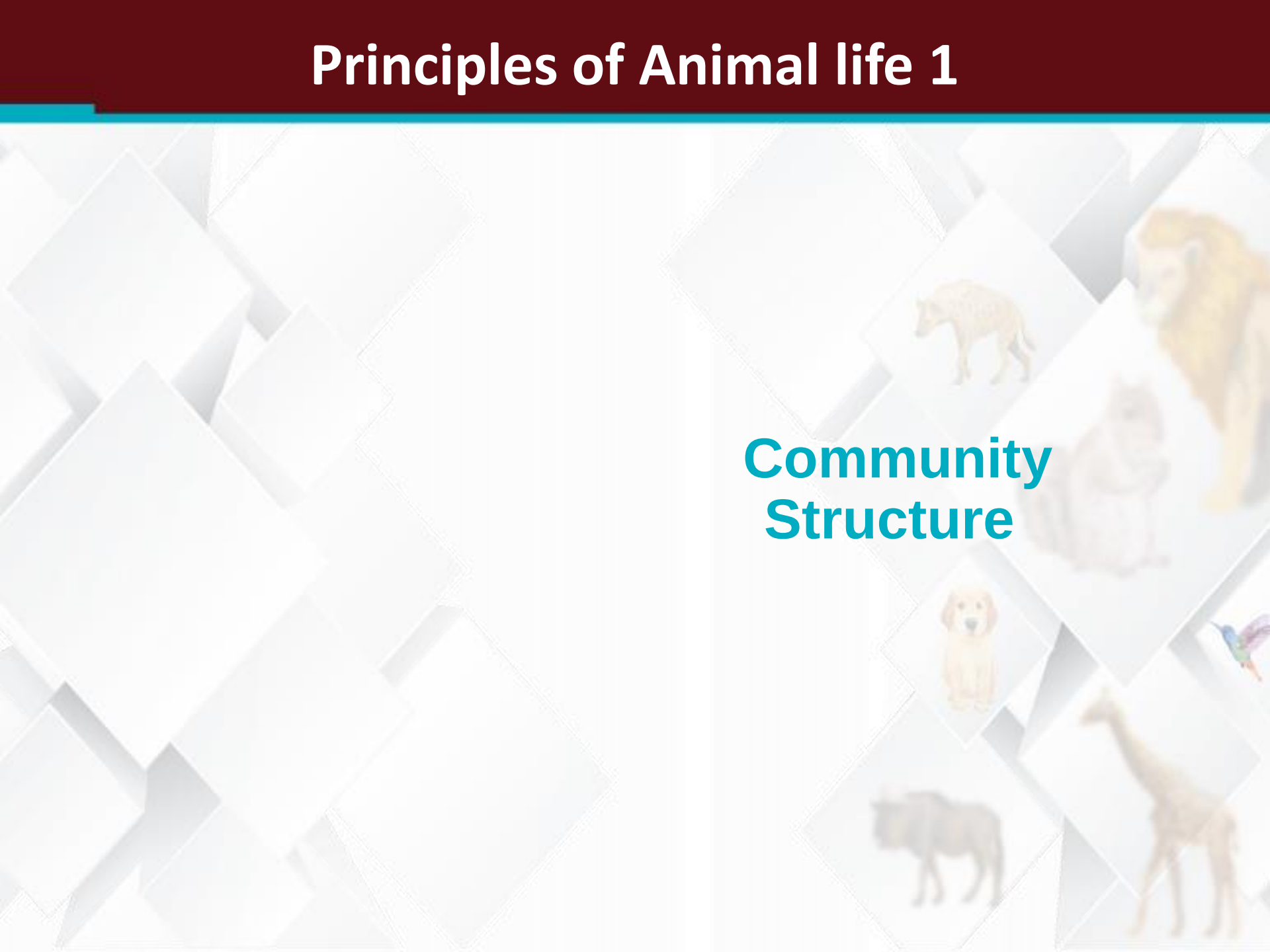
5. Mulerian Mimicry

- When mimic and model both are distasteful.
- If predator eats any one of these, It will not prey any one of them

end

Principles of Animal life 1

Community Structure



Community Structure

Contents

1. Community Structure
2. Dominant Species
3. Keystone Species
4. Temporal Structure
5. Spatial Structure
6. Community Diversity

Community Structure

1. Community Structure

- Communities are not the random mixture
- They are organized entities
- Some members of community have greater importance in organizing community

Community Structure

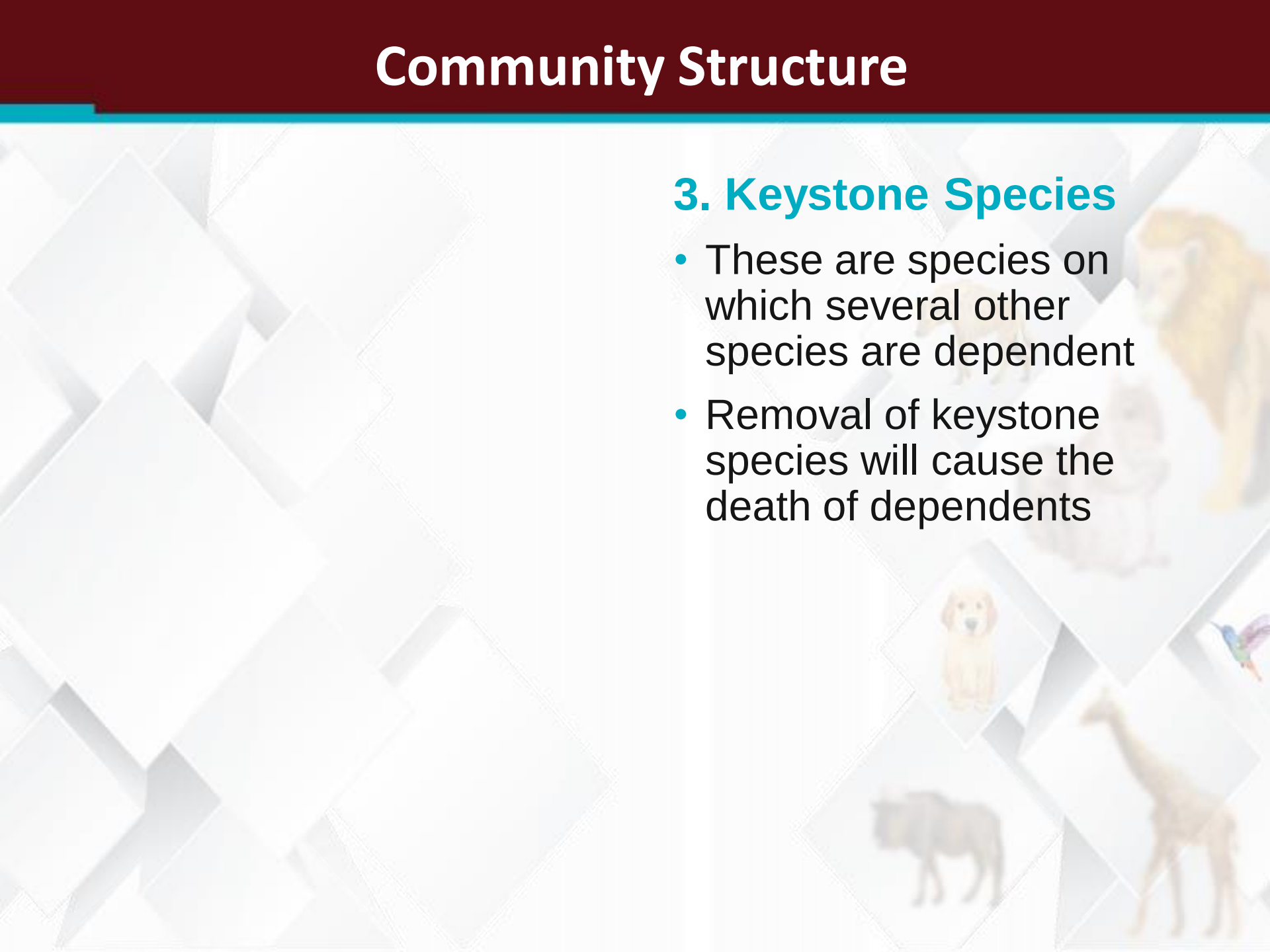
2. Dominant Species

- Species that is responsible for determining community characteristics
- E.g., a stream having a large population of rainbow trout that will determine the population of certain invertebrates

Community Structure

3. Keystone Species

- These are species on which several other species are dependent
- Removal of keystone species will cause the death of dependents



Community Structure

4. Temporal Structure

- Cyclic activities also characterize the community
- Community structure based on cyclic activities of daily or seasonal basis is called temporal structure of community

Community Structure

5. Spatial Structure

- It is obvious in ecosystems where layering is present
- Different populations present at different layers of forests

Community Structure

6. Community Diversity

- Variety of animals present in a community
- It is also called community richness
- Stable climate, more resources, high productivity, moderate predation promote diversity

end

Principles of Animal life 1

Community Stability

Community Stability

Contents

1. Community Stability
2. Succession
3. Pioneer Community
4. Seral Stage
5. Climax Community
6. Primary Succession
7. Secondary Succession

Community Stability

1. Community Stability

- Communities are born and they die
- Some changes in community are not permanent.
- They are replaced by next generation
- Some changes are permanent.

Community Stability

2. Succession

- It is the sequence of community types during maturation of an ecosystem
- It is the change in community structure in a predictable way

Community Stability

3. Pioneer Community

- It is the first community established in an area devoid of life
- E.g. a lake formed by glacier
- The first community establish in this lake will be pioneer community

Community Stability

4. Seral Stage

- During ecological succession and ecosystem is occupied by different communities in a sequence
- Each successional stage is called a seral stage

Community Stability

5. Climax Community

- The final community during an ecological succession is called climax community.
- They are complex and have more species diversity

Community Stability

6. Primary Succession

- The process of succession in an area that was not supporting any life previously is called primary succession.
- Ecological succession in a lake formed by glacier or land exposed by land sliding

Community Stability

7. Secondary Succession

- The process of succession where previous life destroyed by some natural catastrophs (e.g., fire)

end

Principles of Animal life 1

Ecological Niche

Ecological Niche

Contents

1. Ecological Niche
2. Niches overlap
3. Niche Diversification

Ecological Niche

1. Ecological Niche

- It is the role of an organism in an ecosystem
- It include all the attributes of life style of an animal.
- Where it lives
- What it eats
- Where it search for food
- Physical parameters it prefers

Ecological Niche

2. Niches Overlap

- Ecological niches of two species may overlap.
- This will result a competition between them
- Animals will compete for same resources

Ecological Niche

3. Niche Diversification

- Competition between species may be reduced if resources are divided
- One part of resources by one species and second part by other species
- This mechanism is called niche diversification

end

Principles of Animal life 1

Food Chains

Food Chains

Contents

1. Food Chains
2. Grazing Food Chain
3. Detritus Food Chain
4. Significance of food chain

Food Chains

1. Food Chain

- It is the sequence of organisms in an ecosystem where energy moves from one organism to the next organism
- It tells us the sequence to eat and being eaten

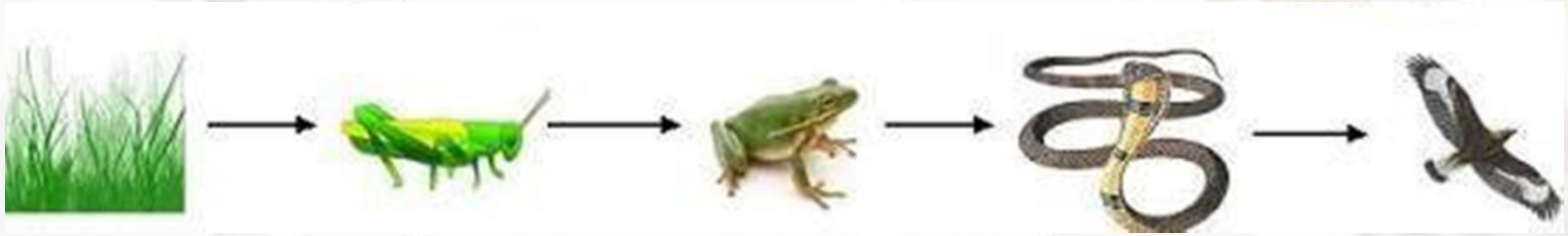
Food Chains

2. Grazing Food Chain

- It is the food chain that starts from the living green plants.
- Green plants are eaten by herbivores which are eaten by carnivores
- Such food chains are directly dependent on sunlight

Food Chains

2. Grazing Food Chain



Ref. bankkita.com

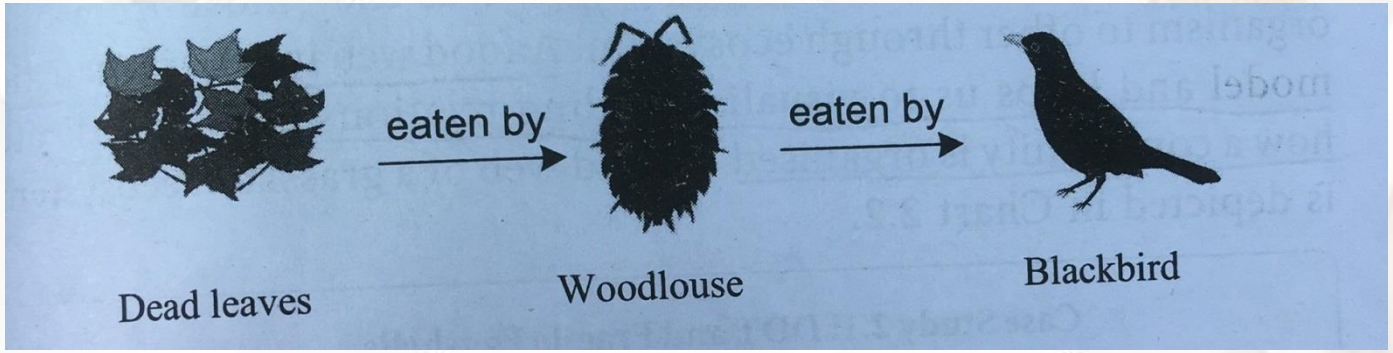
Food Chains

3. Detritus Food Chain

- It is the food chain that starts from dead organic matter to microorganisms and then to organisms
- They are little or not at all dependent on direct solar energy

Food Chains

4. Detritus Food Chain



Ref. .i brainly.in

Food Chains

5. Importance of Food Chain

- To understand the feeding relationships between organisms
- To explore the energy flow mechanisms and matte circulation
- To understand the problems of biomagnification

end

Principles of Animal life 1

Food Webs

Food Webs

Contents

1. Food Webs
2. Grazing Food Web
3. Detritus Food Web
4. Trophic Levels

Food Webs

1. Food Web

- It is a complex network of organisms through which energy and matter is transferred in an ecosystem
- Food web consists of many interconnected food chains

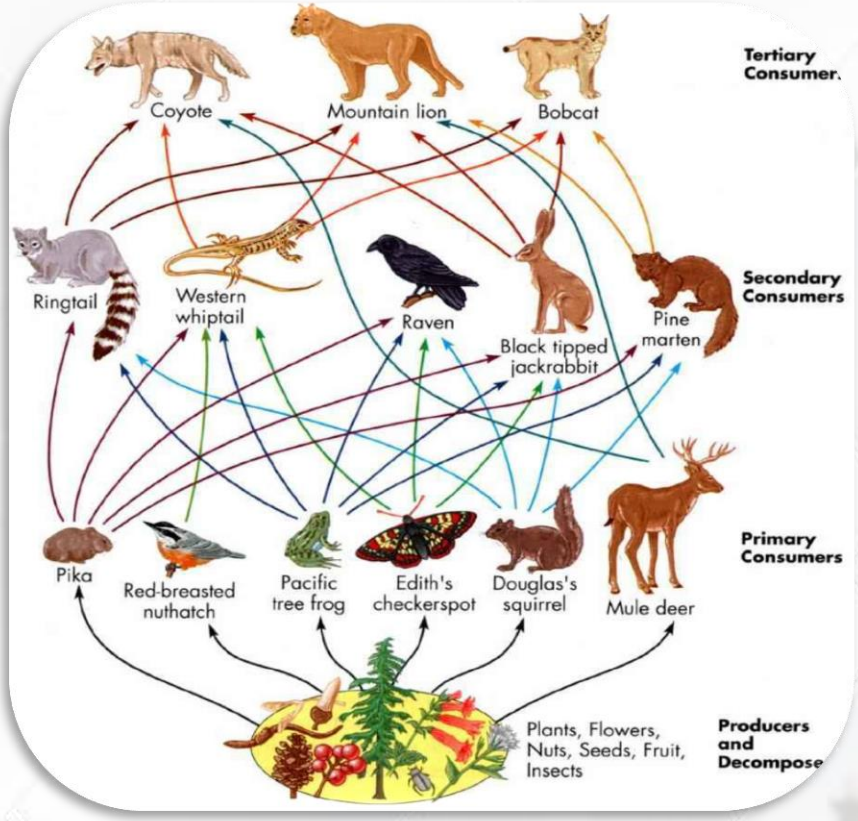
Food Webs

2. Grazing Food Web

- These are the food webs where plants make the starting point.
- Food webs based on algae and green plants

Food Webs

2. Grazing Food Web



Ref. publishyourarticles.net

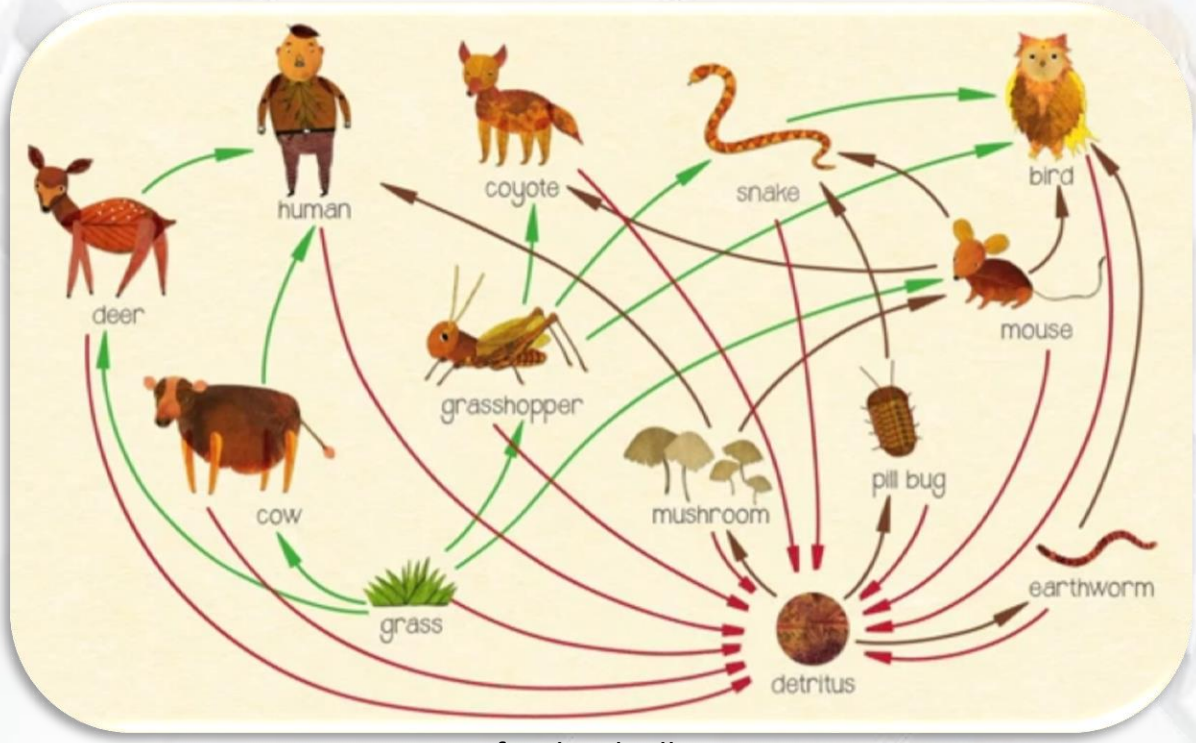
Food Webs

3. Detrital Food Web

- These are the food webs where decomposers make the starting point.
- Bacteria and fungi move the food material to the carnivores

Food Webs

3. Detritus Food Web



Ref. gabrielgellner.com

Food Webs

4. Trophic Levels

- As food webs are very complex
- The animals feeding in a food web are distributed into different trophic levels
- Autotrophs make 1st trophic level

end

Principles of Animal life 1

Trophic Structure: Producers

Trophic Structure: Producers

Contents

1. Producers
2. Photoautotrophs
3. Chemolithotrophs

Trophic Structure: Producers

1. Producers

- They obtain nutrition from inorganic compounds
- Also called autotrophs
- Form the first trophic level of an ecosystem

- Two types:
  - Photoautotrophs
  - Chemolithotrophs

Trophic Structure: Producers

2. Photoautotrophs

- They use sunlight as energy source for the synthesis of food from inorganic compounds
- Green plants, algae, and photosynthetic bacteria

Trophic Structure: Producers

2. Photoautotrophs

- Photosynthesis transform less than 1% of sunlight reaching to earth
- 30% reflects back
- 20% is absorbed by atmosphere
- 50% absorbed by earth

Trophic Structure: Producers

3. Chemolithotrophs

- They obtain energy by oxidizing inorganic compounds
- Usually live in extreme environments
- E.g., sulfur bacteria

end

Principles of Animal life 1

Trophic Structure: Consumers

Trophic Structure: Consumers

Contents

1. Consumers
2. Primary Consumers
3. Secondary Consumers
4. Tertiary Consumers

Trophic Structure: Consumers

1. Consumers

- Organisms which can not synthesize their food from inorganic compounds
- They eat other organisms to get nutrition
- All animals are consumers

Trophic Structure: Consumers

2. Primary Consumers

- They are also called herbivores
- These are the animals which feed on plants
- E.g.,
  - insects
  - grain eating birds
  - reptiles, birds and mammals which feed on leaves and other vegetation

Trophic Structure: Consumers

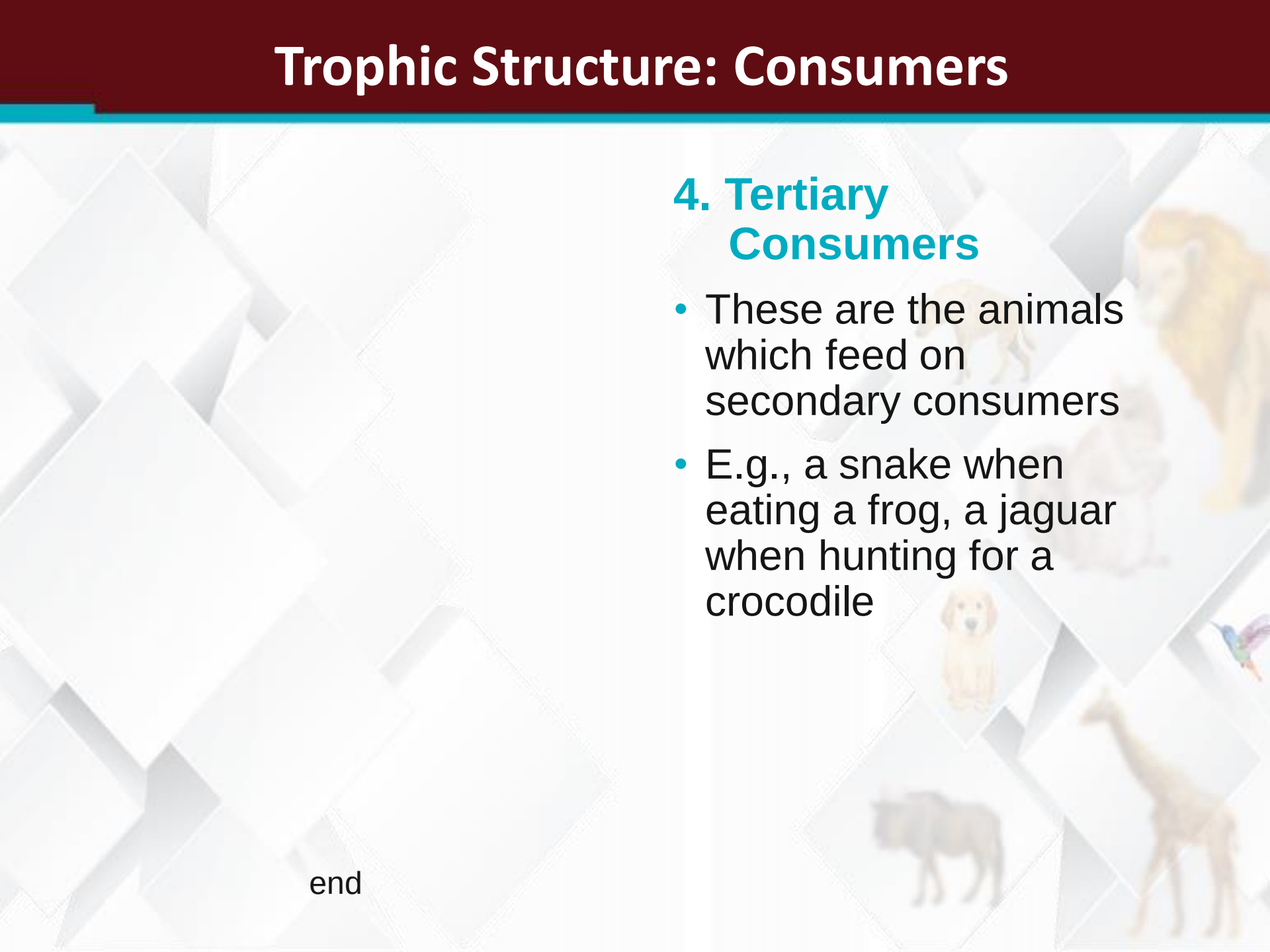
3. Secondary Consumers

- Also called carnivores
- These are the animals which feed on herbivores
- E.g., lion, tiger, frog

Trophic Structure: Consumers

4. Tertiary Consumers

- These are the animals which feed on secondary consumers
- E.g., a snake when eating a frog, a jaguar when hunting for a crocodile



end

Principles of Animal life 1

Trophic Structure: Decomposers

Trophic Structure: Decomposers

Contents

1. Decomposers
2. Bacteria and Fungi as Decomposers
3. Recycling of Nutrients

Trophic Structure: Decomposers

1. Decomposers

- These are the organisms which obtain energy by breaking down dead organic matter into simpler compounds
- Also called saprotrophs

Trophic Structure: Decomposers

2. Bacteria and Fungi as Decomposers

- Chief decomposers of an ecosystem are bacteria and fungi.
- They release extracellular enzymes which breakdown substrate molecules
- Organisms at all trophic level when die are used by decomposers

Trophic Structure: Decomposers

3. Recycling of Nutrients

- If there are no decomposers mounts of dead bodies will be every where
- All useful minerals will remain lock in dead bodies
- There will be little space to live
- Decomposition recycle the nutrients in ecosystem

end

Principles of Animal life 1

Ecological Efficiencies

Ecological Efficiencies

Contents

1. Ecological Efficiencies
2. Efficiencies of Animals

Ecological Efficiencies

1. Ecological Efficiencies

- It is the efficiency by which animal at a trophic level convert food into new biomass
- A large amount of food is not digested by the animals

Ecological Efficiencies

2. Efficiencies of Animals

- Herbivores have lot of plant biomass, but they are less efficient
- Carnivores are more efficient
- At an average 10% biomass conversion efficiency

Ecological Efficiencies

2. Efficiencies of Animals

- Biomass conversion also depends upon the maintenance requirement of animals
- Efficiencies range for less than 1% for most herbivores to more than 35% for ectotherm carnivores

end

Principles of Animal life 1

Ecological Pyramids

Ecological Pyramids

Contents

1. Ecological Pyramids
2. Pyramid of Numbers
3. Pyramid of Biomass
4. Pyramid of Energy

Ecological Pyramids

1. Ecological Pyramids

- It is the graphical representation of relationship between organisms at different trophic levels of an ecosystem
- Three most common types of pyramids:
- Pyramid of number
- Pyramid of biomass
- Pyramid of energy

Ecological Pyramids

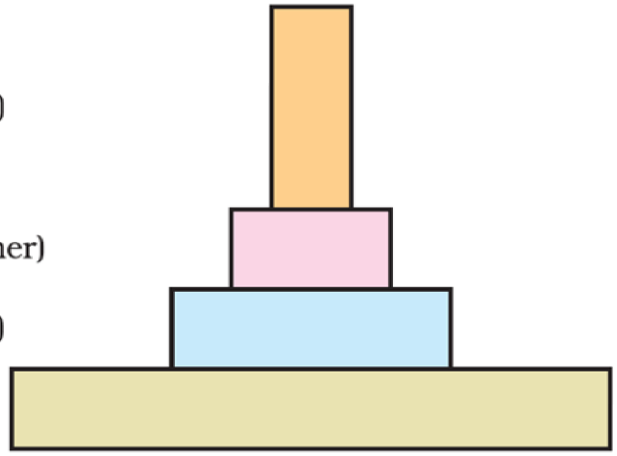
2. Pyramid of Numbers

- It is the graphical representation of total number of different organisms present at each trophic level
- This shows the population of organisms in a food chain

Ecological Pyramids

2. Pyramid of Numbers

Trophic level	Number of individuals
TC (Tertiary consumer)	3
SC (Secondary consumer)	3,54,000
PC (Primary consumer)	708, 000
PP (Primary producer)	5,842,000



Ref. neostencil.com

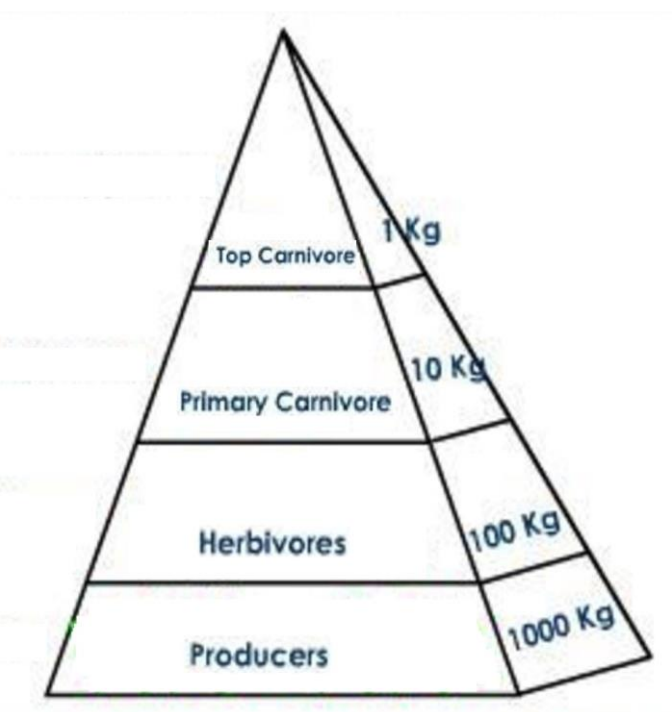
Ecological Pyramids

3. Pyramid of Biomass

- It is the representation of total mass of living matter present at each trophic level.
- Biomass decrease at each ascending trophic level
- Biomass is measured in grams of dry mass per square meter

Ecological Pyramids

3. Pyramid of Biomass



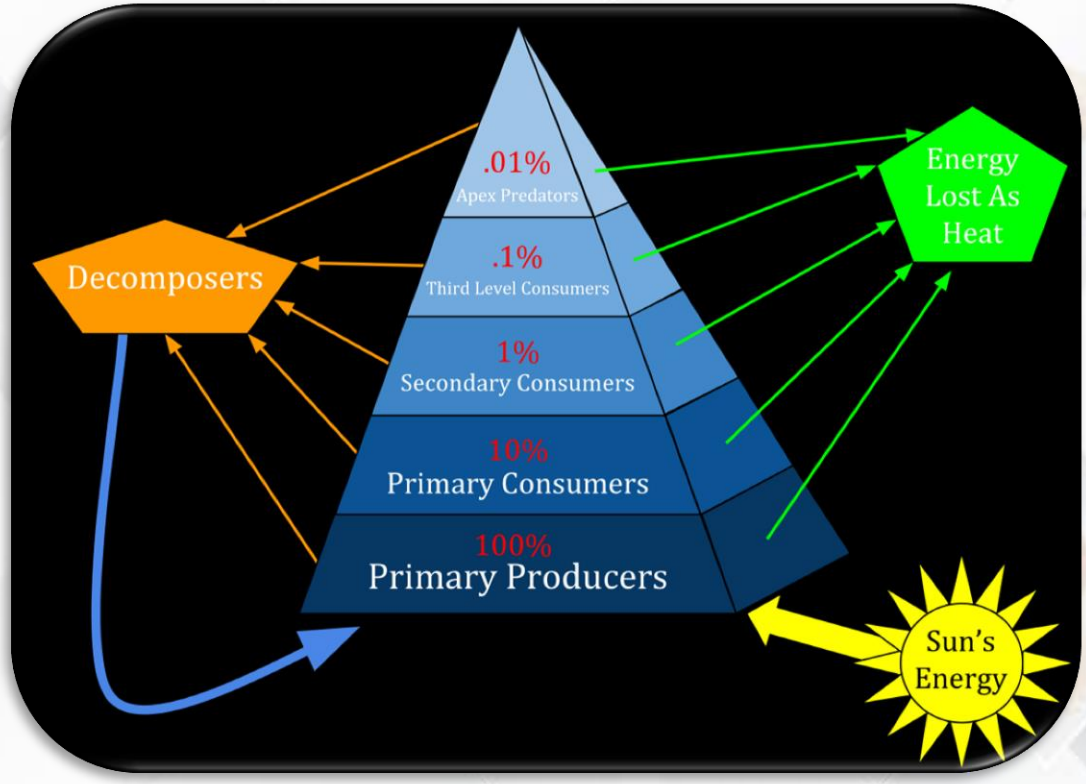
Ecological Pyramids

4. Pyramid of Energy

- It is the representation of total amount of energy present at each trophic level in an ecosystem
- It is always upright
- Energy pyramids help to compare the productivity of ecosystems.

Ecological Pyramids

4. Pyramid of Energy



Ref. .en.wikipedia.org

Principles of Animal life 1

Ecological Pyramids: Examples

Ecological Pyramids: Examples

Contents

1. Pyramid of Numbers Example
2. Pyramid of Biomass Example

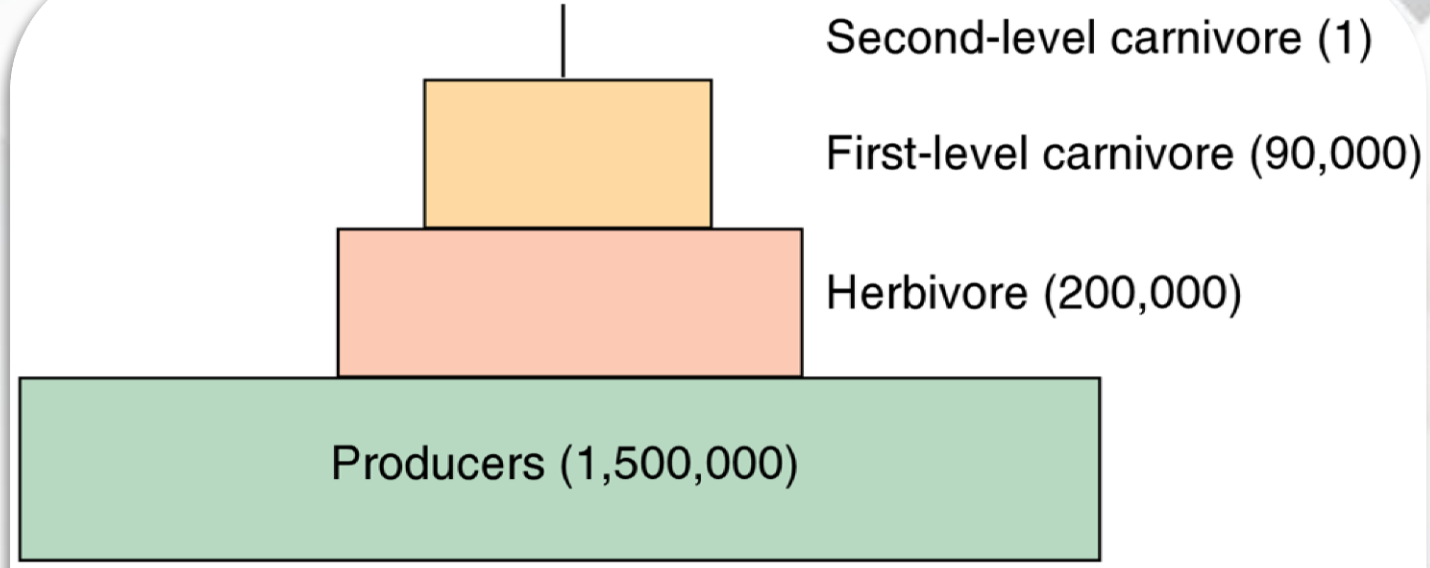
Ecological Pyramids: Examples

1. Pyramid of Numbers Example

- In a grass land the survey was conducted to check the trophic structure.
- Following data was obtained
- Producers: 1.5×10^6
- Primary consumers: 2×10^5
- Secondary consumers: 9×10^4
- Tertiary consumers: 1

Ecological Pyramids: Examples

2. Pyramid of Numbers



Pyramid of numbers (grassland)

Ref. Hickman. Integrated principles of Zoology. 11th ed. p.836

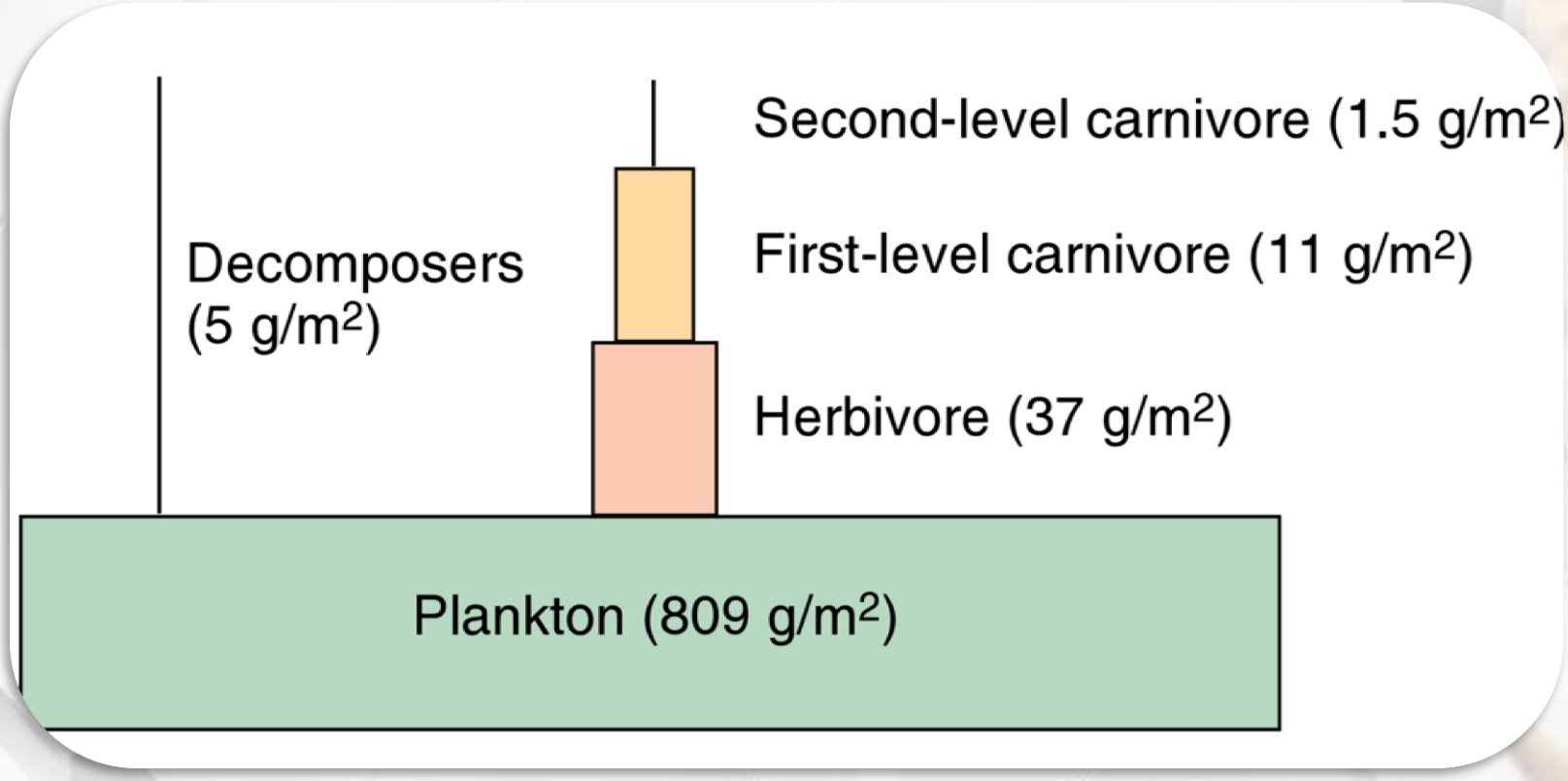
Ecological Pyramids: Examples

3. Pyramid of Biomass Example

- In a pond the survey was conducted to check the trophic structure.
- Following data was obtained
- Producers: 809 g/m^2
- Primary consumers: 37 g/m^2
- Secondary consumers: 11 g/m^2
- Tertiary consumers: 1.5 g/m^2

Ecological Pyramids: Examples

3. Pyramid of Biomass



Ref. Hickman. Integrated principles of Zoology. 11th ed. p.836

Principles of Animal life 1

Energy and Matter Flow in an Ecosystem

Energy and Matter Flow in an Ecosystem

Contents

1. Energy Budget
2. Gross Productivity
3. Net Productivity
4. Flow of energy
5. Flow of materials

Energy and Matter Flow in an Ecosystem

1. Energy Budget

- It is the balance sheet of energy gained and energy expenditure of an organism
- An organism is an open thermodynamic system which exchange energy with its surrounding in three ways i.e., heat, work and potential energy of biomolecules

Energy and Matter Flow in an Ecosystem

1. Energy Budget

- Energy budget allocation varies for ectotherms and endotherms.
- Ectotherms rely on environment as heat source
- Endotherms rely on metabolic heat to maintain body temperature

Energy and Matter Flow in an Ecosystem

2. Gross Productivity

- Total amount of energy assimilated by an organism when an animal eats food
- When an animal eats, food passes through its gut and nutrients are absorbed

Energy and Matter Flow in an Ecosystem

3. Net Productivity

- Most of the energy assimilated is used in metabolic demands which include cellular metabolism and regulation of body temperature
- So, net productivity is the amount of energy store in the form of biomass

Energy and Matter Flow in an Ecosystem

4. Energy Flow

- Organisms require energy in one form or the other
- Energy flow in an ecosystem is major factor for survival for great number of organisms
- Source of energy is sunlight
- Energy flows via food chain and food web
- Flow is by 10% law

Energy and Matter Flow in an Ecosystem

4. Energy Flow

- Energy flow is unidirectional

Energy and Matter Flow in an Ecosystem

5. Flow of material

- All elements essential for life are derived from environment
- They make all form of molecules for life.
- When animals die and their dead bodies are decomposed, these materials are recycled in the ecosystem
- Nutrient flow in ecosystem in a cyclic way

end

Principles of Animal life 1

Carbon Cycle (I)

Carbon Cycle (I)

Contents

1. Importance of Carbon
2. Carbon Reservoir
3. Carbon Source in Aquatic Environment
4. Carbon in Calcium Carbonate
5. Carbon in Fossil Fuels

Carbon Cycle (I)

1. Importance of Carbon

- Carbon is the basic element of all organic molecules
- All biomolecules have a carbon backbone
- It has ability to bond with four other atoms at the same time

Carbon Cycle (I)

2. Carbon Reservoir

- It is plentiful on earth and is rarely limiting factor
- Reservoir for carbon is CO_2 in atmosphere and water.
- When organic matter is completely oxidized, it releases CO_2 into atmosphere

Carbon Cycle (I)

3. Carbon Source in Aquatic Environment

- In aquatic environment, CO_2 combines with water to form H_2CO_3
- When CO_2 level in water decreases H_2CO_3 breaks to release CO_2

Carbon Cycle (I)

4. Carbon in Calcium Carbonate

- Some carbon is tied up in the form of calcium carbonate ($CaCO_3$)
- This $CaCO_3$ is in the part exoskeleton and endoskeleton of mollucs, echinoderms, and other invertebrates
- They ultimately form limestone reservoirs

Carbon Cycle (I)

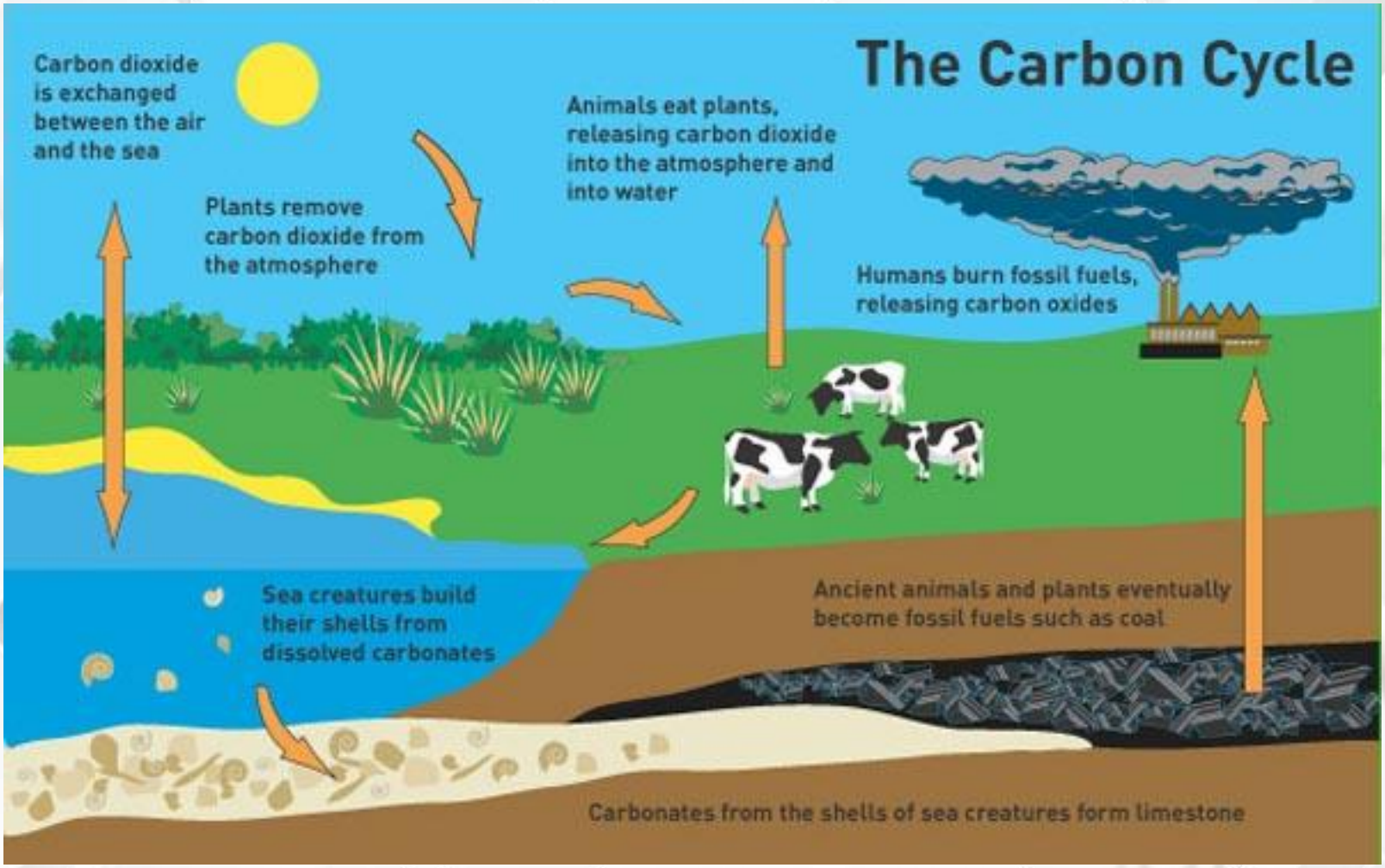
5. Carbon in fossil fuels

- A supply of carbon is also tied up in fossil fuels
- During 65 million year of Carboniferous period (280mya), organic matter got buried and converted into coal, oil and gas.
- When fossil fuels are used, carbon returns to atmosphere.

end

- **LESSON 237-238**
  CARBON CYCLE
  Carbon is the basic element of all organic molecules. All biomolecules have a carbon backbone. Carbon has ability to bond with four other atoms at the same time. Carbon is plentiful on earth and is rarely limiting factor. Reservoir for carbon is CO_2 in atmosphere and water. Photosynthetic plants use this carbon dioxide of air and convert organic compounds. These plants are being eaten by animals. Animals utilize these organic compounds for the synthesis of different biomolecules and also use as a source of energy. When organic matter is completely oxidized, it releases CO_2 into atmosphere. In aquatic environment, CO_2 combines with water to form H_2CO_3.

- When CO_2 level in water decreases H_2CO_3 breaks to release CO_2. Some carbon is tied up in the form of calcium carbonate ($CaCO_3$). This $CaCO_3$ is in the part exoskeleton and endoskeleton of mollucs, echinoderms, and other invertebrates. They ultimately form limestone reservoirs. A large supply of carbon is also tied up in fossil fuels. During 65 million year of Carboniferous period (280mya), organic matter got buried and converted into coal, oil and gas. When fossil fuels are used, carbon returns to atmosphere.


The Carbon Cycle
Carbon dioxide is exchanged between the air and the sea
Plants remove carbon dioxide from the atmosphere
Animals eat plants, releasing carbon dioxide into the atmosphere and into water
Humans burn fossil fuels, releasing carbon oxides
Sea creatures build their shells from dissolved carbonates
Ancient animals and plants eventually become fossil fuels such as coal
Carbonates from the shells of sea creatures form limestone

Principles of Animal life 1

Nitrogen Cycle (I)

Nitrogen Cycle (I)

Contents

1. Importance of Nitrogen
2. Reservoir of Nitrogen
3. Nitrogen Fixation
4. Nitrification
5. Assimilation
6. Ammonification
7. Denitrification

Nitrogen Cycle (I)

1. Importance of Nitrogen

- Nitrogen is an essential component of proteins, nucleic acids and ATP
- It is also found in membrane lipids and modified carbohydrates

Nitrogen Cycle (I)

2. Reservoir of Nitrogen

- Atmosphere is main reservoir
- About 78% nitrogen present in air
- But this is not directly available to plants
- Plants can use nitrogen in the form of NH_3 and NO_3^-
- Animals obtain N from plants

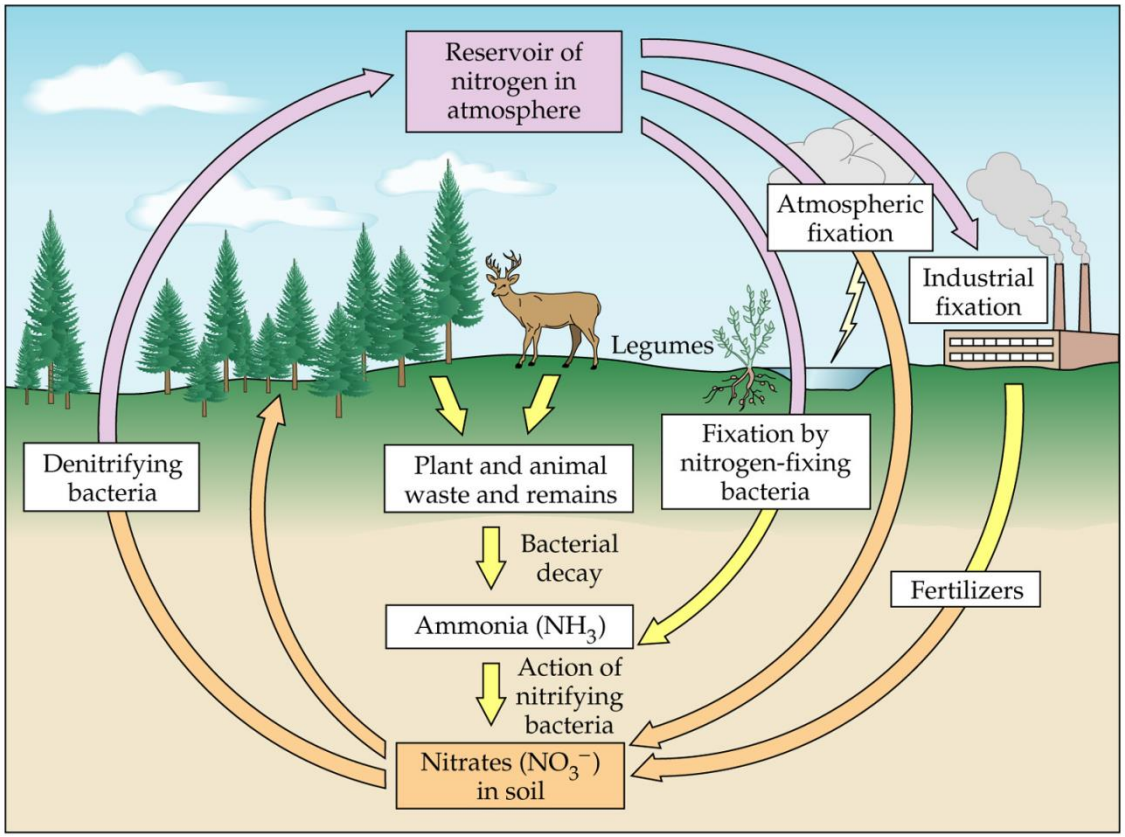
Nitrogen Cycle (I)

3. Nitrogen Fixation

- Gaseous nitrogen converted to NH_3
- Aqueous form of NH_3 is NH_4^+
- Bacteria mainly in the roots of leguminous plants and cyanobacteria do this job
- Lightening
- Industry

Nitrogen Cycle (I)

The Nitrogen Cycle



Ref. http://data.allenai.org

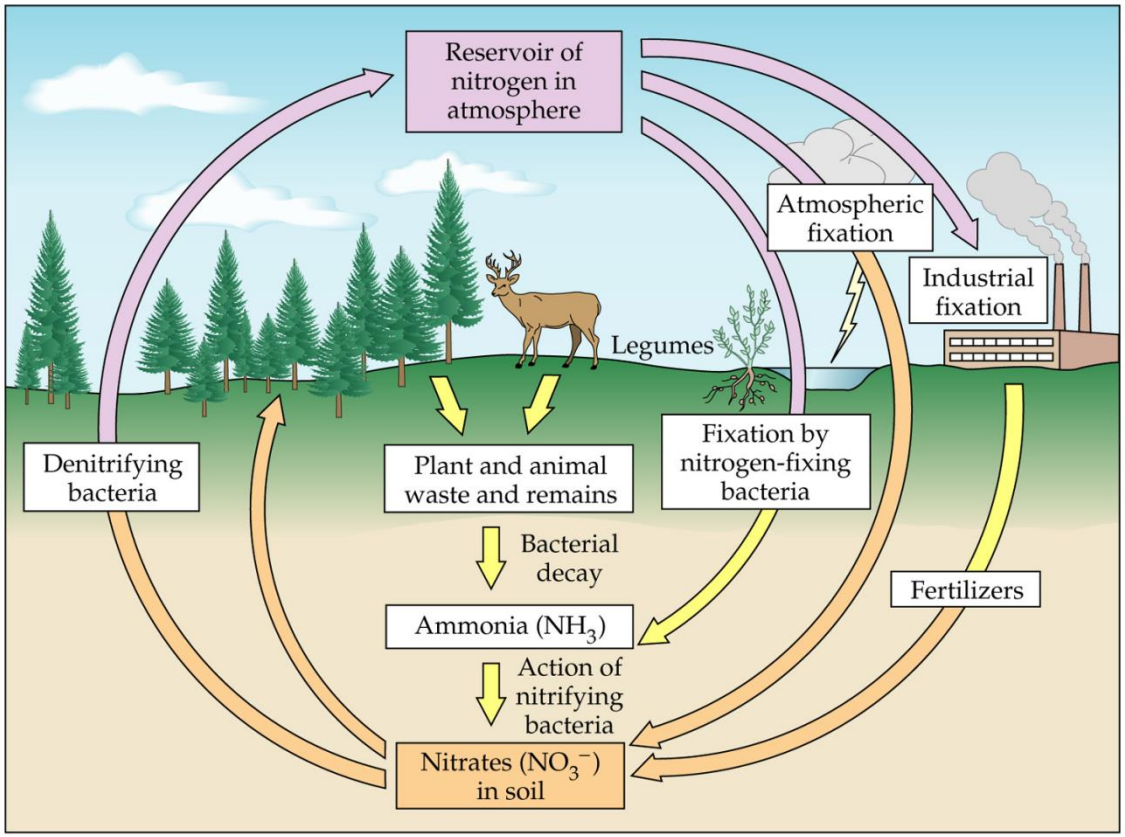
Nitrogen Cycle (I)

4. Nitrification

- It is the process by which NH_3 is converted to nitrate (NO_3^-)
- It is a two step process
- In first step NH_3 is converted to nitrite (NO_2^-) by *Nitrococcus* bacteria
- In second step NO_2^- is converted to NO_3^- by *Nitrobacter*

Nitrogen Cycle (I)

The Nitrogen Cycle



Ref. http://data.allenai.org

- **LESSON 241**
- **PHOSPHORUS CYCLE**
- **1. Importance of Phosphorus**
- Phosphorus is an essential component of DNA, RNA, ATP, Phospholipids, Vertebrate skeleton and Teeth. Sedimentary rocks contains deposits of phosphorus. Extraction of phosphorus takes place by weathering, rain and other erosion processes. Plant, microbes and fungi absorb phosphorus from soil. Animals use phosphorus by utilizing plants.

- Some phosphorus also comes through water. As plants and animals die, phosphorus is return back to the environment via decomposition. Living organisms may again use this phosphorus. Some phosphorus may run along with water and become part of sediments. Human has significant impact by using phosphorus in fertilizers. Agricultural runoff takes phosphorus to water bodies which excessive growth of algae. This phenomenon is called eutrophication.

Principles of Animal life 1

Sulfur Cycle

Sulfur Cycle

Contents

1. Importance of Sulfur
2. Reservoir of Sulfur
3. Mineralization
4. Oxidation
5. Reduction and Incorporation into Organic Compounds
6. The Sulfur Cycle

Sulfur Cycle

1. Importance of Phosphorus

- Essential component of:
- Proteins
- Enzymes
- Vitamins

Sulfur Cycle

2. Reservoir of Phosphorus

- Abundant in nature
- Bright yellow crystalline solid
- Most of it is tied up in rock and buried in depth of oceans
- Atmosphere also contain oxides of sulphur

Sulfur Cycle

3. Mineralization

- It is the process which organic sulfur is converted into inorganic form
- Hydrogen sulfide, elemental sulfur and sulfide minerals are formed

Sulfur Cycle

4. Oxidation

- At this step sulfides and elemental sulfur are converted to sulfates

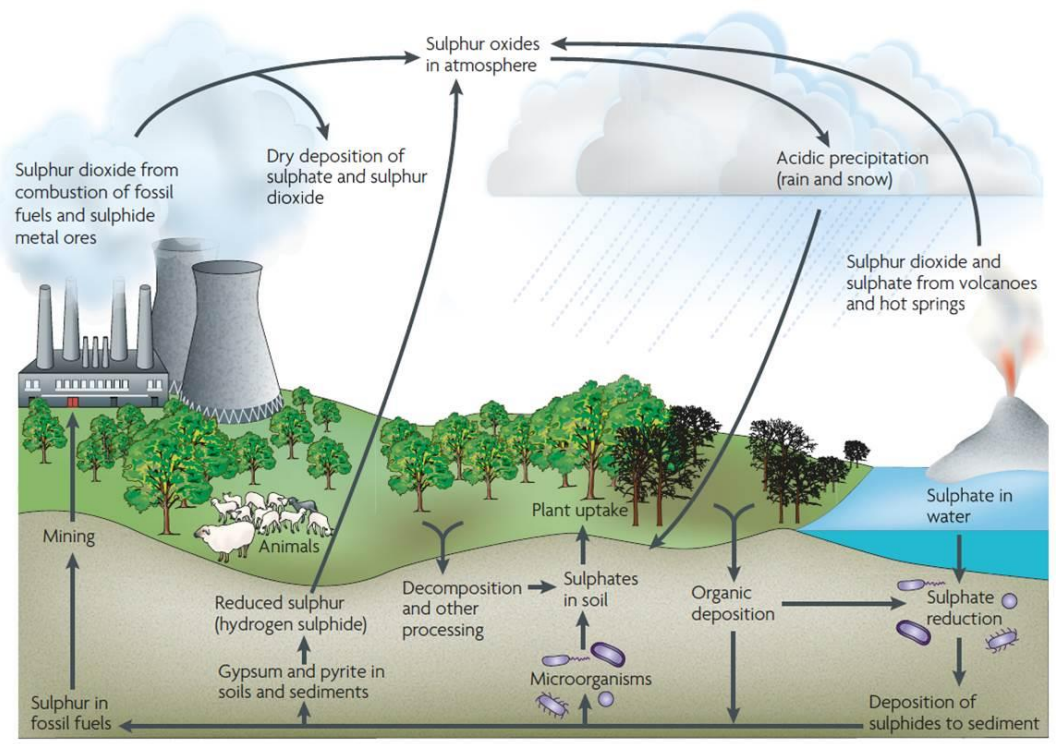
Sulfur Cycle

5. Reduction and Incorporation into Organic Compounds

- Sulfates are reduced by plants, fungi and bacteria and incorporated into organic compound

Sulfur Cycle

6. The Sulfur Cycle



Ref. wur.nl

Principles of Animal life 1

Water Cycle

Water Cycle

Contents

1. Water on Lands
2. Evaporation
3. Precipitation
4. Runoff
5. Infiltration
6. Subsurface Flow

Water Cycle

1. Water on land

- Essential for life
- Life can not exist without water
- 97% water is marine with high salt concentration
- 3% freshwater, mostly glaciers and groundwater
- Less than 1% of water is available to human

Water Cycle

2. Evaporation,

- Water in the oceans, lakes, swamps, rivers and soil when heated is turned into vapors
- Similarly water also evaporates from surfaces of plants and animals
- Transpiration
- Sublimation

Water Cycle

3. Precipitation

- Water vapors move up higher, condense to form clouds
- Clouds are pushed around the world by moving air.
- Further cooling causes precipitation.
- Precipitation may be in the form of rain, hail or snow
- 78% occur over oceans

Water Cycle

4. Runoff

- It is the flow of water across the land
- It includes
  - Surface runoff
  - Channel runoff

Water Cycle

5. Infiltration

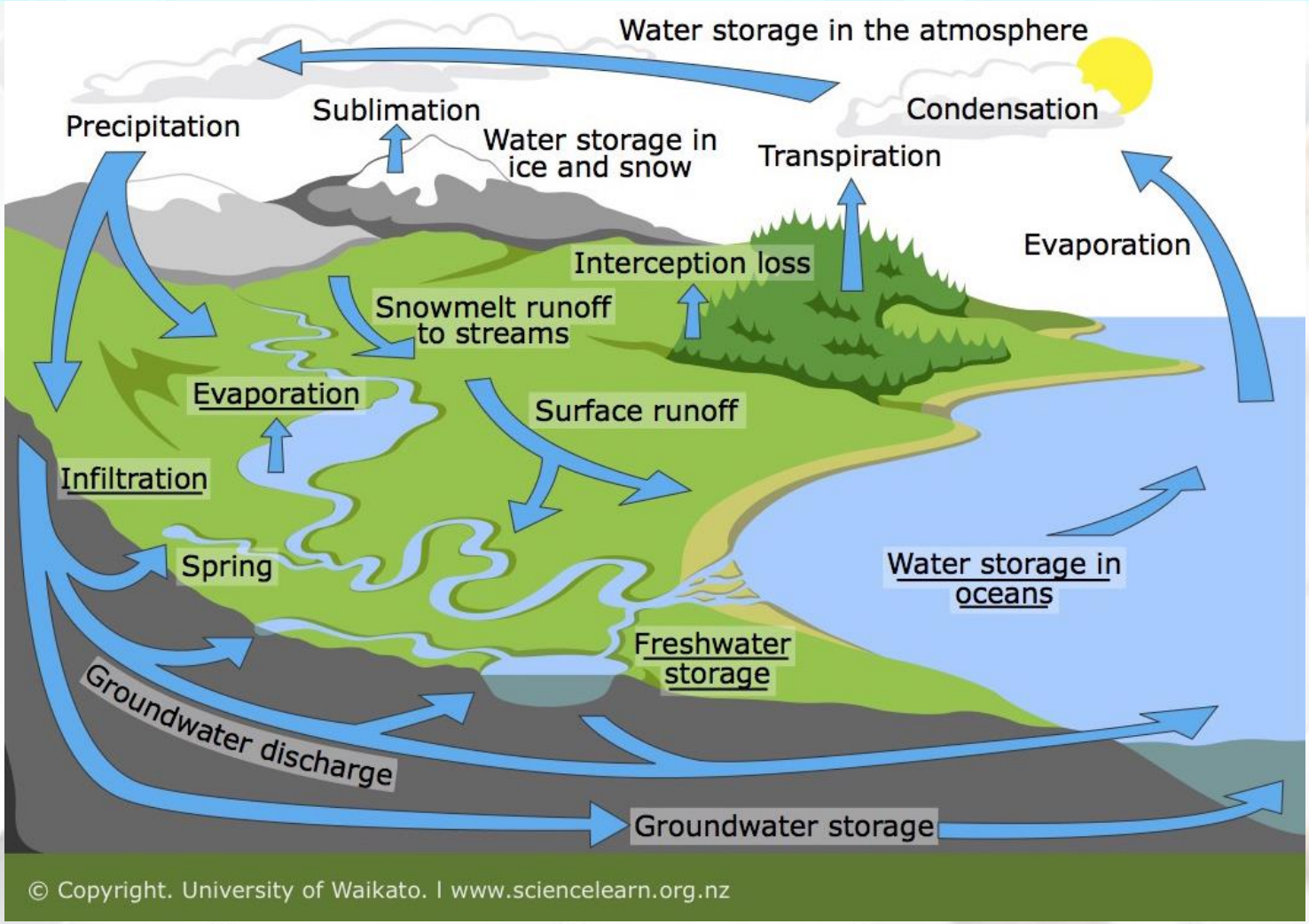
- Flow of water from surface into ground
- It becomes soil moisture or ground water
- Underground water is present in aquifers
- Under ground water may be return to surface as springs or by pumps

Water Cycle

6. Subsurface Flow

- Under ground water moves slowly and may remain in aquifer for thousands of years
- Eventually is reaches to the oceans

Water Cycle



Ref. sciencelearn.org.nz

Principles of Animal life 1

Biosphere and its Subdivision

Biosphere and its Subdivision

Contents

1. Biosphere
2. Subdivisions of Biosphere
3. Lithosphere
4. Hydrosphere
5. Atmosphere

Biosphere and its Subdivision

1. Biosphere

- Thin outer layer of earth which can support life
- It includes all life on earth and the physical environment on which living organisms exist and interact

Biosphere and its Subdivision

2. Subdivision of Biosphere

- Life exists in soil, water and air
- Biosphere is subdivided accordingly
  - Lithosphere
  - Hydrosphere
  - Atmosphere

Biosphere and its Subdivision

3. Lithosphere

- It is the rigid material on the surface of earth
- Most of the terrestrial animals live on lithosphere
- Lithosphere is the ultimate source of all mineral elements required by all living things

Biosphere and its Subdivision

4. Hydrosphere

- It is the water on the surface of earth
- It extends into the lithosphere and atmosphere
- Water is distributed on earth by global hydrological system
- Water returns to oceans by precipitation

Biosphere and its Subdivision

5. Atmosphere

- It the gaseous content of biosphere
- Atmosphere extends up to 3500 km above the surface of earth
- Life in atmosphere is present in lower 8-15km called tropohosphere
- The main gases in troposphere are nitrogen, oxygen and carbon dioxide

Biosphere and its Subdivision

Ref. tourism.euskadi.eus

Principles of Animal life 1

Ecosystem and Biome

Ecosystem and Biome

Contents

1. Ecosystem
2. Biome
3. Biome are Identified on the Basis of Plants
4. Ecocline
5. Coriolis Effect
6. Cells

Ecosystem and Biome

1. Ecosystem

- It is the interaction between living and non living components of the environment

Ecosystem and Biome

2. Biome

- Biome is an area of earth with distinctive association of plants and animals
- Major biomes of the world include:
- Tundra
- Coniferous forests
- Deciduous forests
- Grasslands
- Deserts

Ecosystem and Biome

3. Biomes are Identified on the Basis of Plants

- Botanists divided terrestrial environment long ago into different biomes
- Characterization of biome on the basis of animal life is difficult
- A biome is identified by dominant plant formation

Ecosystem and Biome

3. Biomes are Identified on the Basis of Plants

- Because plants support a particular type of animals, so the biomes also have characteristic type of animals

Ecosystem and Biome

4. Ecocline

- The boundaries of biome are not distinct
- It contains tree of both types of biomes
- This indistinctive boundary having a mixture of plants is called ecocline

Ecosystem and Biome

5. Coriolis Effect

- Biomes are distinct by climate, pattern of rainfall, temperature and solar radiations
- Warm air at equator rises and move towards poles and cold air away from pole
- The earth rotation deflect this air to right and left of the poles
- The effect is called Coriolis effect

Ecosystem and Biome

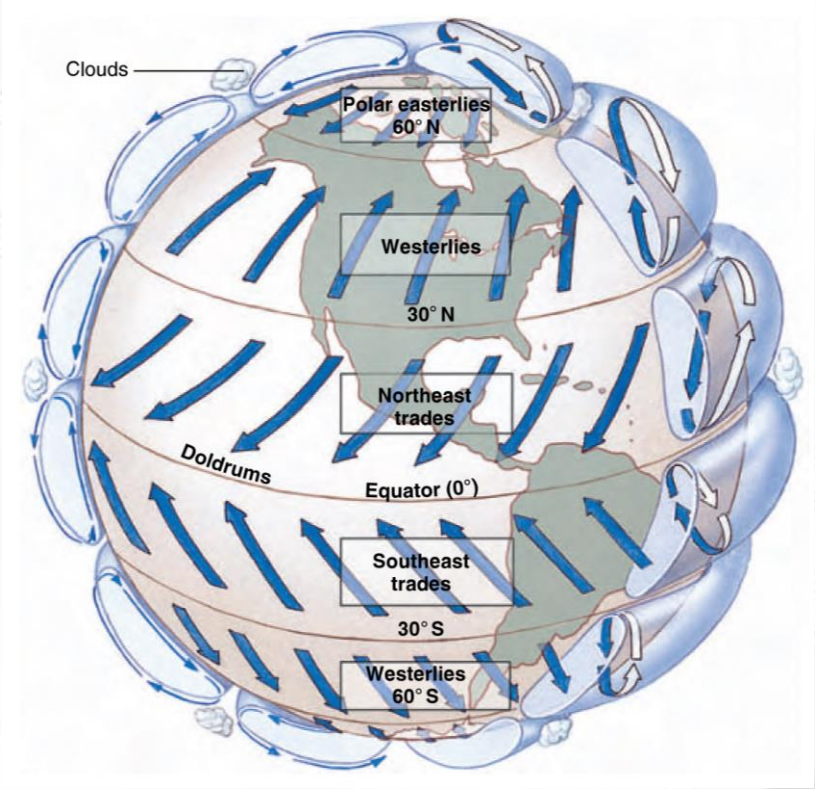
6. Cell

- Air circulation in each hemisphere is broken into three zones from poles

1. Zero to 30^{o} latitude
2. 30^{o} to 60^{o}
3. When cold air move back to poles at high altitude

Ecosystem and Biome

6. Coriolis Effect and Cells



Ref. Hickman. Integrated Principles of Zoology. 11th Ed. p.808

Principles of Animal life 1

Tundra Ecosystem (I) (I)

Tundra Ecosystem (I)

Contents

1. Tundra
2. Characteristics of Tundra
3. Types of Tundra

Tundra Ecosystem (I)

1. Tundra

- Treeless region found in arctic and top of mountains with severe cold climate and permanently frosted subsoil

Tundra Ecosystem (I)

2. Characteristics of Tundra

- **Climate:** a long winter is interrupted by a short summer. Plant can grow in this summer of 60 days
- **Soil:** remain frozen in most of the year. Subsoil permafrost
- Most tundra region is covered with bogs, marshes, ponds and decayed vegetation

Tundra Ecosystem (I)

2. Characteristics of Tundra

- **Average precipitation:** is very low, less than 25cm/year
- **Temperature:** cold environment, average annual temperature is -10^{o}C although it may drop to -50^{o}C in winter and rise to 10^{o}C in summer

Tundra Ecosystem (I)

2. Characteristics of Tundra

- **Plant life:** no trees, lichens, grasses and dwarf woody plants. Growth of plants is very slow due to extreme conditions
- **Animal life:** Rodents such as arctic hare, musk oxen, arctic foxes, owls, weasels and caribou (reindeer)

Tundra Ecosystem (I)

Ref. http://baophapluat.club

Tundra Ecosystem (I)

Ref. en.wikipedia.org

Tundra Ecosystem (I)

3. Types of Tundra

- Tundra exists in two forms:
  - Arctic tundra
  - Alpine tundra

end

Principles of Animal life 1

Tundra Ecosystem (II)

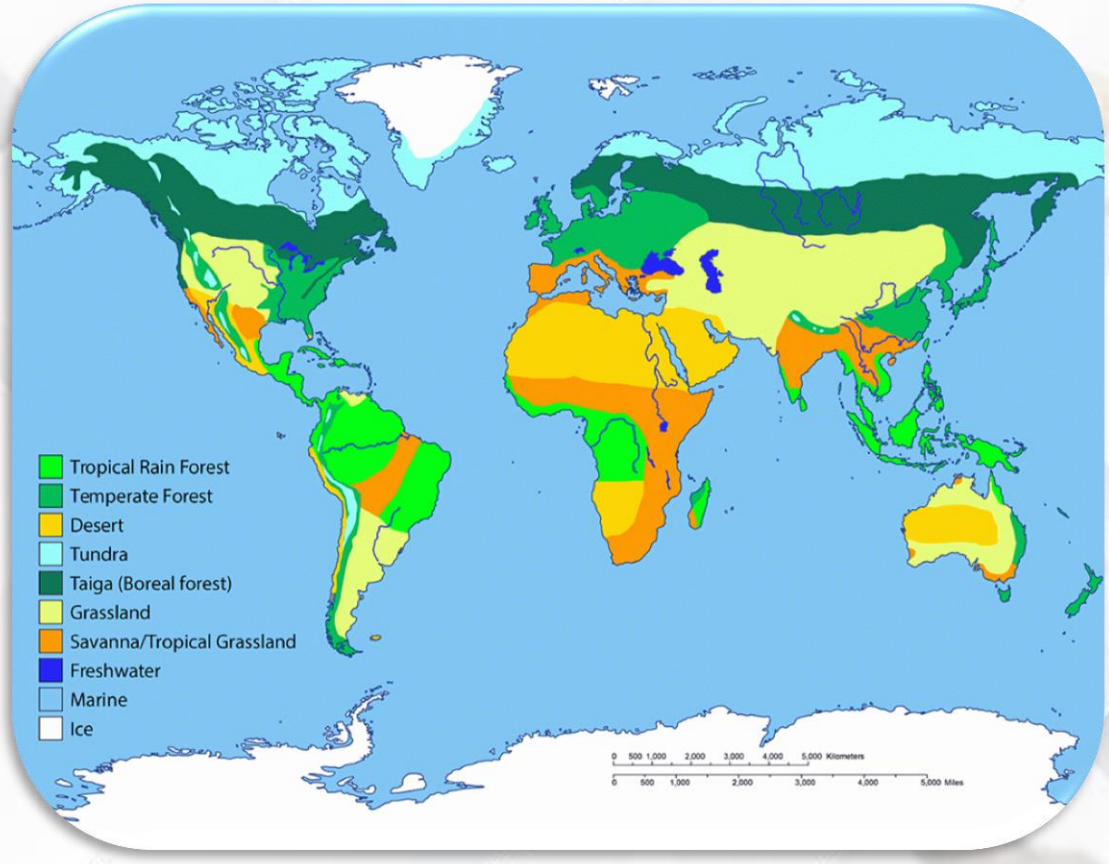
Tundra Ecosystem (II)

Contents

1. Arctic Tundra
2. Alpine Tundra

Tundra Ecosystem (II)

1. Arctic Tundra map



Ref. askabiologist.asu.edu

Tundra Ecosystem (II)

2. Arctic Tundra

- It is present in northern hemisphere
- It includes the areas of northern Russia and Canada
- It is the home of Nganasan and Nenets people
- Temperature ranges form -30^{o}C to 20^{o}C
- Highly vulnerable environment

Tundra Ecosystem (II)

2. Arctic Tundra

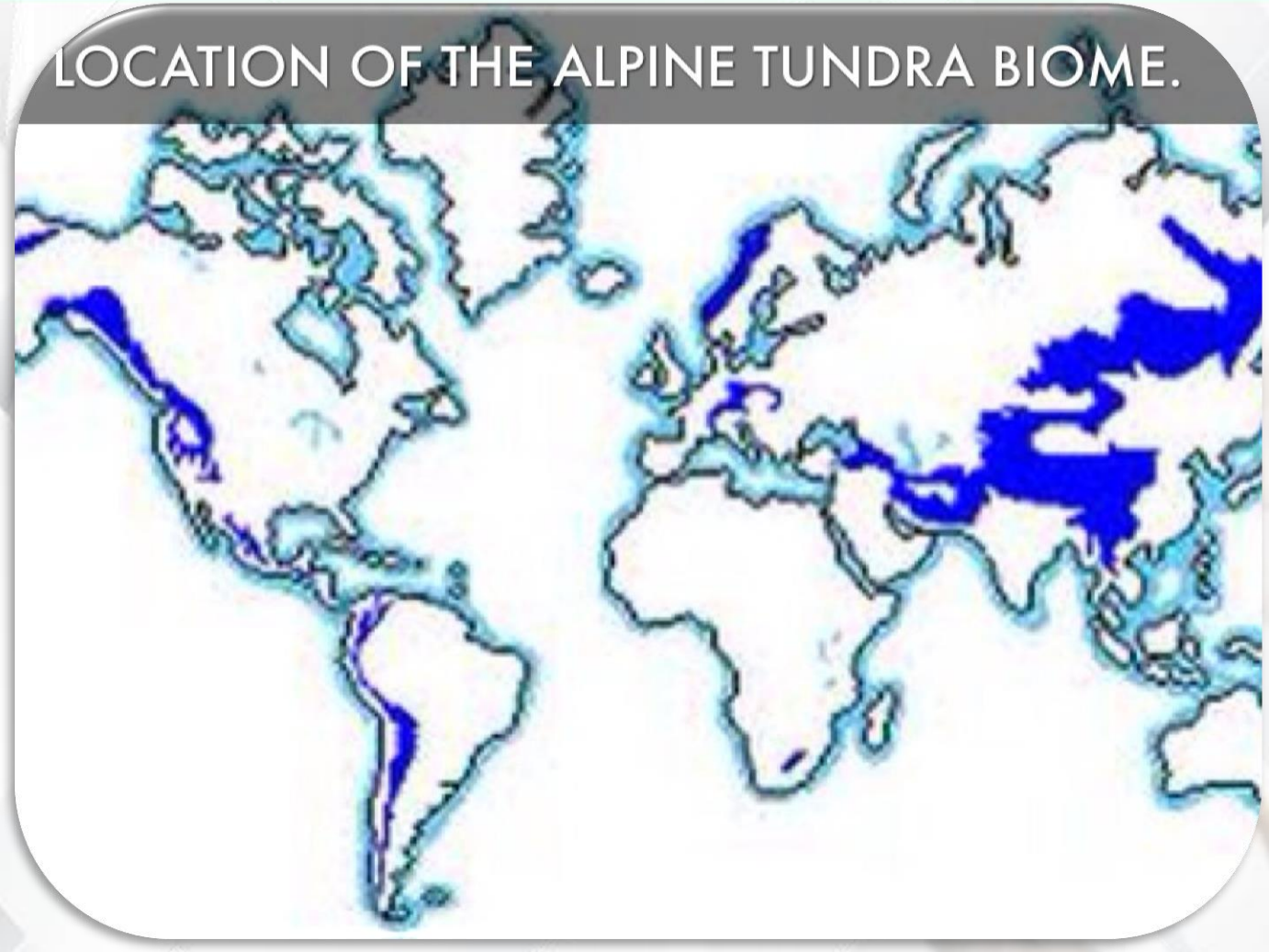
- Animal life include arctic fox, polar bear, gray wolves, caribou, snow geese and musk ox
- Plant life lichen, mosses, grasses and shrubs

Tundra Ecosystem (II)

3. Alpine Tundra

- High elevation are too cold and windy to sport plant life
- North American Cordillera,
  the Alps and Pyrenees of Europe,
  the Himalaya and Kara koram of Asia,
  the Andes of South America, and
  the Eastern Rift mountains of Africa

Tundra Ecosystem (II)



Ref. haikudeck.com

Tundra Ecosystem (II)

4. Flora and Fauna of Alpine Tundra

- Lichens, grasses, shrubs
- They have much more roots and rhizome than shoots which are also storage site during winter.
- Animal life include kea, marmot, mountain goat, bighorn sheep, chinchilla, yak and pika

Tundra Ecosystem (II)

3. Climate Change Impact on Tundra

- Arctic tundra changing dramatically due to global warming
- Animal previously not found in arctic tundra are now competing with native animals
- Shrubs are getting taller and animal like spider are more active.

Tundra Ecosystem (II)

3. Climate Change Impact on Tundra

- Permafrost is melting
- It holds large amount of carbon and methane
- Climate change may change tundra from carbon sink to carbon contributor

end

Principles of Animal life 1

Coniferous Forests (I)

Coniferous Forests (I)

Contents

1. Coniferous Forests
2. Characteristics of Coniferous Forests

Coniferous Forests (I)

1. Coniferous Forests

- Coniferous forest are dominated by cone bearing, needle shape evergreen plants called conifers

Coniferous Forests (I)

2. Characteristics of Coniferous Forest

- **Climate:** cool summer and cold winder, growing season is 130 days

- **Soil:** soil is acidic and relatively infertile

Coniferous Forests (I)

2. Characteristics of Coniferous Forest

- **Average Precipitation:** moderate rainfall, 40-100cm/year. Most precipitation in the form of heavy snow
- **Temperature:** average annual temperature is 10^{o}C although it may drop to -40^{o}C in winter and rise to 20^{o}C in summer

Coniferous Forests (I)

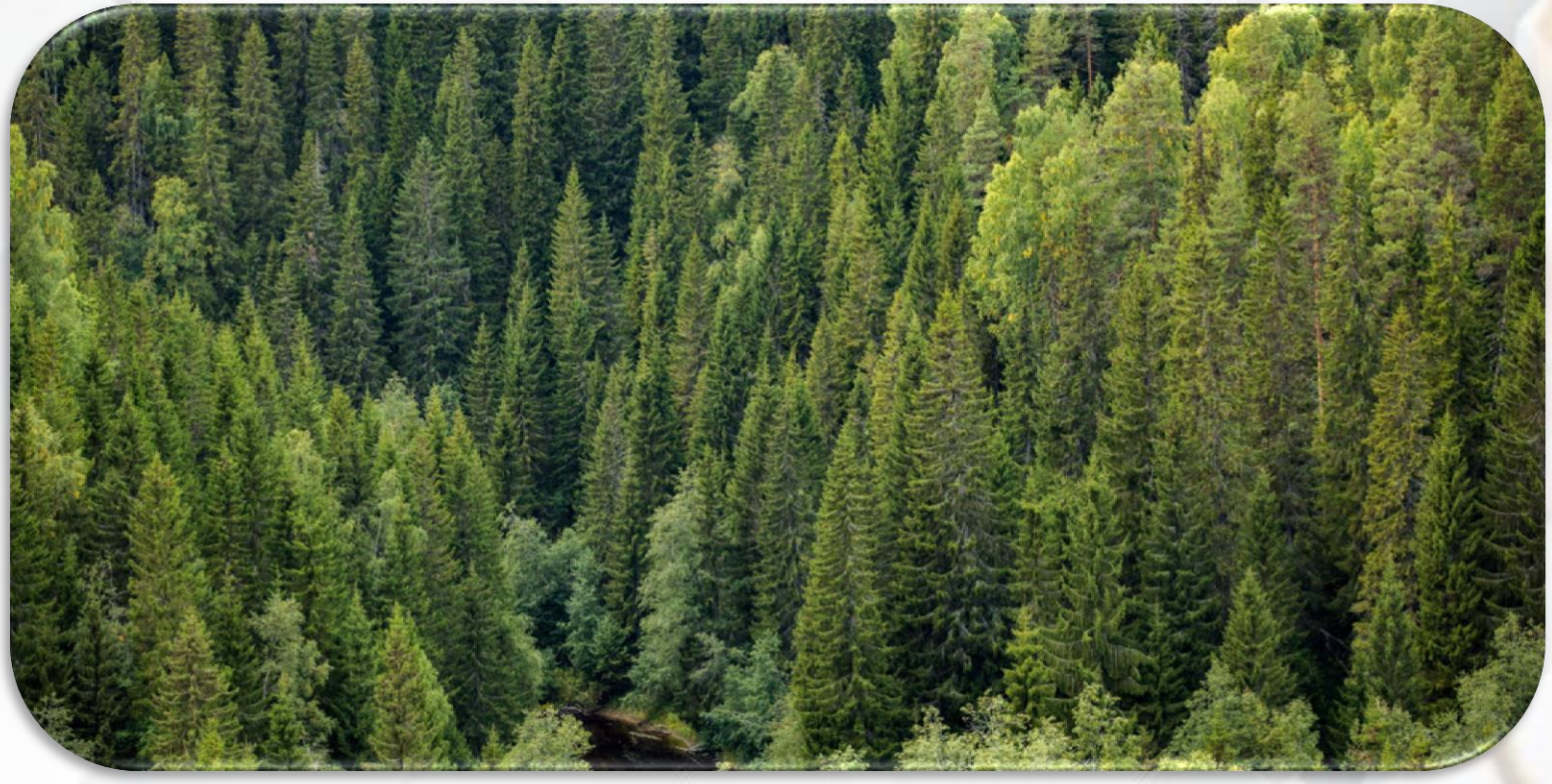
2. Characteristics of Coniferous Forest

- **Plant life:** Spruce, fir and pine. They are evergreen. Some primary production also on warmer summer days
- **Animal life:** low diversity but large population. Snowshoe hare, lynx, wolves, caribou, moose and summer moths

Coniferous Forests (I)

Ref. http://thegreenpowers.com

Coniferous Forests (I)



Ref. sca.com

Principles of Animal life 1

Coniferous Forest (II)

Coniferous Forest (II)

Contents

1. Primary Productivity
2. Recycling of Nutrients
3. Coniferous Forests Location
4. Human Impact

Coniferous Forest (II)

1. Primary Productivity

- It is the main feature of an ecosystem
- Coniferous forests are evergreen, their primary productivity is higher than grasslands but lower than deciduous and tropical forests

Coniferous Forest (II)

2. Recycling of Nutrients

- Decomposition process is slow due to cool environmental conditions
- Nutrients are locked in pine needles that fall from trees
- Fire is important in recycling of nutrients and promoting seed germination

Coniferous Forest (II)

3. Coniferous Forests Location

- It is the largest terrestrial biome on earth
- Taiga: Stretched from North America to Europe and Asia in northern hemisphere
- Temperate Coniferous Forests: They are present worldwide at high altitudes in America, Africa, Europe South Asia

Coniferous Forest (II)

3. Coniferous Forests Location

- In Pakistan they are present at 1000 to 4000 altitude
- Chitral
- Swat
- Dir
- Abbot Abad
- Hilly Areas of Balochistan
- Murree
- Azad Kashmir

Coniferous Forest (II)

4. Human Impact

- Deforestation: Coniferous tree wood is used in buildings and furniture
- Climatic changes:
  - Mild and short winter
  - Acid rain

end

Principles of Animal life 1

Temperate Deciduous Forests

Temperate Deciduous Forests

Contents

1. Temperate Deciduous Forests
2. Climate
3. Soil Conditions
4. Plant Life
5. Animal Life
6. Location
7. Human impact

Temperate Deciduous Forests

1. Temperate Deciduous Forests

- These are forests with moderate climate and well define summer and winter
- They drop their leaves during autumn

Temperate Deciduous Forests

2. Climate

- Four distinct seasons
  - Winter, spring, summer and fall
- Annual precipitation ranges between 75 to 150cm/year
- Temperature varies from -30^{o}C to 30^{o}C

Temperate Deciduous Forests

3. Soil Conditions

- Soil is fertile
- It is brown in colour
- Decomposition of fallen leaves enrich the soil
- Primary productivity is higher than grasslands and coniferous forests

Temperate Deciduous Forests

4. Plant Life

- Broad leaves trees

Oak

Ref. wikipedia.org

Maple

Ref. americanforests.org

Temperate Deciduous Forests

4. Plant Life

- Broad leaves trees

Magnolia

Ref. pinterest.com

Beech trees

Ref. gardenerdy.com

Temperate Deciduous Forests

4. Plant Life

- Broad leaves trees

Fern

Ref. nativeplantspnw.com

Viburnum

Ref. wikipedia.org

Principles of Animal life 1

Temperate Deciduous Forests

Temperate Deciduous Forests

Contents

1. Temperate Deciduous Forests
2. Climate
3. Soil Conditions
4. Plant Life
5. Animal Life
6. Location
7. Human impact

Temperate Deciduous Forests

5. Animal Life

- Great diversity of animal life

Spider

Ref. wikipedia.org

Ants

Ref. sciencing.com

Temperate Deciduous Forests

5. Animal Life

- Great diversity of animal life

White-tailed deer

Ref. animalsake.com

Moose

Ref. wikipedia.org

Temperate Deciduous Forests

5. Animal Life

- Great diversity of animal life

Wolf

Ref. animalsake.com

Raccoons

Ref. pinterest.com

Temperate Deciduous Forests

5. Animal Life

- Great diversity of animal life

Bear

Ref. edu.glogster.com

Tiger

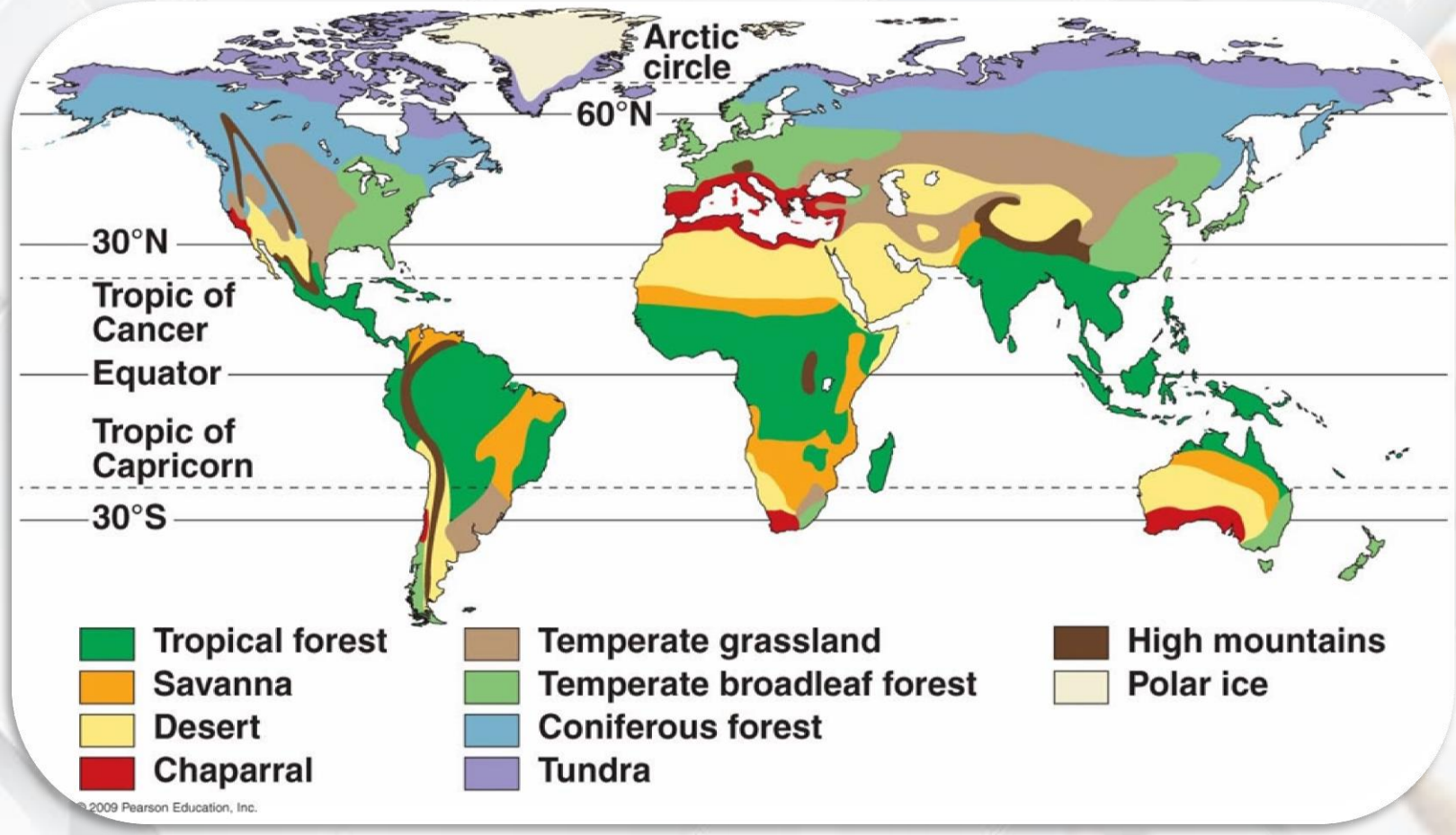
Ref. http://tigers.panda.org

Temperate Deciduous Forests

6. Location

- Eastern USA
- Canada
- Japan
- China
- Some part of Australia

Temperate Deciduous Forests



Temperate Deciduous Forests

7. Human Impact

- Valuable timber
- Mostly cut down from Europe for farming purposes
- Some have been converted to commercial forestry

end

Principles of Animal life 1

The Grassland (I)

The Grassland (I)

Contents

1. The Grassland
2. Climate
3. Soil Conditions
4. Types of grasslands
5. Plant Life
6. Animal Life
7. Location
8. Human impact

The Grassland (I)

1. The Grassland

- These are the areas where vegetation is mainly dominated by the grasses
- They account for 20-40% of world land area
- They lie between forest and deserts

The Grassland (I)

2. Climate

- Grassland is found where there is not enough rain fall
- Precipitation is 25 to 100cm annually
- Temperature ranges from -20^{o}C to 30^{o}C

The Grassland (I)

3. Soil Conditions

- Soil is porous
- The upper layer is nutrient rich making it ideal for agricultural use
- But the low water availability may eventually convert them into deserts

The Grassland (I)

4. Types of Grasslands

- Grasslands of North America are called prairies
- Grasslands of South America and Africa are called tropical savanna

end

Principles of Animal life 1

The Grassland (II)

The Grassland (II)

Contents

1. The Grassland
2. Climate
3. Soil Conditions
4. Types of grasslands
5. Plant Life
6. Animal Life
7. Location
8. Human impact

The Grassland (II)

5. Plant Life

- Grasslands are dominated by different types of grasses.
- Grasses make layering
- Prairies are normally without trees while savanna has scattered trees (less than 30%)
- Clover, salvia, wheat, barley, coneflowers. many species of grasses including, purple needle grass, wild oats, foxtail, ryegrass, and buffalo grass

The Grassland (II)

5. Plant Life

- Plants include clover, salvia, wheat, barley, coneflowers.
- Many species of grasses including, purple needle grass, wild oats, foxtail, ryegrass, and buffalo grass

The Grassland (II)

6. Animal Life

- Grasslands support large populations of herbivores and carnivores
- Insects
- Deer
- Zebra
- Wildebeest
- Giraffe
- Bison
- Lions, tigers, hyenas

The Grassland (II)

Ref. animals.sandiegozoo.org

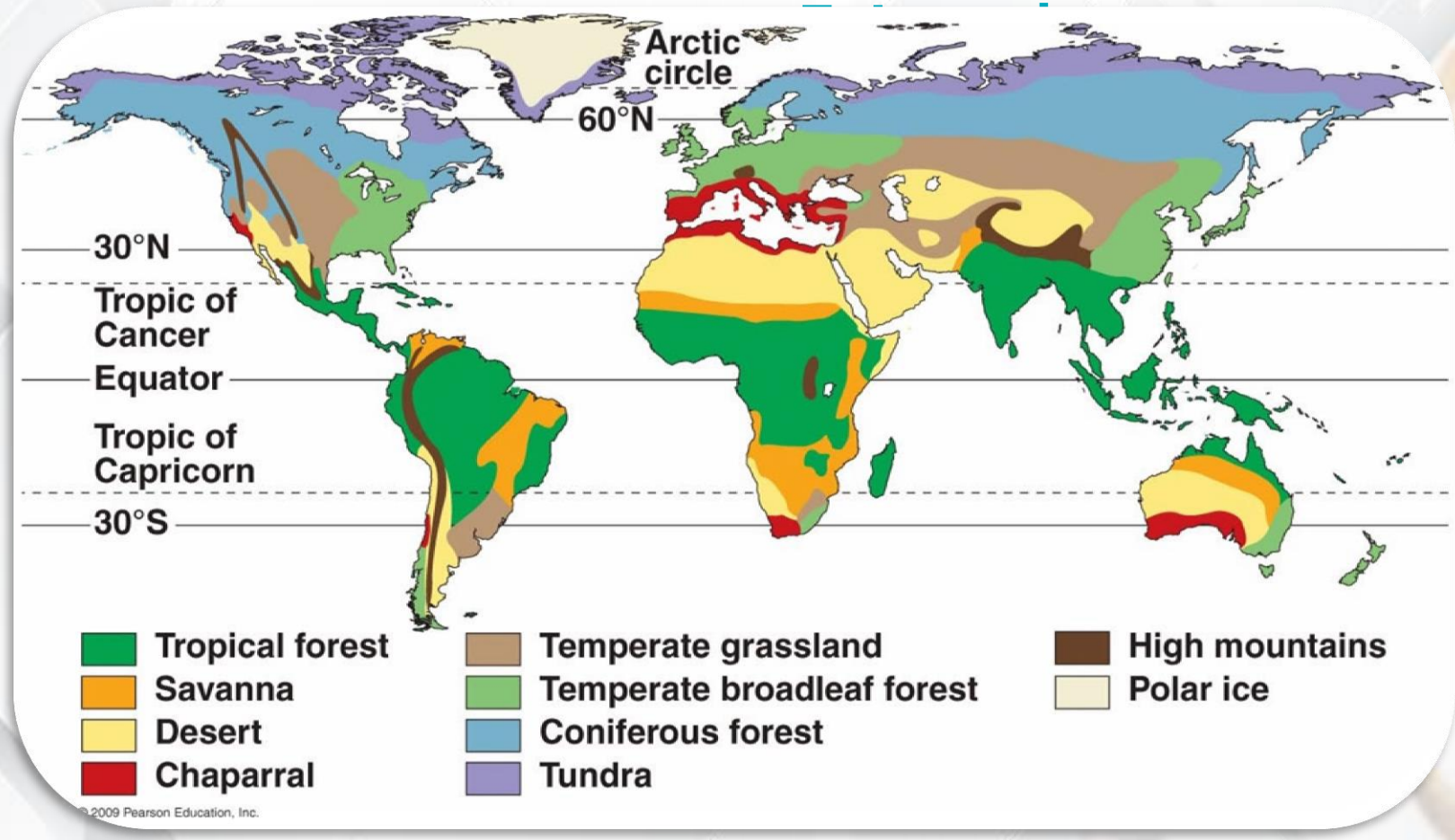
Ref. http://clipartportal.com

The Grassland (II)

7. Location

- Great plains of North America
- Pampas of south America
- Veldt of south Africa
- Steppes of Central Eurasia
- Surrounding the desert of Australia

The Grassland (II)



The Grassland (II)

8. Human Impact

- Continuous global warming
- Conversion to agricultural land
- Urban development

end

Principles of Animal life 1

The Chaparral (I)

The Chaparral (I)

Contents

1. The Chaparral
2. Climate
3. Soil Conditions
4. Plant Life
5. Types of plants
6. Animal Life
7. Location

The Chaparral (I)

1. The Chaparral

- It is the biome fount in small patches in most of the continents
- It is characterized as very hot and dry

The Chaparral (I)

2. Climate

- Winter is very mild and usually about 10^{o}C
- Summer is hot and dry with temperature up to 37.5^{o}C
- Most of the rain occur in winter
- Annual precipitation is 30-75cm

The Chaparral (I)

3. Soil Conditions

- Soil is thin and rocky
- Nutrient poor
- Highly susceptible to erosion
- Soil contain low amount of moisture

The Chaparral (I)

4. Plant Life

- Plants of chaparral are low and shrubby
- They have tough and waxy leaves
- They are broad leaves and evergreen
- Plants have no thick bark so not very resistant
- But recovery is very fast

end

Principles of Animal life 1

The Chaparral (II)

The Chaparral (II)

Contents

1. The Chaparral
2. Climate
3. Soil Conditions
4. Plant Life
5. Types of plants
6. Animal Life
7. Location

The Chaparral (II)

5. Types of Plants

- Two types of plants
- **Sprouters:** dormant meristem, storage and intact root help to regenerate quickly
- **Seeders:** long lived seeds, germination induced by fire
- Plants include toyon, chamise, poison oak, scrub oak, Yucca, Sumac

The Chaparral (II)

Ref. smmtc.org

The Chaparral (II)

6. Animal Life

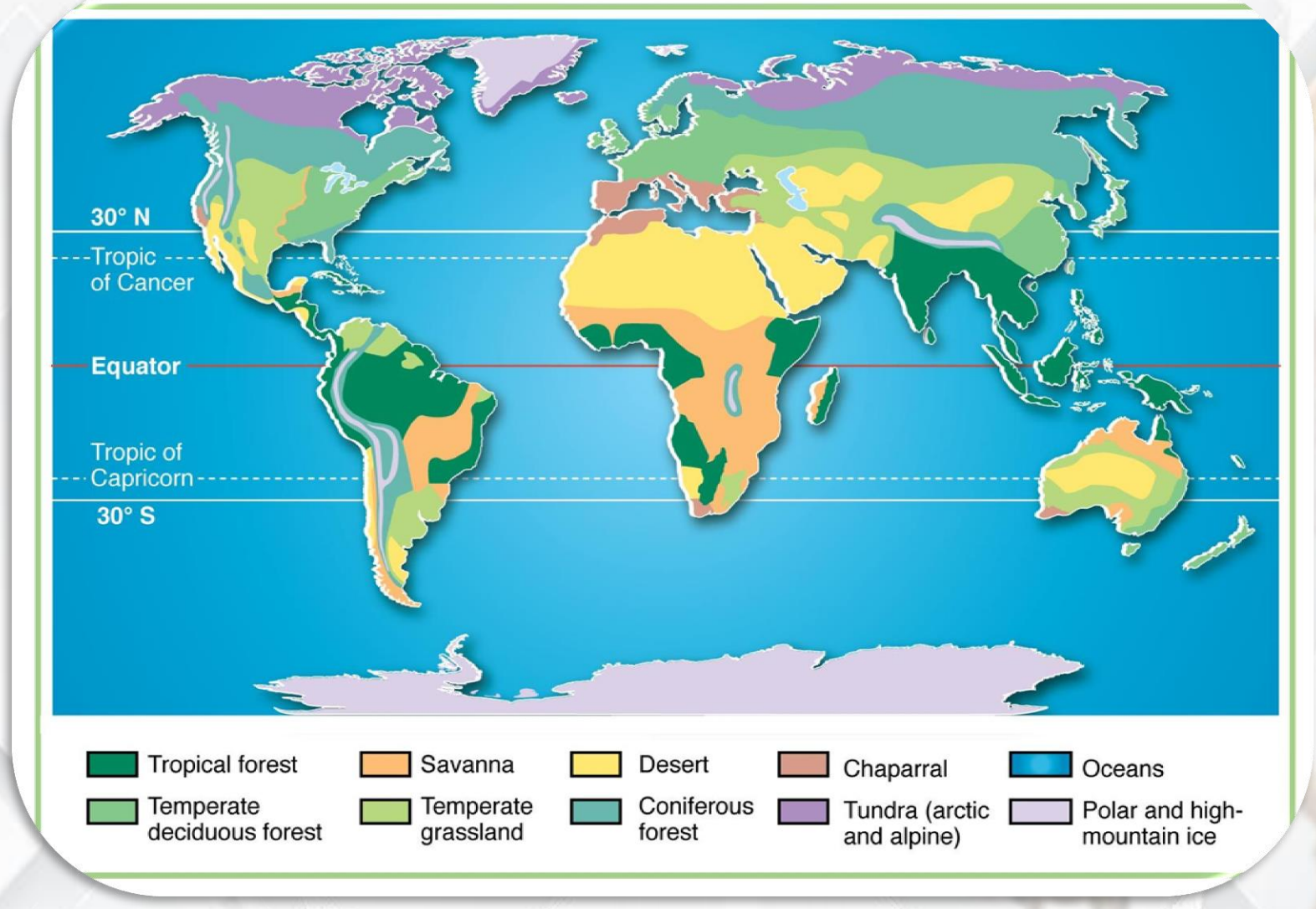
- Animals include
- Insects
- Rodents
- Rabbits
- Lizards
- Snakes
- Mule deer

The Chaparral (II)

7. Location

- West coast of United States and South America
- South Africa
- Western tip of Australia
- Coastal Area of Mediterranianean

The Chaparral (II)



Ref. pinterest.com

Principles of Animal life 1

Desert Ecosystem (I)

Desert Ecosystem (I)

Contents

1. The Desert Ecosystem
2. Climate
3. Soil Conditions
4. Plant Life
5. Types of plants
6. Animal Life
7. Location

Desert Ecosystem (I)

1. Desert Ecosystem

- A region which receives annual precipitation less than 25cm
- They are characterized by scarce vegetation and extreme temperature occilation

Desert Ecosystem (I)

2. Climate

- Extremely dry climatic conditions
- Lowest precipitation in comparison with other biomes
- There go years and years without precipitation
- At day time temperature may be 38 while at night it may be 4^{o}C

Desert Ecosystem (I)

3. Soil Conditions

- Desert soil is dry
- Low organic nutrients
- Low fertility
- Low water holding capacity

Desert Ecosystem (I)

4. Plant Life

- Only those plants can grow which ability to survive in extreme conditions of drought
- Reduced leaves, succulent stems, deep stomata and waxy cuticle are the important adaptations of desert plants

end

Principles of Animal life 1

Desert Ecosystem (II)

Desert Ecosystem (II)

Contents

1. Types of Plants
2. Animal Life
3. Deserts of the World

Desert Ecosystem (II)

1. Types of Plants

- Predominant plants are different species of cacti and thorny shrubs
- They have reduced foliage and drought resistant seeds

Desert Ecosystem (II)

Ref. in-the-desert.com

Desert Ecosystem (II)

2. Animal Life

- Animals of desert include
- Insects
- Other arthropods
- Reptiles: snakes, lizards
- Birds: road runner, cactus wren, turkey vulture, burrowing owl
- Mammals: mule deer, jackrabbit, kangaroo rat, ground squirrel

Desert Ecosystem (II)

Ref. wikipedia.org

Ref. animalsake.com

Desert Ecosystem (II)

Ref. un-habitat.org

Ref. deviantart.com

Desert Ecosystem (II)

Ref. audubon.org

Ref. deviantart.com

Desert Ecosystem (II)

Ref. idfg.idaho.gov

Ref. visitcalifornia.com

Desert Ecosystem (II)

3. Deserts of the World

- Antarctic desert
- Arctic desert
- Sahara desert
- Australian desert
- Arabian desert
- Gobi desert
- Kalahari desert
- Thar desert
- Columbia basin
- Dasht-e-kavir

end

Principles of Animal life 1

Tropical Rainforests (I)

Tropical Rainforests (I)

Contents

1. The Tropical Rain Forest
2. Climate
3. Soil Conditions
4. Plant Life

Tropical Rainforests (I)

1. The Tropical Rainforest

- It is a hot moist biome found near the equator
- They are the richest and most exciting areas of the earth

Tropical Rainforests (I)

2. Climate

- Warm temperature and abundant rainfall
- Average temperature ranges from 20^{o}C to 25^{o}C. It remains constant
- Annual rainfall is more than 200cm
- Little seasonal variation in day length

Tropical Rainforests (I)

3. Soil Conditions

- Soil is relatively nutrient poor
- Most of the nutrients are tied up in organisms
- High moisture and temperature make nutrient recycling rapid

Tropical Rainforests (I)

4. Plant life

- Very diverse plant life
- Thousands of species
- Non of them dominating
- A single hectare typically contains 50 to 70 species
- Forests are highly stratified
- Trees are broad leaved and non-deciduous

end

Principles of Animal life 1

Tropical Rain Forests (II)

Tropical Rain Forests (II)

Contents

1. Plant Types
2. Animal Life
3. Geographic Location
4. Human Impact

Tropical Rain Forests (II)

1. Plant Types

- 50% of plants and animals with 6% land
- Plants grow faster
- Bigger leaves
- Plants include
  - Palm species
  - Rubber tree
  - Bamboo
  - Orchid
  - Tualang
  - Different epiphytes
  - Pitcher plant

Tropical Rain Forests (II)

2. Animal Life

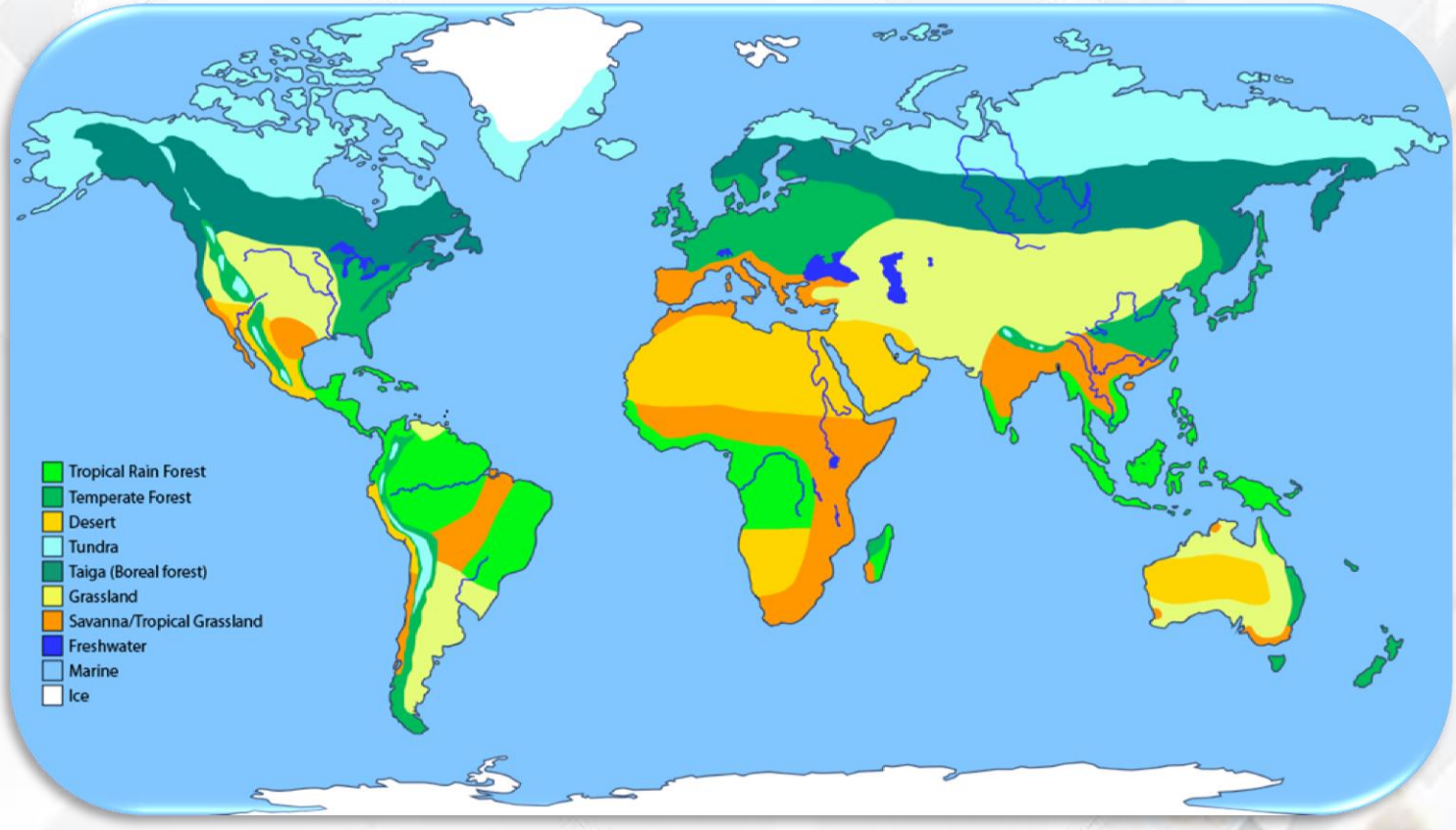
- Stable temperature and high humidity is ideal for both soft bodied and hard bodied endotherms
- Annelids
- Arthropods
- Amphibians
- Reptiles
- Birds
- Mammals

Tropical Rain Forests (II)

3. Geographical Location

- South and central America
- Western Africa
- Southeast Asia
- Australia

Tropical Rain Forests (II)



Ref. socratic.org

Tropical Rain Forests (II)

4. Human Impact

- **Deforestation:** once they were 14% and now they are 6%. It is estimated that in next 40 years the remaining will be lost as well
- **Habitat loss**
- **Extinction**
- **Soil erosion**
- **Climate change**

end

Principles of Animal life 1

Freshwater Ecosystems: Lotic Ecosystem

Freshwater Ecosystems: Lotic Ecosystem

Contents

1. The Lotic Ecosystem
2. Primary Productivity
3. Oxygen Supply
4. Plants and Animal Life
5. Effect of Water Currents

Freshwater Ecosystems: Lotic Ecosystem

1. The Lotic Ecosystem

- It is the aquatic ecosystem of flowing water
- It includes
  - Brooks
  - Streams
  - Rivers

Freshwater Ecosystems: Lotic Ecosystem

2. Primary Productivity

- Attached aquatic plants and algae provide some of the primary productivity
- Most of the primary production comes from the surrounding plants by falling their materials during autumn and rainfall

Freshwater Ecosystems: Lotic Ecosystem

3. Oxygen Supply

- Lotic systems typically have high oxygen level
- Turbulence of water while flowing over rocks, logs and other objects causes oxygenation

Freshwater Ecosystems: Lotic Ecosystem

4. Plants and Animal Life

- Mainly algae and attached aquatic plants are indigenous producers
- Animals include different aquatic invertebrates, larvae and insects
- Fishes, amphibians reptiles birds and mammals are adapted to lotic habitat

Freshwater Ecosystems: Lotic Ecosystem

5. Effects of Water Currents

- It is the most important physical factor of lotic habitat
- Animal adaptation include
- Dorsoventral surface
- Strong swimmers
- Burrowing
- Attachment with substrate

end

Principles of Animal life 1

Freshwater Ecosystems: Lentic Ecosystem

Freshwater Ecosystems: Lentic Ecosystem

Contents

1. The Lentic Ecosystem
2. Vertical Stratification
3. Primary Productivity
4. Oxygen Supply
5. Oligotrophic and Eutrophic Bodies
6. Plant and Animal life

Freshwater Ecosystems: Lentic Ecosystem

1. The Lentic Ecosystem

- This is the ecosystem of standing water
- Lentic ecosystem include:
  - Lakes
  - Ponds

Freshwater Ecosystems: Lentic Ecosystem

2. Vertical Stratification

- Vertical stratification is common especially in temperate region
- In summer, upper layer is warm while lower layer is cold and more dense, so no mixing
- In fall, upper layer is dense so mixing of water occurs

Freshwater Ecosystems: Lentic Ecosystem

3. Primary Productivity

- Photosynthesis mainly in upper layer of water as light is filtered by water
- Phytoplankton are main producers of lentic ecosystem

Freshwater Ecosystems: Lentic Ecosystem

4. Oxygen Supply

- Deeper layers are depleted of oxygen in summer
- In fall the mixing of entire water column supply oxygen to deeper layers

Freshwater Ecosystems: Lentic Ecosystem

5. Oligotrophic and Eutrophic bodies

- Deep, cold, nutrient poor lakes are relatively unproductive called as oligotrophic
- Shallow warm, nutrient rich lakes are highly productive called as eutrophic

Freshwater Ecosystems: Lentic Ecosystem

6. Plant and Animal Life

- Plant life include phytoplankton, periphyton algae, emergents, floating-leaves, submersed and free floating
- Animal life include zooplankton; invertebrates like crabs, crayfish and shrimps; molluscs

end

Principles of Animal life 1

Marine Ecosystems: Estuaries

Marine Ecosystems: Estuaries

Contents

1. The Estuaries
2. Productivity of Estuaries
3. Plants and Animal Life
4. Human Impact

Marine Ecosystems: Estuaries

1. The Estuaries

- Estuaries occur where freshwater meets seawater
- This dilutes the salinity of seawater

Marine Ecosystems: Estuaries

2. Productivity of Estuaries

- Estuaries are highly productive because

I. Streams carry nutrients of freshwater

II. Tides bring nutrients from ocean

Marine Ecosystems: Estuaries

3. Plant and Animal Life

- Producers include phytoplankton, attached aquatic vegetation and emergent vegetation such as cordgrass
- Estuaries are nurseries for crustaceans, molluscs and marine fishes. Feeding grounds for wading and swimming birds

Marine Ecosystems: Estuaries

4. Human Impact

- Human have drained and filled estuaries to subdivision
- Polluted by oil exploration and drilling
- Acids rain have threaten the productivity

end

Principles of Animal life 1

Marine Ecosystems: Intertidal Zone

Marine Ecosystems: Intertidal Zone

Contents

1. The Intertidal Zone
2. Environmental fluctuations
3. Plants and Animal Life

Marine Ecosystems: Intertidal Zone

1. The Intertidal Zone

- Extend from splash zone ocean to low tide marks
- This region is subjected to the action of both tides and waves

Marine Ecosystems: Intertidal Zone

2. Environmental Fluctuations

- It is the harshest and the richest of all marine environments
- Here is pounding surf, sun, wind, rain, erosion, sedimentation and temperature fluction

Marine Ecosystems: Intertidal Zone

3. Plant and Animal Life

- There is bounty of habitat and plenty of nutrients
- Plants are larger. They include seagrass, red algae, magroves
- Motile animals like fishes and crabs move with tides
- Other such as amphipods, isopods, ghost crabs and polychaetes burrow in the sand and mud

Marine Ecosystems: Intertidal Zone

3. Plant and Animal Life

- Water trapped in tidal pools support a rich animal life such as echinoderms, barnacles, limpets and chitons
- Attached animals close shells to avoid drought

end

Principles of Animal life 1

Marine Ecosystems: Neritic Ecosystem

Marine Ecosystems: Neritic Ecosystem

Contents

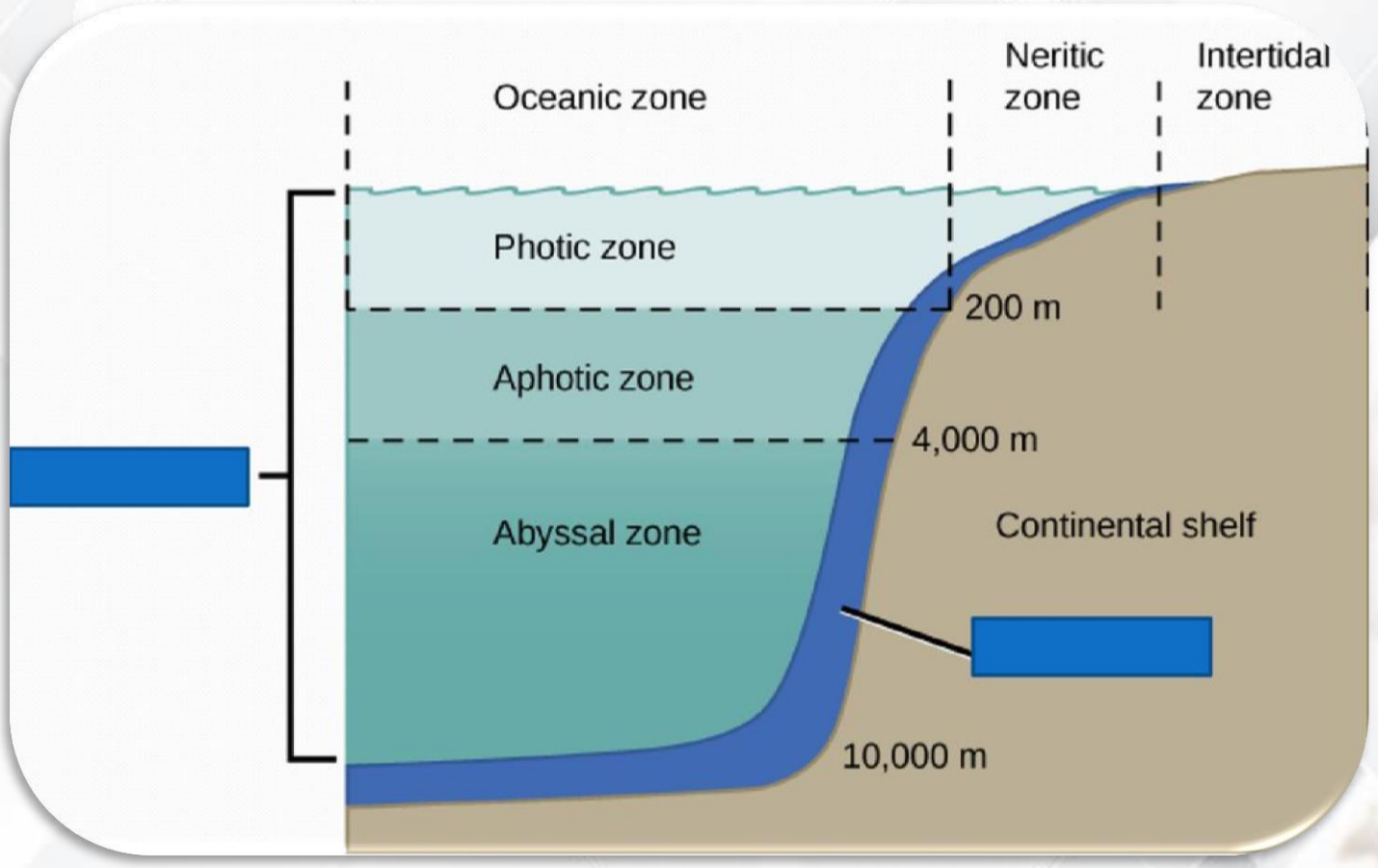
1. The Neritic Ecosystem
2. Productivity
3. Plants and Animal Life

Marine Ecosystems: Neritic Ecosystem

1. The Neritic Ecosystem

- Neritic ecosystem extend from littoral zone to the edge of continental shelf
- It is also called the shallow water

Marine Ecosystems: Neritic Ecosystem



Ref. slideplayer.com

Marine Ecosystems: Neritic Ecosystem

2. Productivity

- Nutrients are delivered by rivers and upwelling at the edge of continental shelf
- This region is more productive than open sea
- Planktonic and attached algae contribute to the primary productivity

Marine Ecosystems: Neritic Ecosystem

3. Plant and Animal Life

- Plants include kelps, algae, seaweeds, seagrasses and phytoplankton.
- Invertebrate animals include crabs, sponges, echinoderms, corals, and polychaetes

Marine Ecosystems: Neritic Ecosystem

3. Plant and Animal Life

- Vertebrate fauna include different types of fishes such as code, tuna, herring, capelin and many other species

end

Principles of Animal life 1

Marine Ecosystems: Oceanic Ecosystem

Marine Ecosystems: Oceanic Ecosystem

Contents

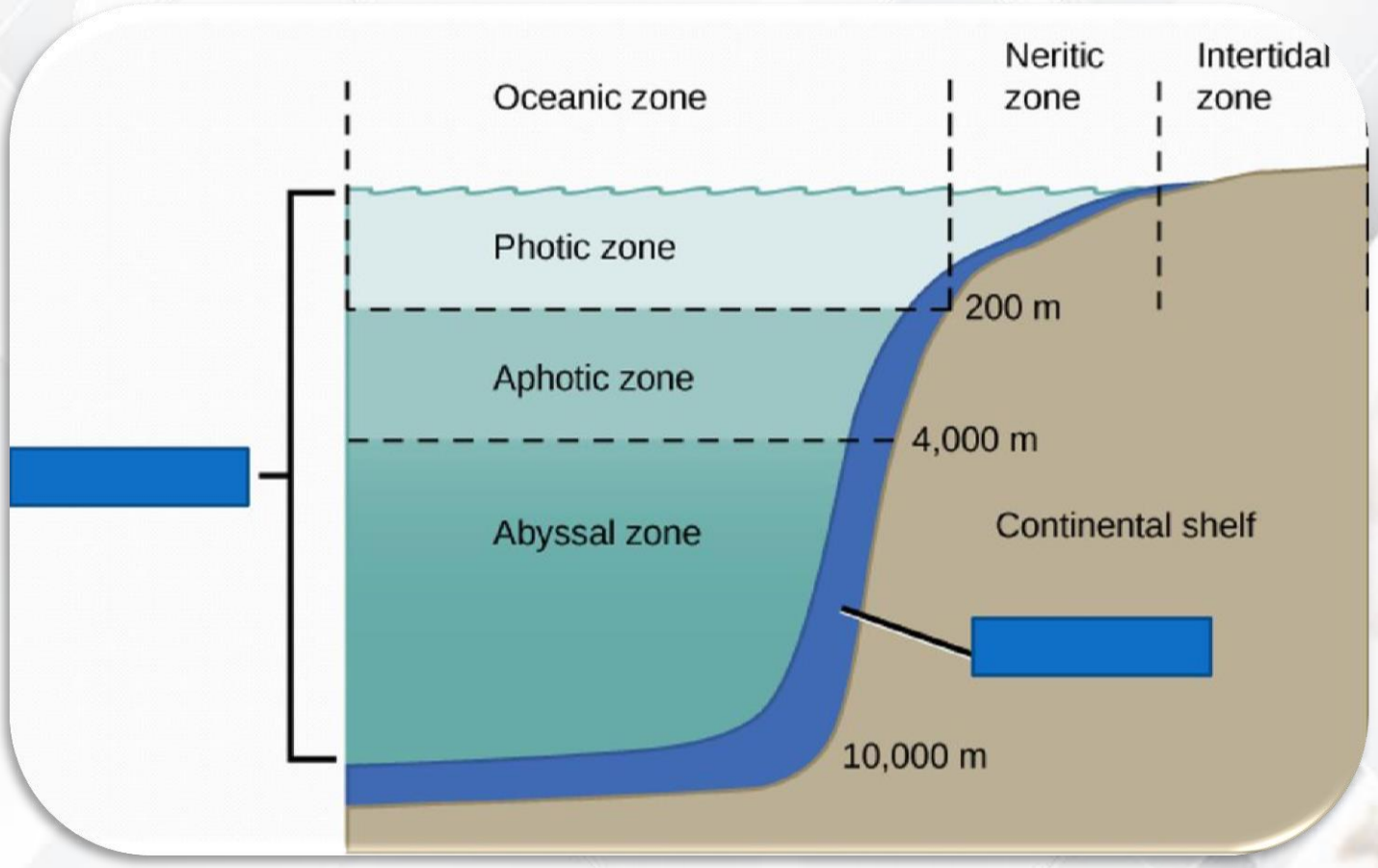
1. The Oceanic Ecosystem
2. Productivity
3. Plants and Animal Life

Marine Ecosystems: Oceanic Ecosystem

1. The Oceanic Ecosystem

- It starts from continental shelf
- It is the open sea

Marine Ecosystems: Oceanic Ecosystem



Ref. slideplayer.com

Marine Ecosystems: Oceanic Ecosystem

2. Productivity

- It is relatively unproductive
- Virtually all primary productivity is in the upper water

Marine Ecosystems: Oceanic Ecosystem

3. Plant and Animal Life

- Phytoplankton in the upper water are the main producers of oceanic system.
- Invertebrate animals include worms, echinoderms, molluscs
- Vertebrates include fishes from small fishes to whales and sharks

Marine Ecosystems: Oceanic Ecosystem

3. Plant and Animal Life

- At depth no light so the animals here have luminescent organs which serve as torch light
- These animals depend on nutrient coming from upper layers

end

Principles of Animal life 1

Marine Ecosystems: Coral Reefs

Marine Ecosystems: Coral Reefs

Contents

1. The Coral Reefs
2. Occurrence of Coral Reefs
3. Rich Biodiversity

Marine Ecosystems: Coral Reefs

1. The Coral Reefs

- It is the underwater ecosystem
- They are formed by colonies of corals. Polyps are held together by calcium carbonate
- They are the association of stony coral and algae
- Coral are members of class Anthozoa
- Algae are zooxanthellae

Marine Ecosystems: Coral Reefs

2. Occurrence

- Reefs are formed over thousands of years in warm, clear and relatively shallow waters
- The greatest depth for their growth is 50-60 m
- Below this light is insufficient to support zooxanthellae

Marine Ecosystems: Coral Reefs

3. Rich Biodiversity

- They are the richest source of biodiversity in oceans
- Coral reefs occupy less than 0.1% of oceans, yet they are home of more than 25% marine species
- Residents include sponges, cnidarians, nematodes, crustaceans, molluscs, echinoderms, tunicates and fishes

Marine Ecosystems: Coral Reefs

Ref. sailingeurope.com

Principles of Animal life 1

Ecological problems: Human Population Growth

Ecological problems: Human Population Growth

Contents

1. Population growth
2. Earth's Carrying Capacity
3. World population

Ecological problems: Human Population Growth

1. Population Growth

- It is the increase in number of individuals in a population
- Expanding population is the root cause of most environmental problems.
- More population means more need for resource: food, urbanization, industrialization

Ecological problems: Human Population Growth

2. Earth's Carrying Capacity

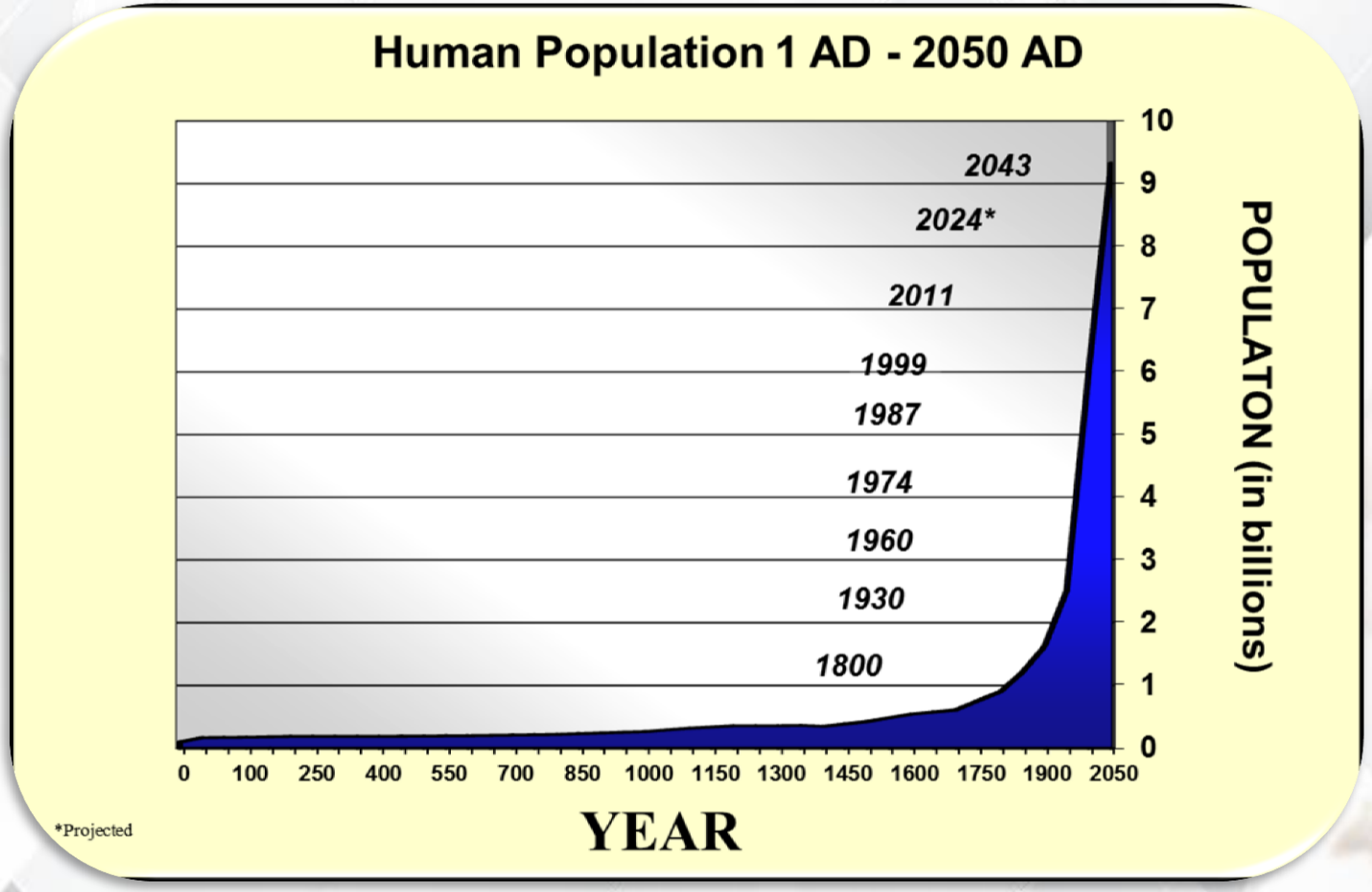
- How many human can be supported by earth?
- Virtually all scientists agree that even 5 billion is more for earth to support with all facilities
- Maximum carrying capacity of earth is may be 9 to 10 billion people

Ecological problems: Human Population Growth

3. Age Structure

- Age structure shows the proportion of individuals at prereproductive, reproductive and postreproductive stage
- Developing countries have more infants means high birthrates
- In contrast to human, in natural population high birthrate is accompanied with high mortality

Ecological problems: Human Population Growth



Ref. sureshemre.wordpress.com

Principles of Animal life 1

Ecological Problems: Pollution

Ecological Problems: Pollution

Contents

1. Pollution
2. Water Pollution
3. Air Pollution
4. Land pollution
5. Biological magnification

Ecological Problems: Pollution

1. Pollution

- Pollution is any detrimental change in ecosystem
- Mostly is results from human activities
- More human population<more demands<pollution

Ecological Problems: Pollution

2. Water Pollution

- Industries generate toxic waste
- Sewage waste
- Agricultural runoff

Ecological Problems: Pollution

3. Air Pollution

- Burning of fossil fuels release CO_2, oxides of sulfur and nitrogen
- Acid deposition by these oxides
- Increased CO_2 causes green house effect
- Global warming

Ecological Problems: Pollution

4. Land Pollution

- Oil spillage
- Plastics
- Pesticides
- Toxic industrial waste
- Garbage
- Buildings leftover

Ecological Problems: Pollution

5. Biological Magnification

- Wastes and poisons enter in food chain and travel to the organisms at highest level of food chain including human
- Thy are not biodegradable and interrupt physiological processes

end

Principles of Animal life 1

Ecological Problems: Resource Depletion

Ecological Problems: Resource Depletion

Contents

1. Natural Sources
2. Depletion of Resources

Ecological Problems: Resource Depletion

1. Natural Resources

- Materials or substances occurring in nature which can be exploited for human benefits
- Important are:
  - Fossil fuels
  - Minerals
  - Air
  - Water
  - Soil
  - Biodiversity

Ecological Problems: Resource Depletion

2. Depletion of Resources

- More population
- Better life
- Overexploitation
  - Fossil fuels
  - Minerals
  - Air
  - Water
  - Land/soil
  - Biodiversity

end

Principles of Animal life 1

Ecological Problems: Biodiversity Loss

Ecological Problems: Biodiversity Loss

Contents

1. Biodiversity of Earth
2. Threats to Biodiversity
3. Solutions

Ecological Problems: Biodiversity Loss

1. Biodiversity of Earth

- More than 1.6 million
- Estimate is 4 to 6 million
- More accurate is between 8.7±1.3million

Ecological Problems: Biodiversity Loss

2. Threats to Biodiversity

- Biodiversity of all natural areas is threatened
  - Acid rain
  - Pollution
  - Urbanization
  - Agriculture
- Human is exploiting 40% of earth net primary production
- Habitat loss

Ecological Problems: Biodiversity Loss

3. Solutions

- Realization about biodiversity
  - 22nd May is World Biodiversity Day
- A system of international ethics that value natural diversity

Ecological Problems: Biodiversity Loss

Ref. sites.google.com